Computing Spacetime

September 2026

Abstract

Four axioms, one cost function, and one integer connect a computational picture of space to particles, forces, and cosmology. Particles are maintained patterns; mass is the cost of keeping them running.

The axioms specify an information speed, conserved compute capacity, the cost function $K(r) = r^{2-D}$, and a resolution floor. Their construction selects $N_c = 3$. In the effective spatial picture, three overlapping lenses and seven internal channels form a Planck-cell transistor. Five operations connect its structure to observables, with zero continuous free parameters and explicitly charged discrete selections.

The paper compares its results with 56 independently measured quantities and specifies 29 falsification tests. The evidence rests on constrained formula choices, selection-adjusted description length, and forward tests; close mass matches alone carry no evidential weight. The parameter comparison ranges from $\sim$2.1$\times$ raw compression to $\sim$1.10$\times$ after selection charges and $\sim$0.72–0.79$\times$ under the most conservative accounting, which favors the Standard Model. The framework's code length also covers cosmology, quantum mechanics, and gravity, with zero continuous free parameters.

General relativity follows under the stated identifications, reversible local equilibrium, and a manifoldlike continuum. The Born rule is exact on fixed finite complete read blocks. The dark sector uses the same cost structure, with its assumptions and measurement tensions stated locally. Constructing the intrinsic exact-six spatial network and its continuum limit remains open. Each result carries its exact, conditional, or structural level of support.

Reading guide. Pages 2–10, The physical picture, explain the proposed mechanism, its scope, and its tests without requiring a physics background. Read these pages for the conceptual account; their links lead to the calculations behind each claim.

The technical argument begins on page 11. Sections 1–6 state the axioms, operations, quantitative results, and limitations. The appendices begin on page 68 and contain the full comparison tables, detailed derivations, mathematical proofs, and numerical qualifications. Code-marked claims map to the supplementary claim index.

The physical picture

Consider space as a medium made of tiny optical instruments. Each grain runs the same machinery: a Planck-cell transistor with an address, three overlapping lenses, and seven channels that a wave can engage. This is the framework's picture of three-dimensional space. Underneath it is a fixed field of possible addresses; how their relations produce this spatial geometry remains an open construction (Appendix O).

A matter pattern at rest still evolves internally. The substrate keeps computing its wave: the lenses turn it, and its engaged channels determine the cost. To persist is to be computed again each tick. What physics calls *mass* is this continuing compute cost.

The three lenses have different jobs. The *source* traps waves, the *gate* couples their internal states, and the *drain* connects the cell to its surroundings. They overlap throughout the cell. The wave carries its spin information; the gate is the lens through which that part of the wave refracts. Figure 1 shows the lenses separately and together.

Separating a source-bound configuration raises its cost; the gate gives equal cost to its accessible states; the drain's cost falls with distance. These three landscapes organize confinement, mixing, and long-range interactions. Section 2.4 gives their common cost function.

Key terms.

- **Substrate**: the underlying field whose active relations acquire the spatial picture above.
- **Cell / transistor**: one Planck-cell compute site in that picture.
- **Channel**: an internal register a pattern can engage.
- **Resonant / transparent**: engaged by the wave / passed untouched.
- **Tick**: one step of the substrate's computation, its smallest unit of time.
- **bE**: the conserved total compute capacity. A pattern draws on it each tick.
- **Mass**: the recurring cost of maintaining a pattern; masses are compared in electron units.
- **Read**: the operation that resolves an origin-to-destination relation.

The counting rule is explained beside the operations in Section 2.1. Technical symbols are collected in Appendix B.4; Section 1.2 explains the levels of support attached to claims.

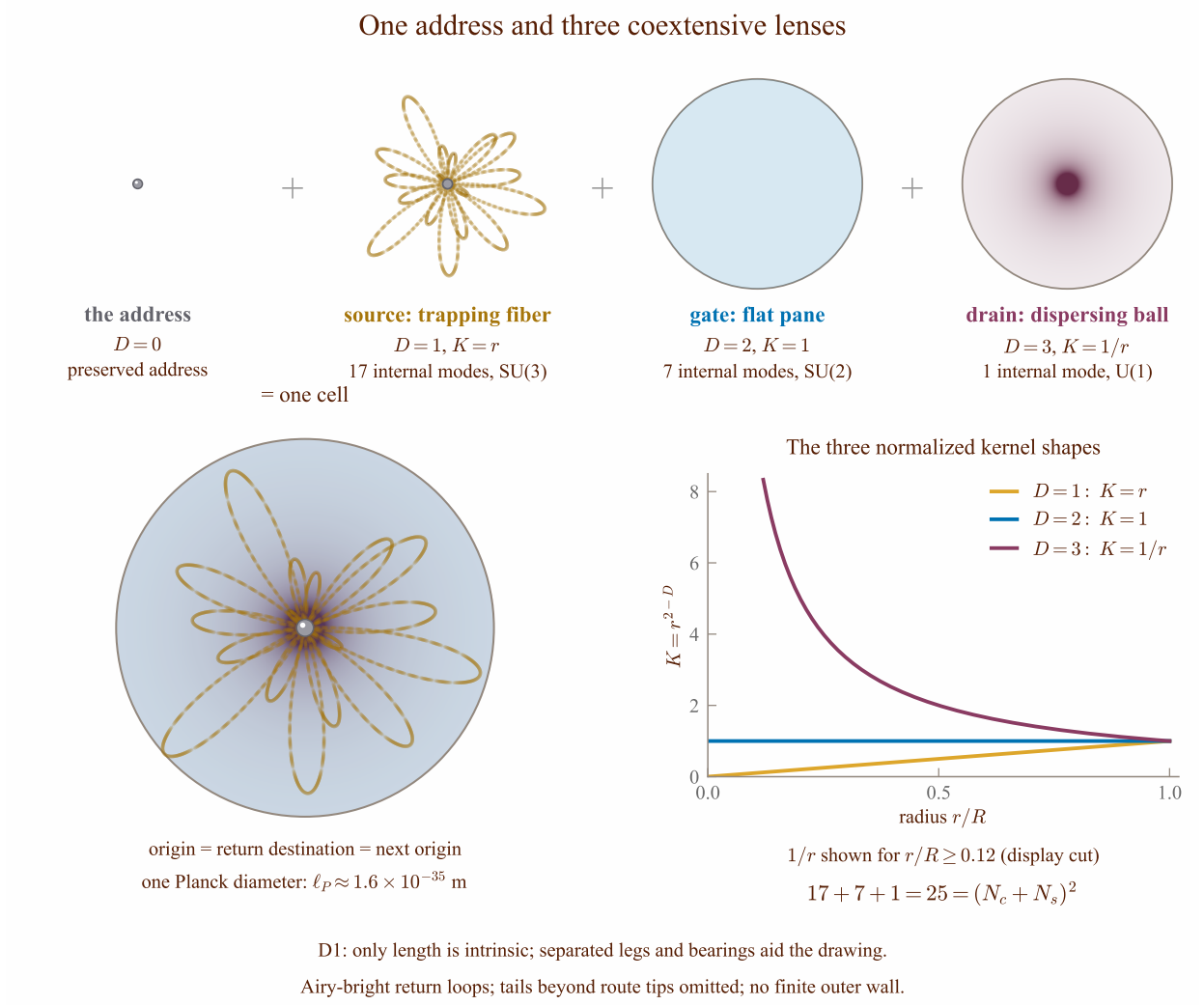


Figure 1: *One Planck-cell transistor: source return loops (gold), gate (blue), and drain (plum), shown separately and together around a fixed address (silver). The lenses overlap throughout the cell. A wave's engaged channels determine which lenses refract it. Loop orientation, separated legs, and marker size are drawing conventions. The right panel shows normalized continuum kernel shapes; the physical radius scale and resolved domain remain to be selected. The* $1/r$ *curve is displayed only for* $0.12 \leq r/R \leq 1$*. Appendix B.2 gives the channel key and full drawing conventions.*

Four axioms. There is a maximum information speed, a conserved total compute capacity, a preference for the least costly computation, and a resolution rule. That last rule sets a floor on distinguishability and requires every computed tick to deliver at least one distinguishable outcome. These are the four axioms. Section 1 states them precisely and introduces the cost function.

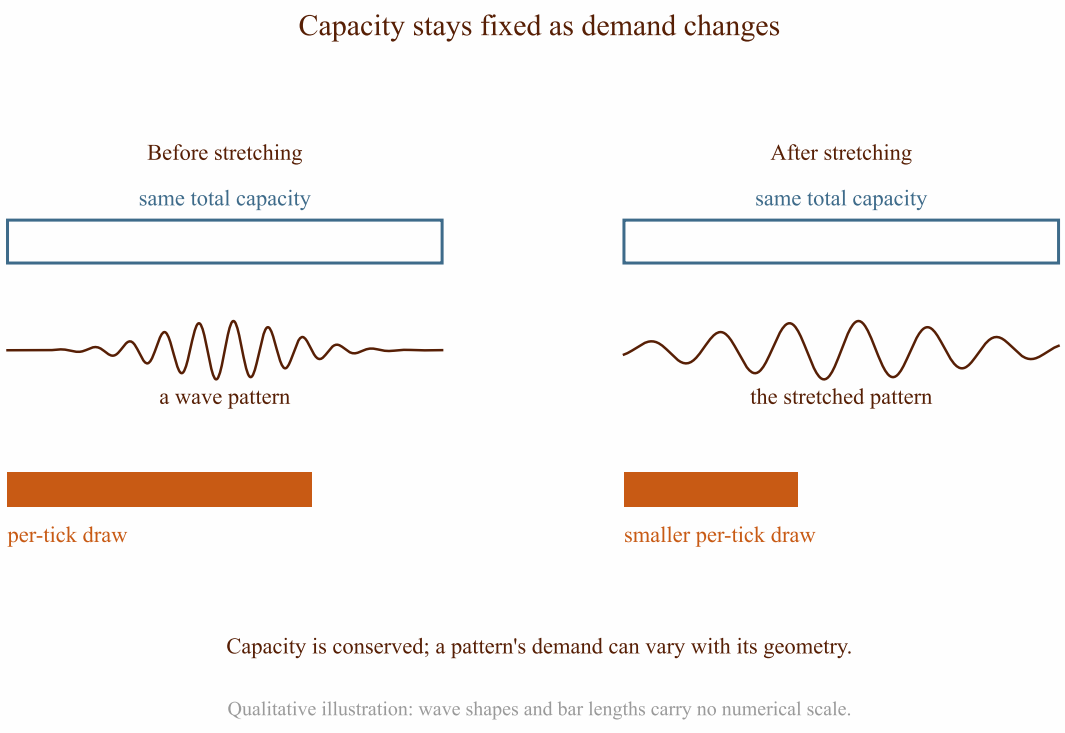


Figure 2: *Capacity and demand. The blue outline represents the same total available compute in both panels. The orange bar represents a pattern's draw on it each tick: stretching the pattern lowers that demand while total capacity stays fixed. Shapes and bar lengths are qualitative. Section 2.5 states the distinction precisely.*

Particle masses. We use the electron's measured cost as the unit; the input table identifies it as the minimum charged-particle mass (Table 1). The proton costs about 1,836 electron units: the work of maintaining its whole standing-wave pattern. Its quarks are configurations within that pattern. Their channel signatures help specify the composite's cost (Section 3.1).

The coupling key in Appendix B.2 groups waves by their engaged lenses; the seven-channel chart resolves each occupied entry (Fig. 30). Detailed counts and family distinctions are in Appendix F.1.

The mass comparison spans 69 hadrons and leptons (Fig. 3); Table 9 gives the formulas and measurement conventions. Shared anchors make these correlated comparisons. The evidence for the formula choices is assessed separately in Section 5.1.

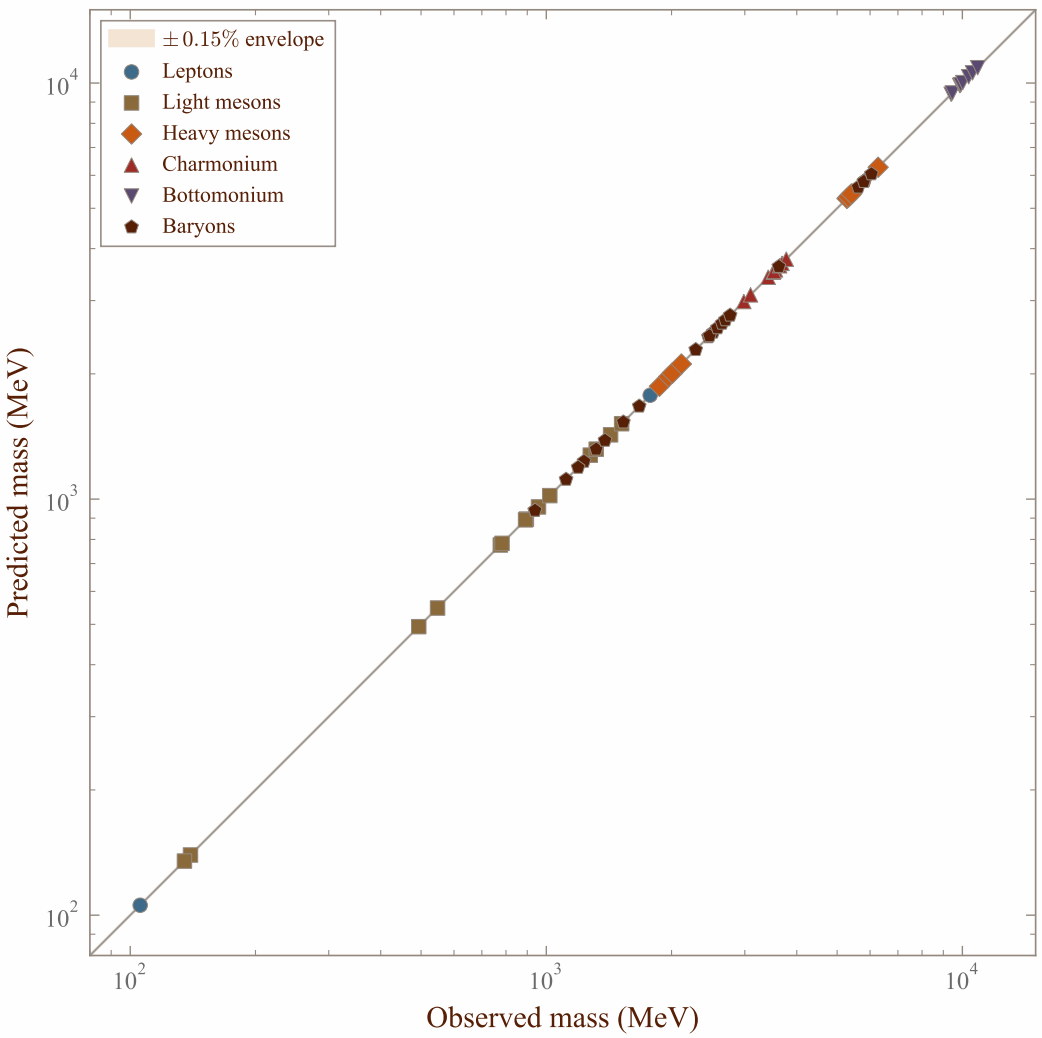


Figure 3: *Predicted vs. observed masses for 69 hadrons and leptons across six sectors (the 68 numbered mass rows plus the neutron, priced by row 12's difference). All points lie within ±0.15% of the diagonal ($y = x$). The tau uses corrected-muon Koide, with the mass-level qualification of Appendix I.5; shared anchors make the plotted rows dependent.*

Light and measurement. A photon connects one event to another. The departing event supplies a reference to its origin. Between emission and absorption, the wave explores possible paths. At absorption, the read completes as a localized event. The photon carries zero internal register state and propagates at the information speed. Its energy is delivered to the absorber; the detailed distinction between propagation and energy is in Appendix E.1.

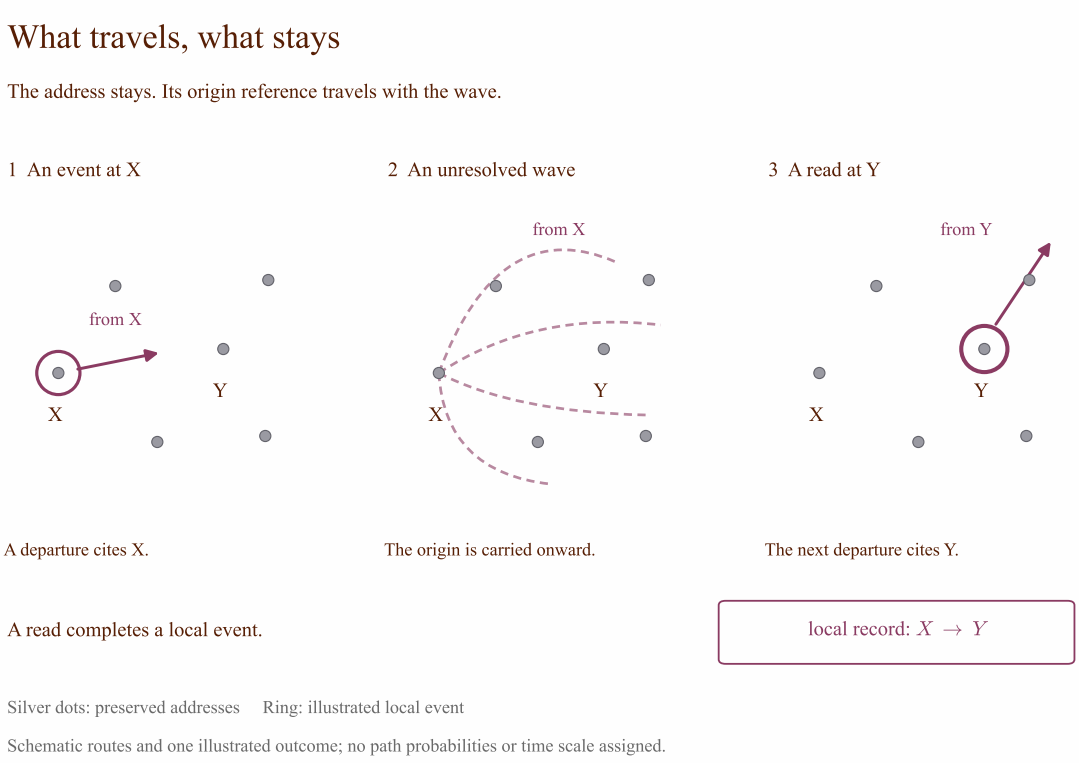


Figure 4: *What travels, what stays. The silver address markers retain their positions in all three frames. An event at X supplies an origin reference; the unresolved wave carries that reference through possible routes. A read at Y completes a localized event recording the X-to-Y relation. A subsequent departure cites Y as its origin. Y is one illustrative outcome. Curves and spacing are schematic: they specify no primitive neighbor graph, path weights, elapsed time, outcome probability or microscopic read instrument. The completed-event symbol denotes a local record; the diagram does not depict an instantaneous update of distant records.*

Quantum alternatives and definite outcomes. A wave can occupy several possible routes at once. Their contributions combine, reinforcing some outcomes and suppressing others through interference. A completed read records one outcome. On a fixed finite set of complete reads, the framework obtains the Born rule: the probability of an outcome is the squared size of its combined wave contribution (Appendix G). Extending this account to the physical read process and an evolving spatial network requires the constructions and conditions in Appendix E.

Motion and time. Each tick both changes a matter wave's internal state and propagates it through neighboring cells. The internal evolution provides the pattern's clock. Motion changes the relation between that clock and the observer's time, as expressed by the right-triangle budget in Fig. 5. Light has no internal-turning share and propagates at the maximum information speed. Section 2.2 gives the corresponding energy–momentum relation.

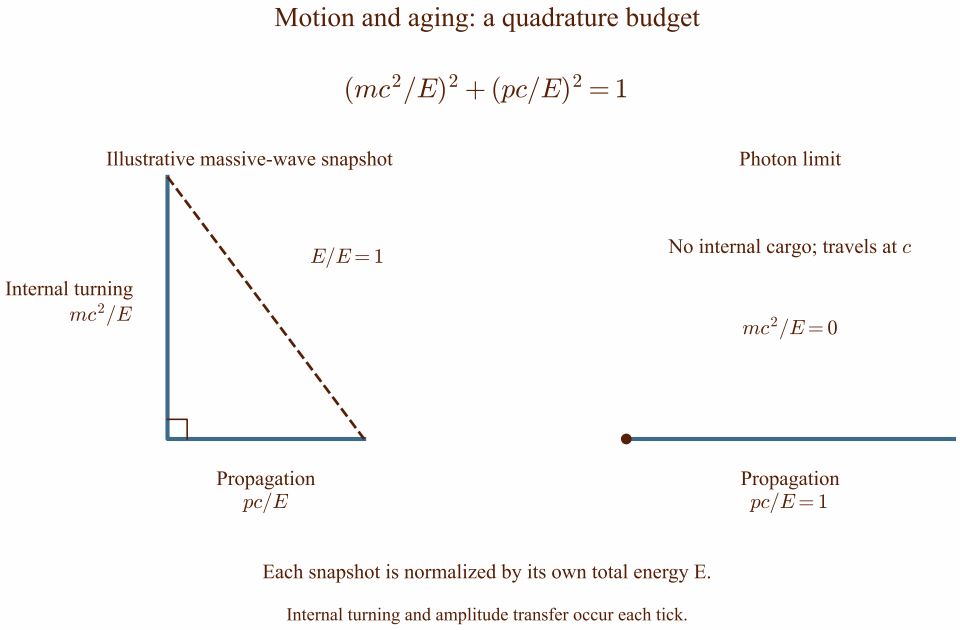


Figure 5: *Motion and aging share a right-triangle budget. Internal turning and propagation are its perpendicular sides; their squares add to the square of the total. Dividing by each snapshot's own energy E gives the unit hypotenuse shown here. Both operations occur each tick. The photon has zero internal-turning share and travels at c. The massive-wave triangle illustrates the relation; its particular proportions are arbitrary.*

Interactions and decay. Mass measures the work of maintaining a pattern. Mixing describes transitions between patterns. Charge counts engaged channels. Decay width describes how readily a pattern ends. Magnetic moment connects charge to spin. The same cells supply all five; Section 2 shows how their counting rules differ.

Gravity as collective refraction.

The glass is packed like wet sand, with no crystal order or preferred direction. Each grain contributes a tiny turning of the wave. A wavelength of starlight spans a number of cells twenty-nine digits long, so their contributions add into one shared wavefront (Fig. 21). Their combined refraction supplies the proposed mechanism for gravitational lensing.

Matter makes the glass busy. Busy regions clock slowly, and waves bend toward them. This is the picture behind gravity. The mathematical recovery of Einstein's equations uses stated thermodynamic identifications and assumes that the event network has a suitable continuum description (Section 4.2).

Black holes. At a horizon, the stationary clock rate seen from outside falls to zero. The paper's boundary rule keeps the register's state while its internal updates stop; correlations can still pass through a supplied boundary coupling. This gives parts of a possible radiation account. A complete account of evaporation still requires those parts to share one physical evolution (Appendix P).

The direction of time. A completed event updates a local record. Its previous content spreads into the surrounding relations, so returning the local pattern to an earlier state does not restore the whole environment. This distinction connects reversible local wave evolution to an accumulating history. It is needed to explain how the framework's reversible update rules coexist with an arrow of time; Section 4.8 gives the information cost and Appendix G its domain.

Dark matter and persistent references. Events keep references to where things last happened. The framework assigns a continuing cost to those references. That overhead gravitates and is invisible to light. Its amount and distribution are the dark-matter claims of Section 4.5.

Dark energy and local clock rates. Fewer standing waves mean less work and faster local clocks. As structure gathers, voids empty. The framework uses that changing clock pattern to describe late-time acceleration. Its implementation choices and observational tensions are stated in Section 4.7.

The three lenses also preserve the universe's earlier eras. The source carries confinement; the gate carries the flat landscape of inflation; the drain opens the present three-dimensional world. On the paper's structural reading, the early era left a frozen record, released much later as the cosmic microwave background, the radiation observed across the sky from the early universe. Section 4.4 follows that history and separates its far-future outlook from its derived results.

Expansion admits stable content into the fixed field of possible addresses. The detailed address notation, activation rule, and two-clock description belong to Appendix O.1; their unresolved geometric construction belongs to Appendix O.

A second comparison: lifetimes. A decay width measures how quickly an unstable pattern ends; a smaller width means a longer lifetime. Figure 6 compares fourteen widths across seven orders of magnitude. The same counting framework used for masses supplies these formulas, with no continuous fitted parameters. Their shared structure makes them correlated tests, and the largest discrepancy is visible: the χ_{c0} width lies 3.34 quoted standard uncertainties below its comparison. Section 3.4 gives the full table and the rule's limitations, including its omission of the effect of available energy on decay rates.

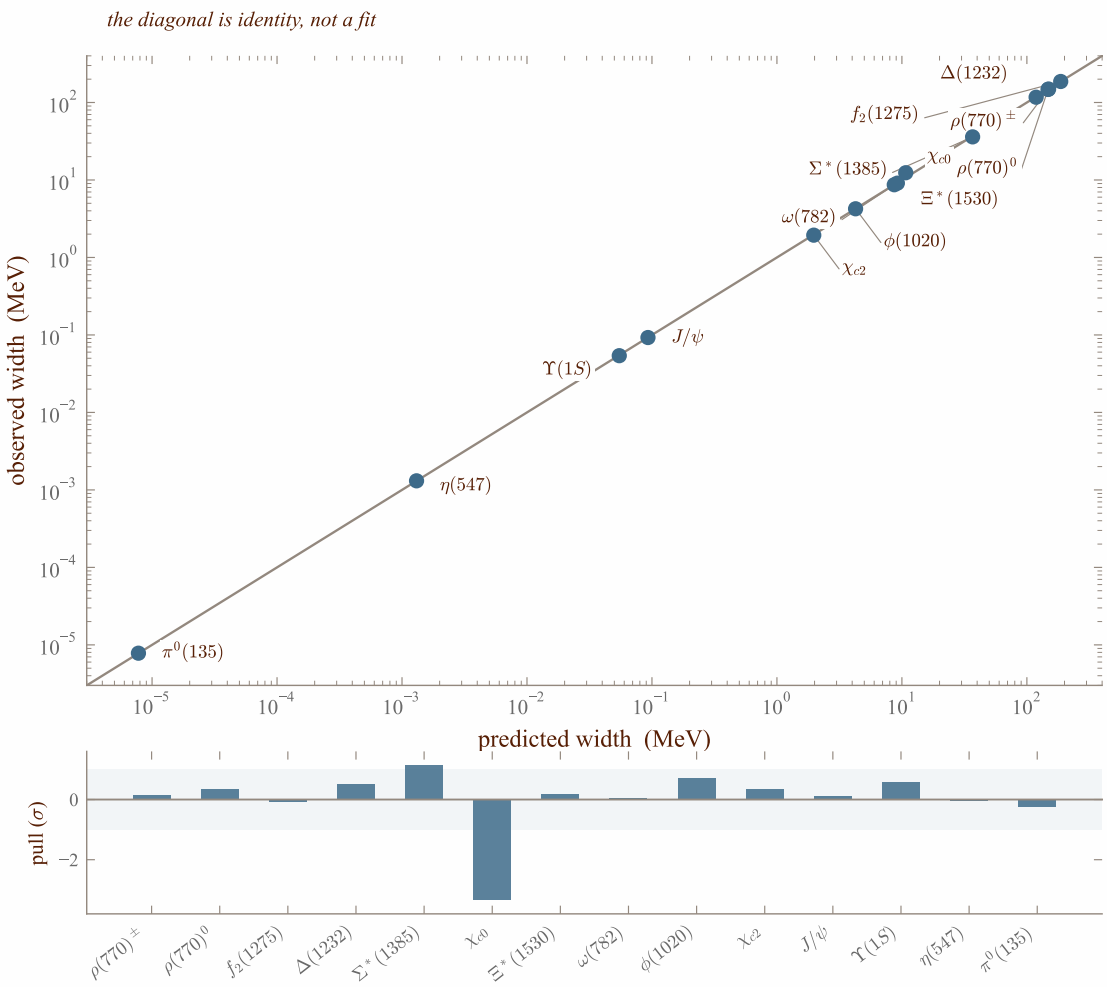


Figure 6: *Fourteen predicted and observed hadron widths across seven orders of magnitude. Each prediction is a kernel rational times the electron mass — decay outcomes counted at the vertex, from* $\Gamma_{f_2} = 365\, m_e$ *down to* $\Gamma_{\pi^0} = m_e/(17 \times 23 \times 168)$ *— and each observed value carries its PDG uncertainty (error bars smaller than the markers throughout). The diagonal is identity, not a fit. Lower strip: each row's pull; the largest absolute pull is* χ_{c0} *at* -3.34σ*. The two charmonium rows use PDG 2026. These are comparisons of the d-versus-d^2 counting, not fourteen independent tests.*

What the comparisons establish. A close numerical match is useful only when the rule selecting it is constrained. The paper therefore distinguishes values fixed by its counting rules, conditional constructions, and structural identifications. Some excitation formulas remain fits within a restricted vocabulary. The mass points share inputs and cannot be counted as independent successes. Section 5.1 assesses the evidence, including the cost of selecting formulas.

That assessment uses description length: how many bits are needed to specify the proposed rules and their choices. The parameter comparison ranges from about 2.1 times compression before selection charges to 1.10 after them, and 0.72–0.79 under the most conservative convention. Values below one favor the Standard Model. The framework's code also covers its cosmology, quantum mechanics, and gravity account, with zero continuous free parameters; measured calibration inputs and discrete selections remain explicit (Table 1, Appendix M).

Tests and limitations. The framework proposes twenty-nine falsification tests, with their predictions and failure criteria in Table 11. Examples include proton stability, the absence of a vacuum-induced rotation of light's polarization, neutrino mixing, and the evolution of cosmic acceleration. Some tests depend on stated conditions. Existing strains include light-polarization measurements, galaxy rotation comparisons, and measurements of the growth of cosmic structure; Table 10 records them alongside the agreements.

The central unfinished construction is the microscopic spatial network: how the same local transistors join into a growing three-dimensional space and reproduce a smooth large-scale geometry. Particle-level open steps include the physical origin of matter–antimatter asymmetry and the response that realizes several mass and mixing rules. The proposed far-future shutdown and restart remains an outlook. These limits qualify the physical picture throughout; Sections 4.2, 6, and Appendix O state the precise obligations.

This completes the conceptual account. The following technical sections give the rules, calculations, comparisons, and proof domains behind it.

Technical argument

Code availability. Every claim, in the main text or the appendices, marked "(supplementary code)" maps to a script in the claim-indexed package `paper_code/`, archived alongside this paper at https://github.com/PenName42/Computing-Spacetime.

1 Axioms and kernel

The formalism describes the effective $D{=}3$ transistor. Its deeper substrate is the causal event network of Section 4.2. Every prediction traces through the structures below.

In Planck units ($\hbar = c = G = 1$) — one resolution quantum per tick ($\hbar{=}1$), one transistor diameter per tick ($c{=}1$), one unit of gravitational coupling ($G{=}1$):

A1. Clock speed. The substrate computes at rate c.[1]

A2. Conservation. The base unit ("bE": bit energy — the quantum of compute capacity) is conserved.

A3. Cost structure. $K(r) = r^{2-D}$ in D spatial dimensions, for $r \geq \ell_P$ (A4's resolution floor; the formula is not evaluated below the grain).

A4. Resolution. $\hbar$ sets the floor on outcome distinguishability. Every computed tick must also deliver at least one distinguishable outcome. The clauses are independent: one bounds resolution; the other excludes a computed tick with no distinguishable outcome. The requirement forces the first spatial channel from the undifferentiated pool and keeps one of the $\Sigma{+}1$ states unmixed at resolution-limited splitting. The floor alone would permit maximal mixing.

Cost flux and shell orientation. A2 requires total cost flux through any closed surface to equal the enclosed source. Cost flux is the kernel's own radial current, $r^{D-1}K'(r) = (2{-}D)$ — r-independent for every D (conservation as scale-freeness), with per-source magnitude $(2{-}D)$ the shell character: $+1$ confining, 0 state-blind (equal cost for every gate label, not absence of the labels themselves; the $D{=}2$ landscape carries no flux because nothing can charge the gate: spin is a state of the standing wave, not a source), -1 dispersive. The cross-era sign check uses that same current rather than an added sign rule: orienting a sourced shell by its character gives $U_D = M(2{-}D)K_D$, hence $U_1 = +Mr$, $U_2 = 0$ up to an irrelevant constant, and $U_3 = -M/r$, with $F_D = -\partial_r U_D = -M(2{-}D)^2 r^{1-D}$ — inward, flat, inward. This is the algebraic bridge between the trapping fibre and the drain potential already used below; why a source force reads the oriented current while an unsourced lens/read uses the positive magnitude K_D remains an operation-typing step rather than a second cost law.

[1] On the Poisson reading of A4 (Section 4.2), c is the emergent signal speed of the causal set; "rate c" is then the effective phrasing, not the primitive.

Three theorems follow. T1 gives the sum over paths: explore every path and keep the cheapest. T2 gives the Born probability rule exactly on fixed finite complete read blocks. T3 places locked bookkeeping energy in the clock-rate gradient at its source. Appendix G states their domains and derivations.

1.1 The kernel and the budget

The kernel is the cell's refractive-index profile; the table below reads its lens shapes against what each D leaves internally, one channel spent as a coordinate at every step.

$$K(D, r) = r^{2-D} \tag{1}$$

D	Kernel	Spatial	Internal (N_c)	Internal structure	Shell role
0	—	0	**4**	All internal	— (address only)
1	r	1	**3**	Color: SU(3)	Source (confining)
2	1	2	**2**	Spin: SU(2)	Gate (state-blind)
3	$1/r$	3	**1**	Charge: U(1)	Drain (dispersive)

Budget: spatial + internal = $D^{\mathrm{s}}_{\max}{+}1 = \mathbf{4}$ at every row. The local gauge algebra is the remainder.

The three landscapes ($K{=}r$, $K{=}1$, $K{=}1/r$) coexist at every interior point. Particle identity specifies which landscapes couple. Their mode spaces form a tensor product, so coupling more landscapes raises mass.

Each spatial dimension converts one budget channel to a coordinate: $N_c(D) = D^{\mathrm{s}}_{\max}{+}1{-}D$. For $D^{\mathrm{s}}_{\max} = 3$: channels $= 3, 2, 1$, giving SU(3)×SU(2)×U(1). $D^{\mathrm{s}}_{\max} = 3$ is fixed by four conditions — one stability theorem and three consistency identities on the structure built at $N_c = 3$, the framework's own closure rather than independent external selection: orbital stability [40], gauge–channel consistency (equivalently, mode-matching: both give $N_c(N_c{-}3) = 0$), gauge-dimension closure ($17{+}7{+}1 = 25 = (N_c{+}N_{\mathrm{spin}})^2$ at $N_c = 3$; root-uniqueness conditional on the disclosed N_{spin} typing, Appendix H), and the transparent-cell count ($T(\Sigma{+}1) = 28 = N_c^3{+}1$, unique positive integer root $N_c = 3$; Appendix H).

Input	Value	Role
c	299,792,458 m/s	Substrate clock speed
$\hbar$	1.055×10^{-34} J s	Resolution floor
m_e	0.51099895069(16) MeV	Absolute mass scale
H_0	67.4 km/s/Mpc	Current expansion rate[§]
m_t	172.76 GeV	electroweak anchor[¶]

Table 1: *Five measured inputs. c and $\hbar$ define the unit system (A1, A4); m_e sets the one irreducible absolute scale (the minimum charged-particle mass).* [§]*Planck ΛCDM value; the conditional common-background calculation gives $H_0 \approx 65.31$ (Section 4.7); particle-sector predictions are insensitive to this difference.* [¶]*Current determinations: PDG 2026 172.60 ± 0.27, LHC combination 172.52 ± 0.33. The seam rows (Section 3.1) tie this anchor to the measured m_W and m_H rather than leaving it free; the window they allow, and what it tests, are given in Section 6.*

Natural mass unit and measured calibration. The framework's minimum-cost normalization takes the positive $D{=}3$ price of a minimally maintained $D{=}2$ neutrino pattern, denoted β_ν, as one rest-cost unit. On the conditional positive-count completion of Section 3.7, this unit is $\beta_\nu = m_\nu/\sqrt{3}$, where m_ν in Eq. (10) denotes the solar-gap scale, not an individual propagation mass. The common work unit is fixed when minimizing the mode counts; realizing those counts and their frequency-bearing mass read remains a microscopic obligation. Changing units alone does not select a spectrum. For numerical comparisons we retain ratios m/m_e, using the more precisely measured electron as the calibration: $m_e = 1$ is the comparison convention, rather than a claim that the electron is the minimum-cost particle. The 2022 CODATA value is $m_e c^2 = 0.51099895069(16)$ MeV, with relative standard uncertainty 3.1×10^{-10}. Calculations retain all published digits of this electron calibration; rounding is reserved for displayed results. Calibration uncertainty is separate from model residuals. The neutrino supplies the natural unit; the electron supplies its measured calibration once their ratio is fixed.

$$B \equiv \frac{m_e}{\beta_\nu} = 2\sqrt{\frac{147}{47}}\left(\frac{63 \cdot 3^{49}}{631}\right)^{1/3} \simeq 1.01872576738 \times 10^8. \tag{2}$$

On this branch, an electron-relative table entry $f = m/m_e$ has natural-unit value $m/\beta_\nu = fB$. The integer requirement applies to the native neutrino work counts; charged masses retain their counted factors and geometrical lens responses.

Table 1 lists the five measured inputs to the core prediction formulas. Downstream checks also use explicitly stated external data and standard solution machinery, including quark thresholds for QCD running and empirical nuclear inputs for neutron-star structure (Sections 3.3, 3.6). H_0 and m_t are precision anchors — derivable in principle but limited by void-clock model uncertainty (H_0, Section 4.7) and m_u experimental uncertainty (m_t, Section 6). G is derived (Section 3.3); $\Omega_b h^2$ is derived (Section 3.7).

1.2 How to read the derivations

The tier words specify what each claim establishes. *Exact* means exact on the stated domain. *Conditional* names a result that depends on declared assumptions or identifications. *Structural* identifies a relation supplied by the architecture at the stated level of support. A *candidate* is a proposed construction, not an established result. *Typing* assigns an object to an operation, register, or domain; an *identification* connects that object to the quantity being interpreted or measured. A *fence* is a stated boundary on a claim. To *cede* a quantity is to leave its determination outside the framework's present claim. These words qualify the local statement; they do not extend it.

Starting structure	Resulting sector	Detailed home
A1–A4 and the channel budget	Kernel regimes and $N_c = 3$	Appendix H
Shells and engaged registers	Five operations and kernel integers	Section 2
Operations on the compute matrix	Masses, mixing, couplings, widths	Section 3
Causal reads and horizon bookkeeping	Conditional gravity	Section 4.2
Dimensional eras and clock rates	Inflation and late-time cosmology	Section 4
Protected counts and their boundaries	Evidence, falsification, open science	Section 5

The map shows dependencies, not equal evidential weight. A numerical code result is identified as such and linked to the supplementary package; no prose proof is implied by its place in the chain.

2 The transistor and its operations

The five operations read different features of one transistor's resonance pattern. They generate the results in Section 3.

2.1 The counting rule

Here d denotes the dimension of the complete register block physically engaged by an operation. Coordinate zeros in a chosen basis do not deactivate a clocked channel. The unresolved wave presents all d labels simultaneously; an empty cell has no engaged block and carries no bill. Operation 1 below derives the same-instant relation count: $d + 2\binom{d}{2} = d^2$ for a free register, and the full d-sector diagonal for a confined register. A later read may return one definite entry without changing the resident block's per-tick cost.

2.2 Five operations

#	Operation	Method	Output	Count
1	Read shell modes	d^2 counting (K regime)	Masses	81-row table
2	Refract between shells	Amplitude $= (d_i/d_j)^{N_c(b)/N_c(a)}$	Mixing, boson masses	CKM, PMNS, m_W, m_H
3	Count resonant channels	Additive (single-quark)	Nucleon charges	g_A=65/51, g_T=50/51
4	Count decay outcomes	A4 on decay pathways	Widths	14 hadrons, 7 orders of mag.
5	Shell-to-shell coupling	$N_c/N_{\text{spin}} \times$ channels$/D$	Magnetic moments	μ_p=39/14, μ_n=−21/11

Each operation reads the transistor differently (Fig. 1): mass measures total refractive drag across coupled landscapes; mixing measures transition amplitude; charge counts resonant channels; width counts gap exits (Fig. 20); moment joins drain charge to gate spin.

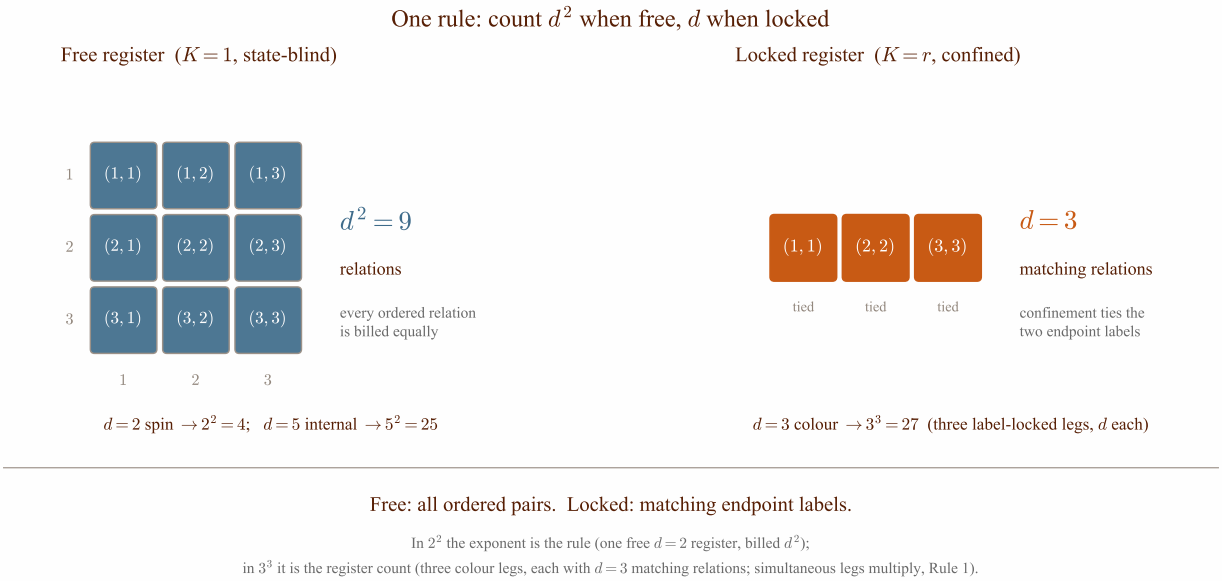


Figure 7: *The d^2-vs-d counting rule (Section 2.1) — the mechanism behind every mass formula. A free register carries all d^2 same-instant component relations; confinement ties the two endpoint labels and, because the common label remains unresolved, leaves the full d-sector diagonal. The proton readouts in Fig. 8 apply the rule to its mass bill.*

Operation 1: Reading shell modes. Mass is a wave's total refractive drag, the product of mode counts across its coupled landscapes. A free landscape (K=1, vacuum refraction) carries d^2 same-instant component relations. A confining landscape (K=r, graded-index refraction) ties endpoint labels by A2 conduction, leaving the d matching relations (Fig. 7).

The bridge is local compositeness: a cell turns the whole resident standing-wave pattern. Each of the complete engaged block's d simultaneous components therefore refracts against that whole d-component block in the same update. Each ordered pair is one primitive relation. At K=1, the state-blind kernel gives every relation equal unit weight; A4 distinguishes endpoint labels; A3 cannot delete a primitive relation without changing the refraction. Thus $d \times d = d^2$. Under confinement, A2 permits only (i, i), while source-shell label symmetry leaves the common label unresolved. The full diagonal has size d.

A2 can normalize relation values, correlating them without deleting their work. The d diagonal updates maintain resident components; they are neither address self-citations nor the return legs excluded by Appendix E.2. Before trace normalization, the optical coherency matrix witnesses the same d^2 real operator-work count. It does not supply the count, and d^2 does not count freely specifiable normalized-state parameters.

The assignment K=1 $\Rightarrow$ d^2, K=r $\Rightarrow$ d applies to resident mass. The strongest-coupled landscape sets that regime: color-carrying hadrons bind at D=1 $\to$ K=r; colorless leptons couple at D=2 $\to$ K=1. Operation 4 instead counts decay-vertex pairings and exits

(strong $\rightarrow$ $K{=}r$; drain-exit $\rightarrow$ $K{=}1/r$), so its powers follow vertex topology. Mixing belongs to a coupling-regime transition.

A2+A3+A4 give the two color-register closures (Appendix I.7): a two-configuration meson closes on a string (δ); a three-configuration baryon closes at a Y-junction (ϵ_{abc}). Their mass and width powers invert (Operation 4). Rule 1 combines simultaneous couplings by multiplication (tensor product) and independent registers by addition (direct sum). Shared resonant cells in the word's matrix row define coupling. The lepton generation space has transparent color cells, hence no such coupling, and enters additively as internal dimension (Appendix F).

Each shell has mode count $N_c(D)^2 + (N_c(D)^2{-}1) = 2N_c(D)^2 - 1$: color states plus gauge generators, giving 17 at $D{=}1$, 7 at $D{=}2$, and 1 at $D{=}3$. Their coupled-mode waveguide equation is $E^2 = (mc^2)^2 + (pc)^2$, with mass as cutoff frequency. The photon ($d{=}0$, no internal modes) has zero cutoff and travels at c. Vacuum is a prism for matter and a window for light: massive-wave speed depends on momentum; the read is achromatic, so ledger addresses do not smear.

Every tick turns a wave's internal state and passes its amplitude to neighboring cells. Light has no internal cargo to turn and crosses one cell per tick. Cargo diverts part of the budget into turning: mass. Internal phase evolution measures proper time. These shares combine in quadrature (Fig. 5), as the dispersion relation above requires; aging is the rest-energy share of the budget, rather than a count of withheld ticks. The maintained pattern runs each tick without a separate scheduler. Only the photon, with zero internal state and zero resident maintenance cost, reaches light speed.

Operation 2: Refracting between shells. A mass-ratio amplitude crossing from $D{=}a$ to $D{=}b$ follows (Appendix I.2)

$$\text{amplitude} = \left(\frac{d_i}{d_j}\right)^{N_c(b)/N_c(a)} \tag{3}$$

Shell ratios and within-shell amplitude normalizations enter separately. The three cases used here are:

- $D{=}1 \rightarrow D{=}1$ (CKM): the shell ratio is $N_c(1)/N_c(1) = 1/1$; gauge symmetry requires a per-axis amplitude, giving exponent $1/2$.
- $D{=}2 \rightarrow D{=}2$ (Koide/PMNS): the within-shell rule gives exponent $1/2$.
- $D{=}1 \rightarrow D{=}2$ (PMNS R_1): the shell ratio $N_c(2)/N_c(1) = 2/3$ combines with the within-shell factor $T_F = 1/2$, giving exponent $1/3$.

This rule fixes a modulus. Appendix I.1 derives complex crossing amplitudes; the argument needed for physical CP remains open (Section 3.2).

Operation 3: Counting resonant channels. The proton (u, u, d) presents $4 + 4 + 5 = 13$ resonant channels. The probe resolves this composite register one slot at a time, so a single-quark operator adds channel counts (Rule 1), giving g_A and g_T. Mass instead bills the joint register (Section 3.1).

Operation 4: Counting decay outcomes. A4 counts the decay vertex's distinguishable outcomes. The strong vertex ($K{=}r$) creates pairings $\rightarrow d^2$, giving $\Gamma_\rho = 17^2 = 289$. A drain

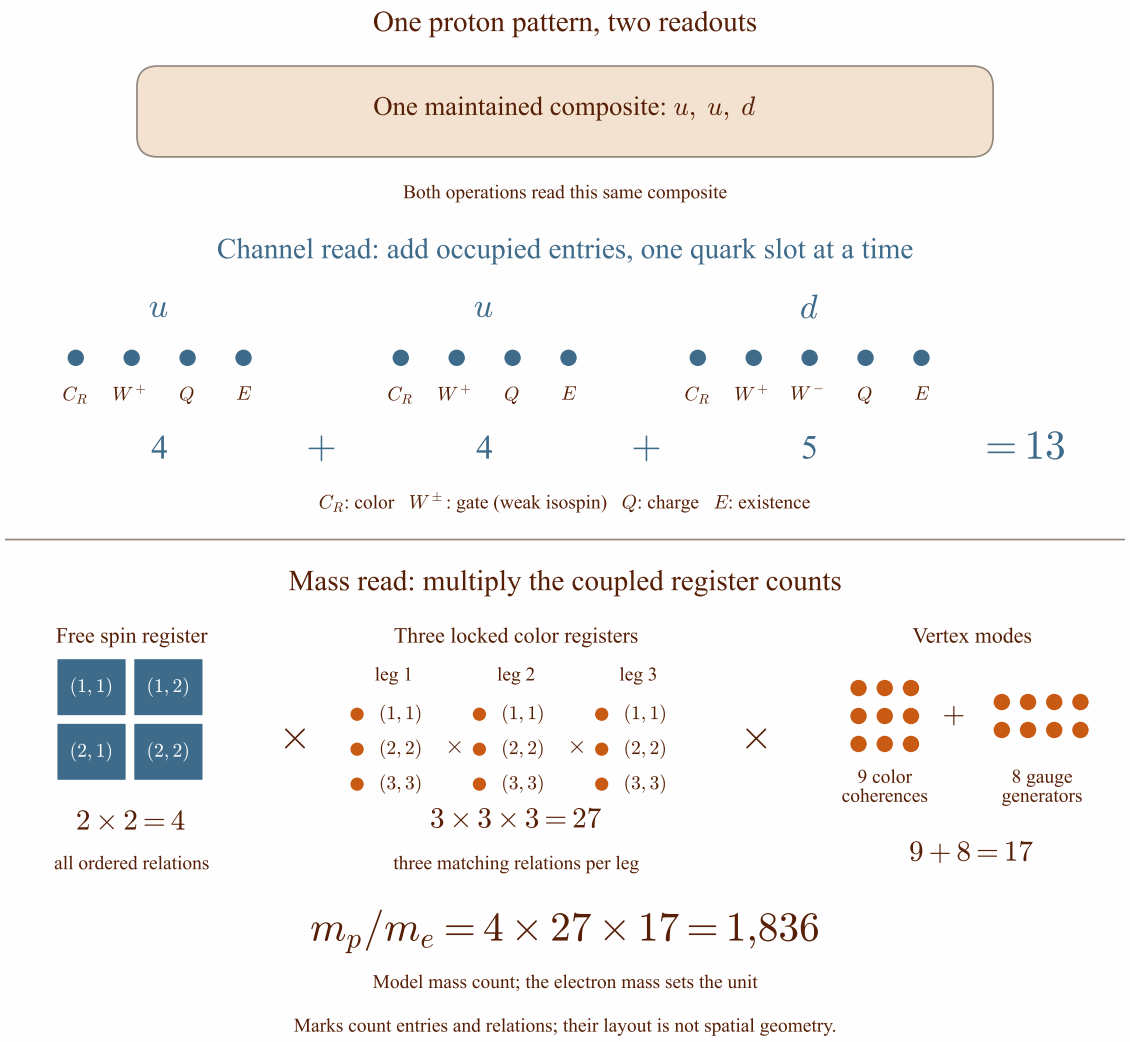


Figure 8: *One proton pattern, two readouts. Channel counting adds the occupied matrix entries of the two u configurations and one d: each u contributes C_R, W^+, Q, E, and d also contributes W^-, giving $4 + 4 + 5 = 13$. A nonzero Q entry counts once; its charge value is not a count weight. Mass multiplies four free-spin relations, three locked color registers with three matching relations each, and seventeen vertex modes (9 color coherences plus 8 gauge generators): $m_p/m_e = 4 \times 27 \times 17 = 1{,}836$ in the model. Marks display the paper's counts and matrix occupations, not spatial positions; E denotes existence.*

exit (K=$1/r$) enumerates individual outcomes $\to d$, giving $\Gamma_\omega = 17$. The latter applies when G-parity or quantum numbers block the strong pairing: the ω's 2π route is G-parity forbidden (1.5% observed), so its width counts the 3π route. Anomaly decays invert the kernel integers. These rules give all 14 widths.

Operation 5: Shell-to-shell coupling. A magnetic moment joins drain (D=3, charge) to gate (D=2, spin): $\mu/\mu_N = \pm\,(N_c/N_{\text{spin}})$ resonant channels$/D$, with charge-dependent denominator D. It is the current-loop identity $\mu = I \times A$, oriented along the spin axis. For the proton (Q=+1), coupling space = Σ+1 = 7; the neutral neutron word (Q=0) resolves through N_c^2+$N_{\text{spin}} = 11$.

How to read the matrix. The five operations specify what a read measures; the matrix-reading rules specify how shell turnings combine (Appendix I.6). Under Rule 1, simultaneously coupled registers have $a \times b$ joint outcomes; separately carried registers add. The proton's $4 \times 27 \times 17$ therefore counts one joint state space. The shells coexist at every interior point (Section 1.1). The optical analogy has the same tensor-product structure: polarization, mode, and frequency are independent properties of one wave.

Every quantity reads a resident's identity in the 11×7 compute matrix: its shell-channel row plus any family coordinate distinguishing repeated signatures (Appendix F). Most operations traverse part of the row. A carried coordinate instead addresses its complete resident family block; for leptons this is $\{\nu, e, \mu, \tau\}$, one neutrino propagation row and three charged-lepton endpoint rows.

A resonant cell refracts the wave; a transparent cell passes it untouched. The traversed region carries locked matching relations at cost d or free component relations at cost d^2, with mass and width powers inverted at the same shell. Mass reads resonant cells at the native power; mixing reads a cross-shell amplitude; charge adds resonant channels; width reads transparent vertex exits; moment reads the drain–gate bridge.

The strong interaction couples through resonant cells; the electromagnetic read travels the transparent region. Its $O(\alpha_{\text{EM}})$ correction is therefore set by the composite's transparent topology (junction, string, or none; Appendix D.4), independently of the leading operation. Its own coupling, $1/\alpha_{\text{EM}} = 17 \times 8 + 1 + 1/28$, likewise reads a resonant×transparent product.

The register grammar adds a separate time axis: frozen $\to$ once, clocked unresolved $\to d^2$, endpoint-locked/diagonal $\to d$, superposed $\to$ span×amplitude (Section 2.5). Region identifies which cells a read traverses; register state determines how their existence is billed.

Operation	Region read	Counting rule	Exemplar
1. Mass	resonant, shell-native power	d (locked) / d^2 (free)	$m_p = 2^2 \times 3^3 \times 17$
2. Mixing	cross-shell amplitude	$(d_i/d_j)^{N_c(b)/N_c(a)}$	CKM, Koide, PMNS
3. Charge	resonant, additive	per-channel count	$g_A = (13/17)(5/3)$
4. Width	transparent, decay vertex	d / d^2, inverted vs. mass	$\Gamma_\rho = 17^2$, $\Gamma_\omega = 17$
5. Moment	drain↔gate bridge	(N_c/N_{spin})×channels$/D$	$\mu_p = 39/14$, $\mu_n = -21/11$
EM residual	transparent only	topology-selected N	junction 21 / string 14 / lepton 5/2 / drain 0

2.3 Two worked examples

Three quarks in a color singlet. Three cost sectors, multiplicative (Rule 1 of the matrix-reading rules, Appendix I.6: coupled shells form a tensor product; Fig. 8): $N_{\text{spin}}^2 = 4$ spin coherences (gate, $K{=}1$, d^2) $\times$ $N_c^3 = 27$ color configurations (source, $K{=}r$, d) $\times$ $(2N_c^2{-}1) =$ 17 gauge modes (source vertex):

$$m_p/m_e = 4 \times 27 \times 17 = 1836 \qquad (\text{obs } 1836.15,\ 0.008\%) \tag{4}$$

The muon illustrates the free-register case (Fig. 9). Its 29-dimensional register carries 29^2 relations; subtracting the resident matter-lepton block's 14 transparent slots and dividing by the electron's four-relation bill gives $(29^2 - 14)/4 = 206.75$. Appendix I.5 gives the register construction, subtraction, and their qualifications.

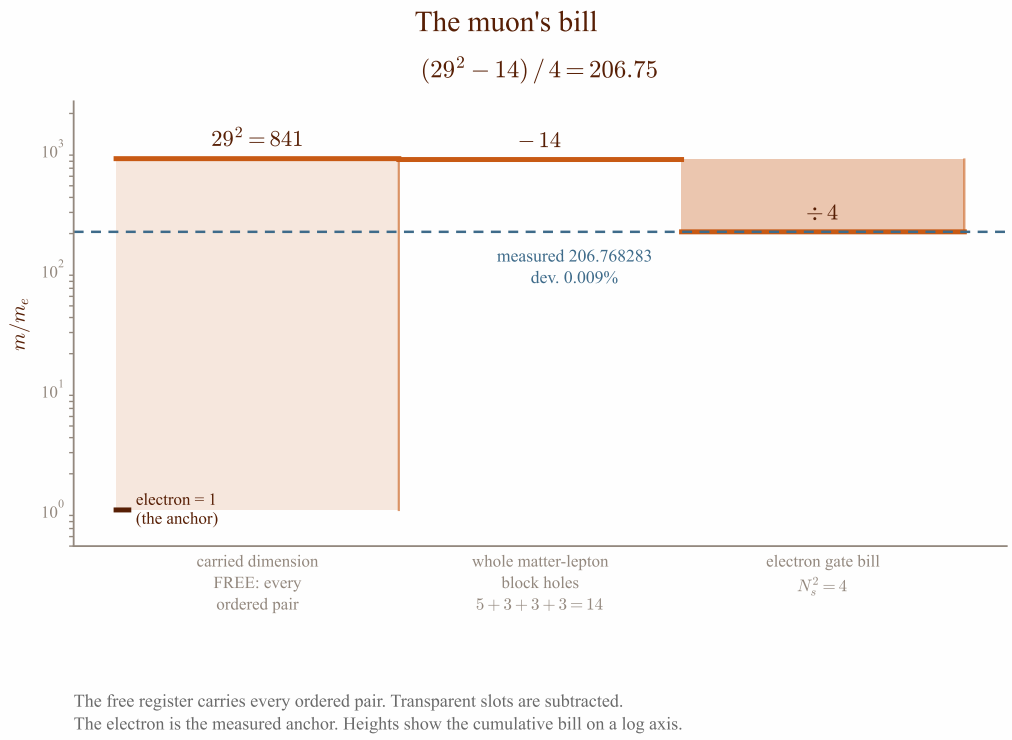


Figure 9: *The muon bill:* $(29^2 - 14)/4 = 206.75$ *(observed* 206.77, 0.009%*). The free register carries* d^2 *relations; the resident matter-lepton block contributes* $5+3+3+3 = 14$ *transparent slots to subtract. Division by* $N_s^2 = 4$ *normalizes to the electron's four-relation gate bill. The electron is the measured anchor. The vertical axis is logarithmic.*

2.4 Kernel integers

Forces are pathfinding costs. One equation, $K(r) = r^{2-D}$, at three values of D:

- $D{=}1$, $K{=}r$ (the V): cost grows with separation, confining the wave. A cost comparison can favor two separate color-closed composites over one increasingly stretched composite. This compares allowed configurations; it does not specify the pair-creation dynamics. The color-closure rule is given in Appendix I.7.

- D=2, K=1 (the plateau): cost is constant. No gradient, no force. The wave propagates freely — space expands for free. This is inflation.
- D=3, K=$1/r$ (the hill): cost is *highest* near the source and *falls* with distance. The wave spreads outward, dispersing as it goes — weakening with range. This is gravity and electromagnetism.

The weak interaction is the gate's own operation: its W^- channel toggles up-type and down-type identity (Section 3.2). The three kernel landscapes govern pathfinding costs. Their shapes also determine their spectra. The curved $1/r$ hill has a hidden SO(4) symmetry forcing integer resonances: hydrogen's spectral series. The flat K=1 plateau is perfectly degenerate: one note, no harmonics. The straight K=r V has no hidden symmetry. Its Airy zeros are transcendental with at most one possible exception; every nonzero zero of their single-Bessel analogues is provably transcendental (Appendix N). Think of a drum's overtones beside a flute's. The same cost function supplies both integer counting and confinement's unprotected spectrum: the framework specifies where its integers stop.

Every kernel integer counts the transistor's available resonant or transparent modes: a cross-section of the glass, fixed by $N_c = 3$.

Kernel integers are gauge-group representation dimensions. The drain (D=3, K=$1/r$, U(1)) has $1/r$ Coulomb n^2 degeneracy from SO(4): n=2 $\to$ 4 = N_{spin}^2, n=3 $\to$ 9 = N_c^2, n=5 $\to$ 25 = $(N_c{+}N_{\text{spin}})^2$. These integers price mode space; a single cell remains an s-orbital, without spatial lobes (Appendix B). The source (D=1, K=r, SU(3)) has symmetric-representation dimensions $\dim(0,N) = (N{+}1)(N{+}2)/2$: N=5 $\to$ 21 = $N_c(\Sigma{+}1)$, N=6 $\to$ 28 = $N_c^3{+}1$. The cumulative count to N=4 is $35 = 49{-}14$.

The source's radial spectrum consists of Airy zeros, with no hidden-symmetry protection (Appendix N) and no established kernel-integer formula for its radial levels. Its radial excitations remain structurally constrained fits, carrying the ∼24-bit measured selection residual (Section 5.1). Bertrand's theorem permits extra dynamical symmetry only for $1/r$ and r^2: the drain has SO(4) $\to$ integer n^2 degeneracies, while r^2 would require D=0, outside the spatial shells. The linear K=r source is unprotected. The established kernel integers belong to the SU(3) gauge algebra; the radial selection residual is measured separately and does not follow from the absence of hidden symmetry. The gauge groups, in turn, follow from $K(D,r) = r^{2-D}$.

The three potentials form source, gate, and drain: D=1, K=r confines color-locked quarks and turns waves back; D=2, K=1 is state-blind and controls parity through W^-; D=3, K=$1/r$ couples charge outward. The total is $(2N_c^2{-}1) + (2N_c(2)^2{-}1) + 1 = 17 + 7 + 1 = 25 = (N_c{+}N_{\text{spin}})^2$, equal to a kernel integer only at $N_c = 3$.

Resonant–transparent duality. The matrix has 10 particle patterns and the all-transparent photon (d=0): an 11×7 grid (Appendix F). Its 77 slots divide into 49 resonant and 28 transparent. Resonant modes produce masses, charges, and the force hierarchy. Transparent modes produce widths, magnetic moments, and electromagnetic coupling: they carry the read through the pattern as if those cells were absent. The photon's all-transparent row carries no resident coupling to the read.

Value	Expression	Name	Matrix cross-section	Contexts
7	$\Sigma+1$	Channel count	W^- col. resonant	a_V, m_u/m_e, V-P
8	N_c^2-1	Gluon count	C_B col. *transparent*	$\alpha_{\rm EM}$, K^{*0}, f_2
9	N_c^2	Color coherences	Q col. resonant	V-P, π^0, f_2'
11	$N_c^2+N_{\rm spin}$	Color-spin states	Matrix rows (10 patterns + γ)	ρ, μ_n, η', Ξ_{cc}
13	$N_c^2+N_c+1$	Pion pair	Matrix D=1 block (resonant)	π, f_2, Ξ^*
14	$N_{\rm spin}(\Sigma+1)$	Spin-channel / lepton ground	Resident matter-lepton block transparent (= resonant at N_c=3)	K, D, B, μ
17	$2N_c^2-1$	Color+gauge	Matrix D=2 block (resonant) = up-type total	α_s, p, $\alpha_{\rm EM}$, widths
19	$(2N_c^2-1)+N_{\rm spin}$	Vertex+spin	Quarks in D=1+D=3	$\psi(3770)$
21	$N_c(\Sigma+1)$	Transparent (no γ)	Transparent cells (no photon)	χ_{c0}, μ_n, β_0
23	$2(\Sigma+1)+N_c^2$	Pair factor	Quarks in D=1+D=2	K, ρ, η width
25	$(N_c+N_{\rm spin})^2$	Color-spin^2	Internal coherences = gauge total	ϕ, Ω_c, β_0, $D^{\rm s}_{\rm max}$
27	N_c^3	Color cube	Gauge shell *transparent*	p, η_c, χ_{c0}
28	N_c^3+1	Cube+1	Full matrix *transparent*	$\alpha_{\rm EM}$, V_{cb}, Υ
29	$N_c^3+N_{\rm spin}$	D=2 scale	Quarks in gauge shell	μ, m_W, D
31	$N_c^3+N_{\rm spin}^2$	Color-spin	Gen-1+gen-2 total	B_c, $\Upsilon(3S)$
35	49−14	Top step	Quark rows total	m_t/m_c
63	$(\Sigma+1)\,N_c^2$	Channel-color	$W^-\times Q$ product	η, ϕ
73	$N_c^4-(N_c^2-1)$	Gauge-physical	Total cells $-N_{\rm spin}^2$	CKM, η', Λ_c
82	N_c^4+1	Participation	N_c^4 generation orbit + the observer's unit (Rule 3b)	V_{us}, B, m_n-m_p
693	$N_c^2(\Sigma+1)(N_c^2+N_{\rm spin})$	Matrix volume	$Q\times W^-\times$rows	strangeness
899	29×31	P-wave	gauge-quarks × gen-1+2	χ_c, χ_b

Table 2: *Kernel integers from $N_c=3$. Each is the minimum integer satisfying its axiom constraint (A3), and each is a mode count of the transistor's gauge groups — equivalently, a cross-sectional count of the 11×7 matrix. Italic* transparent *denotes transparent-mode counts.*

The per-channel counts are $(6,4,3,10,7,9,10)$ resonant and $(5,7,8,1,4,2,1)$ transparent, all kernel integers. Their totals are the structural constants $49=(\Sigma+1)^2$ and $28=N_c^3+1$.

The gate is closed for $N_{\rm spin}^2=4$ patterns without W^- (photon, ν, u, c) and open for $\Sigma+1=7$ (e, μ, τ, d, s, b, t), giving parity violation its geometric origin. The matrix-dimension ratio gives $\phi=(\Sigma+1)/(2\Sigma-1)=7/11=0.6364$, algebraically forced by $N_c(N_c-3)=0$. At A4's floor, each mode supplies one quantum of distinguishable spatial extent: volume is mode count and resonant fraction is filling fraction. The value $7/11=0.6364$ agrees with classic random close packing, $\phi_{\rm RCP}\approx0.637$ [54]. RCP is protocol-dependent; the same review reports higher algorithmic values, 0.64–0.65.

The gauge reads have blind spots: the gluon misses 14 colorless modes; the W misses 12 gate-closed modes; the photon misses the 2 neutrino modes at $Q{=}0$. Their sum $14{+}12{+}2 = 28 = N_c^3{+}1$ equals the transparent total (Fig. 10). The $25 = (N_c{+}N_{\text{spin}})^2$ modes visible to all three reads equal the gauge-dimension total.

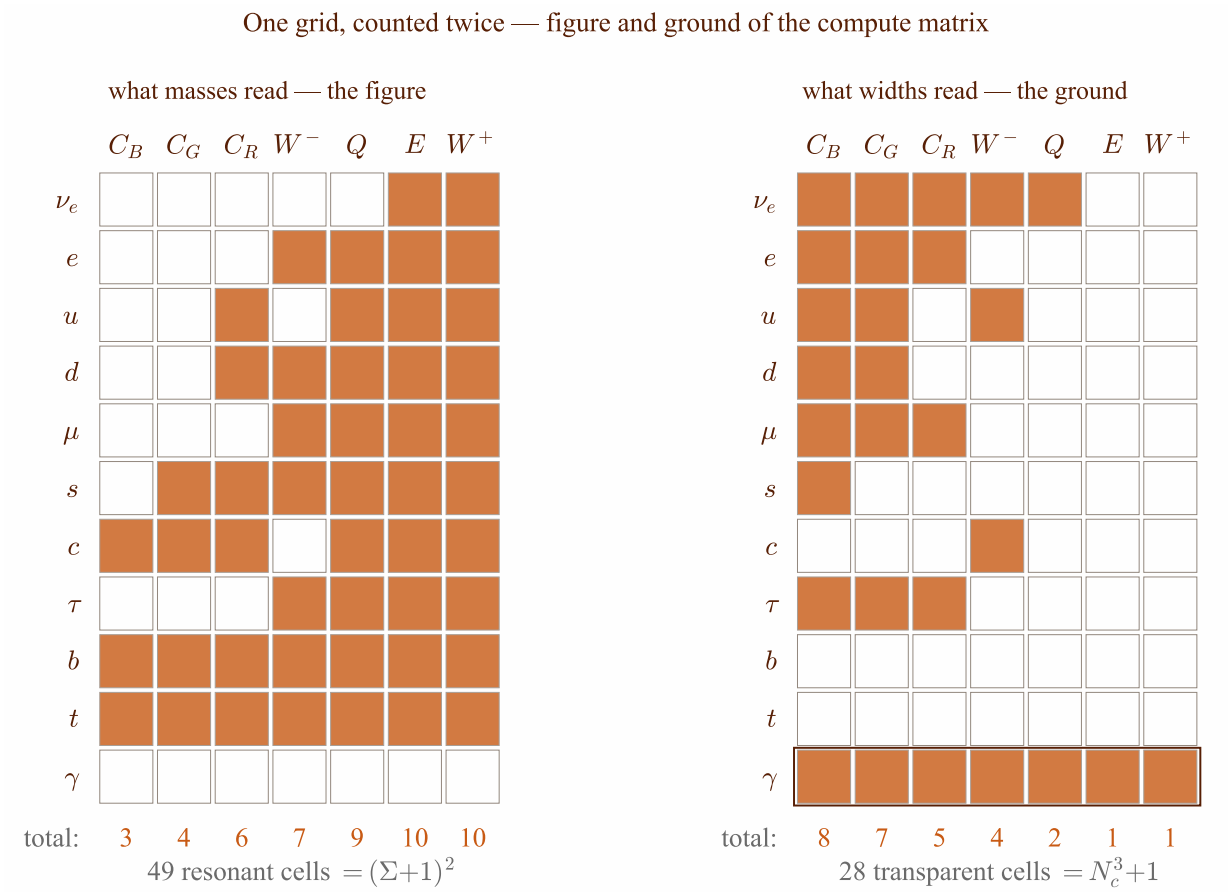


Figure 10: *The compute matrix counted twice: figure and ground. The same* 11×7 *matrix has* $49 = (\Sigma{+}1)^2$ *resonant cells, which masses read, and* $28 = N_c^3 + 1$ *transparent cells, which widths read. The left grid marks channels a wave excites; the right marks those it passes untouched. Every hole on one side is ink on the other:* $49 + 28 = 77$. *The photon row is empty on the left and full on the right: zero internal mass cost, transparent to the read. The resonant count reads refraction, cost, and the force hierarchy; the transparent count reads exit paths, moments, and the electromagnetic coupling. The existence and* W^+ *columns have the same occupancy pattern. The matrix has rank 6; the seventh column adds no independent occupancy pattern. Section 2.5 explains the distinction between existing and being computed with. Columns are sorted by ascending coupling count, with totals below. Fig. 30 colors the same matrix by channel family.*

2.5 Exists $\neq$ computes

A channel may exist without computing: its slot and allocated share persist under A2, while A3 bills only active participation. The channel does not compute when its shell is absent or its charge is invisible to the read. This is the address/reference distinction in Appendix E.1: existing costs nothing; being cited costs bE.

A2 conserves one total capacity, expressed as a norm over the coherence space in view. A particle's share is per-tick demand, not an independently conserved fragment; stretching lowers demand while capacity stands (Fig. 2). The same conservation appears in five spaces: capacity (Appendix B.4), transition probabilities (T1: tickwise conservation is unitarity), the 16-coherence allocation budget (Section 4.5), $(N_c{+}N_{\text{spin}})^2$ quarkonium gauge coherences (Section 3.1), and pairwise position coherences (Appendix I.10). Identifying these spaces as blocks of one register remains structural, not separately derived.

This distinction between existence and computation appears in nine independent sectors:

Sector	What exists	What does not compute	Consequence
Parity violation	Right-handed fermions at $D{=}2$	Through $K{=}1$ (W^- absent)	$V-A$ structure
Dark-matter equilibrium	Bookkeeping bE	Nonlinear self-sourcing through a second Φ^2_{DM}	Φ_b^2 (traces baryons)
Photon	$d{=}0$ (zero internal state)	Any internal evolution	Massless, $v = c$
Neutrino EM	ν at $Q{=}0$	Through EM sector	47/49 factor (EM-blind)
Nuclear surfaces	Gauge ports without neighbors	Exchange with absent partner	Surface energy a_S
Superconductivity	Spin doublet at $K{=}1$	Cross-relations while the electrons are separate residents (d diagonals)	Cooper pairing (one composite, d^2 relations)
Inflation	Q channel at $D{=}2$	Through absent $D{=}3$ shell	30/29 factor (A_s)
Spatial relation bill	Oriented relations between occupied baryonic addresses	Diagonal address self-citations	$N(N{-}1) \to N^2$ macroscopically
Void-clock flux	Accumulated clock boost	As dark-energy source (A3 per-tick)	$w(z)$ phantom crossing

The register grammar (structural). A clocked unresolved register refracts its complete engaged block each tick, giving Operation 1's same-instant d^2 count. Zero coefficients in a chosen representation do not shrink the block. A frozen register keeps its slot and share under A2, but is billed once. The rule is frozen $\to$ once; clocked unresolved $\to d^2$; endpoint-equality/diagonal support $\to d$. Clocking bills the whole register unless a read superselects a sector: resolution and projection differ, and only projection creates blocks.

A frozen register is a latent image, held unchanged and billed once at development; a clocked register is film still being exposed. The quantitative effects run in two directions. Redistributing a frozen share over computing modes enhances their weight ($30/29$ in A_s); resolving only the readable part of an existing set suppresses it ($47/49$ in Δm^2_{21}).

A blocked channel can force a third state, the superposed register. It bills the blocked direction's full coherence span through the gauge-physical amplitude once: the η' premium 11×73 (Section 2.7, Appendix K.4). Without blockage, A3 resolves flavor-pure and this premium is absent (ω, ϕ). The three states are allocated but unscheduled; scheduled every tick; trapped and emulated through the one open channel.

Within flavored rotations of the three light confined quarks, the exhaustive check finds one blocked direction: every relative re-pairing of u, d, s routes, while the fully symmetric trace does not (supplementary code). The η' premium is therefore unique within this scope. This structural rule unifies the rows; it is not separately falsifiable beyond them.

Two sectors give quantitative predictions. For the **47/49 EM-blind factor**, the matrix's 49 occupied cells carry equal A4 mode units (Appendix F). Only the neutrino's two cells (W^+, E) are EM-transparent (Q empty). A second-order $1 \leftrightarrow 3$ intermediate resolves relationally at density $47/49$ when its own cells cannot answer the read, as for neutrinos. When they can, as for quarks, resolution is local with no factor. This gives Δm^2_{21} at 0.2–0.7σ against post-JUNO fits and θ_{13} at 0.3σ (Appendix J). For the **30/29 inflationary factor**, the $D{=}2$ era's Q slot cannot fluctuate without the $D{=}3$ shell. A2 redistributes its frozen share over 29 active states, multiplying power by $30/29$ and giving A_s at 0.1σ (Appendix J).

The redistribution principle. The $30/29$ and $47/49$ factors instantiate one optimization result. Cost minimization (A3) concentrates a conserved quantity (A2) in the priced side of a discrete asymmetric partition (A4), with enhancement n/k for n total and k active components. This is a corollary of complementary slackness (Karush–Kuhn–Tucker, KKT, 1951). The observable divides conserved total by active count; it makes no claim about allocation within that sector, where a linear objective can otherwise concentrate freely.

The framework supplies the active/inactive partition through exists $\neq$ computes and identifies the instances: inflationary $30/29$, EM-blind $47/49$, and dark-matter $\Omega_{\rm DM}/\Omega_b = (N_c{+}1)^2/N_c = 16/3$.

2.6 $K{=}1$ universality

At the gate's inflection point of $K(r) = r^{2-D}$, $K{=}1$ and geometry dominates energy. Suppressing source or drain leaves the state-blind gate; any system at effective $D_{\rm eff} = 2$ accesses $K{=}1$:

System	Route to K=1	K=1 consequence
Inflation	Cosmic D=2 phase	Flatness, uniformity
Quark-gluon plasma	Deconfined above T_{QCD}	$\eta/s = 1/(4\pi)$ (KSS bound)
Color superconductivity	Fermi surface	CFL with $d = N_c^3 = 27$
Black hole horizons	$D_{\mathrm{eff}} = 2$ at $d\tau/dt = 0$	$S_{\mathrm{BH}} \sim A/\ell_p^2$; boundary correlations
FQHE	Magnetic field confines to 2D	$\nu = p/(2p{+}1)$ Jain hierarchy
Flat-band superconductors	Band geometry (MATBG)	Cooper pairing from quantum metric
TI surface states	D=2 boundary	Topological protection

In the fractional quantum Hall effect (FQHE), K=1, $N_{\mathrm{spin}} = 2$ spin channels, and A2's existence channel give the Jain hierarchy $\nu = p/(N_{\mathrm{spin}}p + 1) = p/(2p{+}1)$. Identifying "2 flux quanta" with N_{spin} and "+1" with existence remains substrate-tier. The registers belong to the Planck-sphere transistor (Section 1.1), and the existence line is one occupancy record; the hierarchy is therefore predicted to ignore carrier spin polarization, as observed in fully polarized systems.

The principal 2- and 4-flux families share $\nu = p/(\mathrm{span} \cdot p \pm 1)$. The liquid's p internal states are locked: one matched-endpoint diagonal per state, giving the d count. The gate supplies the span at the d^2-vs-d fork: locked N_{spin} gives $2p{\pm}1$; clocked N_{spin}^2 gives $4p{\pm}1$. The occupancy record's ternary sign supplies ±1; its register-wide flip is particle–hole conjugation. This locking is an operation-specific diagonal typing, not K=1 confinement, and remains a substrate-tier state-count identification. Even span plus odd record gives odd denominators. The 5/2 class is unreachable and ceded to its differently typed paired mechanism (supplementary code).

The cell's full channel structure supplies higher spans. Engaging source and drain alongside the gate gives $\Sigma = N_c{+}N_{\mathrm{spin}}{+}1 = 6$, hence $p/(6p{\pm}1)$. Adding the gate coherence excess gives $\Sigma{+}N_{\mathrm{spin}} = 8$, hence $p/(8p{\pm}1)$. These are the only even extensions at their engagement levels, selected by A3's minimum; odd spans violate parity. Source confinement (K=r) and drain dispersion (K=$1/r$) resist the collective K=1 state, making these developing states weaker than full plateaus.

The 4/11 and 4/13 states apply the hierarchy recursively to composite fermions (CFs) on the same K=1 landscape. At span 2, $4/11 = (4/3)/(11/3)$ uses $\nu^* = 1{+}1/3$, with a principal FQHE at partial filling; $4/13 = (4/5)/(13/5)$ uses $\nu^* = 1{-}1/5$, with a secondary hole state. These are continued fractions of the same spans. Each level is weaker because the inner gap is a fraction of the outer one. A3 breaks accidental higher symmetry back to the universal gate dimension $N_{\mathrm{spin}} = 2$.

The 2025 magic-angle twisted bilayer graphene (MATBG) superfluid-stiffness measurement is consistent with a substantial quantum-geometric contribution [25]; the geometric/conventional split remains experimentally unresolved (Section 4.8). At baryon densities $N_c\rho_0$ to $N_c^2\rho_0$, the source changes from K=r to K=1 in two steps: first $N_c = 3$ color configurations saturate (partial deconfinement), then all $N_c^2 = 9$ (full CFL). The intervening quarkyonic regime is A3's optimal confined/deconfined mix, mapping onto McLerran–Pisarski without importing it.

2.7 Chiral structure from the ledger

Pions are light because rotating the vacuum's opposite-sense light-quark pairs costs almost nothing. The mismatch parks in a transparent matrix cell; the lightest mesons measure that parking cost. A3 gives the Gell-Mann–Oakes–Renner (GMOR) relation $m_{\mathrm{PS}}^2 = B\,(m_{q_1}+m_{q_2})$ (Appendix L); the relation is derived from the cost structure rather than imported from QCD.

The flavor-singlet rotation is blocked: no massive row has a transparent W^+ or E cell. Its ledger therefore exits through the photon row, giving the $\eta' \to \gamma\gamma$ anomaly. One-loop composition with $f_\pi = 180\,m_e$ closes the K and η_8 deviations at the conventional scale $0.78 \pm 0.06\,\mathrm{GeV}$ with no further fitted constants. This is consistency with standard chiral perturbation theory; the framework does not pin that scale. Both the condensate scale B and loop scale are flows (Section 5.3).

3 Particle and interaction results

Masses, couplings, mixing angles, and widths are outputs of five operations on one transistor. The following sections group them by the standard classification for comparison.

Mass comparisons use m_e as the unit. Observed values are from the Particle Data Group review (PDG 2026) [1] unless noted.

3.1 Mass spectrum

Mass is the total refractive drag a standing wave accumulates coupling to the transistor's landscapes.

A3 fixes each kernel integer as a minimum cost. Rule 1 fixes how costs combine: coupled layers multiply (tensor product); independent layers add (direct sum). The integers decompose into the structural constants of Table 2 (Section 2.4); composite factors beyond that principal set are decomposed in Appendix K. Radial-excitation offsets remain structurally constrained fits (Section 5.1). Section 2 gives the rules; Appendix I gives the detailed derivations.

The full comparison is Table 9 in Appendix A, with every formula, uncertainty, scheme qualification, and shared anchor retained. The prose below explains the result classes and their boundaries.

The quark ladder and measured labels. The generation ladder ($m_s/m_d = (9/2)^2$, $m_b/m_s = (27/4)^2$, $m_c/m_u = (73/3)^2$, $m_t/m_c = (35/3)^2$, $m_u/m_e = 30/7$; Appendix F) supplies internal coordinates. It is not scored against lattice/PDG quark-mass labels. Confinement ($K{=}r$) admits no lone-quark standing wave: a quark is a register configuration, while the computed and weighed objects are composites. The top decays before confinement closes a composite; its measured resonance parameter belongs to its computed decay products. That parameter's pole-mass ambiguity of order Λ_{QCD} expresses the same category boundary. The framework uses it as the measured anchor in Table 1.

A lattice label is a swap slope: a hadron's response to flavor substitution through the full multi-loop calculation, a flow (Section 5.3). A one-loop bridge cannot identify it with the ladder (supplementary code). The ladder matches the one-loop-tier inversion of observed mesons to 0.03% at the standard NLO fit; lattice labels include beyond-one-loop contributions. The jointly-forced combination $m_s/m_{\rm ud} = 1053/38 = 27.71$ differs from lattice averages (27.23–27.42) by a 1–2% beyond-one-loop truncation remainder. This is the remainder class also measured in the $\eta \to 3\pi$-versus-lattice discrepancy in the ellipse parameter Q.

The quark-sector comparisons are the hadron rows and swap prices: $K - \pi = 693\, m_e = F$ exactly; the strangeness steps ($289/238/207$, $F/2$; Section D.3); and the isospin pair $m_d - m_u = 5\, m_e$ at MS-bar 2 GeV, $m_d/m_u = 13/6$. The last ratio is also corroborated by $\gamma = \arctan(13/6)$ at 0.4σ. Convention-dependence cancels in the ratio and is pinned by the stated scheme in the difference.

The equations $m_d - m_u = 5\, m_e$ and $m_d/m_u = 13/6$ uniquely give $m_u = 30/7\, m_e$, $m_d = 65/7\, m_e$ at that scale. These are solved coordinates, not additional inputs. The compute matrix gives the same base: $m_u/m_e = (13{+}17)/7 = 30/7$ is the $D{=}1$ and $D{=}2$ resonant block totals over the $\Sigma{+}1$ channels (Table 2); the quark-row total gives $35/7 = 5$ (supplementary code). The ladder supports the compute-matrix column (Appendix F), the m_t self-consistency chain (Section 6), and the chiral stocks (Appendix L). Table 3 collects the prices; their sums also give the fitted constituent-quark masses (Section 3.5).

Swap	Word context	Price	Verified against	Off
$u \leftrightarrow d$	any	$5\, m_e$ (MS-bar, 2 GeV); ratio $13/6$	isospin block above; $\gamma = \arctan(13/6)$	0.4σ
$u/d \to s$	string (δ)	$693\, m_e = N_c^2(\Sigma{+}1)(N_c^2{+}N_{\rm spin})$	$K - \pi = 354.11$ MeV	0.004%
$u/d \to s$	junction, Λ-type anchor	$F/2 = 346.5\, m_e$	$\Lambda - p$; $\Xi_b - \Lambda_b$ ($\Xi_c - \Lambda_c$: 2.4%)	0.2%
$u/d \to s$	junction (Σ/decuplet), no heavy	$289\, m_e = 17^2 = (2N_c^2{-}1)^2$	decuplet steps (surcharges $297/289/276$)	0.3%
$u/d \to s$	junction, one heavy	$238\, m_e = 14 \times 17$ ($J{=}3/2$ base; family 229–248 by spin/step)	eight one-heavy pairs (Section D.3)	0.02–0.85%
$u/d \to s$	junction, two heavy	$207\, m_e = 9 \times 23 = N_c^2(2(\Sigma{+}1){+}N_c^2)$	Ω_{cc}	1.1σ

Table 3: *Swap prices: the cost of one register substitution, read from mass differences of computed composites (string/junction closure: Appendix I.7). "Off": σ where measurement uncertainty dominates; % for splitting residuals (the convention of Table 9). These are the quark sector's convention-free comparisons — relations among rows already priced (the string-word price is the difference of two printed integers, $966 - 273 = 693$; the junction steps follow the vertex rule of Section D.3) — not additional predictions; the one untested relation in this family, $\Omega_{bc} - \Xi_{bc} = 207\, m_e = 106$ MeV, is counted in Table 12, not here. The anchor family (Λ-type, isoscalar light diquark) prices at $F/2$ where the relation is formula-level (the light and bottom members, 0.2%; the charm difference sits 2.4% high); the Σ/decuplet ladder prices by heavy count. The lattice quark-mass labels are swap* slopes *in a chosen scheme; these are swap* prices *between computed states, and the two are different categories of object (the paragraph above).*

The seam rows (m_W, m_H). The W and Higgs are within-$D{=}2$ layer-crossing transmissions (Section 2.2), bridging the continuous and discrete sectors. They are the table's only

masses priced in radians: the vector W uses the full SO(2) circumference 2π, normalized to $d_\mu = 29$; the scalar Higgs uses π, normalized to $\Sigma = 6$ (Fig. 11).

Counts self-resolve under A4. A continuous SO(2) measure supplies no record (Appendix I.1), so each crossing resolves against the EM register at a fee $\pm\alpha_{\mathrm{EM}}$. This applies the 47/49 readable-density census of exists $\neq$ computes, confirmed by JUNO, to the seam as a structural identification (Appendix J). A charged crossing writes one emitted quantum, the A3 minimum ($-\alpha_{\mathrm{EM}}$, the W). A neutral crossing cannot write and must read ($+\alpha_{\mathrm{EM}}$, the Higgs). A bare crossing would leave a computed event without a distinguishable record, excluded by the grammar. The m_t-free ratio sits at $+0.15\sigma$; HL-LHC tests the three-way relation in Table 11.

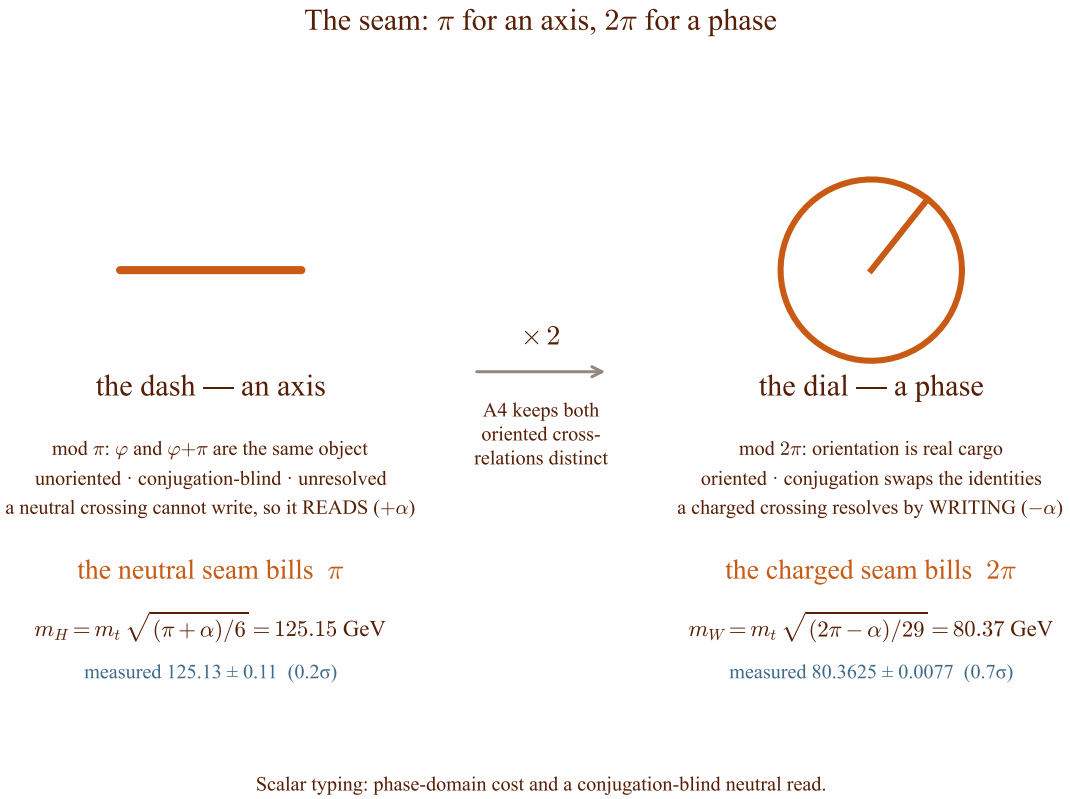


Figure 11: *Dash and dial: the seam split, conditional on the scalar-seam typings of Appendix I.1. Dash — the mod-π axis (the neutral seam, reads, $+\alpha_{\mathrm{EM}}$); dial — the mod-2π phase (the charged seam, writes, $-\alpha_{\mathrm{EM}}$); the factor of two between them is the free register's two oriented cross-relation components, kept distinct by A4. The m_t-free check:* $m_W/m_H = \sqrt{((2\pi-\alpha_{\mathrm{EM}})/29)/((\pi+\alpha_{\mathrm{EM}})/6)} = 0.642148$ *against the measured* $80.3625/125.13 = 0.642232$ (1.3×10^{-4}).

Spectrum and scope. The 81 numbered rows apply mass as distinguishable outcomes per tick: d^2 for free states at $K{=}1$, d for confined states at $K{=}r$. All hadron masses match within 0.15% (Fig. 3); $\eta(547)$ and $D^{\pm}$ lie at 0.11–0.13%. The residuals have 0.027% standard deviation and no drift by sector, mass scale, or constituent count (Fig. 12). This flatness is the signature of composites priced as wholes; constituent-mass binding corrections would vary in relative size by orders of magnitude. Isospin ratios, CKM elements,

and gauge couplings lie within 1.3σ of their quoted comparisons; $1/\alpha_{\mathrm{EM}}$ is structural, with the $O(\alpha_{\mathrm{EM}}^2)$ residual in Section 3.3.

Only 18 of the 81 rows are independent. The rest share anchors and kernel integers: anchors follow the five operations, while excitation offsets are structurally constrained but not individually forced (Appendix K). Including couplings, mixing, cosmology, nucleon charges, widths, nuclear binding, and QCD phases gives 56 predictions of independently measured quantities.

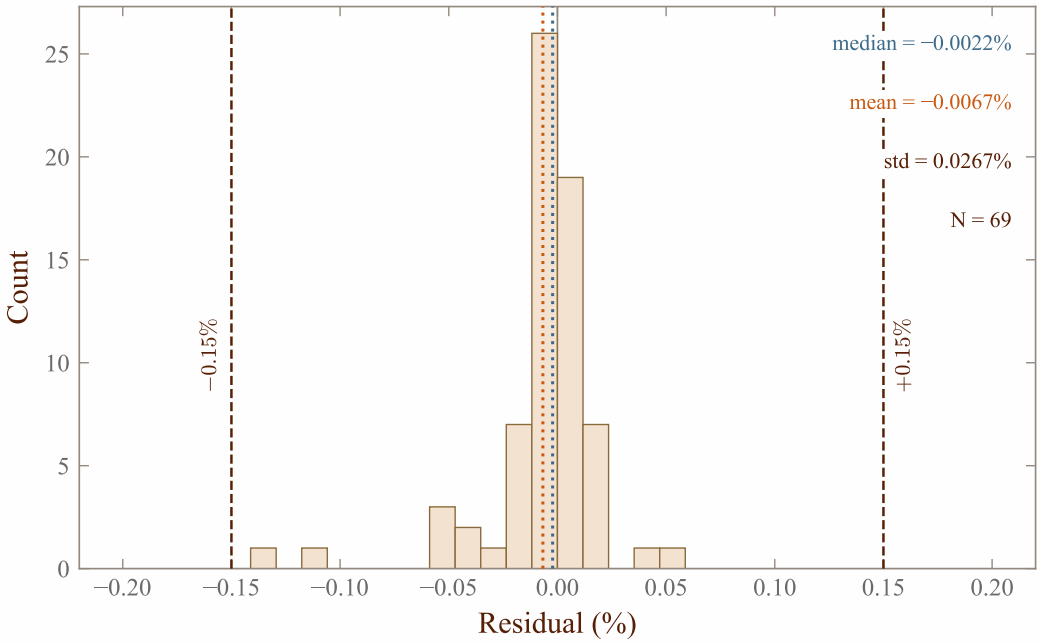


Figure 12: *Distribution of residuals (predicted − observed)/observed. Median* −0.002% *(blue dotted), mean* −0.007% *(orange dotted), standard deviation* 0.027%*. The dashed lines mark the* ±0.15% *envelope. All 69 hadron and lepton values fall within it; the largest residuals are* η *at* −0.11% *and* $D^{\pm}$ *at* −0.13%*. Shared anchors make these correlated comparisons.*

The numerical envelope alone supplies no evidence. The constrained null comparisons, uniqueness conditions, and selection charges are stated in Appendix A; the evidence summary is in Section 5.1.

3.2 Mixing angles

Mixing angles are coupling-transition amplitudes — the refraction coefficient when a wave's dominant landscape changes.

CKM mixing. Within the $D{=}1$ shell, a CKM element is the amplitude for the flavor substitution whose price appears in the mass table ($K - \pi = 693\, m_e$) and whose slope the lattice labels measure. The row patterns give exact change sets for all nine swaps: $u \to d$ adds W^-; $c \to s$ exchanges C_B for W^-; $t \to b$ changes no occupied cell. These are endpoint signatures, not magnitude or phase rules, and unchanged occupancy does not imply absent weak action (supplementary code).

The generation-one steps $d - u$ and $e - \nu$ share the move ($+W^-$, $\Delta Q = -1$). Thus $d + \nu = u + e$ holds cell-for-cell in register values. Exact-placement nulls make this identity non-generic, with the charge-cell equality ceded to ordinary charge conservation. Generations 2 and 3 differ from the shared step by $-C_B$ and $-W^-$ respectively: their departure from the balanced generation-one books is the mixing census (supplementary code). V_{us} follows the within-shell rule $V \sim \sqrt{m_i/m_j}$, exponent 1/2; V_{cb} is a gauge-completion amplitude; V_{ub} is their chain product (Table 4).

Element	Formula	Predicted	Observed	Pull
$\lvert V_{us}\rvert$	$(N_{\text{spin}}/N_c^2)(82/81)$	0.2250	0.2243 ± 0.0008	0.8σ
$\lvert V_{cb}\rvert$	$(N_c/73)(28/27)$	0.04262	0.0420 ± 0.0005	1.2σ
$\lvert V_{ub}\rvert$	$V_{us}\,V_{cb} \times 5/13$	0.003688	0.00375 ± 0.00020	0.31σ
$\sqrt{\bar\rho^2 + \bar\eta^2}$	$5/13$	0.385	0.3890 ± 0.0075¶	0.5σ
γ	$\arctan(m_d/m_u) = \arctan(13/6)$	65.2°	$66.4^{+2.7}_{-2.8}{}^\circ$	0.4σ
$\bar\rho$	$(5/13)\cos\gamma$	0.161	$0.1576^{+0.0092}_{-0.0091}$	0.4σ
$\bar\eta$	$(5/13)\sin\gamma = 5/\sqrt{205}$	0.349	$0.3556^{+0.0071}_{-0.0069}$	0.9σ
$\sin 2\beta$	derived	0.710	0.709 ± 0.011	0.1σ
J (Jarlskog)	$\lvert V_{us}\rvert\lvert V_{cb}\rvert\lvert V_{ub}\rvert \sin\gamma \times \sqrt{(1-V_{us}^2)(1-V_{cb}^2)\,(1-V_{ub}^2)}$	3.12×10^{-5}	$(3.16^{+0.13}_{-0.11}) \times 10^{-5}$ [1]	0.3σ

¶Central value and error propagated (independent-error approximation, matching the treatment of the two rows above) from the PDG 2026 CKMfitter values of $\bar\rho$ and $\bar\eta$; PDG does not separately quote this radius.

Table 4: *CKM predictions. Observed V_{us}: PDG 2024 $K_{\ell 3}$ average (the cleanest single-channel determination). Factors 82/81 and 28/27: A4 resolution floors at the crossing scale (the +1 is the observer's participation in the measurement). The 5/13: the $N_c+N_{\text{spin}} = 5$ color and spin channels projected against the pion pair cost 13. J: the standard rephasing-invariant measure of CP violation [74], computed here from the four rows above with no new input — the only comparison in this table that is convention-free by construction rather than by scheme-matching.*

The structural gate angle is $\gamma = \arctan(m_d/m_u)$ (Fig. 13): a two-state amplitude ratio gives its arctangent, and the W^- occupancy gives the cost ratio $m_d/m_u = 13/6$. Its numerical agreement with CKM supplies no physical CP carrier. A common phase on the u- and c-source W^- writes is removable gauge, and $|W^-\rangle = i\,|W^+\rangle$ denotes one ray rather than a second gate state (supplementary code).

The two 5/13 entries in rows 3 and 4 cannot both be exact under unitarity: they differ by $V_{ud}V_{us}/V_{cd} \approx 0.975$. The chain-exact branch gives radius 0.375, 1.9σ against 0.3890 ± 0.0075; the radius-exact branch changes the chain by the same ∼2.5%. Measurement must distinguish them. Their J values differ by ∼2.6%, below the current ∼4% precision, so the table's chain-exact J is presently a consistency check. The $\sin 2\beta$ row uses the radius-exact branch (0.710, 0.1σ); the chain-exact matrix gives 0.695, or 1.3σ against 0.709 ± 0.011 (supplementary code).

The CP boundary. An equal phase on all six W^- writes or all nine weak-port transitions is removable row/column rephasing gauge (supplementary code). The primitive route

triangle has one independent cycle; the unconstrained nine-edge graph has four. A single physical phase follows only after complete $U(3)$ closure and its gauge quotient. Its carrier is the rephasing-invariant closed-loop quartet J (Table 4); a gate-angle argument such as γ is not invariant. Cell phases are exactly gauge, and equal same-instant write phases cancel around every charge-conserving route. Thus the static, order-blind class cannot generate the carrier.

Deriving the physical CP carrier from A1–A4 remains open because A2 updates are monotonic and tick-ordered. A $D{=}1$ source excursion is a complete departure–return loop: turnaround writes nothing, the books close at the read, and the source phase advances between completed reads (Fig. 1). The remaining axiomatic candidate is a non-separable, route-dependent ordered product on the full nine-edge map. It must give $J = 0$ for equal-phase controls, survive row/column rephasing, and transform correctly under $W^+ \leftrightarrow W^-$ conjugation. Until that construction exists, J is measured, not derived.

The static exclusion also covers color-keyed and cross-seam fees. Only color cells have the occupancy variance needed for a loop phase, but the color-identical c and t rows make every color-keyed phase vanish on (c, t) quartets. Unitarity makes $|\mathrm{Im}|$ quartet-universal, so measured $J \neq 0$ excludes a color-only carrier. The phase's character, $e^{i\gamma}$ algebraic but not a root of unity, also excludes every structural-angle fee exactly (supplementary code). For cross-seam occupancy pairs, the pair-key map is the identity on the quartet basis. Fees then become fixed integer combinations of measured unitarity-triangle angles: a re-parametrization with no mechanism (supplementary code). A measured $J \neq 0$ therefore requires the open ordered-route carrier or physics outside the present billing rule; the theory does not yet decide which.

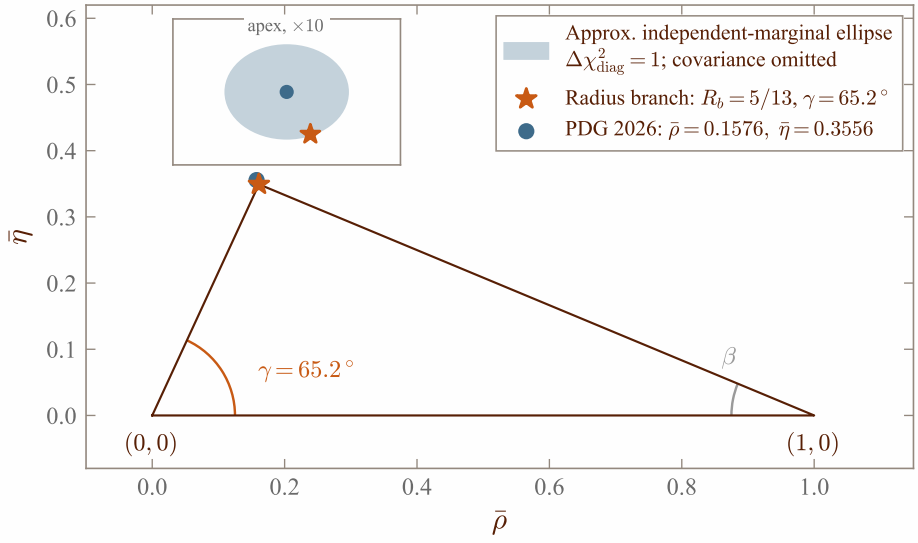


Figure 13: *CKM unitarity triangle, using the radius-exact branch:* $(\bar\rho, \bar\eta) = (5/13)(6, 13)/\sqrt{205}$ *and* $\gamma = \arctan(13/6)$. *The star lies inside the blue independent-marginal ellipse, centered on the quoted PDG values with symmetric display semiaxes* $(0.0092, 0.0070)$. *This approximation supplies no joint covariance or two-dimensional confidence level. The inset magnifies the apex* $10\times$*; coordinates are calculated before rounding. The rephasing-invariant CP carrier remains the separate open step described in the text.*

PMNS mixing. PMNS transmits from $D{=}1$ quarks at the weak vertex to the neutrino's $D{=}2$ shell. The cross-shell exponent on the mass ratio is $1/3$. A 3×3 Hamiltonian built from quark mass ratios and the $47/49$ EM-blind factor gives θ_{23} and θ_{13} directly (Appendices J.3, J). At the minimum splitting Δm^2_{21}, A4 overrides that Hamiltonian: $\sin^2(2\theta)$ saturates at $\Sigma/(\Sigma{+}1) = 6/7$, giving $\theta_{12} = 33.9°$.

The floor sits at $+0.25\sigma$ against the JUNO-including NuFIT 6.1 fit [9], $33.81^{+0.35}_{-0.46}$, or $+0.3\sigma$ against its with-SK value $33.76^{+0.42}_{-0.41}$. The raw Hamiltonian gives 32.2°, disfavored at 3.5–3.8σ across those values and 2.9σ against JUNO's first result alone. JUNO's 6-year precision $\pm 0.10°$ will decide (Fig. 14).

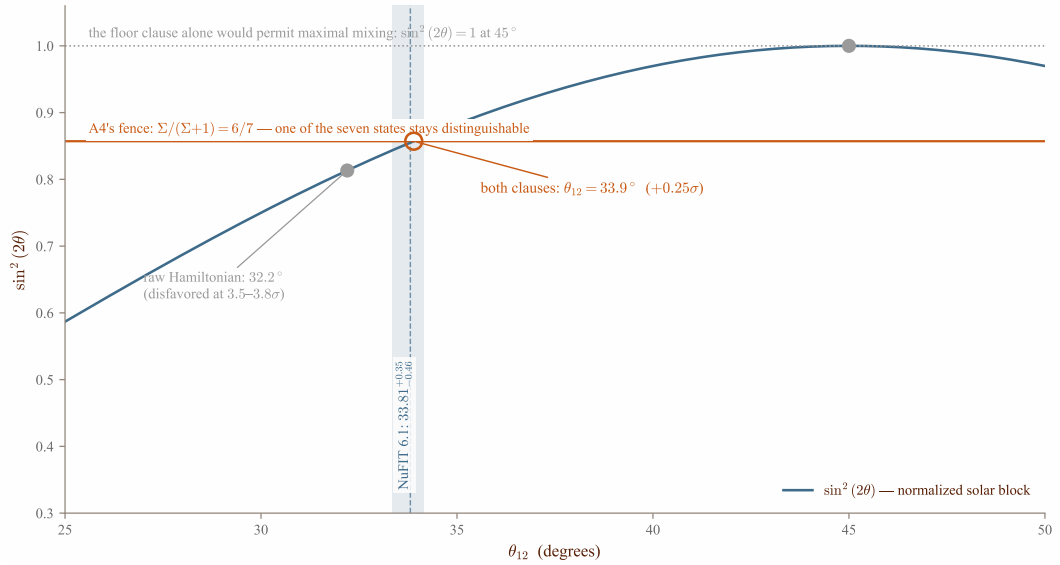


Figure 14: *Both clauses of the resolution axiom, priced at the resolution-limited splitting. The normalized solar-block transition probability* $\sin^2(2\theta)$ *saturates at A4's fence* $\Sigma/(\Sigma{+}1) = 6/7$ *— one of the seven accessible states stays distinguishable — giving* $\theta_{12} = 33.9°$*,* $+0.25\sigma$ *against the JUNO-including NuFIT 6.1 fit's* $33.81^{+0.35}_{-0.46}$ *(band). The floor clause alone would permit maximal mixing (45°); the raw Hamiltonian value 32.2° is disfavored at 3.5–3.8*σ *(text). JUNO 6-year (*$\pm 0.10°$*) is definitive.*

The predicted angles are $\theta_{12} = 33.9°$, $\theta_{23} = 49.4°$, and $\theta_{13} = 8.59°$. NuFIT 6.1 [9], for normal ordering with SK atmospheric data, gives $33.76° \, ^{+0.42°}_{-0.41°}$, $43.29° \, ^{+0.96°}_{-0.79°}$, and $8.62° \pm 0.11°$. The first and third pulls are $+0.3\sigma$ and -0.3σ. NuFIT 6.0 [5] gave $33.68° \pm 0.70°$, $48.5° \, ^{+0.7°}_{-0.9°}$, and $8.52° \pm 0.11°$, with pulls $+0.3\sigma$, $+1.3\sigma$, and $+0.7\sigma$.

The θ_{23} strain is octant-dependent. NuFIT 6.1's global minimum is lower-octant 43.3°, but the upper-octant minimum 48.0° lies only $\Delta\chi^2 \approx 1$ higher. The 49.4° prediction is 2.1–2.3σ from the global minimum and 1.9–2.1σ from the same-octant minimum: a strain, not an octant exclusion. JUNO's 59-day result in 2025 [8] gives $\theta_{12} = 33.78° \pm 0.54°$ (0.2σ); its 6-year $\pm 0.04 \times 10^{-5}$ precision will test $\Delta m^2_{21} = 7.55 \times 10^{-5}$.

Leptonic CP. The PMNS route uses the $D{=}1$ source projection, a disclosed operation-specific identification. Its sequential $1 \to 2 \to 3$ route is a tree. If that route is the literal T1-composition of its primitive legs, every leg phase is removable and $J_{\mathrm{CP}}^{\mathrm{PMNS}} = 0$ exactly (supplementary code). An unmatched loop phase would instead shift the mixing angles

jointly and give $J \neq 0$. The theory has not selected a branch: $J_{\rm CP} = 0$ is exact conditional on composition, and DUNE tests that condition.

CKM already selects the unmatched branch. Composing the V_{ub} chain makes the $(u, c; s, b)$ quartet real, while measurement gives $\gamma = 65^\circ$ on that plaquette (supplementary code). DUNE therefore tests whether the lepton route differs from the quark route. NuFIT 6.1 [9] gives $\delta_{\rm CP} = 212^\circ\,^{+26^\circ}_{-36^\circ}$ with SK atmospheric data, or $207^\circ\,^{+23^\circ}_{-20^\circ}$ without. The value $\pi = 180^\circ$ lies 0.9–1.4σ below the best fit and within the 3σ range.

Koide and T_F. The SU(2) gate fixes $T_F = 1/2$ through its fundamental normalization $\mathrm{Tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$. This follows from the budget-forced group $N_c(2) = 2$, with no separate input. The Koide relation [49] follows from $K(D{=}2) = 1$ and Z_3 symmetry: the masses form a Z_3-symmetric circulant in $\sqrt{m}$ space,

$$\frac{M}{S^2} = \frac{\sum m_i}{(\sum\sqrt{m_i})^2} = \frac{1 + 2\,T_K}{3} = \frac{2}{3} \tag{5}$$

exactly (obs 10^{-5}). The closing mass level remains unresolved. Against PDG-2024's $m_\tau = 1776.93 \pm 0.09$, solving for m_τ from the bare muon gives -1.2σ; the $O(\alpha_{\rm EM})$-corrected muon gives $+0.4\sigma$, the value in row 8 of Table 9. The $O(\alpha_{\rm EM})$ correction is amplified $\sim$15$\times$ ($\approx m_\tau/m_\mu$). A precision $\sigma(m_\tau) \lesssim 0.05\,\mathrm{MeV}$ distinguishes the levels at 3σ.

3.3 Gauge couplings

Gauge couplings are shell-to-shell mode ratios.

The force hierarchy emerges from how many shells participate in each coupling:

Force	Operation	Scope	Coupling	Scale
Strong	Ratio within $D{=}1$	1 shell	$\alpha_s = 2/17 \approx 0.12$	$\sim 10^1$
Electromagnetic	Product, summed across shells	bridges	$1/\alpha_{\rm EM} = 137$	$\sim 10^2$
Gravity	Power over all	all $\Sigma{+}1$	$m_P/m_e \sim 3^{49}$	$\sim 10^{45}$

More shells = weaker = more universal. Coupling topology sets the arithmetic: a within-shell ratio, products summed across bridging shells, or a power over the full transistor. Each shell contributes its resonant $\times$ transparent product; $1/\alpha_{\rm EM} = 17 \times 8 + 1 + 1/28$ adds sub-leading terms to the dominant shell. Thus ratio $<$ product $<$ power is the force hierarchy.

$$\alpha_s(M_Z) = \frac{2}{2N_c^2 - 1} = \frac{2}{17} = 0.11765 \qquad (\text{obs } 0.1180 \pm 0.0009,\ 0.39\sigma) \tag{6}$$

$$\frac{1}{\alpha_{\text{EM}}(0)} = (2N_c^2 - 1)(N_c^2 - 1) + 1 + \frac{1}{N_c^3 + 1} = 137.0357 \quad (\text{obs } 137.035999;\ O(\alpha_{\text{EM}}^2)) \tag{7}$$

$$\sin^2\theta_W = \frac{N_c^2(N_c^2+N_c+1)}{N_{\text{spin}}(N_c^2+N_{\text{spin}})(2(\Sigma+1)+N_c^2)} = \frac{117}{506} \qquad (\text{obs } 0.23122(6),\ 0.1\sigma) \tag{8}$$

$$\frac{m_P}{m_e} = \frac{3^{49} \times 63}{631} \qquad (\text{obs } 2.389 \times 10^{22},\ \sim 9\,\text{ppm}) \tag{9}$$

The observed m_P/m_e comes from measured G through CODATA [7]. Gravity services all $49 = (\Sigma+1)^2$ resonant slots at minimum cost $N_c = 3$ per slot (one color-shell traversal), giving N_c^{49}. The prefactor $63 = (\Sigma+1)N_c^2$ is the internal structure on which that exponent acts. The denominator's $630 = 7 \times 9 \times 10$ multiplies the colorless-channel resonant censuses W^-, Q, $W^+{=}E$ (Appendix F); $+1$ identifies the observer as one additional distinguishability unit, as in 82/81 and 28/27. The integer 631 is unique within the row's window: one unit changes the result by $\sim$1600 ppm, versus the $\sim$9 ppm match. That $+1$ remains a selected identification, charged in the description-length audit (Appendix M); the 9 ppm agreement is not a choice-free derivation.

The same ratio expresses the grain in laboratory units: $\ell_P = \hbar/(m_P c) = \bar{\lambda}_e \times 631/(63 \times 3^{49}) = 1.616 \times 10^{-35}\,\text{m}$. This restates m_P/m_e, not an additional prediction. The grain is definitional in the axioms' Planck units (Section 1); one measured mass fixes the absolute scale (Table 1). The row locates the fixed point where a standing wave's gravitational deformation reaches its resolution: at m_P, the Schwarzschild radius is twice the reduced Compton wavelength. Counting places that mass $3^{49} \times 63/631$ above the lightest charged standing wave (Fig. 15).

The resonant–transparent split gives each coupling its operation:

- $\alpha_s = 2/17$: transparent / resonant, a ratio within one shell.
- $1/\alpha_{\text{EM}} = 17 \times 8 + 1 + 1/28 = 137.036$: resonant × transparent. The dominant product $17 \times 8 = 136$ gains the photon $+1$ and drain $1/(N_c^3{+}1) = 1/28$ terms.
- $m_P/m_e \sim 3^{49}$: N_c raised to the full resonant count, with prefactor 63/631 in the exact expression above.
- $\sin^2\theta_W = 117/506$: cross-shell refraction (Operation 2, Section 2.2). The neutral wave engages gate and drain together; the $D{=}2 \to D{=}3$ amplitude exponent $N_c(3)/N_c(2) = 1/2$ gives a bare ratio of joint mode counts. Which two counts the wave presents remains open (Appendix M).

The $1/\alpha_{\text{EM}}$ formula is structural. Its observed-minus-structural residual is $+2.85 \times 10^{-4}$, of order $O(\alpha_{\text{EM}}^2)$; Fig. 12 uses the opposite sign. The absence of $O(\alpha_{\text{EM}})$ is itself a structural prediction: the no-self-reference rule that gives $N = 0$ for widths forbids the

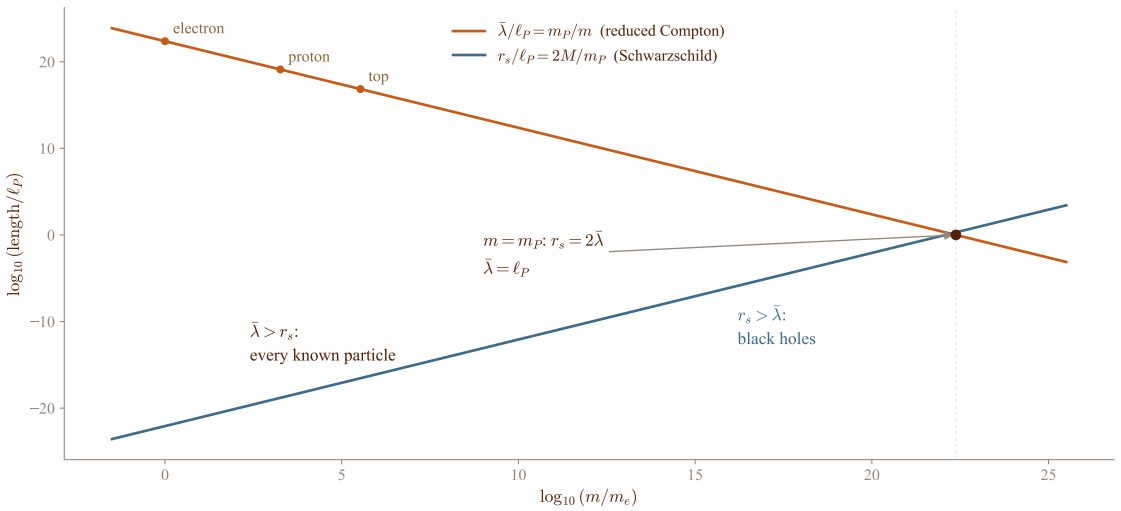


Figure 15: *Two rows of this paper, drawn to scale. The reduced Compton wavelength falls with mass ($\bar{\lambda}/\ell_P = m_P/m$, this section); the horizon radius rises ($r_s/\ell_P = 2M/m_P$ in cell units, Section 4.8); the marked point is this section's fixed point: at $m = m_P$ the Schwarzschild radius equals twice the reduced Compton wavelength, and $\bar{\lambda} = \ell_P$. (The two lines themselves cross nearby, at $m = m_P/\sqrt{2}$, where $\bar{\lambda} = r_s = \sqrt{2}\,\ell_P$ — a distinct point 0.15 dex away, indistinguishable at this scale and carrying no separate claim.) Every known particle sits far up the wavelength line; black holes lie on the horizon line, one cell diameter of radius per half Planck mass accreted.*

read from correcting its own coupling (Appendix D.4). Even the smallest kernel-integer $O(\alpha_{\rm EM})$ candidate is excluded at $>300\sigma$, while the measured residual is at the $\alpha_{\rm EM}^2$ scale. Its $O(\alpha_{\rm EM}^2)$ coefficient remains an undetermined flow. Only $\alpha_{\rm EM}$ is measured precisely enough to expose a residual; α_s and $\sin^2\theta_W$ sit at 0.4σ and 0.1σ against looser data.

With measured quark thresholds as inputs, $\alpha_s(M_Z) = 2/17$ reproduces the QCD running curve from 1.8 to 1500 GeV through standard two-loop RGE, at $\chi^2/\text{dof} \approx 1.0$ and without further free parameters. The thresholds and RGE are external; the β coefficients decompose into kernel integers at every threshold (Appendix I.9). Identifying $2/17$ as the MS-bar coupling at $\mu = M_Z$ remains unforced: the framework selects neither scheme nor scale. The running check inherits that identification, as does the $\sin^2\theta_W$ comparison (Appendix M); the Thomson-limit referent $\alpha_{\rm EM}(0)$ is convention-free.

3.4 Widths

Decay width measures how rapidly a maintained pattern terminates. The width rule counts the distinguishable outcomes available at its decay vertex. In the effective sphere picture, gaps carry no addressable resident computation below the resolution floor. Their absent resident bill alone specifies no decay trajectory. A read propagates through the drain's $K{=}1/r$ modes, connecting events across the lattice.[2]

The available openings set the standing wave's lifetime: decay is cavity leakage, and mass divided by width gives its quality factor.

[2]The spheres are the $D{=}3$ effective appearance of a pre-geometric event network, ordered by causal relations (Section 4.2).

Mass counts outcomes per tick that maintain a state; width counts those that lose it. Both use A4 and the m_e unit. Width's counting is set by the vertex kernel:

Meson	Decay outcomes ($\times m_e$)	Predicted	Observed	Off
$\rho(770)^{\pm}$	$(2N_c^2{-}1)^2 + N_c = 292$	149.2 MeV	149.1 ± 0.8	0.1σ
$\rho(770)^0$	$(2N_c^2{-}1)^2 = 289$	147.7 MeV	147.4 ± 0.8	0.3σ
$\omega(782)$	$2N_c^2{-}1 = 17$	8.687 MeV	8.68 ± 0.13	0.05σ
J/ψ	$N_{\text{spin}}/(N_c^2{+}N_{\text{spin}}) = 2/11$	92.9 keV	92.6 ± 2.5	0.1σ
$\Upsilon(1S)$	$N_c/(N_c^3{+}1) = 3/28$	54.8 keV	54.0 ± 1.3	0.6σ
$\eta(547)$	$1/(17 \times 23) = 1/391$	1.307 keV	1.31 ± 0.05	-0.06σ
$\pi^0(135)$	$1/(17 \times 23 \times 168)$	7.78 eV	7.81 ± 0.12	-0.3σ
χ_{c0} ($J{=}0$)	$N_c(\Sigma{+}1) = 21$	10.73 MeV	12.4 ± 0.5	-3.34σ
χ_{c2} ($J{=}2$)	$N_c^3/(\Sigma{+}1) = 27/7$	1.97 MeV	1.94 ± 0.09	$+0.34\sigma$
$\Delta(1232)$	$(N_c^2{-}1)(N_c^3{+}N_{\text{spin}}) = 232$	118.6 MeV	117 ± 3	0.5σ
$\Sigma^*(1385)$	$(N_c^2{-}1)N_c^2 = 72$	36.8 MeV	36.0 ± 0.7	1.1σ
$\Xi^*(1530)$	$N_{\text{spin}}\, N_c^2 = 18$	9.2 MeV	9.1 ± 0.5	0.2σ
$\phi(1020)$	$(N_c{+}N_{\text{spin}})^2/N_c = 25/3$	4.26 MeV	4.249 ± 0.013	0.72σ
$f_2(1275)$	$(N_c{+}N_{\text{spin}})(N_c^4{-}(N_c^2{-}1)) = 365$	186.5 MeV	186.7 ± 2.2	-0.08σ

Table 5: *Fourteen hadron widths spanning seven orders of magnitude (7.78 eV to 187 MeV). Zero free parameters. The χ_{c0} and χ_{c2} comparisons use PDG 2026 [1]. K=r (strong vertex) creates pairings $\to d^2 = 17^2$ (ρ); K=1/r exit (G-parity blocks the two-body K=r pairing $\omega \to 2\pi$, so the ω leaves by the three-body route $\omega \to 3\pi$, 89% of its width) enumerates outcomes $\to d = 17$ (ω); anomaly decays invert the kernel integers. Cross-check: $\Gamma_\rho/\Gamma_{J/\psi} = 17^2 \times 11/2 = 1589.5$ (obs 1592, 0.14%).*

The cavity quality factor is mass divided by width: $Q_\rho \approx 5$ with all strong channels open; $Q_{J/\psi} \approx 33{,}000$ with $N_{\text{spin}} = 2$ spin exits out of $N_c^2{+}N_{\text{spin}} = 11$; $Q_\Upsilon \approx 172{,}000$ with $N_c = 3$ exits out of $N_c^3{+}1 = 28$; and $Q_p = \infty$ when baryon number closes every decay route. OZI (Okubo–Zweig–Iizuka) = most channels sealed; sealing exits increases Q. The leakage spectrum runs from broad resonances through quarkonium to exact stability.

The formulas contain no phase-space factor: near-threshold $\phi \to K\bar{K}$ and far-above-threshold $\rho \to \pi\pi$ use the same kinematics-blind count. This is a structural limit of the grammar, not a derived cancellation.

The χ_{c0} prediction is 3.34 quoted-average standard deviations below the PDG 2026 width 12.4 ± 0.5 MeV, whose quoted uncertainty includes scale factor 2.8; χ_{c2} is 0.34 above 1.94 ± 0.09 MeV. The small kernel vocabulary saturates the width range: on the reference comparison corpus, a size-matched random vocabulary matches 8–9 of the 14 widths on average (supplementary code). The fourteen comparisons test the d-versus-d^2 structure, with the same blocks pricing mass and width (Fig. 6); they are not fourteen independent statistical tests.

Setting Γ_{π^0} equal to the standard Adler–Bell–Jackiw (ABJ) result $\alpha^2 m_\pi^3/(64\pi^3 f_\pi^2)$ gives $f_\pi = 180\, m_e = N_{\text{spin}}^2 N_c^2 (N_c{+}N_{\text{spin}}) = 91.98$ MeV (obs 92.22 ± 0.14 from $f_{\pi^\pm} = 130.41 \pm 0.20$ (PDG 2025), 1.7σ).

3.5 Nucleon charges and magnetic moments

Nucleon charges count resonant channels; magnetic moments couple the drain shell to the gate.

Quantity	Formula	Components	Predicted	Observed	Pull
g_A	$\frac{13}{17} \times \frac{5}{3}$	$\frac{\text{coupled channels}}{\text{gauge vertex}} \times \frac{N_c+N_{\text{spin}}}{N_c}$	$65/51 = 1.2745$	1.2753 ± 0.0013	0.6σ
g_T	$\frac{10}{17} \times \frac{5}{3}$	$\frac{N_c^2+1}{\text{gauge vertex}} \times \frac{N_c+N_{\text{spin}}}{N_c}$	$50/51 = 0.980$	0.989 ± 0.033	0.3σ
μ_p/μ_N	$+\frac{3}{2} \times \frac{13}{7}$	$\frac{N_c}{N_{\text{spin}}} \times \frac{\text{channels}}{\Sigma+1}$	$39/14 = 2.786$	2.793	0.26%
μ_n/μ_N	$-\frac{3}{2} \times \frac{14}{11}$	$-\frac{N_c}{N_{\text{spin}}} \times \frac{\text{channels}}{N_c^2+N_{\text{spin}}}$	$-21/11 = -1.909$	-1.913	0.21%
μ_p/μ_n	$-143/98$		-1.4592	-1.45990	0.049%

Table 6: *Nucleon charges and magnetic moments. g_A counts* all *active computation (13 coupled channels); g_T counts* static *structure ($N_c^2+1 = 10$). Magnetic moments: the Q column (D=3) couples to $W^\pm$ columns (D=2). Proton (Q=+1): coupling space = $\Sigma+1 = 7$; neutron (Q=0): the neutral word resolves through $N_c^2+N_{\text{spin}} = 11$. The ratio μ_p/μ_n improves on SU(6)'s $-3/2$ by a factor of 56. $g_T = 50/51$ is a prediction; JLAB 12 GeV and the EIC will test it.*

Proton spin fraction: of the 13 coupled channels, 4 belong to D=2 spin (W^+, W^-) and 9 to color/charge/existence. Equal angular-momentum weight per channel (A3) gives $4/13 = 0.308$. COMPASS [29] finds $\Delta\Sigma \in [0.26, 0.36]$ at $Q^2 = 3\,\text{GeV}^2$, containing this value. Its range is dominated by gluon-helicity uncertainty and is not Gaussian. Since $\Delta\Sigma$ depends on scheme and scale, the comparison is bound to COMPASS's MS-bar convention and Q^2 (Section 5.3). The ratio $\Delta\Sigma_p/\Delta\Sigma_n = 56/65 = 0.862$ is testable at the EIC.

Neutron lifetime: the standard weak-decay formula takes g_A to the bare/corrected alternatives 880.3 and 878.22 s. The nuclear/QCD table gives their 2025–26 comparisons; Nab decides.

The moment classes. A constituent mass is a sum of swap prices. Each light configuration carries $m_p/3 = 612\,m_e$: the totally antisymmetric ϵ junction gives equal thirds, with no distinguished leg. The strange configuration adds its Λ-anchor price, $612 + F/2 = 958.5\,m_e$, selected from the table's strange prices by data. Its magnetic contribution is $\mu_q = Q_q\,(m_p/\text{share})\,\mu_N$, the imported Dirac moment form evaluated on these derived shares.

Three binary boundaries classify ground-state baryon moments: generation span, spin alignment, and letter identity.

Single-generation distinct letters (J=1/2) use Operation 5's whole-word totals (39/14, $-21/11$; $+O(\alpha_{\text{EM}})$: 15 ppm). *Mixed-generation words* (J=1/2) itemize their charge circulation. The anti-aligned odd letter is the return arm, tolled -1/route. The route = $N_c+N_{\text{spin}} = 5$ when the gen-2 pair congests the color road, or $N_c(N_c+N_{\text{spin}}) = 15$ when the gen-2 letter travels: Rule 1's addition versus multiplication. The aligned pair carries the remainder, and the weights sum to $T = (2N_c^2-1)/(N_c(N_c+N_{\text{spin}})) = 17/15$. This gives $\Xi^0 = (4/3)\mu_s - \mu_u/5$ at 0.1σ, Ξ^- at 0.3σ, Σ^- at 0.1σ, and Σ^+ at 1.5σ, the one strain,

below the α_{EM} scale. The Λ's frozen isosinglet pair leaves one arm damped by a frozen coherence of $(N_c{+}N_{\text{spin}})^2$: $\mu_s \times 24/25$, at 0.0σ.

All-aligned distinct letters ($J{=}3/2$) have no recoupling or loop: their magnetic contributions add. The five hyperon values are forecasts (Table 12). *Identical letters* cannot be itemized, so the read bills the register, $Q \times N_{\text{spin}}$: Ω^- gives -2 against -2.02 ± 0.05 (0.4σ), and Δ^{++} gives $+4$ (Table 12). In Fig. 27, identical letters occupy pure corners, mixed-generation words the strange edges, single-generation words the light edge, and the Λ the doubled center.

There are no continuous parameters; the discrete assignments are priced (supplementary code). The T normalization $T = (2N_c^2{-}1)/(N_c(N_c{+}N_{\text{spin}})) = 17/15$ divides junction source modes by the mixed-generation route's color-spin modes, both independently motivated kernel integers. It holds at all four mixed-generation sites. Standard $T{=}1$ spin coupling misses the same constituent-mass comparisons by 19–53σ (supplementary code).

3.6 Nuclear and QCD

Nuclear binding and QCD phases arise at shell boundaries. The cheapest two-nucleon bond is $B_d = 13/3\, m_e$: pion-pair cost $N_c^2{+}N_c{+}1 = 13$ divided by color $N_c = 3$. Bulk coupling to $\Sigma{+}1 = 7$ channels gives $a_V = 7 \times B_d$. The QCD transition temperature, string tension, and quark–gluon plasma (QGP) viscosity combine kernel integers at the $D{=}1/D{=}2$ boundary.

Neutron star structure. A three-phase equation of state uses $a_V = 15.50\,\text{MeV}$ in the nuclear phase, $\sqrt{\sigma} = 438\,\text{MeV}$ and $c_s^2 \to 1/3$ in the conformal phase, plus empirical saturation density, incompressibility, and symmetry energy. Its thresholds are $N_c\, n_0 = 3\, n_0$ for quarkyonic onset and $N_c^2\, n_0 = 9\, n_0$ for CFL (color–flavor locking). It gives $M_{\text{max}} = 2.47\, M_\odot$ and $R_{1.4} = 12.7\,\text{km}$. NICER gives 12.35 ± 0.75 (0.4σ); PSR J0952$-$0607 at $2.35 \pm 0.17\, M_\odot$ gives a consistent lower bound on M_{max}, not a two-sided pull. The sound-speed peak at $3\, n_0$ matches model-agnostic Bayesian inference. The N_c and N_c^2 thresholds are falsifiable structural predictions; empirical inputs leave about $\sim 0.15\, M_\odot$ uncertainty in the absolute values (supplementary code).

3.7 Neutrino masses and CMB

The neutrino is the $D{=}2$ era's native fermion; the CMB is its frozen register (Section 4.4).

Applying the shell-crossing rule twice (Appendix I.2) gives the solar-gap scale $m_\nu \equiv \sqrt{\Delta m_{21}^2}$:

$$m_\nu = \frac{m_e \cdot (m_e/m_P)^{1/N_c}}{N_{\text{spin}}} \times \sqrt{\frac{47}{49}} = 8.69\ \text{meV} \qquad \Delta m_{21}^2 = m_\nu^2 = 7.55 \times 10^{-5}\ \text{eV}^2 \tag{10}$$

The exponent $N_c(3)/N_c(1) = 1/N_c$ measures the full shell stack traversed by a $D{=}3$ read of a $D{=}2$ native; N_{spin} counts the neutrino's gate modes at the crossing. The three eras

Quantity	Formula	Predicted	Observed	Pull
B_d (deuteron)	$(N_c^2+N_c+1)/N_c\, m_e = 13/3\, m_e$	2.214 MeV	2.225 MeV	0.46%
a_V (nuclear vol.)	$B_d \times (\Sigma+1)$	15.50 MeV	15.56 MeV	0.4%
a_V/B_d	$\Sigma+1 = 7$	7	6.99	unique
κ_2 (QCD phase)	$N_{\rm spin}/(2(N_c^2-1)^2) = 1/64$	0.01563	0.0153 ± 0.0018 [52]	0.2σ
η/s (QGP)	$1/(4\pi)$ [53]	0.0796	0.08–0.16 (heavy-ion extractions)	at bound
$T_{\rm QCD}$	$\sqrt{\sigma}/\sqrt{N_c^2-1}$	155.0 MeV	156.5(15)–158.0(6) (lattice) [85, 52]	1–5σ
$\sqrt{\sigma}$	$N_{\rm spin}N_c(N_c^2+N_{\rm spin}) \times 13\, m_e$	438 MeV	440 ± 10 MeV	0.2σ
α' (Regge)	$1/(2\pi\sigma)$	$0.83\,\text{GeV}^{-2}$	$\sim$0.87 (ρ trajectory)	$\sim$5%‡
τ_n	$(65/51)(1+\alpha_{\rm EM}/5)$ downstream	878.22 s	$877.83^{+0.30}_{-0.28}$ (UCNτ); 878.3 ± 0.4 (PDG 2026)	1.3–1.4σ†

Table 7: *Nuclear and QCD predictions.* †τ_n*: downstream from the* $O(\alpha_{\rm EM})$*-corrected* $g_A = (65/51)(1+\alpha_{\rm EM}/5)$ *(Appendix D.4); the bare* $g_A = 65/51$ *gives* 880.3 s *(3.8–8.8*σ *against the bottle values — and at the* g_A *level itself, bare* 65/51 *sits* 0.6σ *from the PDG average but* 3.4σ *from PERKEO III [13]). Conditional on resolution of the bottle–beam discrepancy (Section 3.5). The 2025–26 trend narrows the escape branch: UCN*τ *reports* $877.83 \pm 0.22\,(\text{stat})^{+0.20}_{-0.17}\,(\text{syst})$ *s [119]. Adding the uncertainties in quadrature gives the displayed rounded* $+0.30/-0.28$ *s; pull ranges use those two sides as diagnostics, and the summary census retains their upper endpoint. A J-PARC electron-counting* beam *result is consistent with bottles [120], and NIST finds a significant molecular-hydrogen charge-exchange bias unlikely [121]; with PDG-2024's direct* $V_{ud} = 0.97367$ *the bare chain gives* 881.5 s *(the corrected chain uses the CKM-unitarity* $V_{ud} = 0.97436$*; Appendix D.4). Nab (taking data now) decides: PERKEO-class* λ *confirms the corrected value and excludes the bare one. The independently measured ratio* $a_V/B_d = 6.99$*, uniquely 7 among small integers.* ‡α'*: the* $\sim$5% *gap is Coulomb curvature at low* J*; removing the* $1/J$ *term from the* ρ *trajectory recovers an asymptotic slope of* $\sim$0.84*, within* 1.5%*.*

persist as coextensive shells (Section 4.4). The neutrino's resident computing participation is its D=2 gate, accompanied by the free E mark; its present-day spatial wave is three-dimensional. The D=3 read prices its rest mass; the gap scale alone does not select an absolute mass eigenvalue. A conditional native-plane response permits changing orientation while retaining three-dimensional travel (Eq. (35)).

Phase length and the read. The electron's reduced phase length is $\bar{\lambda}_e/\ell_P = m_P/m_e$; its rest phase advances one radian in $\bar{\lambda}_e/c$. This fixes a phase scale rather than an envelope radius. Writing $m_{\nu,0} = m_e(m_e/m_P)^{1/N_c}$ and $a = \sqrt{47/49}/N_{\rm spin}$, Eq. (10) gives the gap scale $m_\nu = a\, m_{\nu,0}$, with the correction applied once. The inverse scale $\hbar/(m_\nu c)$ is not itself an eigenmode's reduced phase length. On the conditional positive branch below, the lightest mode has $\bar{\lambda}_1 = \hbar/(\beta_\nu c) = \sqrt{3}\,\hbar/(m_\nu c)$; the other two lengths are $\bar{\lambda}_1/2$ and $\bar{\lambda}_1/10$. Their numerical values inherit the common calibration and the physical mass-read premise. A constant attenuation of an already evolving wave leaves its normalized phase rate unchanged; the physical mass read must specify the frequency-bearing operator (Appendix J; supplementary code).

NuFIT 6.1 [9] gives $7.530^{+0.096}_{-0.097} \times 10^{-5}$ (0.2σ); JUNO's 59-day result [8] gives $7.50 \pm 0.12 \times 10^{-5}$ (0.4σ). JUNO's 6-year precision $\pm 0.04 \times 10^{-5}$ will be decisive. The splitting ratio is $\Delta m^2_{32}/\Delta m^2_{21} = N^2_{\rm spin}(N_c^2-1) = 32$, against 32.45 ± 0.49 [9] (0.9σ).

The sign-selected PMNS branch reads its ordered eigenvectors as normally ordered

propagation modes (Appendix J.3); the generator's eigenvalue magnitudes are not masses. For a conditional positive completion, let their masses be $m_i = \beta_\nu n_i$ with one fixed work unit $\beta_\nu > 0$ and distinct positive integer work counts $n_1 < n_2 < n_3$. These are billed counts, not coherent dimensions d_i to be squared again. Retaining $(n_3^2 - n_2^2)/(n_2^2 - n_1^2) = 32$ gives $n_1 \geq 1$, $n_2 \geq 2$ and $(n_2 - n_1)(n_2 + n_1) \geq 3$, hence $n_3^2 \geq 4 + 32 \times 3 = 100$; $(1, 2, 10)$ attains all three bounds. A3 therefore selects this minimum if the modes are physically admissible and the other required work is fixed. Matching the retained solar gap gives

$$\begin{gathered} \beta_\nu = \frac{m_\nu}{\sqrt{3}}, \qquad (m_1, m_2, m_3) = \beta_\nu (1, 2, 10) \\ (m_1, m_2, m_3) \simeq (5.016, 10.032, 50.161)\ \text{meV}, \\ \sum_i m_i = 13\beta_\nu \simeq 65.2\ \text{meV}. \end{gathered} \tag{11}$$

This minimum is exact within the supplied integer-count class; the recurring native responses that realize its counts remain to be derived. The integers label propagation modes, not the masses of three separately emitted flavors. A weak emission prepares a coherent combination of these modes, and their squared gaps remain $\Delta m_{21}^2 = m_\nu^2$ and $\Delta m_{32}^2 = 32 m_\nu^2$. JUNO+global data favor normal ordering at 2–3σ. The conditional sum is testable by DESI/Euclid at $\sigma \sim 20$ meV. Neutrinos are Dirac: ν occupies W^+, its A2 antiparticle occupies W^-, and the structural distinction $\nu \neq \bar{\nu}$ forbids $0\nu\beta\beta$. (supplementary code).

CMB parameters.

$$\eta = \left(\frac{m_e}{m_P}\right)^{2/5} \times \frac{N_c!}{N_c^2 + N_{\text{spin}}} = \left(\frac{m_e}{m_P}\right)^{2/5} \times \frac{6}{11} = 6.10 \times 10^{-10} \tag{12}$$

(obs $6.10 \pm 0.04 \times 10^{-10}$, Planck CMB-derived [2]). Big Bang nucleosynthesis (BBN) with the derived η through the standard calibrated nucleosynthesis fitting formulae (η is the framework's contribution; the yield functions are not): $Y_p(\text{He-4}) = 0.247$ (obs 0.2449 ± 0.0040, 0.5σ; the 2025 EMPRESS XV value $0.2387^{+0.0036}_{-0.0031}$ sits 2.3σ low, and a 2026 reanalysis moves it to 0.2402 ± 0.0040, 1.7σ — the He-4 value moves with source and reanalysis); $\text{D/H} = 2.51 \times 10^{-5}$ (obs $2.527 \pm 0.030 \times 10^{-5}$, 0.6σ; best single 2024 measurement 2.533 ± 0.024, 1.0σ).

$$A_s = \frac{4N^2}{3\pi \times 7^{14}} \times \frac{30}{29} = 2.10 \times 10^{-9} \quad (\text{Planck: } 2.10 \pm 0.03 \times 10^{-9},\ 0.1\sigma) \tag{13}$$

The D=2 configuration density matrix has 7^{14} entries: S_7 symmetry gives Hilbert space 7^7. Appendix J gives the chain from $\rho = m_P^4/(8\pi \times 7^{14})$ to the base amplitude.

The 30/29 factor applies exists $\neq$ computes to the 2D era: Q receives a share but cannot fluctuate without the D=3 shell, so A2 redistributes it over 29 computing states. Appendix J gives the derivation and the degenerate-pair alternative. For S_8, the conditional

positive spectrum gives $S_8 \approx 0.835$ in the common-background CAMB/clock calculation (Section 4.7).

The adiabatic null. With equal S_7 shares, a spatially frozen Q slot, and one A4 admission per e-fold, the era's only species-relative local variable is admission time: the residual stagger δN (Section 4.4). Everything inherits one offset in the same history under the imported separate-universe identification. At this structural tier the primordial scalar record is therefore purely adiabatic, with zero isocurvature. Every single-clock model shares this null, so only a confirmed primordial isocurvature detection discriminates. Such a detection first tests the open post-transition occupation chain (Section 4.3); if that chain closes, it tests the shared history itself (Table 11).

The low-multipole deficit. The fully degenerate $D{=}2$ register (S_7) passes no spatial pattern into the third dimension. The A_s fluctuations arise as admission-completion stagger at the transition (Section 4.4); without a preferred direction, the fossil is isotropic on every scale. Its largest-angle feature is its edge, set by the first admission (the era-total forward map at $\ell \sim 35$; supplementary code). A mode enters the frozen register only after being clocked and leaving it, so larger scales carry no record.

The observed large-angle power puts that first admission at or beyond $\ell \approx 5$, requiring more than $\sim$61 e-folds. A per-multipole Gamma-form score calibrated on the Planck points' asymmetric errors excludes a record starting at $\ell \gtrsim 8$ and permits one at $\ell \lesssim 5$ (supplementary code). This is an internal approximation, not the official foreground-marginalized likelihood. Planck low-ℓ EE permits $\ell \lesssim 3$ and excludes $\ell \gtrsim 13$ under every tested likelihood form. At $\ell \approx 5$, the answer depends on that form: the published symmetric bars cross zero and force no unique score (supplementary code).

A2 fixes the first admission's amplitude. Every ingredient of A_s—density, departure rate at full capacity, S_7 configuration space, and 30/29 redistribution—is conserved through $D{=}1 \to D{=}2$.

Amplitude, scale, and transfer. A2 derives the first admission's amplitude across the transition; the sky bounds its scale through the era total. Reprocessing uses a standard causal-cosmology transfer, downstream of the derived era geometry just as BAO peak shapes are downstream of the expansion history (supplementary code).

Bounded domains. Across the sampled sizes, aspects, boundary conditions, and observer positions, every bounded-domain template at the Sachs–Wolfe tier suppresses intermediate multipoles too strongly and scores below the unsuppressed spectrum (supplementary code).

Parameter	Framework formula	Predicted	Planck	Tension
$\Omega_b h^2$	η (above)	0.0223	0.02237 ± 0.00015	0.5σ
$\Omega_{\rm DM} h^2$	$(N_c{+}1)^2/N_c \times \Omega_b h^2$	0.119	0.1200 ± 0.0012	0.9σ
n_s	$1 - 2/57$	0.9649	0.9649 ± 0.0042	0.0σ
A_s ($\times 10^{-9}$)	$4N^2/(3\pi \times 7^{14}) \times 30/29$	2.10	2.10 ± 0.03	0.1σ
ℓ_1	acoustic-scale projection from fw $H(z)$	220.1	220.0 ± 0.5	0.2σ

Table 8: *Four CMB parameters derived with zero free parameters ($\Omega_b h^2$ restates η through the standard baryon-to-photon conversion), plus the θ^*-constrained anchor ℓ_1. With the conditional positive neutrino spectrum and the supplied relic-density conversion, $\Omega_m h^2 = \Omega_b h^2 + \Omega_{\rm DM} h^2 + \Omega_\nu h^2 \simeq 0.0223 + 0.119 + 0.0007 = 0.142$ at the displayed precision. The separate fixed-CDM CAMB reference with three positive masses gives peak ratios $H_2/H_1 = 0.45316$ and $H_3/H_1 = 0.44191$, retaining the displayed 0.453 and 0.442. This reference holds its background fixed; the θ^*-anchored void projection supplies ℓ_1 separately (supplementary code). ℓ_1 is not an independent prediction: it constrains H_0 via θ^*. The common-background calculation gives $H_0 \approx 65.31$ (Section 4.7); its acoustic constraint does not recompute these temperature peak ratios.*

4 Cosmology and gravity

The ledger that prices particles also runs the cosmos. Dark matter is the cost of one event knowing where another happened; dark energy is the cost function dissolving where particles have left. The counting rules determine the dark-matter amount and profile. The conditional late-time mechanism computes expansion, growth, baryon acoustic oscillations (BAO), and S_8 from one evolving state (Section 4.7). Local reads, $H_{\rm local}$ and a_0, retain separate endpoint and response choices.

4.1 The cost channels

Dark energy and local clock rates. Fewer standing waves means fewer computations means faster local clocks. The clock-rate gradient pushes matter wallward. The acceleration is the cost function dissolving where particles have left. Compute drag creates structure; structure creates voids; voids undo the drag.

Gravity, dark energy, and dark matter (DM) are three regimes of one cost function:

$$\mathcal{D}(\mathbf{x}) = \underbrace{-M_{\rm total}/r}_{\text{drain shell (all bE)}} + \underbrace{\tfrac{3}{4}\,\delta_v(\rho)}_{\text{void clock}} \tag{14}$$

Here $d\tau/dt = \sqrt{1 + 2\,\mathcal{D}}$ is the clock rate. Dense regions contain more transistors carrying standing waves: more compute activity per volume, slower clocks. A3 draws things together because nearby events communicate more cheaply. Their jobs remain distinct; each relation is maintained for fewer ticks because its read loop closes sooner. This clock gradient is gravitational time dilation.

Sparse regions have fewer active transistors and faster clocks: dark energy. Throughout, $\delta_v \geq 0$ is fractional underdensity read as a positive *depth*. The bookkeeping term

$C_{\text{book}} = -1/3$ shifts all clocks equally and has no force-producing gradient of its own. Its energy nevertheless enters the drain's M_{total} alongside baryonic bE (T3), where it gravitates (Section 4.5).

Only clock ratios are observable. Since a read is its completed route (Section 4.2), void pacing accumulates along that path; endpoints alone do not price the read used in late-time cosmology (Section 4.7). The two terms form a resource monitor: kernel-weighted resident load, including baryonic and bookkeeping bE, plus local idle share relative to the mean. The rate $d\tau/dt$ is its throttle: computing lowers the frequency of completed exchanges.

A3 bills the resident load through the drain. Idle share is not an additional resident load: it changes local pacing, and its activation-flux response enters the gravitational potential (Appendix C.4). In perturbations this response acts through that potential, not through a separate local Hubble-friction term (Section 4.7).

4.2 General relativity

Dense regions compute more and clock slowly. Waves bend toward them as light bends through glass. Consistency of the horizon bookkeeping yields Einstein's equations as an equation of state, within the identifications and continuum conditions stated below. Three standard theorems enter under these conditions.

The excitation field deforms the substrate: dense regions contract it and sparse regions relax it. That deformation is the metric. The substrate adjusts to the local frame, preserving local Lorentz invariance; the average defines a cosmological rest frame, as the CMB does in general relativity. The Planck length is the average resolution grain, rather than a fixed rod contracted by boosts.

A4 bounds distinguishability, not separation: events may be closer than ℓ_P while remaining indistinguishable. The fundamental event-read network is a causal set. Classical sequential growth (CSG) [56] supplies causal-order vocabulary and covariance tests; the physical update need not be a classical CSG draw (Appendix O). Birth-order is gauge, the discrete general covariance condition, so it selects no fundamental frame.

A3's economy identifies histories by causal relations and channel content. Addresses specify dynamically fixed positions, channels specify types, and provenance chains are causal relations. None supplies an additional event token. An identity marker doing no derivational work would be an uncosted extra. Two webs agreeing on every relation and channel content are therefore one history. Birth-order gauge extends from poset relabeling to isomorphism.

A2's ordered ledger maps to CSG internal temporality; A3's stateless cost evaluation maps to the Markov property. K on causal relations is a covariant poset observable. A1 confines source evaluation to the past light cone, supplying Bell causality. Close-packed spheres at $\phi = 7/11$ are this network's coarse-grained $D{=}3$ appearance. Their statistical CMB rest frame belongs to the cosmological solution.

The gravitational derivation is thermodynamic. Its response must be tensorial: $K(D{=}3) = 1/r$ couples universally to bE (A2) and refracts the read. A scalar Nordström field bends no light, so it cannot supply that response. The framework therefore requires a tensor metric response; Fierz–Pauli supplies the standard spin-2 context [45].

Jacobson's equation-of-state argument [46] requires two ingredients, each with its stated identifications. First, $K{=}1$ state-blindness on $D_{\text{eff}}{=}2$ surfaces where $d\tau/dt \to 0$ gives horizon entropy $S \sim A/\ell_p^2$ (Section 2.6). Consistency with the independently derived G fixes the coefficient $1/4$ with zero freedom (Section 3.3): a theorem of existing structure (supplementary code).

Second, A1+A4 assign uniformly accelerated worldlines a thermal distinguishability spectrum: a local Unruh temperature. The thermal form follows in the continuum limit at the T1/T2 tier. The all-zero vacuum register (Section 4.8), A2's non-negative quanta, and A3's minimum supply cost positivity. The thermal argument uses positive-frequency free continuum read modes together with the boost kinematics specified here. Read transparency ($d{=}0$) supplies free modes, a falsifiable commitment (Section D.2). A4 distinguishes the two oriented cross-relation components, doubling the gate circle's half-turn monodromy to exactly 2π. The global tick selects a unique vacuum: α-deformations erase thermality rather than changing its temperature (supplementary code).

The continuum and A1→SR kinematics share one condition: the causal order must be *manifoldlike*, faithfully embeddable at Poisson density in a Lorentzian four-manifold. Poisson density is the A4 reading. Under this condition, the Poisson process gives local Lorentz invariance [55]; the embedding supplies the thermal argument's boost kinematics. For the uniformly sampled chirp, finite-tick corrections obey the explicit bounds of Appendix E.5; the thermal ratio follows in the controlled continuum and regulator limits (supplementary code).

Continuum uniqueness is the causal-set Hauptvermutung; whether growth produces manifoldlike orders remains open [41]. The framework fixes $D^{\text{s}}_{\text{max}} = 3$ (Appendix H), supplies exact local transport identities, and has finite packing and graph evidence. The intrinsic reciprocal exact-six atlas, indefinitely refinable amorphous gluing, and continuum homogenization remain one joined construction (Appendix O). The dispersion and birefringence tests constrain its outcome.

The growth boundary. The thread/splice surrogate has a covariant amplitude witness and a unique budget-neutral critical allowance, $k = 3$. Tested fixed-population families do not grow width and depth together. These results constrain a growth law without selecting microscopic neighbours or constructing a $D = 3$ atlas.

At $D{=}2 \to 3$, every old address opens a complete isotropic local $D = 3$ neighbourhood on the same tick. The $D = 2$ state supplies no inherited edge or normal. Equal content must glue those local charts into one space. Appendix O separates exact results, finite evidence, and the open atlas–transport theorem.

The horizon functional also supplies Clausius. Unresolved alternatives count A4 distinguishability (Shannon entropy). Crossing bE has an unreadable microstate beyond the causal boundary: heat, by A1+A2. A3 minimizes cost subject to horizon entropy, giving $Q - TS$. These follow from the axioms on a causal boundary. The constrained minimum

is Gibbs, and its equilibrium variations obey $\delta Q = T\,\delta S$: the Clausius relation $\delta Q = T\,dS$ as first-order optimality of the identified functional (supplementary code).

Applying that balance to every local Rindler horizon gives the full Einstein equations as an equation of state, conditional on reversible local equilibrium and the open manifoldlike continuum. Λ enters as an integration constant; the substrate supplies no corresponding object because nothing bills where nothing acts (Section 4.7), so it is zero.

In a static vacuum exterior, the drain alone contributes to $\mathcal{D}$; the void is a separately typed late-time channel (Section 4.1). Then $d\tau/dt = \sqrt{1+2\mathcal{D}}$ gives the exact Schwarzschild component $g_{00} = 1 - 2GM/rc^2$.

Within the derived Einstein equations' domain, GR light deflection, Shapiro delay, and perihelion precession follow without separate inputs. The deflection combines throttled clocks (Section 4.1) with the deformed fabric. For a null wave the tensor response makes these entries equal and aligned: clocks alone give $0.87''$, together they give $1.75''$. A scalar's equal, anti-aligned fabric contribution cancels the clock contribution, giving the Nordström zero.

Exactly 2 tensor-transverse GW polarizations and zero scalar breathing mode are predicted, testable by LIGO/ET. FRW symmetry gives the Friedmann equations. Reducing the void-clock acceleration (Section 4.7) to backreaction from inhomogeneous clocks remains open, as in mainstream averaging [57].

GW250114 tests the Kerr ringdown spectrum with a first-overtone amplitude nonzero at 3.5σ [89], and the Hawking area theorem at 4.8σ [90]. These 2025–26 results bear on the thermodynamic ingredients. LHAASO's GRB 221009A limit, $E_{\mathrm{QG},1} > 10\,E_{\mathrm{Planck}}$ [84], excludes preferred-frame Planck-scale discreteness while leaving the Lorentz-invariant class used here untouched [55].

4.3 Inflation

Inflation is the era of the flat pane. Expansion runs on the gate's $K{=}1$ landscape: a constant kernel leaves no gradient to deflect a wave and nothing to charge the stretch.

The dimensional lifecycle follows the axioms. A4 forces $D{=}0 \to D{=}1$: the undifferentiated bE pool cannot satisfy distinguishability, so one channel becomes spatial, giving $K{=}r$. This transition is the origin of time. A4's distinguishable outcomes create ordering; without them there is no sequence or before-and-after. A3 then drives $D{=}1 \to D{=}2$ because $K{=}1$ is cheaper than $K{=}r$. The $D{=}2$ gate phase is state-blind and force-free. Its flatness is exact.

The slow-roll parameter is $\varepsilon_1 = 1/(4N^2)$. At $K{=}1$, energy partitions equally over the $D^{\mathrm{s}}_{\max}{+}1 = 4$ budget channels, spatial + internal (Section 1.1). The departing channel ($D{=}3$) carries fraction $1/4$. Its energy cannot fluctuate spatially because the shell does not yet exist; A2 also forbids its disappearance. A2 therefore redistributes it over the era's remaining computing capacity. This is the principle behind the $30/29$ factor (Appendix J.2), applied here to the admission register rather than the substrate census.

Each e-fold freezes one distinguishable unit out of the register (A4). The frozen record neither rebalances nor receives, so the carrier depletes. Departure accumulates at $1/N_{\text{rem}}$ per e-fold on a conserved load. Thus $\varepsilon_1 = (1/4)(1/N_{\text{rem}})^2$ is the channel's share times its squared rate.

The map from ε_1 to observables imports the standard perturbation-theory slow-roll relations. Their per-mode N-convention is identified with the remaining capacity at the pivot. This $\sim$0.001-in-n_s identification has different second-order consequences under the two readings; the running is therefore not claimed (supplementary code).

The remaining capacity at the pivot mode is $N = 57$, with one state per e-fold (A4). Two observations select it: weakly n_s, and sharply the A_s normalization, which gives $N = 57.0 \pm 0.4$ (Section 3.7). The largest recorded sky angles put the era total above a lower bound of $\sim$61 e-folds (Section 3.7; Fig. 16). The spectral tilt is:

$$n_s = 1 - 2/N = 0.9649 \qquad (\text{obs } 0.9649 \pm 0.0042,\ 0.0\sigma) \tag{15}$$

Tensor-to-scalar ratio: $r = 4/N^2 = 1.23 \times 10^{-3}$ (LiteBIRD $\sim$2036).

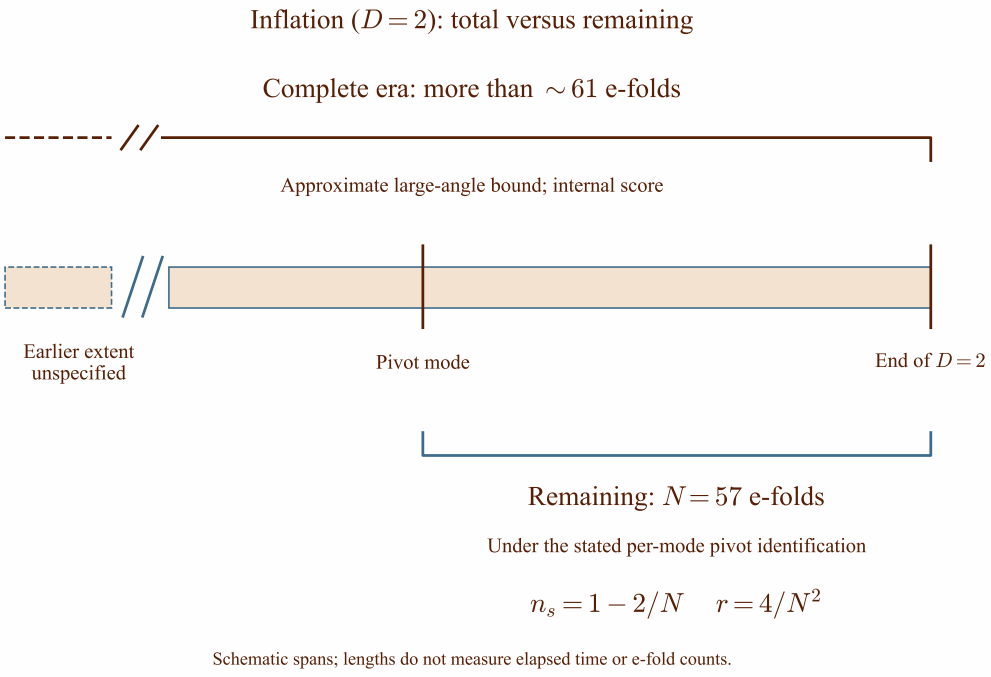


Figure 16: *Two spans of inflation. The full D=2 era extends beyond the approximate lower bound of $\sim$61 e-folds from the internal large-angle score in Section 3.7. The $N = 57$ used for the quoted CMB predictions is the capacity remaining at the pivot, the reference mode, under the stated per-mode identification. The broken left end leaves the earlier extent unspecified. An e-fold measures a factor of e in expansion; the drawn lengths are schematic.*

The channel opens in one tick (A1 + discrete D): K=1 has no spatial structure to propagate through. The potential scale is $V^{1/4} \approx 6.1 \times 10^{15}$ GeV, from $V = 3\pi^2 A_s\, r\, \bar{m}_P^4/2$ with framework-derived A_s and r. Here $\bar{m}_P = m_P/\sqrt{8\pi}$ is the reduced Planck mass used in the slow-roll normalization.

The instantaneous-equilibration reference is $T_{\rm rh} = (30/\pi^2 g_*)^{1/4} V^{1/4} \approx 2.5 \times 10^{15}\,\text{GeV}$ at $g_* = 106.75$. The species factor changes this reference only as $g_*^{-1/4}$. A delayed thermal endpoint also depends on the intervening demand dilution and expansion (Section 4.4).

The local typicality convention counts distinguishable configurations equally at fixed allocated demand, sphere by sphere. A shareable mode admits any nonnegative integer occupation; Pauli exclusion limits an identical resident mode to 0 or 1 (Appendix I.1). For the zero-chemical-potential relativistic inventory, their mean occupations are $1/(e^x - 1)$ and $1/(e^x + 1)$, where $x = \epsilon/(k_B T)$ for mode energy ϵ. The D=3 energy measure $x^3\,dx$ gives relative weights 1 and 7/8. The identified 28 shareable and 90 excluded modes therefore give $g_* = 28 + (7/8)90 = 106.75$.

The assignment of mode types is an identification, not a derivation: gauge and Higgs modes are reads; residents are the matrix's ten rows with their color, spin, and antiparticle copies, the neutrino at one helicity. The sharing rule computes the weights under this counting convention (supplementary code).

4.4 The history of the universe

Habitability requires no tuning in this construction: $N_c = 3$ is the only permitted value, and places the couplings, mass ratios, and perturbation amplitude in the range allowing chemistry and structure.

The three shells are also epochs (Fig. 1). The confining, flat, and dispersive landscapes coexist within each cell and describe the universe's successive kernel regimes. Each era retains a role in the present architecture.

D=0: before space, before time. Total bE exists without coordinates, pathfinding, or sequence. An undifferentiated pool cannot meet A4: distinguishable outcomes require ordering. One budget channel becomes a coordinate, beginning both space and time.

The D=1 era: the V. With one spatial dimension, $K = r$ makes separation costly and confines everything. Three internal channels remain, with SU(3) symmetry. That structure survives as each transistor's source shell. Budget conversion re-forces the color channels' $K = r$ every tick (Section 1.1); every proton interior is the era's fossil.

The D=2 era: inflation. A3 finds $K = 1$ cheaper than $K = r$. The second dimension opens onto a flat landscape with no force, no gradient, and perfect degeneracy: space expands for free. The S_7 cost symmetry shuffles all seven internal channels, each receiving an equal share throughout. A4 admits one new distinguishable state per e-fold. After the era's last e-fold, its information capacity is full. The largest recorded sky scales require more than ∼61 e-folds (Section 3.7).

Admission has a physical face at structural tier. Each e-fold carries a band of modes across the horizon, beyond the reach of anything still clocking. The modes leave the clocked register and freeze, billed once and persisting by A2 (Section 2.5). On this reading, the CMB is that frozen register: a latent image exposed before the universe had an optic

to focus with. Recombination develops it by releasing light, read thirteen billion years later.

The neutrino is the era's only native fermion. Its channel pattern (W^+ and existence) needs only the gate shell, making it the era's one complete native pattern. This is typing; continuous operation throughout the era is not separately derived. Its present-day mass scale is the cross-shell toll of a D=2 native measured in the later D=3 world: Eq. (10) supplies the solar-gap scale, and Section 3.7 gives its conditional positive propagation spectrum. The 47/49 EM-blind factor measures readable density among the register's occupied cells (Section 2.5). It is the local census rule of Appendix J, not a universe-wide information fraction.

The fossil record of this era is the CMB: the tilt $n_s = 1 - 2/57$, the amplitude A_s (Section 3.7), and the tensor ratio $r = 4/N^2$.

The D=3 transition: one tick. With capacity full, the third dimension opens in a single tick (A1 + discrete D). The S_7 degeneracy lifts into three distinct gauge costs, SU(3)×SU(2)×U(1), with instantaneous-equilibration reference $T_{\rm rh} \approx 2.5 \times 10^{15}\,\text{GeV}$. During D=2, K=1 permits no interactions or thermodynamics: bE is compute capacity, and temperature does not yet exist. Thus $T_{\rm rh}$ evaluates the standard formula on a pre-thermodynamic budget; no pre-existing temperature is lowered.

At the tick, the equal partition over $D^{\rm s}_{\rm max}+1 = 4$ channels becomes four cost sectors (Section 4.3). If these shares become thermal energy densities $\rho_i = V/4$ in the same volume, the conserving temperature map is $T_i/T_{\rm rh} = [g_*/(4g_i)]^{1/4}$, where $\sum_i g_i = g_*$. Equal mode shares $g_i = g_*/4$ therefore give equal temperatures $T_i = T_{\rm rh}$.

The expansion comparison also counts the interval before thermalization. Let $a_{\rm re}$ be the stretch from the opening tick to the thermal endpoint, whose temperature is $T_{\rm re}$. At fixed endpoint species inventory and with subsequent adiabatic cooling, the inherited cosmological matching gives

$$\Delta = \ln\frac{T_{\rm rh}}{T_{\rm re}} - \ln a_{\rm re}. \tag{16}$$

The four-channel candidate $\Delta = \ln 4 \simeq 1.39$ agrees with the N–$V^{1/4}$ target 1.36 ± 0.40 within 0.1σ. Its physical condition is $T_{\rm re} = T_{\rm rh}/(4a_{\rm re})$, or equivalently $\rho_{\rm re} = V/(256a_{\rm re}^4)$. Thus the total wave demand must dilute beyond the ordinary $1/a$ decrease per relativistic wave, which by itself gives $\Delta = 0$ after the volume and elapsed expansion are included. The sector-to-occupation and demand-dilution law selecting this endpoint remains to be derived (supplementary code).

Wave population and demand density. At the same zero-chemical-potential relativistic thermal endpoint, number counting uses $x^2\,dx$ rather than the energy measure $x^3\,dx$. The excluded-mode number weight is 3/4, giving

$$n_{\rm th} = \frac{\zeta(3)}{\pi^2}\frac{191}{2}\left(\frac{k_B T}{\hbar c}\right)^3, \qquad \bar{\epsilon} = \frac{\rho}{n_{\rm th}} = \frac{\pi^4}{30\zeta(3)}\frac{427}{382}k_B T \simeq 3.01938\,k_B T. \tag{17}$$

Here $191/2 = 28 + (3/4)90$ is the number-weighted mode inventory; $n_{\rm th}$ counts expected thermal wave occupations per volume, including read/radiation modes. Let $\mathcal{N}_{\rm ref} =$

$n_{\rm th}(T_{\rm rh})\mathcal{V}_0$ refer to hypothetical instantaneous $D = 3$ equilibrium in the opening volume, and $\mathcal{N}_{\rm re} = n_{\rm th}(T_{\rm re})a_{\rm re}^3\mathcal{V}_0$ to the thermal endpoint. At fixed inventory, $\mathcal{N}_{\rm re}/\mathcal{N}_{\rm ref} = e^{-3\Delta}$ and $\bar{\epsilon}_{\rm re}/\bar{\epsilon}_{\rm ref} = e^{-\Delta}/a_{\rm re}$. Thus $\Delta = \ln 4$ requires population and mean-demand ratios $1/64$ and $1/(4a_{\rm re})$, whose product gives the complete demand ratio $1/(256a_{\rm re})$. The reference is a hypothetical thermal state, not the actual prethermal $D = 2$ population; these endpoint relations do not select an annihilation or dilution history (supplementary code).

"One tick" names only the universal opening of the drain operation: every transistor admits a complete isotropic local D=3 ball-coordinate neighbourhood on that tick, without a coordinating read or a transfer of bE between addresses. The $D = 2$ state is not a finite sheet embedded in a prior $\mathbb{R}^3$ and carries no normal into the transition; the overlapping local neighbourhoods must identify equal address content to form one boundaryless spatial atlas. The statement that volume grows as r^3 refers to graph-ball volume around an address after this gluing, not to separate physical balls or an outer spherical boundary.

Synchrony follows from the era's gradient-free K=1 landscape. Every cell runs the same zeroth-order history, completing simultaneously at zeroth order in the cosmological solution's rest frame (Section 4.2). The fluctuations measured by A_s are the residual stagger (Section 3.7). The imported standard separate-universe δN identification gives $\delta N \sim \sqrt{A_s} \approx 5 \times 10^{-5}$ e-folds. Every use of the tick here is too coarse to resolve that stagger.

The bE and register content are not deposited everywhere by that opening; they remain carried by the standing waves, which resolve at successive destinations by the origin-address rule above.

The lifting has no assignment choice within the theory. Budget conversion assigns kernels timelessly (Section 1.1); the D=2 era already witnesses the assignment, with Q frozen against its unopened shell (Appendix J.2). The tick therefore lifts a degeneracy without assigning channels to shells.

Labels add no structural choice. The existence channel predates the era and stays fixed. Every assignment of the six era-active channels to shell structure $(3, 2, 1)$ is a relabeling of every other: one code-verified orbit (supplementary code). No structurally distinct alternative boot exists.

The entire D=2 population initially shares one position. Pauli exclusion drives immediate decrowding into position. At candidate tier, this neutrino-led spread is parameter-free, $L \approx 1.43/T_{\rm rh}$: the era's only complete natives move first, with lighter patterns spreading farther (supplementary code).

At the same transition the W^- cost asymmetry supplies the exact $13/6$ ratio and its structural gate angle. Matter wins conditionally if A2's ordered ledger promotes that asymmetry to the still-open rephasing-invariant loop carrier (Appendix I.8). The resulting three-dimensional regime supports gravity, stable orbits, and chemistry. The present era is the hill: $K = 1/r$, cost highest near the source and falling with distance — the dispersive landscape in which structure forms and light travels (Fig. 26 compares the ring spacings of the two kernels).

The far future, and the restart. Expansion stretches free waves below A4's read floor, where their relations cease to be computed with. This is derived for the free sector, longest wavelengths first. Bound patterns retain their internal refraction at their residents' clocks. The far-future continuation additionally assumes that their evolving response allows their demand to disperse into the free sector or fall below the external read floor.

The evaporation branch returns bound matter through black-hole boundary radiation to free waves that redshift in turn. Its horizon temperature follows the conditional thermal read of Section 4.2; the boundary coupling and complete-evaporation endpoint retain the conditions stated in Appendix P.

Isolated baryons that escape every horizon are exactly stable (Section 4.8). If no photon remains above A4's read floor, they supply no completed photonic read. Their internal refraction continues at each local tick while their photonic records receive no further completed update ("the books are put in order at the read, not along the way").

Their patterns remain written on addresses that subdivide on the void clock. Following this growth is a physical evolution of the bound wave and its refracting operator. Appendix B.2 gives the coefficient changes needed for a dilated Airy eigenprofile. The proposed ending requires such bound patterns eventually to spread below the external read floor.

The universe empties into one void, with fewer active transistors per volume and faster clocks. When no relation remains readable, no standing wave distinguishes one place from another. A3 then closes the empty channels in a sudden phase transition: $D{=}3 \to 2 \to 1 \to 0$. The drain loses charge to disperse, the gate spin to couple, the source color to confine. At $D{=}0$, the total bE drawn on by the last Hubble volume still exists, unchanged by A2, without a location. No clock runs, so no interval separates closing from reopening. A4 acts on the whole undifferentiated pool; one channel becomes spatial, and the entire bE is at the one point from which the computation restarts. This is an outlook, not a derived lifecycle, with the conditional boundary stated below.

Conditional caveat: this reads A4's "distinguishable outcome" as externally distinguishable, requiring the photonic read; the paper's treatment of the photon as the unique read supports this, but internal gluon exchanges are not explicitly excluded.

4.5 Dark matter

Dark matter is the drain's charge for referencing the source: $C_{\text{book}} = -c^2/D = -1/3$ at $D{=}3$, one charge-shell state referencing three color-shell states at each event. This bookkeeping bE is conserved by A2's per-event balance and gravitates (T3). A reference draws the same demand at any geometry, so its comoving amount is fixed even when waves stretch (the capacity/demand distinction in Appendix B.4). The matrix separates two billing moments. Its linearly dependent existence cell (rank 6, Appendix F) marks an occupied address for free; a later event pays to reference that address. The resulting perturbation equations are those of pressureless gravitating matter:

$$\delta'_{\text{DM}} = -\theta_{\text{DM}}, \qquad \theta'_{\text{DM}} = -aH\,\theta_{\text{DM}} - k^2\Phi \tag{18}$$

At the linear CMB epoch ($\delta \sim 10^{-5}$), bookkeeping bE grows perturbations like CDM. Its density ratio uses the relation rule's allocation block, distinct from the resident mass bill. Before geometry, the $D{=}0$ pool has four indistinguishable internal budget labels. Their complete allocation block contains four population slots and both orientations of six cross-channel relations: $4 + 2\binom{4}{2} = 16$. A2 conserves the bE allocated over this once-written block; particles do not own separate capacities. The $D{=}0$ slot remains as existence without computing as space (exists $\neq$ computes), leaving $D^{\mathrm{s}}_{\max} = 3$ active spatial dimensions (A3). Allocation relations per active dimension give:

$$\frac{\Omega_{\mathrm{DM}}}{\Omega_b} = \frac{(D^{\mathrm{s}}_{\max}{+}1)^2}{D^{\mathrm{s}}_{\max}} = \frac{(N_c{+}1)^2}{N_c} = \frac{16}{3} = 5.33 \tag{19}$$

giving $\Omega_{\mathrm{DM}}h^2 = (16/3)\,\Omega_b h^2 = 0.119$ (Planck 0.1200 ± 0.0012, 0.9σ).

At nonlinear scales ($\delta \gg 1$), the relation rule acts on distinct spatial addresses. This read excludes self-citations, unlike the muon's resident bill (Appendix I.10):

$$\rho_{\mathrm{DM}}(\mathbf{x}) = A\,\Phi_b(\mathbf{x})^2 \qquad \text{(virialized halos)} \tag{20}$$

Here Φ_b is the baryon-only gravitational potential. The baryonic Tully–Fisher relation (BTFR) fixes $A = \sqrt{a_0}/(4\pi G^{5/2} M_b^{3/2})$, leaving zero free parameters per galaxy. For occupied-address amplitudes ϕ_i, define $N_{\mathrm{eff}} \equiv \Phi_b^2/\sum_i \phi_i^2$. The exact oriented cross-sum is $\sum_{i\neq j}\phi_i\phi_j = \Phi_b^2(1 - 1/N_{\mathrm{eff}})$; macroscopic support $N_{\mathrm{eff}} \gg 1$ gives Φ_b^2 with the self term absent. The ratio $16/3$ and profile Φ_b^2 apply the same grammar to different objects: conserved allocation fixes how much DM, active spatial relations fix where it goes. BTFR connects them through $a_0 = cH/\Sigma$, which encodes ω_m.

Profile shape. An extended baryonic disk of scale length R_d has $\Phi_b(r) \approx -GM_b/\sqrt{r^2 + a^2}$ ($a \approx 1.2\,R_d$), giving $\rho_{\mathrm{DM}} = A\,G^2 M_b^2/(r^2 + a^2)$: the **pseudo-isothermal (pISO) profile** $\rho_0/(1 + r^2/r_c^2)$. Flat-weight fits to an exponential disk's Φ_b^2 give $r_c \approx 1.2\,R_d$, stable across fit ranges. Alternative weightings and an enclosed-mass match span 0.97–$1.27\,R_d$ (supplementary code). This core scale also marks where $|\Phi_b|$ falls to $1/\sqrt{2}$ of its central depth; the disk's rotation-curve peak is separately at $2.15\,R_d$. T3 equilibrium thus derives the empirically preferred pISO form [18]. Its core follows the baryons, $r_c \propto R_d$. Self-interacting dark matter instead sets $r_c \propto r_s \times f(\sigma/m)$ through the halo scale radius r_s: a direct discriminator.

Cusp–core prediction. Baryonic morphology sets the inner density slope: $\to 0$ for smooth dwarfs and low-surface-brightness (LSB) galaxies, $\to -2$ for concentrated bulges. This gives diverse slopes without feedback tuning. For mass-weighted baryonic scale length R_{eff}, $\rho_0 r_c = v_{\mathrm{flat}}^2/(4\pi G\,R_{\mathrm{eff}})$ varies with galaxy mass. The reported universal value, $\rho_0 r_c \approx 140\,M_\odot/\mathrm{pc}^2$ [19], is therefore a selection effect.

The lensing comparisons and their observational assumptions are collected in Appendix C.

4.6 Two regimes

Gravitational coherences (Section 4.5) are the quantum substrate's ground state (T1/T2): they require no separate formation process, but their distance-weighted relations carry the active demand whose clock-rate gradient is gravity. Collisions destroy coherence (Fig. 17). Scattering randomizes a standing wave's gravitational phases and removes off-diagonal coherency terms, as scattering depolarizes light. Collisionless baryons retain Φ_b^2 when stellar relaxation time $t_{\rm relax} \gg t_{\rm age}$; repeatedly scattered hot or turbulent gas loses it when collision time $\ll t_{\rm age}$. Cold gas in ordered rotation counts as collisionless on the relevant timescale. The criterion is binary per component because stellar and gas collision rates differ by dozens of orders of magnitude.

The substrate maintains d^2 coherences with the wavefunction, so Φ_b^2 tracks collisionless baryons at each instant without slow dynamical relaxation. Its apparent equilibration during galaxy assembly is the growing fraction of gas that settles into ordered rotation or becomes stars. Thus $f(r)$ is the local collisionless-baryon fraction, fixed by assembly history. Restricting it to stars severely degrades gas-rich-dwarf rotation curves and collapses their cored slopes (Appendix I.10; supplementary code).

Mass-budget continuity. A2 fixes one conserved bookkeeping amount per comoving region through the 16/3 allocation (Section 4.5); environment changes its profile. The T3 equilibrium profile Φ_b^2 is available precisely when the host baryons are collisionless: references persist until the next read, so coherences settle where reads are rarer than settling. Without such a host, bookkeeping bE retains its CDM-like profile from linear growth and moves ballistically under gravity. It remains dynamically independent of gas and can collapse independently below the atomic-cooling threshold, accompanied only by uncooled gas. The two profiles share a fixed allocation, $\rho_{\rm DM} = f\,\rho_{\Phi^2} + (1-f)\,\rho_{\rm prim}$, with f the collisionless baryon fraction; nothing is created or destroyed at the boundary.

- *Galaxies* ($f \approx 1$): pure Φ_b^2 — all rotation-curve and lensing results unchanged.
- *Clusters* (dominated by the intracluster medium (ICM), $f \approx 0.15$): primordial-profile dominated (NFW).
- *Groups*: expected to interpolate; clusters define a distinct relation at $\sim$17$\times$ the galaxy acceleration scale [28]; a direct group-scale measurement is outstanding.

BTFR fixes the interior amplitude. Assigning each halo its own $(16/3)M_b$ budget is excluded: the BTFR-normalized profile reaches that amount at $r \approx (16/3)\sqrt{GM_b/a_0} \approx 22$–$57$ kpc in the far field, with extended-profile shifts of $\sim$10–25%. Lensing requires the profile or correlated mass at Mpc scales, $\sim$20–36$\times$ larger; both boundary-truncation candidates fail against KiDS (supplementary code). The 16/3 allocation of the global density matrix constrains the cosmic budget. Mpc stacks also contain correlated ambient bookkeeping, whose contamination is computed without stack-fitted parameters in Appendix C. A stellar-dominated system has $f \approx 1$ and no primordial reservoir: DF44-type ultra-diffuse galaxies therefore directly test Φ_b^2 (Section D.1).

System	Dominant baryons	Collisionless?	DM profile
Galaxies (all types)	Stars, cold gas in rotation	Yes ($t_{\rm relax} \gg t_{\rm age}$)	Φ_b^2 (pISO cores)
Clusters	Hot ICM (85%)	No ($\lambda/R \sim 1/65$)	Primordial CDM (NFW)
Galaxy groups	Mixed (stars + hot gas)	Transitional	Intermediate
Merging clusters	ICM shocked; galaxies pass through	Mixed	CDM-like (NFW)

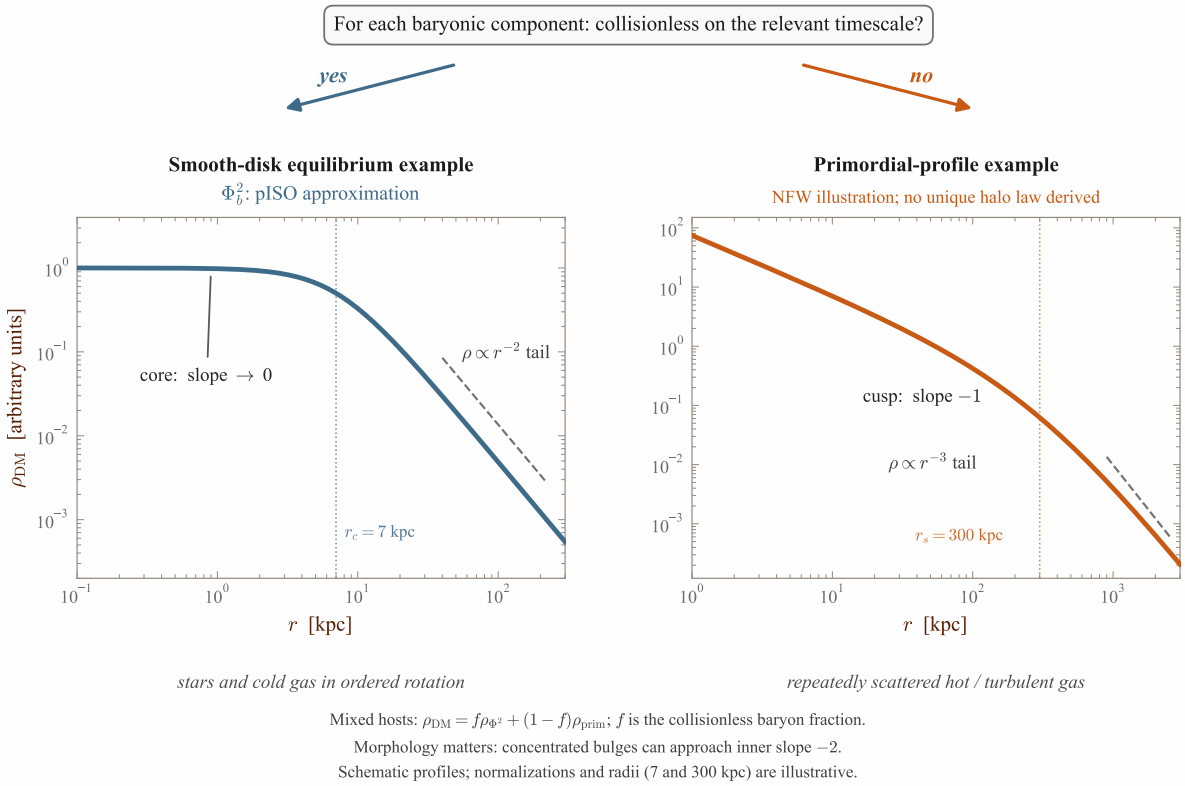


Figure 17: *Illustrative equilibrium and primordial profiles. The smooth-disk pISO curve approximates equilibrium Φ_b^2; other baryonic morphologies can give steeper inner slopes. The NFW curve illustrates a primordial component. The collision criterion applies separately to each component: collisionless components can equilibrate, while collisional components retain their primordial history, and both can coexist in one host. The radii and normalizations are illustrative.*

The galaxy, group, cluster, merger, and satellite comparisons retain their individual regimes and limitations in Appendix C.

4.7 Dark energy

Voids have less compute activity and faster clocks: $\gamma_{\rm void} = \sqrt{1 + \frac{3}{2}\delta_v}$. This factor expresses regional pacing relative to the cosmological calculation's reference clock; the fully idle substrate tick is specified in Appendix O.1. This inner 3/2 follows from A1+A3 locality: each cell reads only sources inside its own past lightcone, including the centered critical-bE load inside its apparent horizon. The outer 3/2 below is separately a

finite-depth ansatz. The exact growing spherical control gives $\operatorname{div} q = h + (4/21)h^2 + \ldots$, reaching 3/2 only for an empty void (supplementary code). Ordinary dynamics drains matter wallward while the faster interior clock accumulates more proper evolution per background time. Their joint monotonic deepening is summarized by:

$$\frac{d\delta_v}{dz} = -\frac{3}{2}\,\delta_v(1-\delta_v)\,\sqrt{1+\tfrac{3}{2}\delta_v}\;/\;(1+z) \tag{21}$$

The clock factor diagnoses read rate on a geometry; it does not determine that geometry. Identical clock and load data admit different constraint-satisfying expansion and shear (supplementary code). Appendix C.4 supplies a source-to-geometry response: activation flux sources an effective density with a gauge-verified Poisson relation to the void potential. The snapshot map from one-point load statistics to clock boost is unique to 0.01% across constraint-satisfying completions (supplementary code). Pacing remains the identification described below. Volume-weighted clock-rate differences produce a phantom crossing (Fig. 18), with the displayed reference calculation's Chevallier–Polarski–Linder (CPL) summary $(w_0, w_a) \approx (-0.70,\ -0.74)$: $w > -1$ at low redshift and $w < -1$ above. This is the quadrant DESI DR2 prefers over ΛCDM at 2.8–4.2σ across supernova samples [3].

The reference CPL point is within $\sim 1\sigma$ of Union3 and DES-SN5YR combinations, while Pantheon+'s w_0 marginal is $\sim 2.5\sigma$ lower. Joint w_0–w_a consistency is unevaluated; the marginals are strongly correlated. Every void-clock implementation crosses, but implementation freedom spans 0.45–0.73 in redshift. The analytic-transfer reference chain gives $z = 0.73 \pm 0.05$ from its census-window sweep; the common CAMB state below gives $z \approx 0.72$ as a central value. Depth implementations and fit windows shift w_0 by ~ 0.05 and w_a by ~ 0.10; the width–flux pacing family spans more (supplementary code). Falsification (Section D.2) tests crossing existence and the w_0 bound, not crossing redshift. The reference prediction uses zero fitted parameters and incurs no Bayesian complexity penalty, compared with three for $w_0 w_a$CDM. Non-parametric 2025–26 binned-$f_{\rm DE}$ reconstructions find $\rho_{\rm DE}(z)$ peaking within $0.5 < z < 1$, with $f_{\rm DE} > 0$ at $\sim 2\sigma$ across low-redshift bins [66]. Their tentative crossing at $z \approx 0.6$–0.8 [112] brackets 0.73 ± 0.05, whereas CPL projections prefer $z \approx 0.4$–0.5. The DESI 5-year test addresses this parameterization-dependent split.

Flux sourcing. A3 evaluates $K \times$ source at each location and tick, without memory. Accumulated time dilation exists by A2, but the source is computation freed this tick: flux. Flux gives a crossing; total departure from homogeneity, or stock, does not. In the paired canonical-flux reference control, the θ^* constraint independently excludes stock sourcing: its only solution has $H_0 \approx 49$ and $\Omega_m \approx 0.60$. Flux gives $H_0 = 65.6$ (supplementary code), and DESI DR2 prefers its crossing.

The effective density obeys $\dot\rho_{\rm DE} + 3H(1+w)\rho_{\rm DE} = Q$, with Q sourced by structure growth. There is no independent field equation or fundamental scalar kinetic term. The crossing occurs in Q, evading the standard field-Lagrangian phantom-crossing no-go. The common-background numerical implementation represents the resulting density shape by an effective conserved-fluid $w(a)$ and a supplied parameterized

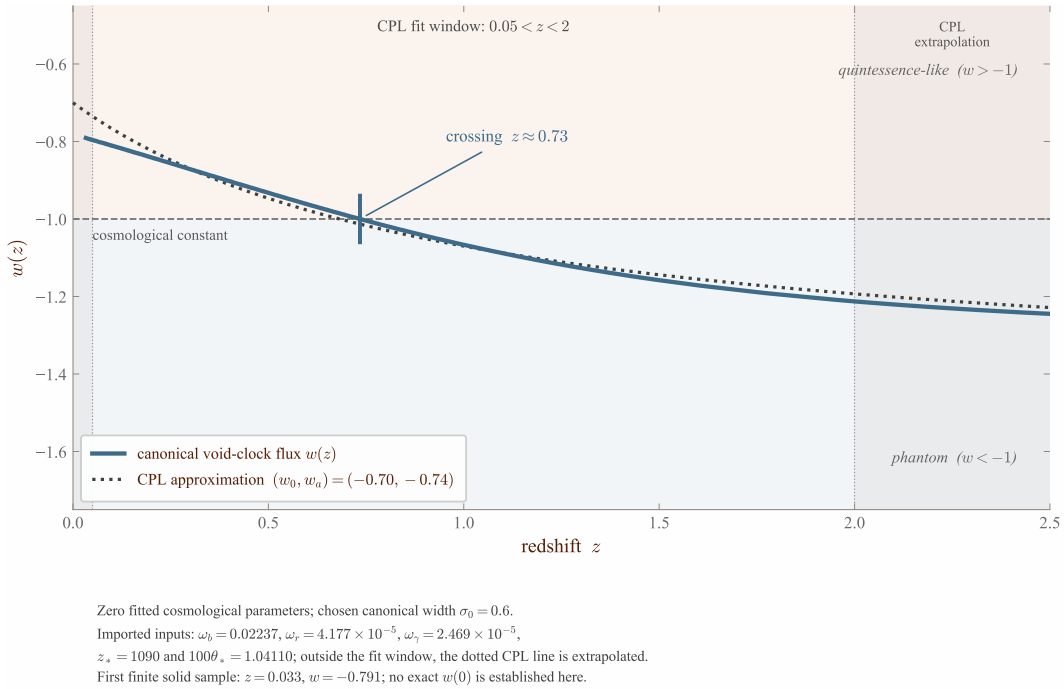


Figure 18: *Canonical-flux reference equation of state. The solid canonical-flux calculation uses $H_0 = 65.6$ fixed self-consistently by θ^* (supplementary code) and crosses $w = -1$ at $z \approx 0.73$. It uses zero fitted cosmological parameters, the implementation's chosen width $\sigma_0 = 0.6$, and the imported inputs specified in the text and code. The dotted CPL fit $(w_0, w_a) = (-0.70, -0.74)$ uses $0.05 < z < 2$; its continuation outside that interval is extrapolation. Its w_0 is the falsification-table quantity. The lowest finite solid-curve sample is $w \approx -0.79$ at $z \approx 0.033$, without fixing an exact $w(0)$. The curve is quintessence-like below the crossing and phantom above. The separate analytic-transfer reference chain's $z = 0.73 \pm 0.05$ census-window sweep (Section 4.7) is not an uncertainty band on this curve or on the common CAMB state.*

post-Friedmann (PPF) perturbation prescription; this representation does not derive the microscopic source Q.

Amplitude from N_c. The ratio $\omega_c/\omega_b = 16/3$ (Section 4.5) gives $\omega_{cb} = (19/3)\,\omega_b = 0.1412$ from derived $\omega_b = 0.0223$; present total matter is $\omega_m = \omega_{cb} + \omega_\nu$, about 0.142 including the conditional positive spectrum and its supplied relic-density conversion (Section 3.7). Exact flatness gives $\Omega_{\rm DE} = 1 - \Omega_m - \Omega_r$, reducing to $1 - \Omega_m$ when late-time radiation is neglected. The void clock supplies this complementary sector's mechanism. Canonical flux needs no further normalization: its absolute coefficient cancels against the flatness complement. The distinct hybrid implementation calibrates η_0 through $\gamma_w(0) = 1/\sqrt{\Omega_m}$ (supplementary code). Both have $\Omega_m = \omega_m/h^2$, dependent on H_0.

Radiation. The photon background enters through the imported standard density $\rho_\gamma \propto a^{-4}$, while massive neutrinos pass from relativistic to nonrelativistic behavior through the supplied thermal evolution (Appendix I.10). Established radiation machinery takes over here, as for BBN yields (Section 3.7). A2 conserves capacity and per-event balance; ρ_γ measures demand. Stretch lowers each wave's per-tick draw by the endpoint law $1+z$, so the a^{-4} dilution leaves the conserved stock intact.

Appendix C gives the local-clock reconstruction and geometry-dependent comparisons.

Common background and growth. The conditional positive rest masses $\beta_\nu(1,2,10)$, with $\sum_i m_i = 65.2\,\mathrm{meV}$, enter one CAMB background and linear total-matter transfer calculation. Iterating the volume-weighted clock flux on this same expansion history gives $H_0 \approx 65.31\,\mathrm{km/s/Mpc}$, $\Omega_m \approx 0.334$, $\sigma_8 \approx 0.792$, $S_8 \approx 0.835$ and a crossing near $z \approx 0.72$ (supplementary code). The calculation retains the reference $\omega_b = 0.02237$, $\omega_c/\omega_b = 16/3$, the stated primordial inputs, $100\theta^* = 1.04110$, and supplied equal thermal populations with $N_{\mathrm{eff}} = 3.044$. Each rest mass is converted to its present thermal energy density; neutrinos add to cold and baryonic matter. The reference rule $\sigma_{\mathrm{ref}}(0, R^*) = 0.6$ at $H_0 = 65.3$ gives $R^* \approx 11.90\,h^{-1}\mathrm{Mpc}$. CAMB's synchronous total-matter variance, including cold matter, baryons and massive neutrinos, supplies the width of the Eulerian mean-one log-normal clock PDF. This PDF and slicing identification, the flux normalization at $z = 0.01$ and constant continuations, the thermal prescription and PPF response remain conditional inputs. Numerical convergence supplies central values, not a physical uncertainty budget or a joint observational fit.

S_8. Against the common-state $S_8 \approx 0.835$, Combined CMB (Planck+ACT+SPT), $0.836^{+0.012}_{-0.013}$, differs by 0.1 of its quoted standard uncertainty [61]; KiDS Legacy, $0.815^{+0.016}_{-0.021}$, by 1.3 [31]; and Planck PR4, 0.819 ± 0.014, by 1.1 [32]. DES Y6 gives 0.789 ± 0.012 [33], a 3.8σ central-value discrepancy using its quoted uncertainty alone. These comparisons do not include a theory covariance; a joint likelihood must propagate the background and transfer assumptions. DES quotes 1.8σ against Combined CMB. ACT DR6 lensing alone reports $S_8^{\mathrm{CMBL}} = \sigma_8(\Omega_m/0.3)^{0.25} = 0.818 \pm 0.022$ [118]; this differs from the conventional $S_8 = \sigma_8(\Omega_m/0.3)^{0.5}$ used above and requires a separate comparison. The separate fixed-background, fixed-total-matter CAMB control gives $\sigma_8 = 0.7888$, $\Omega_m = 0.3285$ and $S_8 = 0.8254$, reducing CDM to compensate the neutrino density. This fixed-total-matter control and the common state use different background and density conventions; their difference does not isolate the neutrino spectrum. The void clock enters growth through background $H(z)$; its activation-flux density enters perturbations through gravitational potential, not local Hubble friction (Section 4.1; Appendix C.4). The separate local-friction substitution violates A2 perturbatively (supplementary code) and gives $S_8 \geq 0.89$ under the tested weightings (supplementary code).

BAO and the common background. Distances and the drag horizon from this same state give official-covariance $\chi^2 \approx 15.5$ on 11 DESI DR2 points [3], and 15.8 on all 13 points including the quasar pair. The corresponding 11-point diagonal diagnostic is 14.6. The two-parameter fitted ΛCDM reference gives $\chi^2 \approx 9.6$ under the 11-point official covariance (supplementary code), about 5.9 below the common-state central score. These are conditional model comparisons, without a joint fit or model-evidence calculation. Planck-anchored ΛCDM gives 32.5 under that official covariance. The common calculation uses volume-averaged flux; the separate analytic-transfer mass-weighting control is excluded at $\Delta\chi^2 > 100$.

The separate width–flux pacing control evolves the window field by A3's wallward evacuation drift (Section 4.1) on its own background and returns $\sigma(0) = 0.62$ with a

crossing. Unrelated seeds, including pure matter, converge to that control's fixed point (supplementary code). The pacing identification uses the perturbation equations' linear-growth rate (Section 4.5) for the window demand. A3 bills an additive load gradient per tick, with a source-linear kernel; density nonlinearity enters the volume law. Putting it into the rate again double-counts that law and fails the loop's consistency conditions (supplementary code). Pacing by the $\sqrt{\Omega_m}$ share or GR growth index brackets the linear rate and preserves the width and crossing existence, while shifting the crossing.

In the common CAMB state the present window width is $\sigma(0, R^*) \approx 0.591$. LCDM and reference-flux starting histories reach the same numerical endpoint at the tested tolerance. The undamped expansion-map residual is below 10^{-6} on a dense redshift grid; increasing CAMB accuracy changes S_8 by about 7×10^{-5} (supplementary code). Global uniqueness and the physical uncertainty budget remain unestablished. The depth equation and $H_{\rm local}$ are distinct implementations. With the snapshot map fixed and the depth equation's $3/2$ exact only for an empty void, their remaining difference is pacing (supplementary code).

The source-to-geometry response is the activation-flux density of Appendix C.4 (supplementary code); the whole late-time sector, $w(z)$, BAO, S_8, $H_{\rm local}$, and a_0, remains conditional on Appendix O's open manifoldlike atlas and homogenization limit.

In the common state, BAO and the crossing measure the same $H(z)$ deviation through distances and effective equation of state. One θ^* constraint couples the crossing, BAO, S_8 and the central $H_0 \approx 65.31$; they cannot be adjusted independently. Table 1's separate H_0 anchor applies to particles. This comparison is θ^*-anchored: at fixed r_d, BAO alone remains ΛCDM-consistent [59].

Vacuum energy. The grammar has no Λ-shaped object: nothing bills where nothing acts. Its effective replacement has amplitude $\Omega_{\rm DE} = 1 - \Omega_m - \Omega_r$, flux-sourced redshift shape, and the activation-flux geometric response of Appendix C.4, with pacing still identified. A continuum integration constant permits Λ; the substrate supplies no such degree of freedom. The 10^{122} discrepancy confuses occupation counts with variance: the all-zero register has no occupation to bill, while zero-point energy is variance.

The MOND scale:

$$a_0 = \frac{cH_{\rm local}}{\Sigma} = \frac{cH_{\rm local}}{6} \tag{22}$$

Here $\Sigma = N_c(1)+N_c(2)+N_c(3) = 3+2+1 = 6$, deriving $a_0 \sim cH$ [48] from channels. BTFR is $v^4 = GM_b\, a_0$ [22]. Model-dependent KBC $H_{\rm local} \approx 67$–69 gives $a_0 \approx 1.09$–$1.12 \times 10^{-10}\,\mathrm{m/s^2}$, compared with $\sim 1.2 \pm 0.26$ [21] at 0.3–0.4σ. The measured quantity is the community RAR fit's $g^\dagger$, whose interpolation differs from the theory's curve. Both approach $\sqrt{a_0\, g_{\rm bar}}$ when $g_{\rm bar} \ll a_0$, so the comparison is convention-matched only in that deep limit, as for $\sin^2\theta_W$ in Section 5.1. Even there, the theory over-predicts SPARC $g_{\rm obs}$ by mean 0.04 dex: 5.9–7.5σ under galaxy bootstrap, with 0.04–0.11 dex across all accelerations. Agreement is asymptotic, not exact. This local scale uses $H_{\rm local}$ rather than the common-background central value $H_0 \approx 65.31$. Faster underdense clocks give $H_{\rm local} > H_0$ (Section 4.7).

Billing follows each host's local clock, so a_0 varies with environment. At the current modeling tier this gives an intrinsic RAR scatter floor of $\sim$0.01–0.02 dex, below every mea-

sured bound, and a $\sim$0.03 dex deep-regime void/wall offset (supplementary code). The latter is flat with radius, unlike an external-field effect's growing decline; no published analysis has tested this split.

The far-field relation is $g_{\rm obs} = g_{\rm bar} + \sqrt{a_0\, g_{\rm bar}}$, with $g_{\rm DM} = \sqrt{a_0\, g_{\rm bar}}$ from the enclosed-mass integral of Φ_b^2. Rotation curves use that integral, not its limit: LSB galaxies can reach $g_{\rm bar} \ll a_0$ before $r \gg 1.2\, R_d$. The prediction also differs measurably from MOND's simple $\mu(x) = x/(1+x)$. For 175 SPARC galaxies [4], $\chi^2/N = 30.1$ for the theory, 28.3 for MOND, and 45.4 for a fixed cored-isothermal halo in the same calculation (supplementary code). On the shared fit domain, the framework and MOND perform comparably. Both framework radius regimes use the same A from Section 4.5, hence one $\rho_{\rm DM} = A\Phi_b^2$ normalization. A paired spherical-profile calculation separates the 0.1472 dex mean RAR-offset spread into 0.1317 dex from the finite-radius/geometry substitution and 0.0155 dex from an outer-amplitude rescale (Appendix I.10). This outer-normalized control is distinct from the implementation used in the preceding χ^2 benchmark. The median disk mass-to-light ratios 0.34 and 0.43 belong to minimizing each galaxy's absolute mean log offset for the algebraic curve and MOND; the finite-integral control gives 0.68 on the same points. These are nuisance-grid diagnostics, not a determination of a_0.

An independent test uses 14 LITTLE THINGS dwarfs outside the Spitzer Photometry and Accurate Rotation Curves (SPARC) 175-galaxy sample, with primary mass models [18] and overlap removed by NED aliases. Models, scoring, and exclusion rules were fixed independently of these data (supplementary code). With no parameter changes, pooled $\chi^2/N = 25.5$ for the theory and 30.3 for MOND, with a 7–7 head-to-head tie. IC 1613 supplies 89 of 295 points and reverses the pooled ordering: without it, MOND gives 21.7 and the theory 25.6. At $n = 14$ the result is an out-of-sample tie with zero refitting. BTFR scatter is predicted as $\delta_v^{\rm BTFR} = \frac{1}{4}\log_{10}(M_b^{\rm peak}/M_b)$, a velocity residual in dex distinct from void depth δ_v. TNG100-1 gives slope 0.261 ± 0.004 versus 0.250, a 2.75σ difference, for centrals with $M_b^{\rm peak}/M_b > 1.02$ using a dispersion-based velocity proxy. The supplementary selection, API query, fit, and output are provided; rerunning requires a free TNG API key.

The normalization-free ratio $a_0(z)/a_0(0) = H(z)/H(0)$ distinguishes evolution from constancy. At $z \approx 2$, the theory and ΛCDM's similar $H(z)$ give $\approx 3\times$, while MOND holds a_0 constant. The 2026 MUSE-DARK III sample of 79 galaxies at $0.33 < z < 1.44$ [30] shows rising $a_0(z)$, disfavoring constant-a_0 MOND: $a_0(z{\sim}1) = 2.38^{+0.12}_{-0.10} \times 10^{-10}\,{\rm m/s^2}$. The $H(z)$-scaled prediction is $\sim$15–20% lower, within the $\sim$35% systematic floor from beam smearing and stellar-mass calibration. Evolution is confirmed; its amplitude remains undecided. ELT/HARMONI ($\sim$2031) will test at percent precision.

4.8 Structural consequences

The following structural consequences explain known results; they are not novel predictions.

The photon. As the matrix's read operation, $d{=}0$ carries zero internal state, proper time, and compute cost; $v{=}c$ spends the full A1 budget on space. Amplitude crosses cells

between encoding events, but no internal cargo travels: the photon is the encoding event. T1 explores connections and A3 selects the null geodesic. With no internal state to divide, a photon cannot be halved. Truncating its wave packet instead initiates a new query, with A2 producing a superposition of $0, 1, 2, \ldots$ photons [37].

Black holes. A horizon is where the stationary exterior clock rate vanishes: $\mathcal{D} = -1/2$. Under the supplied frozen-boundary rule, the register keeps its state while its internal updates stop. The exterior reads mass, charge, and angular momentum; the supplied Schwarzschild family has $r_s/\ell_P = 2M/m_P$.

A frozen register can transfer correlations through its boundary without resuming internal updates. Conditional constructions give a Page-like entropy turnover, complete escape from specified finite networks, and thermal flux on a supplied exterior. These establish distinct parts of a possible evaporation account. They require a common physical carrier, boundary coupling, evolving geometry, and energy balance before describing complete evaporation. Finite exterior curvature does not settle the interior, and a macroscopic shrinking law does not establish its endpoint. Appendix P gives the constructions and their precise domains.

Arrow of time. Each encoding event creates $(D-1)/D$ passive connections. The derived past hypothesis is $S_{\text{book}}(D{=}0) = 0$; at $D{=}3$, $1/3$ is active and reversible, $2/3$ passive and irreversible. Their dispersal accumulates as the second law. A3 consults only the present, so each pattern retains only its latest record. Every update overwrites, costing at least $k_B T \ln 2$ per bit (Landauer, from A3+A4). Superseded content spreads through environmental relations at read speed, never reassembled: A1 moves it forward and A4 requires one distinguishable outcome per tick. Holding the present is free; recovering the past means catching what left at read speed.

Ternary register. Seven channels with three inscriptions each give $N_c^{\Sigma+1} = 3^7 = 2187$ formal assignments. Removing the antiparticle sense from existence leaves an unsigned-E catalogue of $2 \cdot 3^6 = 1458$ channel inscriptions, before occupancy constraints. These are channel descriptions, not a particle census or a count of independent spin states; Appendix F.1 applies the existence constraint. Independent shells add (Rule 1): $N_c^3 = 27$ configurations from $K{=}r$ and $N_{\text{spin}} = 2$ gate states from $K{=}1$ give $27 + 2 = 29 = d_\mu$ (Appendix I.5). This counts content space, not the fewer realized matrix rows. Since 29 is prime and exceeds either shell's census ceiling, independent-shell products cannot give it; mixing is forced. The particular configuration-plus-dimension accounting follows the counting rules (supplementary code). The all-zero register is vacuum, $d{=}0$, indistinguishable from the photon; in amplitude basis it supplies the GR derivation's thermal vacuum (Section 4.2). Within this 29-content account, the $28 = N_c^3{+}1$ non-vacuum labels equal the matrix's transparent-cell count: substrate-level resonant–transparent duality.

Other structural results. A *glueball* is a source-free closed color-flux loop, the topology theorem's third legal case, with neither endpoints nor junction (Appendix I.7). It is an eligible maintained standing wave, but has no discrete register to count; gluon spin is a continuous SO(2) phase (Appendix I.1). Every mass, including the ground state, is therefore a vibrational eigenvalue of the $K{=}r$ landscape. The theory supplies $\sqrt{\sigma} = 438\,\text{MeV}$ (0.2σ, Section 3.6) and cedes J^{PC} multipliers to lattice QCD, as for radial Airy excitations

(Section 5.3). A confirmed multi-state pattern of simple kernel-integer multiples of $\sqrt{\sigma}$ would falsify this cede.

Higgs self-coupling: $\kappa_\lambda = 1$ because m_H is an SO(2) geometric ratio at K=1, rather than a potential minimum; beyond-Standard-Model (BSM) Higgs is forbidden. HL-LHC tests at ∼50%.

Annihilation: particle–antiparticle D-channel labels cancel; T1 erases their joint record, and A2 releases bE into cheapest photons (d=0).

Casimir forces: plates supply boundary-dependent mode counts, without vacuum energy density.

Superconductivity. At K=1, the $N_{\text{spin}} = 2$ spin channels cost equally while retaining distinguishable labels. Separate electron residents contribute d inter-electron diagonals; one Cooper-pair block computes its cross-relations and contributes d^2, independently of either resident mass bill. The same local-compositeness rule gives lepton Koide. Breaking that block into separate diagonals costs the energy gap; macroscopic pair-phase coherence gives the Meissner effect. Which metals possess an attractive pairing channel remains an import from Bardeen–Cooper–Schrieffer (BCS) theory. In conventional superconductors, phonons supply effective K=1 state-blindness within a three-dimensional metal's Debye energy shell. Other family members realize it through literal $D_{\text{eff}} = 2$ geometry (Section 2.6); flat-band geometry supplies it without phonons through the quantum metric, as in MATBG.

Any $D_{\text{eff}} = 2$ system with $N_{\text{spin}} = 2$ accessible spin states realizes a K=1 collective state: geometric pairing or another family member, including FQHE-class liquids (Section 2.6). Interaction structure chooses the state, with the attractive-channel condition still imported. A 2025 spin- and valley-polarized quarter metal measurement shows flat-band chiral superconductivity without an accessible spin doublet [86]; the doublet route is sufficient, not necessary. For spin-degenerate systems, the operational discriminator is superfluid stiffness above the single-band Fermi-liquid estimate (Table 11): conventional intraband stiffness vanishes in an ideal isolated flat band. MATBG exceeds its conventional estimate by an order of magnitude, consistent in size with quantum-geometric theory [25]. The geometric/conventional split remains experimentally unresolved; the excess is the test.

Ferromagnetism. Fully ordered spins need one macroscopic register ("all up," $d = 2$); fully disordered spins need one statistical statement (independent, 50/50). Partial order carries the cost of a detailed correlation pattern, with boundary spins computed against both domains. At the Curie point, diverging correlation length makes demand span every scale. The specific-heat divergence marks this cost peak.

A3 favors order below T_c and randomness above. At T_c itself, the equilibrium state is critical; the steep cost landscape gives a singular transition rather than a broad crossover. Minimizing the total domain-wall cost gives standard surface-energy minimization.

The three kernels determine three fluctuation regimes when transformed in their own dimensions; the upper-critical-dimension rule $d_{\text{uc}} = 2\sigma$ for a $\sim 1/k^\sigma$ propagator is imported. Dispersive $K = 1/r$ gives $\sim 1/k^2$ and $d_{\text{uc}} = 4$, one above $D^{\text{s}}_{\text{max}} = N_c = 3$. Crit-

ical phenomena are thus non-trivial but perturbatively accessible ($\varepsilon = 1$), specifically at $N_c = 3$. Distance-independent K=1 is the infinite-range, mean-field-exact limit, so BCS is mean-field-exact, with Ginzburg parameter $\sim 10^{-14}$. In its own dimension, K=r also gives $\sim 1/k^2$ and $d_{\text{uc}} = 4 > 1$: confining strings fluctuate strongly. Nambu–Goto is accurate as a fluctuating two-dimensional field theory, including the Lüscher term, rather than a mean-field theory (supplementary code).

Derived known results. (The K=1 family — FQHE, QGP viscosity, CFL pairing — is catalogued with its mechanism in Section 2.6.)

Result	Framework contribution	Value	Observed
Cornell potential	$K(D{=}1) + K(D{=}3)$	$\sigma r - \alpha_s/r$	Lattice QCD
Gottfried violation	W^- as $1/(\Sigma{+}1)$ of gate modes	$\int (\bar{d} - \bar{u})\, dx = 1/7$	NMC (deep inelastic scattering, DIS): 0.147 ± 0.039 (0.1σ); E866 (Drell-Yan): 0.118 ± 0.012 (2.1σ)
R-ratio	derived $N_c \times$ derived charges	$N_c \sum Q_q^2 = 2,\ 10/3,\ 11/3$	matches by threshold[¶]

[¶]LO; with the ceded QCD correction series (Section 5.3) evaluated at $\alpha_s = 2/17$, indistinguishable from standard QCD against published continuum R measurements from 2.2 to 34 GeV — the two α_s values differ by <3% of the experimental uncertainty at every tested energy (supplementary code).

Structural predictions.

- **Proton stability**: baryon number is a topological invariant at K=r; proton exactly stable.
- **No monopoles, no unification**: gauge group from $N_c(D) = 4{-}D$; D is integer, so $N_c(D)$ cannot run.
- **Strong CP**: P-even axioms $\rightarrow \theta_{\text{eff}} = 0$; no axion. The dimensional structure localizes parity violation to D=2 alone: the effective spacetime dimensions of the three shells (1+1, 2+1, 3+1) alternate even–odd–even; chiral symmetry (the Weyl decomposition of spinors) exists in even spacetime dimensions, constraining the D=1 and D=3 gauge couplings to be vectorial. At D=2 the constraint is absent, permitting the $V{-}A$ structure that A1's clock selects.
- **Higgs hierarchy**: m_H is geometric (SO(2)), not a potential minimum.
- **No graviton**: the framework treats the Einstein equation as an equation of state, using Jacobson's thermodynamic argument [46]; GW polarizations: 2 tensor-transverse, 0 scalar.
- **No stochastic GW background from phase transitions**: m_H geometric (crossover EWPT) + matrix completeness (no BSM sector) $\rightarrow$ LISA sees only astrophysical sources.
- **EW vacuum absolutely stable**.
- **Lepton universality**: $R(K) = 1$ exactly; $R(D^*)$ at its SM value.
- **Muon** $g{-}2$: no new particles (resolved 2025, residual $\sim 0.5\sigma$).

- $d_L^{\rm GW}/d_L^{\rm EM} = 1$: no modified GW propagation.
- **Dimensionless constants exactly constant.**
- **Boltzmann brains**: every record change is a billed encoding event, and at $D{=}0$ nothing acts to pay — no fluctuation writes a brain.
- **Inhomogeneous dark energy**: ISW–galaxy cross-correlation ∼20% stronger in voids ($S/N \approx 3$ with SO × DESI ∼2028).
- **Quantum advantage**: a quantum register at $K{=}1$ (coherent) bills d^2 density-matrix coherences; decoherence transitions it to $K{=}r$ (d configurations) — the d^2-vs-d counting rule applied to information. The same mechanism that confines color also collapses quantum computers. Decoherence here is the event Appendix E names: once the environment writes a relation distinguishing the branches, they no longer share one deduplicated ledger entry, and the d^2 coherence count falls to d.

5 Tests and boundaries

The comparisons below distinguish agreement, strain, falsification, and the boundary between counts and flows. Every "(supplementary code)" claim maps to the claim-indexed `paper_code/` package (p. 11).

5.1 What works

The results comprise the 81-row table (69 hadron and lepton masses within <0.15%, isospin, CKM, and $n_s/\ell_1/\Delta m^2$), four gauge couplings, PMNS, 14 widths over 7 orders of magnitude, two nucleon charges, two moments, nuclear binding, QCD phase curvature, QGP viscosity, and four derived CMB parameters with the θ^* anchor (Section 3.7). Counting shared anchors once gives 56 independently measured predictions from 21 kernel integers (Table 2), five measured inputs, and zero continuous free parameters. Of 39 σ-scored comparisons,[3] 34 lie within 1σ, 36 within 1.3σ, and 37 within 2σ. The χ_{c0} width differs by 3.34σ; $T_{\rm QCD}$ differs by 1–5σ across lattice determinations (Section 3.6). Other exceptions retain their local tiers: τ_n conditional on g_A (Section D.1), B_c^* at 2.3σ against its V-P family (Section D.3), and θ_{23} near $\sim 2\sigma$ against NuFIT 6.1's global minimum (Section 3.2). The birefringence null is the closest falsification threshold: joint ACT+Planck gives $\sim 4.8\sigma$, or 3.5σ after dust-EB robustness cuts [87].

The photon ($d{=}0$) has no substrate-mode coupling, so vacuum dispersion, momentum diffusion, and birefringence are exactly zero at all energies. All 15/15 sign and ordering checks agree, including $K^{*0} > K^{*\pm}$ versus $\pi^{\pm} > \pi^0$. The 39-pull distribution, using the rounded table pulls, has $\chi^2/{\rm dof} = 0.61$ using the nearest $T_{\rm QCD}$ lattice value, or

[3] The code census counts each physical quantity once across six tables: eleven prediction-table pulls, CKM, fourteen widths, two nucleon charges, nuclear/QCD, and CMB (supplementary code). Duplicate entries are $|V_{us}|$, $|V_{cb}|$, $|V_{ub}|$, and the $\bar\rho$–$\bar\eta$ radius across prediction/CKM tables, and n_s, ℓ_1 across prediction/CMB tables. Percent-scored masses are excluded; interval-scored rows use their upper endpoint for this census. Prose-scored θ_{23}, the V_{cb} chain–radius fork, B_c^*, the Σ^+ moment, Ω_{cc}, and the η' width remain local comparisons; swap-price cells relate quantities already counted.

1.23 using the farthest. Under independent standard-normal pulls, the lower-tail probability for the former is 0.027 (supplementary code). Correlation among predictions alone does not lower the expectation: independent unit-variance measurement noise gives expected $\chi^2/\mathrm{dof} \geq 1$ for data-independent predictions (supplementary code). We assign no evidential weight to the narrowness; formula selection is charged in Appendix M. Data membership is fixed by the census above; disagreeing measurements remain included or disclosed. The 14 width formulas assign zero $O(\alpha_{\mathrm{EM}})$ correction to drain-mode ratios (Appendix D.4); the χ_{c0} discrepancy limits their agreement with the comparisons (Section 3.4).

Description length. For 23 SM+ΛCDM parameters independently predicted by the theory, excluding quark-mass labels on both sides (Section 3.1), the code lengths are ∼164 versus ∼344 bits for the Standard Model's 19 continuous parameters. The raw ratio is 2.1×, falling to ∼1.10× after machine-enumerated selection charges. These include three grammar-open anchor-plus-offset terms at the uncovered-row rate and radial fits at their measured ∼24-bit cost. The harshest constructible parameter-ledger accounting gives ∼0.72–0.79×, favoring the Standard Model, by charging kernel integers derived from $N_c = 3$ as free selections. These conventions bracket the comparison; Appendix M gives both ledgers and their arithmetic. The underlying ∼164-bit code covers cosmology, quantum mechanics, and gravity as well as particle parameters; selection charges increase its assessed length under the conventions above. The framework retains zero continuous free parameters throughout. A separate comparison covering the same 69 masses restores lattice QCD's quark-mass inputs; it favors the theory centrally and gives ∼0.79× at its adversarial floor.

Forcing tiers. Ground-state mass anchors follow the counting rules, with local identifications retained. Couplings have their stated qualifications; the neutral-wave counts underlying $\sin^2\theta_W$ remain open (Appendix M). Each mass anchor's 0.15% window contains one grammar-licensed integer (code enumeration; supplementary code). Orbital excitation formulas have source × gate decompositions and cross-sector sums 137 and 64; the common microscopic rule selecting each J state's participating relations remains open (Appendix K.4). Radial excitations are structurally constrained fits: the Airy spectrum has no established kernel-integer derivation, and the present fit grammar carries a measured ∼24-bit selection residual (Appendix N). Appendix M details the $\sin^2\theta_W$ ranking and three qualifications, grammar-rule perturbations, and N_c=4 comparison.

5.2 Tensions and decisive tests

Table 10 gives every strained comparison and its current status. Table 11 specifies the prediction, experiment, timeline, and falsifying observation for each test. Both tables, their qualifications, and the additional predictions are in Appendix D. The tests carry unequal risk; agreement with a structural count is distinct from an independent measurement test.

5.3 Demarcation: counts and flows

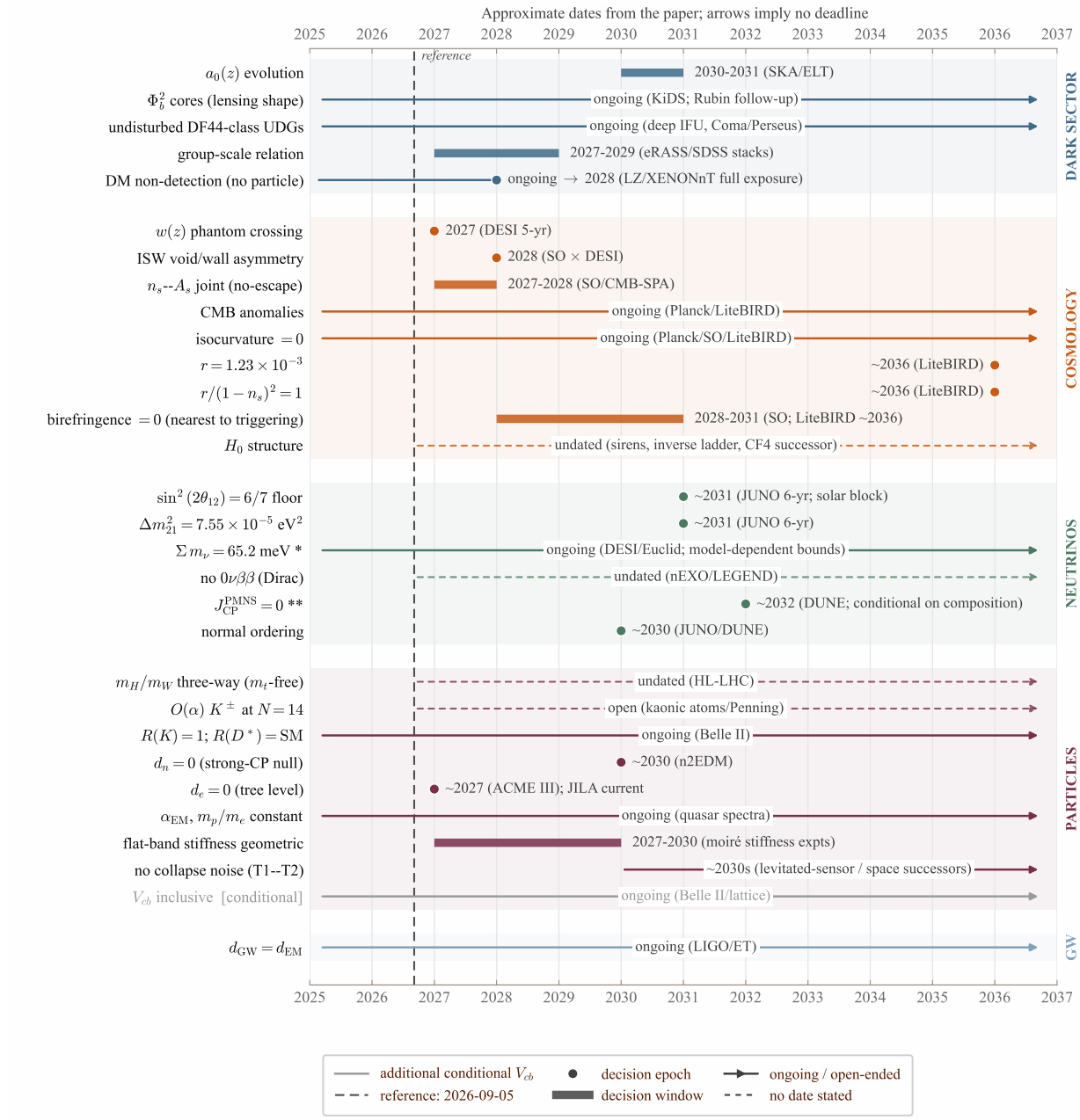
Protected counting marks the theory's boundary. Mode numbers, channel occupations, and coherence dimensions emerge as kernel integers. Quantities that run, integrate, or bind are accumulations: unprotected, and transcendental in every proven case. The single-Bessel result is exceptionless; Airy zeros permit at most one exception, with full exceptionlessness conjectural (Appendix N). These flows pass to standard solution machinery seeded by the derived constants. The boundary separates Airy radial excitations (Fig. 33, the measured ∼24-bit selection residual of Section 5.1) from SU(3) angular integers (Section 2.4); running Λ_{QCD} inherits all its evidence from same-scheme $\alpha_s = 2/17$ at 0.4σ. Heavy-quark radial excitations remain structurally constrained fits. Other ceded flows are the bracketed chiral scale B (Section 2.7), exotic bindings (Section 6), and nuclear many-body structure beyond $A = 2$. The deuteron's $B_d = 13/3$ and $a_V/B_d = 7$ are the last nuclear counting results. Downstream, standard quantum field theory (QFT) supplies RG running, the Bohr radius, $a_e = \alpha_{\mathrm{EM}}/(2\pi)$, weak decay rates, $v_{\mathrm{EW}} = 2m_W/g$, branching ratios, and chemistry.

This demarcation predicts which quantities can be integers. In the optical picture, a count describes the substrate's available modes, while a flow accumulates along a wave's particular passage.

The operation-dependent $O(\alpha_{\mathrm{EM}})$ residual structure and its limitations are stated in Appendix D.4.

Evolution and convention. Cosmological predictions are constraint-selected endpoints. Linear growth is an attractor: in matter domination, the decaying mode falls relative to growth as $a^{-5/2}$. The void functional is stateless (A3; Section 4.7); distances are quadratures of $H(z)$; H_0 solves the present θ^* constraint. The tested common-state starting histories reach one numerical endpoint, and growth's recombination mode content is the attractor output over the upstream ∼60 e-folds (supplementary code). A data comparison also requires a convention-free object summarized identically on both sides. DESI and the canonical-flux reference compare the same CPL summary of $w(z)$, as isospin ratios share conventions (Section 3.1); Λ_{QCD} inherits its same-scheme evidence from $\alpha_s = 2/17$ (Table 10). Lattice quark-mass labels retain scheme and matching dependence without a convention-free referent even within one scheme, so no values are claimed.

Four boundaries. A priced prediction must pass four independent tests. *Eligibility*: it is a computed object; lone-quark labels and sourceless vacuum variance are excluded (Section 4.7). *Integrality*: it is a protected count, while accumulations are ceded to standard machinery. *Convention*: a convention-free object underlies a comparison expressed identi-



* 65.2 meV: conditional positive-count spectrum $(1, 2, 10)$; cosmological bounds depend on the model.

** PMNS null conditional on the sequential route being the T1 composition of its primitive legs.

Figure 19: *The 29 falsification tests of Table 11, plus the additional conditional V_{cb} row (gray), grouped by domain. Points and bars show the approximate dates and windows stated in the table; arrows retain ongoing or open-ended status without implying deadlines. Dashed arrows identify tests with no stated date. The neutrino-mass sum is conditional on the positive-count completion and its mode admissibility; the PMNS null is conditional on the composed route. Both qualifications are marked below the plot. The vertical dashed line marks 5 September 2026.*

cally on both sides. *Evolution*: the result is a constraint-selected endpoint rather than an initial-data-dependent trajectory. None replaces the others. The convention test also limits the theory's use of "community label": quark-mass labels (Section 3.1) and continuum

effective-field-theory (EFT) parameters (Appendix E) require an exhibited scheme, basis, or summary ambiguity. Difficulty deriving a quantity is insufficient.

6 Limits and open science

Electroweak precision. Anchored on observed m_t, m_W and m_H move linearly by 0.047 and 0.072 GeV per 0.1 GeV of anchor. The ± 0.0077 GeV m_W measurement requires $m_t = 172.749 \pm 0.017$ GeV; m_H permits 172.73 ± 0.15. These agree at 0.15σ and lie 0.47–0.69σ above direct central values, using PDG 2026 172.60 ± 0.27 and LHC 172.52 ± 0.33 errors. Fixing the anchor at 172.60 instead gives m_W at 80.293 GeV, 9.0σ low. A direct measurement at tens-of-MeV precision below $\sim$172.7 GeV would strain the row, beyond current or planned collider precision. Once m_W fixes the anchor, the independent test is m_t-free m_H/m_W, currently at $+0.15\sigma$.

The internal ladder gives $m_t^{\text{chain}} = 176.5$ GeV, 2.2% from the anchor, through three ratios (Appendix F). Each is within $\sim 1.4\sigma$ of its PDG label, but these convention-bound comparisons are scaffolding rather than prediction rows (Section 3.1). Direct $m_t/m_c = (35/3)^2$ gives 172.86 GeV. The accumulated difference localizes to m_u and closes at $m_u = 2.141$ MeV, $\sim 0.3\sigma$ below PDG-2024 2.16 ± 0.07 MeV. Convention-independent comparisons show no drift: $V_{ub} = V_{us}V_{cb} \times 5/13$ is 0.3σ from the cited exclusive measurement, while d_μ is exact and d_τ remains a consistency label (Appendix I.5). Normal ordering retains the sign-selected mixing-operator identification; the conditional positive-count completion has $m_1 = \beta_\nu > 0$ (Section 3.7, Appendix J.3).

Mixing and mass remain distinct operations. On the conditional positive-count and composed branches, one floor-completed propagation operator gives the positive spectrum and all three final angles. The common work unit, mode admissibility, and absolute gap scale remain separately typed; Appendix J.3 gives the construction and distinguishes its two $1/3$ exponents.

The void-clock average still requires its stated effective stress-energy interpretation and continuum reduction. Appendix C retains the local averaging, curvature, and finite-model qualifications. The field-theory and transport constructions are collected in Appendix E: exact fixed-graph results do not close the full spinor, source, or growing-atlas obligations.

For the common CAMB/clock state's $H_0 \approx 65.31$ km/s/Mpc, the reference ± 1.5 km/s/Mpc spans implementations, including canonical flux at 65.6, rather than physical uncertainty for that state (Section 4.7).

Exotic hadrons. The complete topology invariants are δ ($q\bar{q}$) and ϵ_{abc} (qqq), so multiquark singlets are networks of legal composites, without a new primitive vertex (Appendix I.7). A3 favors junction-poor separated hadron pairs. Every confirmed narrow below-threshold multiquark, $\chi_{c1}(3872)$, T_{cc}^+, the three $\Sigma_c \bar{D}^{(*)}$ pentaquarks, and $P_{cs}(4459)$, lies within $\lesssim N_c \times B_d \times 3 \approx 20$ MeV of a two-hadron threshold; distant above-threshold multiquarks are ordinary resonances. Binding energies are ceded QCD flows, without

kernel-integer predictions. A narrow state isolated from every two-hadron threshold would falsify the minimum-cost reading. Lattice T_{bb}, bound $\sim$80 MeV below BB^*, is precisely such a test: observation would realize the legal compact ϵ–$\bar{\epsilon}$ chain against that reading; non-observation supports it.

Five predictions face unresolved measurement splits: V_{cb} inclusive/exclusive, τ_n bottle/beam with the nested unitarity/direct-V_{ud} fork, m_W CDF-II/ATLAS–CMS, H_0 SH0ES/TRGB, and T_{QCD} across lattice results. Each value follows its stated structure without a split-specific dial. V_{cb} uses route typing (Table 9, note ‡). Neutron lifetime follows PERKEO g_A and exact unitarity; no stated path reaches beam values, and the comparison uses the harsher UCNτ datum. The electroweak seam uses the combination whose 97% compatibility criterion excludes CDF-II. The measured-depth void band cannot reach SH0ES. The full T_{QCD} spread includes its 5σ endpoint. Identification costs, chiefly $\sim$1–2 bits for V_{cb} routing, are charged in Appendix M; the five sides are not free selections. Their joint exposure remains: adverse resolutions would give V_{cb} 6.2σ, τ_n $\sim 5\sigma$ against the beam average, m_W $\sim 7\sigma$ against CDF-II, and T_{QCD} 5σ (supplementary code). The 2026 local-ladder result already strains H_0 (Table 10).

Total bE is the one irreducible initial condition: the compute capacity with which the substrate begins. The Planck length is the transistor diameter (Section 3.3); m_e separately anchors absolute mass scale (Table 1). The axioms fix the machinery; initial bE fixes how much of it there is.

7 Conclusion

Four axioms, one transistor: three shells, seven channels, five operations. Gauge structure, generations, mixing, hadron masses, quantum mechanics, general relativity, and the dark sector describe one machine, each at its stated tier. Its constants follow from $N_c = 3$, with zero continuous free parameters and no fine-tuning or anthropic selection. The counts–flows boundary limits its reach (Section 5.3); twenty-nine falsification tests, most within the decade, expose it to experiment (Section 5).

A Detailed predictions and comparison conventions

The following table and notes contain the full particle and cosmological comparison. The interpretation of the particle classes is in Section 3.

Table 9: *Framework predictions. "Off": σ where measurement uncertainty dominates; % for precision mass matches. Notes give the comparison conventions; the following text states the evidential limits.*

#	Quantity	Formula	Predicted / Observed	Off
	CKM moduli and CP comparison (derivation tiers in Section 3.2)			
1	$\lvert V_{us}\rvert$	$(N_{\text{spin}}/N_c^2)(82/81)$	0.2250 / 0.2243 ± 0.0008	0.8σ
2	$\lvert V_{cb}\rvert$	$(N_c/73)(28/27)$	0.04262 / 0.0420 ± 0.0005‡	1.2σ
3	$\lvert V_{ub}\rvert$	$V_{us}V_{cb} \times 5/13$	0.003688 / 0.00375 ± 0.00020 (excl.)§	0.31σ
4	$\sqrt{\bar{\rho}^2+\bar{\eta}^2}$	$(N_c{+}N_{\text{spin}})/(N_c^2{+}N_c{+}1) = 5/13$	0.385 / 0.3890 ± 0.0075	0.5σ
	Isospin structure (within-generation swaps)			
5	$m_d - m_u$ (MS-bar, 2 GeV)	$5m_e$	2.555 / 2.51 ± 0.07 MeV	0.64σ
6	m_d/m_u	$13/6$	2.167 / 2.165 ± 0.094	0.02σ
	Leptons			
7	m_μ/m_e	$(29^2-14)/4$	206.750 / 206.768	0.009%
8	m_τ/m_μ	Koide	16.818 / 16.818	0.002%
	Electroweak (EW; anchored on observed m_t)			
9	m_W	$m_t\sqrt{(2\pi-\alpha_{\text{EM}})/29}$	80.37 / 80.3625 ± 0.0077 GeV	0.7σ
10	m_H	$m_t\sqrt{(\pi+\alpha_{\text{EM}})/6}$	125.15 / 125.13 ± 0.11 GeV	0.2σ
	m_Z	$m_W/\sqrt{1-\sin^2\theta_{W,\text{OS}}}$	91.19 / 91.19 GeV	$\sim 0\%^*$
	Proton and neutron			
11	m_p/m_e	$4 \times 27 \times 17$	1836 / 1836.15	0.008%
12	$m_n - m_p$	$(5m_e/2)(82/81)$	1.293 / 1.293 MeV	0.006%
13	r_p	$4\hbar/(m_pc)$	0.841 / 0.841 ± 0.0004 fm	0.04%
	Pseudoscalar mesons			
14	$\pi^\pm$	$3 \times 7 \times 13 = 273$	139.50 / 139.57	0.049%
15	π^0	$\pi - N_c^2 = 264$	134.90 / 134.98	0.054%
16	$K^\pm$	$3 \times 14 \times 23 = 966$	493.62 / 493.68	0.011%
17	η	$17 \times 63 = 1071$	547.28 / 547.86	0.106%
18	η'	$\eta + 11 \times 73 = 1874$	957.61 / 957.78	0.018%
19	$D^\pm$	$3 \times 14 \times 87 = 3654$	1867.2 / 1869.7	0.132%
20	D_s	$D + 198 = 3852$	1968.4 / 1968.4	0.001%
21	$B^\pm$	$3 \times 14 \times 246 = 10332$	5279.6 / 5279.4	0.004%
22	B_s	$3 \times 14 \times 250 = 10500$	5365.5 / 5366.9	0.027%
	Vector mesons			
23	ρ	$3 \times 2 \times 11 \times 23 = 1518$	775.7 / 775.3	0.056%
24	ω	$\rho + 14 = 1532$	782.9 / 782.7	0.024%
25	ϕ	$2K + 63 = 1995$	1019.4 / 1019.5	0.002%
26	K^{*0}	$3 \times 8 \times 73 = 1752$	895.3 / 895.6	0.031%
27	$K^{*\pm}$	$K^{*0} - 7 = 1745$	891.7 / 891.7	0.003%
	Tensor mesons			
28	f_2	$3 \times 64 \times 13 = 2496$	1275.5 / 1275.5	0.004%
29	a_2	$f_2 + 84 = 2580$	1318.4 / 1318.2	0.013%
30	K_2^*	$f_2 + 297 = 2793$	1427.2 / 1427.3	0.006%
31	f_2'	$\phi + K + 9 = 2970$	1517.7 / 1517.4	0.018%
	Heavy mesons and V-P splittings			
32	$D^{*\pm}$	$D + \pi + 7 = 3934$	2010.3 / 2010.3	0.001%

Table 9: continued

#	Quantity	Formula	Predicted / Observed	Off
33	D^{*0}	$D^* - 7 = 3927$	2006.7 / 2006.9	0.008%
34	D_s^*	$D_s + \pi + 9 = 4134$	2112.5 / 2112.2	0.013%
35	B^*	$B + 87 = 10419$	5324.1 / 5324.7	0.012%
36	B_s^*	$B_s + 87 + 9 = 10596$	5414.5 / 5415.4	0.016%
37	B_c	$36 \times 341 + 3 = 12279$	6274.6 / 6274.5	0.001%
	Charmonium ($c\bar{c}$: anchor $\eta_c = 3{\times}27{\times}72{+}25/3$, then excitations)[†]			
38	η_c	$3 \times 27 \times 72 + 25/3$	2984.4 / 2984.1	0.010%
39	J/ψ	$3 \times 28 \times 72 + 25/2$	3096.9 / 3096.9	<0.001%
40	χ_{c0}	$J/\psi + 621$	3414.2 / 3414.7	0.014%
41	χ_{c1}	$J/\psi + 810$	3510.8 / 3510.7	0.004%
42	χ_{c2}	$J/\psi + 899$	3556.3 / 3556.2	0.004%
43	h_c	$(\chi_{c0} + 3\chi_{c1} + 5\chi_{c2})/9$	3525.4 / 3525.4	0.001%
44	$\eta_c(2S)$	$\psi(2S) - 95$	3637.5 / 3637.5	0.001%
45	$\psi(2S)$	$N_c \cdot 28 \times 86 - 21/2$	3686.1 / 3686.1	<0.001%
46	$\psi(3770)$	$\psi(2S) + 171$	3773.5 / 3773.7	0.006%
	Bottomonium ($b\bar{b}$: same structural constants, different anchor)			
47	η_b	$3 \times 14 \times 438$	9400.3 / 9398.7	0.017%
48	$\Upsilon(1S)$	$3 \times 14 \times 441$	9464.7 / 9460.3	0.047%
49	χ_{b0}	$\eta_b + 899$	9859.7 / 9859.4	0.003%
50	χ_{b1}	$\eta_b + K$	9894.0 / 9892.8	0.012%
51	h_b	$(\chi_{b0} + 3\chi_{b1} + 5\chi_{b2})/9$	9900.4 / 9899.3	0.011%
52	χ_{b2}	$\eta_b + K + 36$	9912.4 / 9912.2	0.002%
53	$\Upsilon(2S)$	$\Upsilon + 1092$	10022.7 / 10023.3	0.005%
54	$\Upsilon(3S)$	$\Upsilon(2S) + 651$	10355.4 / 10355.2	0.002%
55	$\Upsilon(4S)$	$\Upsilon(3S) + 438$	10579.2 / 10579.4	0.002%
56	$\Upsilon(5S)$	$\Upsilon(4S) + 598$	10884.8 / 10885.2	0.004%
	Strange baryons (Λ anchors at $F/2$; Ξ adds 50.5 Pauli cost; the corner step $\Omega{-}\Xi$ is full F)			
57	Δ	$p + 575$	1232.0 / 1232.0	0.002%
58	Λ	$p + F/2$	1115.3 / 1115.7	0.038%
59	Σ	$\Lambda + 150$	1191.9 / 1192.6	0.062%
60	Ξ	$\Lambda + 397$	1318.1 / 1318.3	0.012%
61	Ω	$\Xi + F$	1672.2 / 1672.5	0.012%
62	Σ^*	$3 \times 30 \times 30 + 25/3$	1384.0 / 1383.7	0.019%
63	Ξ^*	$3 \times 13 \times 73 + 150$	1531.5 / 1531.8	0.022%
	Charmed baryons			
64	Λ_c	$p + 73 \times 253/7$	2286.4 / 2286.5	0.001%
65	Σ_c	2×7^4	2453.8 / 2454.0	0.006%
66	Σ_c^*	$\Sigma_c + 126$	2518.2 / 2518.4	0.008%
67	Ξ_c	$3 \times 23 \times 70$	2468.1 / 2467.7	0.017%
68	Ξ_c'	$\Xi_c + 217$	2579.0 / 2578.8	0.008%
69	Ξ_c^*	$23 \times 225 + 2$	2645.4 / 2645.5	0.003%
70	Ω_c	$3 \times 7 \times 250 + 25$	2695.5 / 2695.3	0.008%
71	Ω_c^*	$\Omega_c + 138$	2766.0 / 2765.9	0.005%
72	Ξ_{cc}	$28 \times 253 + 4$	3622.0 / 3621.6 [114]	0.011%
	Bottom baryons			
73	Λ_b	$B + F - 29$	5618.9 / 5619.6	0.012%
74	Σ_b^+	$\Lambda_b + 374$	5810.1 / 5810.6	0.009%
75	Σ_b^-	$\Lambda_b + 384$	5815.2 / 5815.6	0.008%
76	Σ_b^{*+}	$\Sigma_b + 39$	5830.0 / 5830.3	0.006%
77	Ξ_b^-	$\Lambda_b + F/2$	5796.0 / 5797.0	0.017%
78	Ω_b	$N_c(N_c^2{-}1)(2N_c^2{-}1)(N_c^3{+}N_{\text{spin}})$	6046.1 / 6045.8	0.005%

Table 9: continued

#	Quantity	Formula	Predicted / Observed	Off
Cosmology and neutrinos				
79	n_s	$1 - 2/57$	0.9649 / 0.9649 ± 0.0042	0.0σ
80	ℓ_1	acoustic-scale projection from fw $H(z)$	220.1 / 220.0 ± 0.5	0.2σ
81	$\Delta m^2_{32}/\Delta m^2_{21}$	$N^2_{\text{spin}}(N_c^2 - 1) = 32$	32 / 32.45 ± 0.49 [9]	0.9σ

Table notes. Note: pentaquark binding rows are excluded — exotic binding energies are flows (Section 5.3), not kernel-integer counts. m_W, m_H: the seam rows; see the seam discussion in Section 3.1. $^*m_Z$ uses on-shell $\sin^2\theta_W = 0.2232$ from the framework's MS-bar 117/506 via standard EW radiative corrections (a converted value — distinct from the pole-mass on-shell referent $1 - m_W^2/m_Z^2 = 0.22333$ of Appendix M); the 3.5% correction is not a free parameter. *Scheme note (isospin rows):* light-quark labels are MS-bar at $\mu = 2\,\text{GeV}$; the ratio m_d/m_u is renormalization-group invariant, so row 6 is convention-free; the difference $m_d - m_u$ runs with the quark-mass anomalous dimension, so row 5 is a scheme-matched comparison at the MS-bar 2 GeV reference, not a convention-free one (errors propagated from PDG-2024/FLAG-2024). The quoted 2.165 ± 0.094 is the PDG 2026 listings average ($m_u/m_d = 0.462 \pm 0.020$); the lattice-only determination (0.473 ± 0.017) gives 2.114 ± 0.076, moving the row's pull from 0.02σ to 0.7σ. The full isospin cell structure across the baryon table (flat + Coulomb + magnetic terms, every constant printed elsewhere in this paper) is reproduced at zero free parameters by a model included in the supplementary package (`zero_dial_isospin`); the two rows above are its comparison entries (the ratio convention-free; the difference scheme-matched, as noted). The generation ladder itself is internal scaffolding, not a comparison row — see the quark-ladder discussion in Section 3.1. ‡PDG 2026 *inclusive* determination (the PDG average mixes schemes). Inclusive (OPE, operator product expansion): 0.0420 ± 0.0005 (1.2σ); exclusive (form-factor): 0.0395 ± 0.0005 (6.2σ; against the tightest lattice-only exclusive fits, up to 8.7σ). The inclusive/exclusive V_{cb} puzzle is an open problem in quantum chromodynamics (QCD), and the 2026 tables sharpen both sides of it; the framework predicts the inclusive determination is closer to truth: the element prices the register substitution itself (Section 3.2), and the exclusive route reaches it through a single composite's form factor — a flow (Section 5.3) entering at full strength — while the inclusive sum over final states suppresses the composite-shape dependence to power corrections. §PDG 2026 average 0.00389 ± 0.00016: 1.3σ. †Alternative $\eta_c = 8\times10\times73$ fits *closer* (0.005% against the printed form's 0.010%) but lacks the $3\times N\times72$ structural link to J/ψ; A3 selects the formula that shares decomposition with the vector partner — structure over raw precision, disclosed. *Note (Ξ_{cc} row):* the observed value is LHCb's precision Ξ_{cc}^{++} measurement, $3621.55 \pm 0.23 \pm 0.30$ [114] (the 2017 observation [6]: 3621.4); LHCb's Ξ_{cc}^{+} sits at $3620.0 \pm 0.9\,^{+1.9}_{-1.3}$ [62]. *Note ($D^\pm$ row):* the printed value 3654 coincides with the D isospin-multiplet centroid $(m_{D^\pm} + m_{D^0})/2$ to 0.003% — disclosed as a numerical coincidence ($\sim 1/30$ within the row's own envelope), not read as a mechanism: the π, K, and B anchors all track the charged member, so a centroid reading for D alone would be special pleading. *Note (h rows):* the two P-wave spin singlets are the outcome-weighted centroids of their χ_J rows ($2J+1 = 1, 3, 5$ distinguishable outcomes). The singlet's spin register exists and does not compute (Section 2.5); in a P-wave that register is joint with the pair's bearing, so the state-blind gate bills every joint outcome at the counting convention's equal weight (Appendix F) — where the S-wave singlet, with no joint to trace, keeps its own bill (η_c against J/ψ). An identification, fenced to heavy quarkonium (a light P-wave computes with the gluon coherences and no gate factor: $f_2 = 3 \times 8^2 \times 13$); the unprinted $\chi_b(2P)$ centroid sits 0.35σ from $h_b(2P)$ (supplementary code). *Note (η row):* the table's one shell-wide-priced state. The η superposes words from *different* string-price classes ($u\bar{u} + d\bar{d} - 2s\bar{s}$ spans the 273- and 966-class words), so no letter cells exist to refract through,

and the wave bills the source shell's full span ($17 = 2N_c^2-1$) across the channel–color coherence space ($63 = (\Sigma+1)N_c^2$) — the one formula in the meson table with no leading N_c string factor. Same-class superpositions keep letter pricing (π^0, ω, K^0). The ϕ's additive 63 is the same span billed once atop its dominant $s\bar{s}$ pairing — span billing is amplitude-independent (the superposed register, Section 2.5) — triggered by its small non-ideal-mixing admixture. Structural reading; the η residual (-0.11%) is undressed, and 1071 is the unique licensed value in its 0.15% window. Normalization factors: $25/3=(N_c+N_{\text{spin}})^2/N_c$ (pseudoscalar), $25/2=(N_c+N_{\text{spin}})^2/N_{\text{spin}}$ (vector) — A2 conserves $(N_c+N_{\text{spin}})^2 = 25$ gauge coherences; A3 costs per active shell (color for spin-0, spin for spin-1).

Formula constraints. A4 counts the joint outcomes of simultaneously coupled registers as a product (Rule 1, Appendix I.6). Every hadron anchor is therefore a pure product by construction; Appendix M compares alternative arithmetic forms. A3 fixes each factor as the minimum count satisfying its constraint. For example, $13 = N_c^2+N_c+1$ is the minimum quark-antiquark pair cost: the compute matrix's $D=1$ cross-section contains exactly 13 quark color cells (Table 2). A 12-outcome read merges types, violating A4; the exhaustive census contains no fourteenth outcome; A3 selects the floor (supplementary code).

Dependence on N_c. At $N_c = 4$, the same rules give $m_p/m_e = 17{,}856$ (obs 1836) and $\alpha_s = 0.065$ (obs 0.118). At $N_c = 2$, the spin doublet, $\mathbb{R}$-vs-$\mathbb{C}$ argument, and lepton sector degenerate. Only $N_c = 3$ produces the observed spectrum. This is rigidity against this universe's data: the rules cannot be retuned to another N_c. It is not confirmation from data independent of the grammar's comparison set.

Among the ten kernel-table forms without forced algebraic factorization (e.g. $2N_c^2 - 1$, $N_c^3 + N_{\text{spin}}$, $N_c^4 + 1$; Table 2), 9 of 10 are prime at $N_c = 3$, against the per-form $1/\ln(n)$ expectation of 3.4. Over $N_c \in [2, 200]$, $N_c = 3$ alone reaches that count; the next highest is 7/10 under the same ladder, or 6/10 with $N_{\text{spin}} = 2$ fixed. The other eleven entries are forced powers (N_c^2, N_c^3, $(N_c + N_{\text{spin}})^2$) or explicit products and give 0/11 primes. The tenth form, N_c^4+1, has $+1$ on an odd count and is even at every odd N_c, so compositeness at $N_c = 3$ is forced. This is an algebraic correlate, not a derivation mechanism (supplementary code).

The nine primes are the vocabulary's atoms: each is one constraint's minimum register count; composite counts join registers under Rule 1. With N_{spin}, N_c, and N_c+N_{spin}, they include every prime through 31, then 73. Primes without a transistor register (37, 41, 43, 47, ...) enter only as an observer unit on a joint ($137 = 17 \times 8 + 1$, 631, 82), a blind subtraction ($47 = 49 - 2$), a difference of priced rows, or a radial fit. They supply no independent primitive register factor. Whole observer-completed blocks may contain them arithmetically, as $82 = 2 \times 41$ and the B bill's $246 = 3 \times 82$. Every hole below 1000 in the stated bounded product grammar contains an outside-atom prime; an outside-atom factor alone does not imply a hole (supplementary code).

Limits of numerical coverage. Products of 2–4 factors from Table 2 hit 32/37 PDG mass-ratio targets within 0.5% and 21/37 within 0.15%. These window counts provide no evidence for the specific integers. Random vocabularies of the same size and range give only 0.8 ± 1.3 hits at the tighter tolerance, but vocabularies also matched in expression den-

sity reproduce the counts down to 0.01% (supplementary code). The statistic measures smallness, which A3 already predicts through minimum counts.

The bare mass envelope likewise has no standalone evidential force. Above ∼170 MeV, any mass lies within 0.15% of some integer multiple of m_e, by the rounding bound $0.5\,m_e/M$. A log-uniform scan over $[273, 20{,}000]\,m_e$ (pion to 10 GeV) finds 1,207 unique products of 2–4 factors from the 21 integers, or 4,974 across the full mass-table span. They cover 33.6% of 0.15%-wide windows, rising to 36.3% in the 10^3–$10^4\,m_e$ decade. At the success criterion's ±0.15% half-width, twice that window width, coverage is 53.6% and 59.9% respectively (supplementary code). At the proton's 0.008% precision it falls to 2.1% (supplementary code). Bare recurrence across masses, widths, couplings, and moments also carries no separate weight: any dense small vocabulary reproduces it under matched formula freedom.

Related arithmetic constructions. Lenz's $m_p/m_e \approx 6\pi^5$ dates to 1951 [103]; $273\,m_e$ for the pion belongs to the pre-QCD integer-multiple literature, including Nambu's empirical $137\,m_e/2$ spectrum [104]. Eddington [105] and Wyler [106] constructed arithmetic expressions for α_{EM}. Brannen [107] exhibited the Koide family phase $\delta = 2/9$ used in Appendix I.5 as a fitted parametrization. These precedents show why numerical agreement alone is insufficient. The comparisons here therefore include null tests, per-window uniqueness enumerations, and selection-adjusted description-length accounting.

Evidence beyond coverage. The distinct checks are per-window uniqueness (one grammar-licensed expression within each anchor's window; Section 5.1), description length including selection costs (Appendix M), the N_c structure tests above, and the untested rows of Section D.3. A further generative-class test compares the vocabulary with a thousand vocabularies built from the same structural forms at random seeds. The kernel's mass-table coverage lies $\sim 8\sigma$ outside that family: 46 of 69 against the best rival's 32. Internal correlation makes geometry-generated vocabularies generically poor at tiling the spectrum; the kernel is the exception (supplementary code).

B Cell geometry and the dimensional eras

B.1 The effective cell and its surroundings

In its $D{=}3$ effective description, the substrate is a glass of close-packed Planck-diameter spheres, with no crystal order or preferred direction. The underlying field of possible addresses is fixed and pre-geometric. The $D{=}2 \to 3$ transition supplies its spatial atlas; the packing becomes meaningful then. Each sphere runs one cost function, $K(r) = r^{2-D}$, pricing influence across distance r in D spatial directions. A crystalline atlas would give light a preferred axis; none has been measured.

The cost function supplies three coextensive lenses: $D{=}1$ confines ($K{=}r$, color); $D{=}2$ is flat ($K{=}1$, spin); $D{=}3$ disperses ($K{=}1/r$, charge). All three are coextensive across the cell's resolved spatial domain (Fig. 1; Section 1.1).

At each center sits the preserved $D{=}0$ address: a point with no extent, facing, or internal sequence. It remains fixed while the surrounding wave and registers evolve. Void and resident ticks index those updates; the address has no clock of its own. Its permanence is the fixed-label structure of Appendix O.1. The spatial kernels are priced only for $r \geq \ell_P$; the point address is their reference, not another spatial shell.

In its radial description, a single transistor is an s-orbital. Its source computes no angular structure (Fig. 22), its gate is uniform, and its drain has one internal mode with a radial ladder. These radial profiles are spherically symmetric; the source-loop orientations drawn below specify no computed bearing. Directional lobes (p, d, f) arise only when neighboring cells supply a preferred axis.

B.2 Transistor drawing and mode counts

Transistor figure and channel key.

One Planck-cell transistor: an address (silver), source return loops (gold), gate (blue), and drain (plum), shown separately and superposed. The three lenses overlap throughout the cell; each wave engages the channels listed in the key. The key uses the matter-row convention of Fig. 30; antiparticles exchange W^+ and W^-. The free occupancy marker E is separate from these computing lenses. The photon carries no resident channel bill and interacts with charged patterns at a read. A composite is one maintained wave: its constituent configurations specify the engaged couplings, even when its net charge is zero. Each source loop returns to the same frozen address; its orientation and separated legs are drawing conventions. The address marker's size is symbolic. Appendix B.2 gives the mode counts and drawing conventions. Here D labels the kernel regime. The right panel shows normalized continuum kernel shapes, not a calibrated spatial interior. The finite curves $K = r$ and $K = 1$ are extended to zero for illustration; $K = 1/r$ is undefined there and is displayed only for $0.12 \leq r/R \leq 1$. The physical radius scale and resolved domain remain to be selected (Appendix B).

Resident channel engagement

Wave / configuration	Source ($K = r$)	Gate ($K = 1$)	Drain ($K = 1/r$)
Photon γ	Transparent	Transparent	Transparent
Neutrino ν	Transparent	W^+	Transparent
Charged leptons e, μ, τ	Transparent	W^+, W^-	Q
Quark configurations u, c	Occupied color entries	W^+	Q
Quark configurations d, s, b, t	Occupied color entries	W^+, W^-	Q

Figure 1 shows one address and three coextensive lenses. The $D{=}0$ address is the budget's first row: all internal, address only (Section 1.1). Each spatial lens acts across the same resolved domain; the cell is their superposition. The kernel plots show their shapes, with the continuum extensions and display cutoffs stated in the caption.

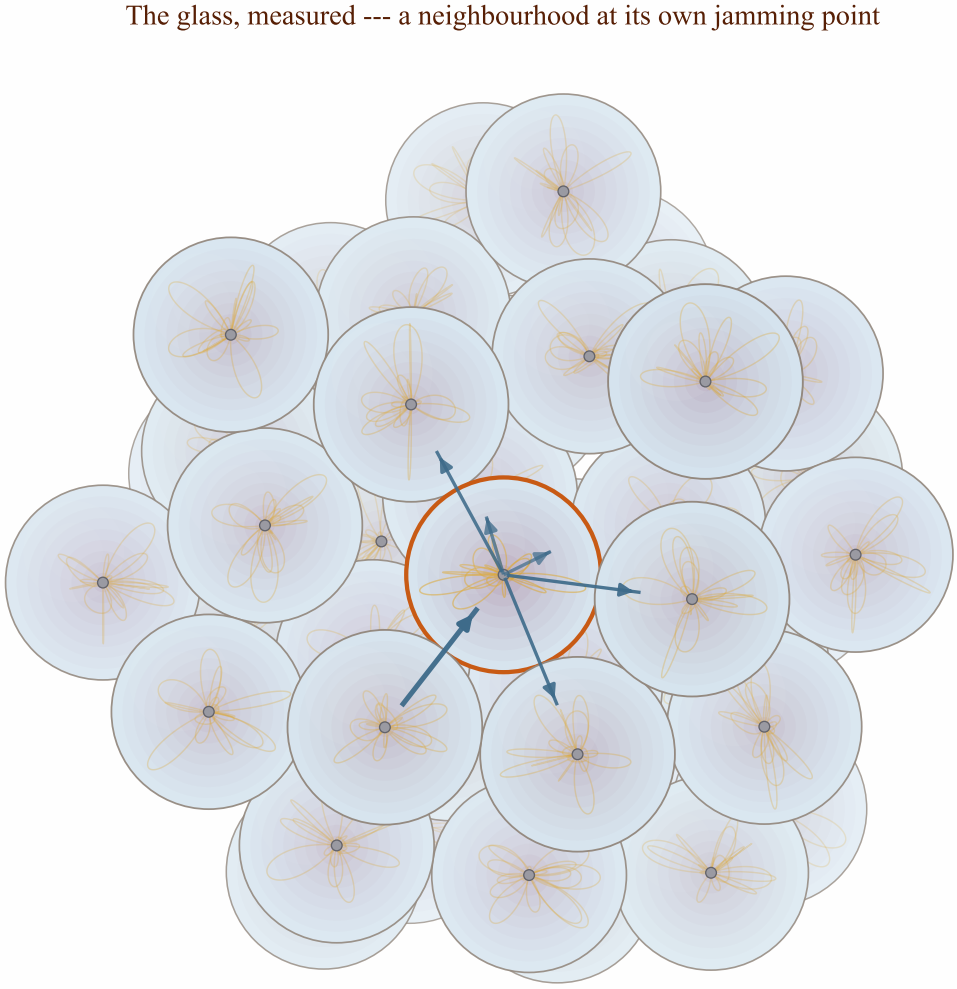


Figure 20: *A neighbourhood of cells and the loops inside one, in the D=3 effective picture.* (a) *a cut of whole cells from a periodic jamming control with* $N = 1000$ *equal Planck-diameter spheres. The measured packing has* $\phi_J = 0.6370$*, inside the random-close-packing window* 0.634–0.640 *and* 0.10% *from* $(\Sigma+1)/(2\Sigma-1) = 7/11$*; after rattlers are removed its mean rigid-backbone contact count is* $z = 6.05$*, near the isostatic value* $2D = 6$*, and its pair correlations show a split second peak with no* $\sqrt{2}$ *crystal line. Independent* $N = 1000$ *and* $N = 2000$ *controls remain inside* $\phi_J = 0.634$–0.637 *and* $z = 6.00$–6.05*. This mechanical contact graph is evidence for an amorphous effective geometry, not the literal address graph: each address has six primitive reciprocal neighbour relations, but those relations need not be sphere tangencies. The arrow overlay illustrates one exact-six address star and its amplitude hand-off. It reuses six contact positions as a display convention; this does not identify mechanical contacts with primitive address relations.*

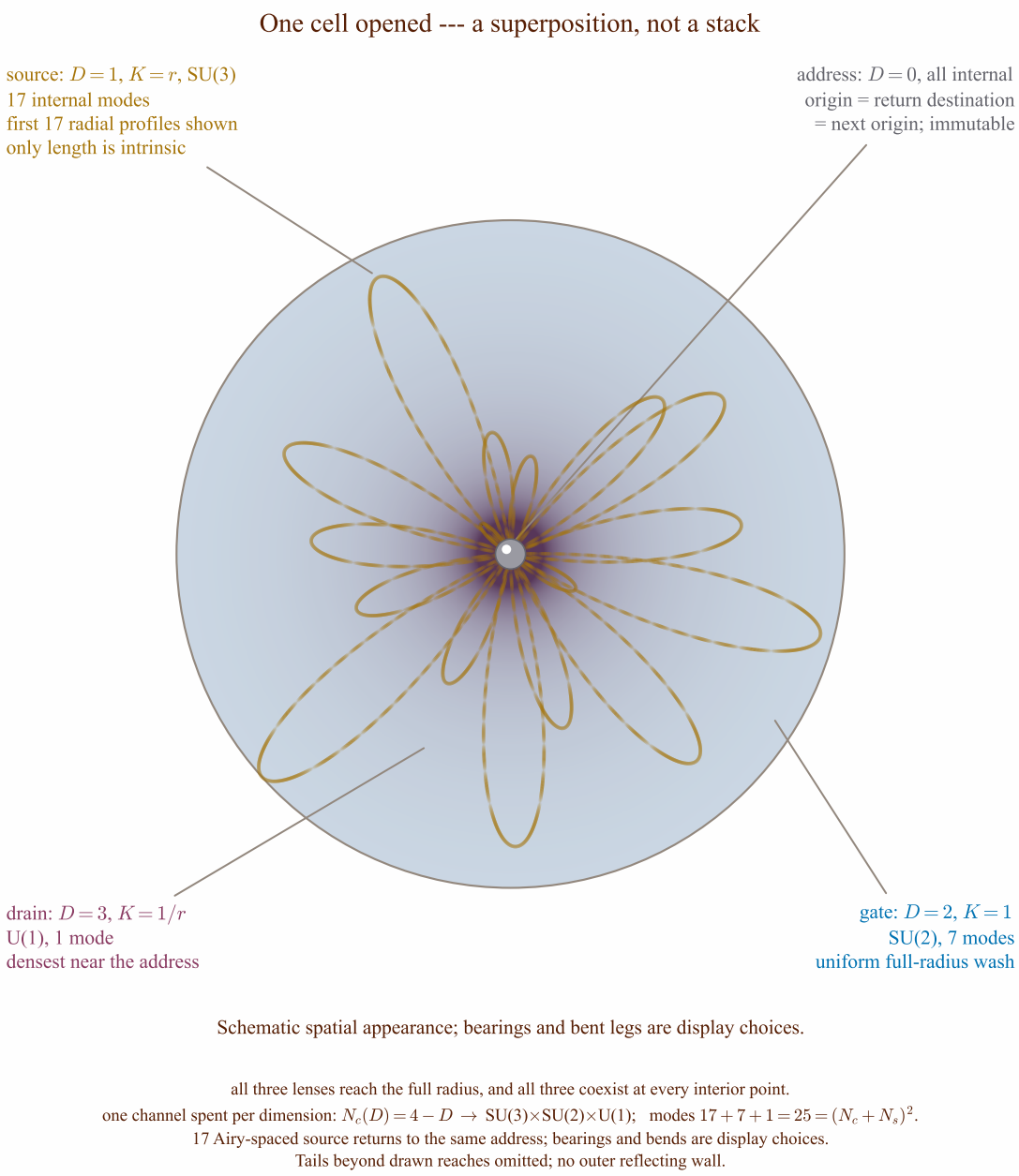


Figure 20: *A neighbourhood of cells and the loops inside one (continued).* (b) *one cell opened as in Fig. 1, with seventeen source returns to the same address; their drawn bearings and bends carry no extra D=1 information. The three lenses are coextensive, and the source-shell radial ladder continues beyond the drawn reaches (Section 2.4). The deeper substrate is pre-geometric; the figure shows its D=3 spatial appearance.*

On a supplied resolved interval, a fixed C^2 radius calibration $r = f(x) > 0$, with $f'(x) > 0$, carries the radial measure to $f^{D-1}f'\,dx$ and its operator to

$$\frac{1}{f^{D-1}f'}\partial_x\left(\frac{f^{D-1}}{f'}\partial_x\right), \qquad \frac{f^{D-1}}{f'}\partial_x f^{2-D} = 2 - D. \tag{23}$$

The second identity preserves kernel flux. The reduced one-dimensional wave instead uses measure $f'\,dx$; for kinetic coefficient A its physical current is $(2A/\hbar)\,\mathrm{Im}(\bar{U}\,\partial_x U)/f'$. Requiring the plotted source and drain power-law shapes to remain exact permits only $r = Lx$, with $L > 0$. This fixes neither L nor the physical resolved interval (supplementary code).

Active widening has a separate operator cost. For a normalized eigenmode u of the half-line source $H = -A\partial_r^2 + Br$, $A, B > 0$, with the Dirichlet origin and decaying tail of Appendix N, set $u_a(r) = a^{-1/2}u(r/a)$. Its expectation is $T/a^2 + aV$, where $T = A\int |u'|^2 dr$ and $V = B\int r|u|^2 dr$. Stationarity at $a = 1$ gives $V = 2T$, so every radial level E_n obeys

$$\frac{\langle u_a, Hu_a\rangle}{E_n} = \frac{a^{-2} + 2a}{3} = 1 + \frac{(a-1)^2(2a+1)}{3a^2} \geq 1.$$

This prices a changed profile against the retained source lens. The widened profile is an eigenmode of $-A'\partial_r^2 + B'r$ precisely when $A'/A = a^3B'/B$, with $E_n'/E_n = A'/(Aa^2) = aB'/B$. At fixed A, this requires $B' = B/a^3$ and gives $E_n' = E_n/a^2$. A changing instantaneous eigenprofile additionally requires its actual wave transport. These radial-operator relations select neither the complete proton mass nor the physical source-strength history (supplementary code).

Transport also constrains an exact profile change. On a fixed radial geometry, a complete unitary of propagation range L obeys

$$\|P_{[R,\infty)}U\psi\|^2 \leq \|P_{[\max(0,R-L),\infty)}\psi\|^2.$$

Here P_I projects onto radii in I. An Airy profile has positive tail weight on every open outer interval, so its exact normalized dilation by $a > 1$ cannot be reached at finite L. A supplied smooth bounded-displacement continuum transport can nevertheless dilate its core while retaining and translating the tail; its density error in total variation is bounded by the original tail probability beyond that core. These statements compare the same fixed physical radius; changing geometry requires its own causal identification between slices (supplementary code).

The internal mode census is $2N_c(D)^2 - 1$: 17 for the SU(3) source, 7 for the SU(2) gate, and 1 for the U(1) drain, totaling $25 = (N_c + N_{\mathrm{spin}})^2$. These are representation counts. The Airy radial index is separate: the figures use the first seventeen profiles of the unbounded radial ladder to illustrate source returns. Their sample size does not derive a correspondence between radial levels and the seventeen internal modes (Section 2.4).

Each illustrated return keeps the same origin and destination address. Length is its spatial datum; the local elastic rule is $a_{\mathrm{out}} = -a_{\mathrm{in}}$ at that frozen point. The source problem uses one address-centered distance for every mode (Appendix N, Eq. 51); its profiles

share exactly one radial node, the address (Proposition 16). Other dark nodes belong to individual excited profiles.

The drawings separate outgoing and returning legs into visible loops. Their width, bearing and bends are display choices; the Airy coordinate is the distance from the address, not the arclength of the drawn bend. Soft-turning reaches guide the icons without imposing an outer node or wall. The decaying tails continue beyond those reaches. Squared Airy amplitude sets brightness, with faint outlines carrying the route through its nodes. The address marker has symbolic size; it represents no interior. The shared colors are silver address, gold source, blue gate and plum drain. A single cell computes no preferred bearing; directional structure requires relations with neighboring cells (Appendix B).

B.3 Collective refraction

No single cell bends starlight. A wavelength spans a number of cells twenty-nine digits long, and their microscopic contributions sum into one clean refraction (Fig. 21). Matter makes the glass busy. Busy glass clocks slowly and bends waves toward it. A slower clock draws less per tick; things drift together because together is cheaper to compute. That is gravity. Light bends by twice Newton's amount because the wave's clock and the substrate's void clock contribute equally. Appendix O.1 spells out those two factors.

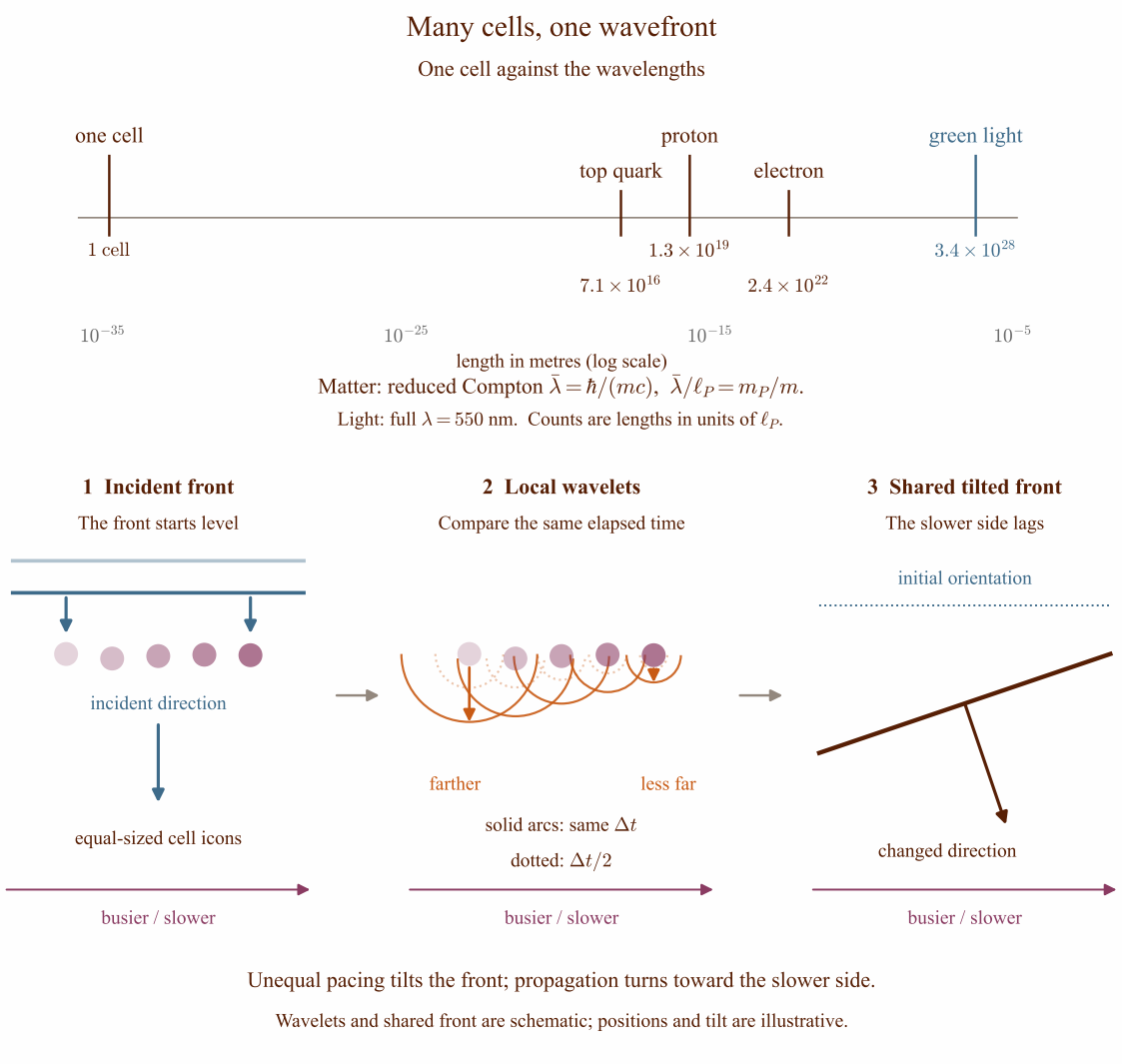


Figure 21: *Many cells, one wavefront. Top: lengths and their counts in Planck units. Matter uses the reduced Compton wavelength $\bar{\lambda} = \hbar/(mc)$, with the exact identity $\bar{\lambda}/\ell_P = m_P/m$; the electron's framework count $3^{49} \times 63/631 \approx 2.4 \times 10^{22}$ follows the gravity row (Section 3.3). The chosen green light uses its full wavelength, 550 nm, spanning 3.4×10^{28} cells. The twenty-nine-digit statement refers to that full wavelength; division by 2π gives a reduced count of 5.4×10^{27}. Bottom: an initially level front, local wavelets compared at equal elapsed time (dotted arcs at half that time), and a shared tilted front whose propagation turns toward the slower, busier side. Equal-sized cell icons, wavelets, positions and tilt are schematic. This illustrates collective refraction; it does not calculate an envelope or a gravitational bending angle.*

B.4 Symbols and technical vocabulary

- **bE** (bit energy): the universe's total compute capacity, conserved (A2) — the one value the universe begins with, worked with through its entire history (Section 6). A particle's share is a per-tick draw on that capacity: demand, not a fragment — a stretched wave draws less per tick, and the capacity is untouched.

- Σ: the channel sum across the three shells, $N_c(1)+N_c(2)+N_c(3) = 3+2+1 = 6$; with the existence register, $\Sigma+1 = 7$ — the seven channels above.
- **Kernel integer**: a mode count forced by the axioms. The principal set: 7, 8, 9, 11, 13, 14, 17, 19, 21, 23, 25, 27, 28, 29, 31, 35, 63, 73, 82, 693, 899 — each a function of $N_c = 3$ (Table 2).
- **e-fold**: one factor-of-e ($\approx 2.7\times$) stretch of space. During inflation (D=2), the universe expanded by more than $\sim$61 e-folds — each one nearly tripling the size of space — for a total stretch above $e^{61} \approx 3 \times 10^{26}$.
- **CKM / PMNS** (Cabibbo–Kobayashi–Maskawa / Pontecorvo–Maki–Nakagawa–Sakata): the tables of odds that a quark (CKM) or a neutrino (PMNS) of one family converts to another.
- $w(z)$: dark energy's pressure-to-density ratio over cosmic time; $w = -1$ is a constant vacuum, "phantom" is below -1. S_8: a single number for how clumpy matter is today. **ΛCDM**: the standard cosmological model — cold dark matter plus a cosmological constant (Λ). **MOND** (modified Newtonian dynamics): a rival approach that modifies gravity at low accelerations instead of adding dark matter.

B.5 Shell views and eras

These views resolve the shell structure used in the opening picture. Their renderings, mode counts, and display conventions are stated in the captions.

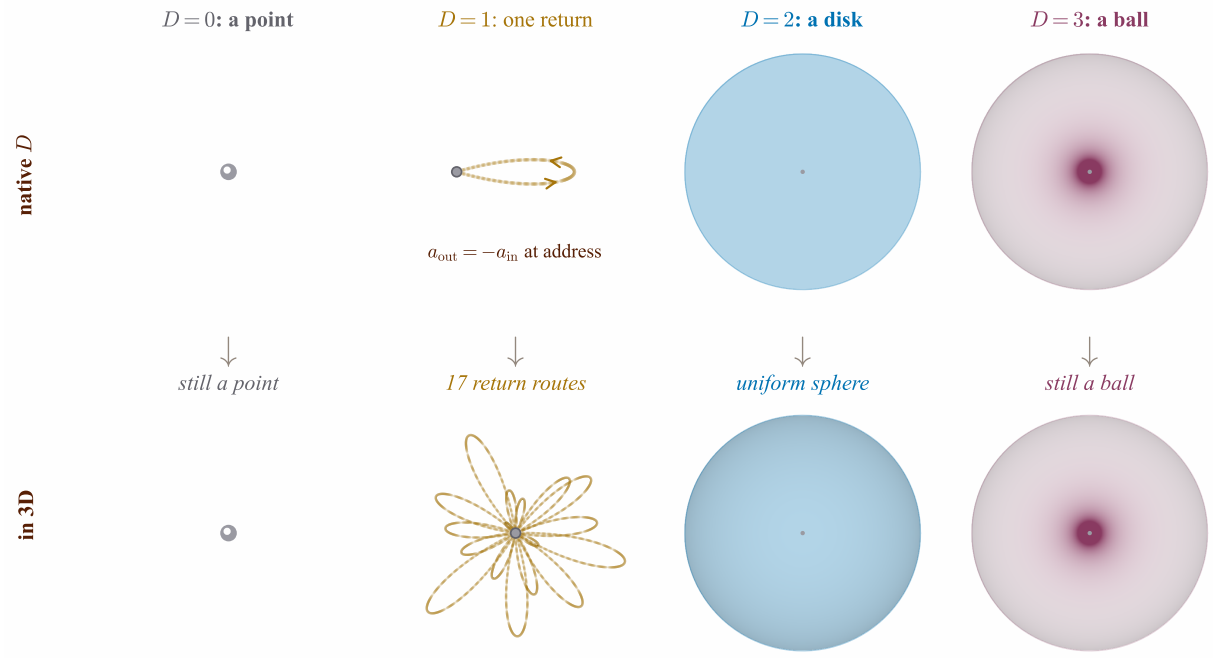


Figure 22: *Each shell in its native description (top) and its spatial appearance (bottom). D=0: a point—the address has no extent. D=1: one return (mode 17), with the local sign reversal $a_{\text{out}} = -a_{\text{in}}$ at its address; below, all seventeen returns on the Airy length ladder. Their bearings and the separation of outgoing and returning legs are drawing choices, not extra information computed in D=1. D=2: a flat disk (K=1, no radial structure); in 3D, every direction is equivalent, so it fills the volume uniformly—a sphere with no gradient. D=3: already three-dimensional; the $1/r$ gradient is unchanged. The bottom row separates the three lenses of Fig. 1. The source icons omit the decaying tails beyond their turning reaches, as in Fig. 23.*

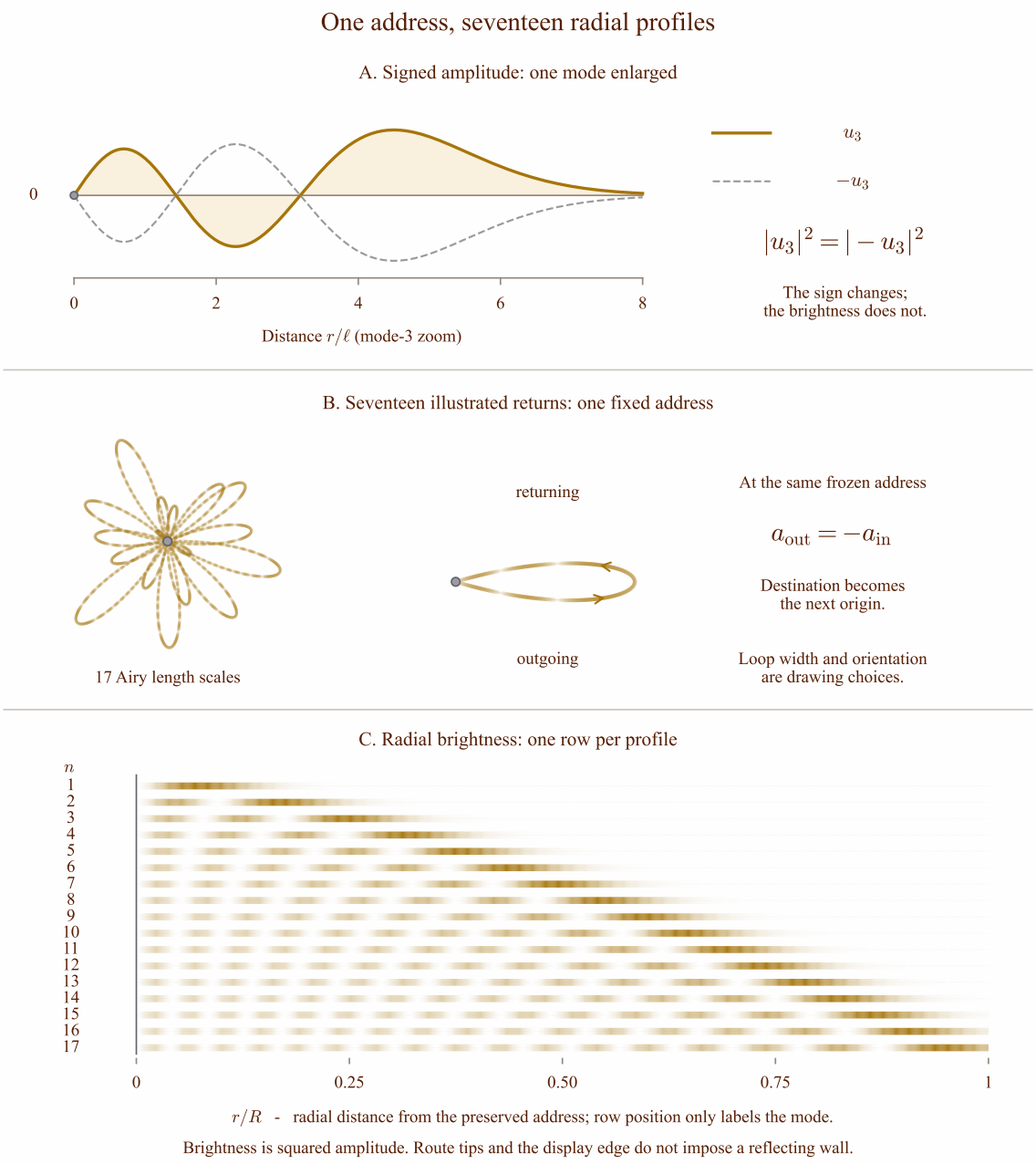


Figure 23: *One address, seventeen illustrated radial profiles (Appendix B.2). (a) The signed mode-3 profile and its negative have the same squared brightness; this overall-sign comparison is distinct from the local incoming/outgoing relation in (b). (b) All seventeen illustrated returns preserve one address and their Airy length scales. An enlarged route shows departure and return; at the frozen address $a_{\text{out}} = -a_{\text{in}}$, so the destination becomes the next origin (Appendix N). Loop orientation, curvature and the separation of the legs are drawing choices: $D{=}1$ computes length, not a bearing. Soft-turning reaches guide the icons without imposing finite outer nodes. (c) The unchanged radial brightness profiles $|\mathrm{Ai}(r/\ell - a_n)|^2$, one row per mode, with $\ell = R/a_{17}$. Each row uses its own brightness scale and mode n has n antinodes. The origin is their only shared radial node (Proposition 16); row position only labels the radial mode. The display ends at R; the half-line profiles continue with decaying tails.*

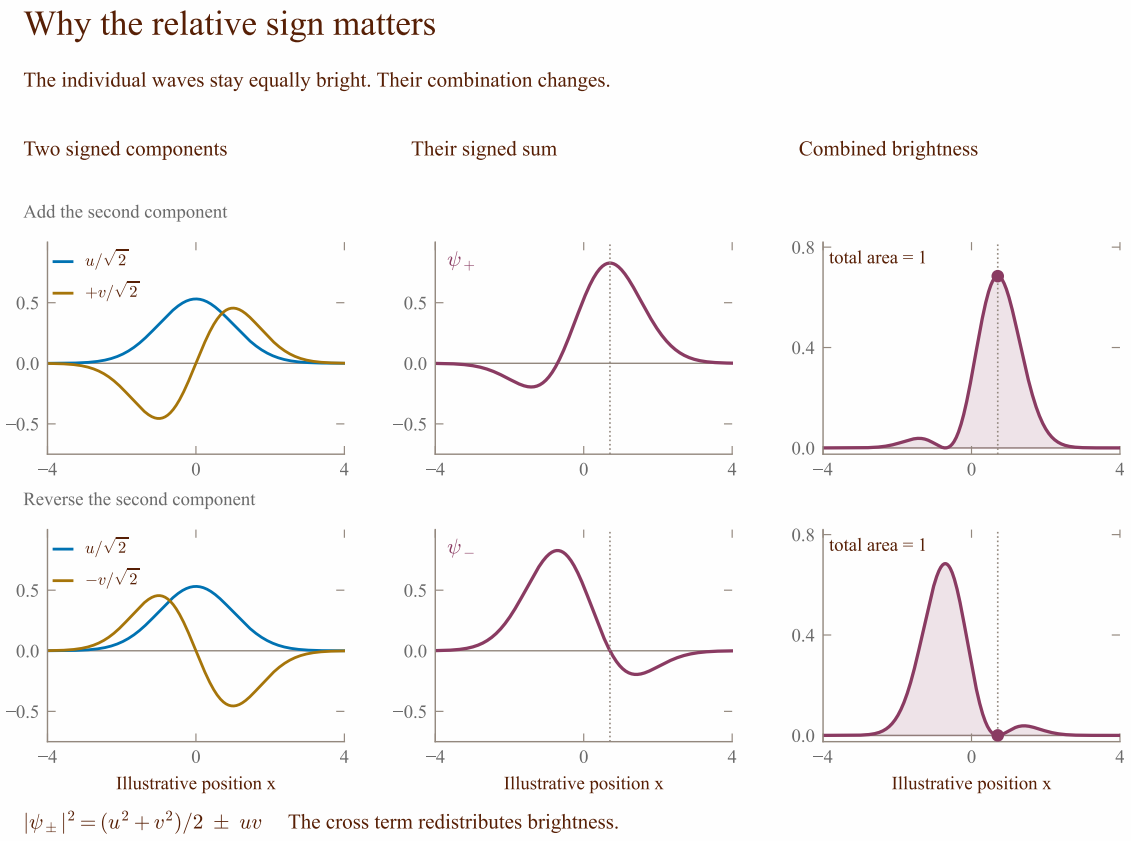


Figure 24: *Relative sign and coherent brightness. The prescribed orthonormal profiles* $u(x) = \pi^{-1/4}e^{-x^2/2}$ *and* $v(x) = \sqrt{2}xu(x)$ *give* $\psi_\pm = (u \pm v)/\sqrt{2}$*. Both states integrate to unit brightness; the cross term redistributes it. At* $x = 1/\sqrt{2}$ *the components add or cancel. Tails continue beyond the displayed range. This example assigns no physical occupation to the source modes and is distinct from the overall-sign comparison in Fig. 23.*

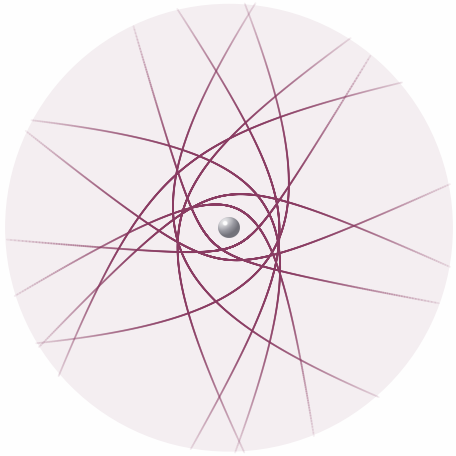


Figure 25: *The dispersive drain in the effective $D = 3$ picture. Unbound Coulomb scattering trajectories about the address illustrate the escape branch of $K = 1/r$; range fading is display opacity. The routes are schematic and do not specify a microscopic transport history. The coextensive source and gate are shown in Fig. 22.*



Figure 26: *One machine, two kernels. (a) Source ($K = r$): seventeen Airy departure–return icons, each closing at the same address. Their turning reaches follow the Airy ladder ($a_n \sim n^{2/3}$), while their bearings and bends are drawing choices. Squared Airy amplitude sets the brightness along each route, normalized per mode; the decaying tails beyond the tips are omitted. SU(3) representation counts are internal, not spatial lobes. (b) Drain ($K = 1/r$): six Coulomb s-orbital radial profiles, $n = 1, \ldots, 6$, rendered as concentric cross-sections through $42a_0$. Brightness represents $r^2|R_{n,0}|^2$, peak-normalized per mode, rather than the source panel's amplitude-squared measure. These are six levels of the single drain channel; its ladder is unbounded. Their characteristic radii grow as n^2; the SO(4) representation count is also n^2, while a single isotropic cell realizes only the s-wave (Appendix B). Opacity compression and faint route guides aid visibility; the panels do not share a quantitative density scale. The two length ladders illustrate the confining and dispersive kernels. Color key as in Fig. 1.*

C Detailed cosmological comparisons

C.1 Lensing profiles

Galaxy–galaxy lensing. KiDS-1000 excess-surface-density stacks of 259,000 isolated galaxies [23] test Φ_b^2 at 0.035–2.6 Mpc. These measure lens–matter cross-correlation, including missed satellites and extended baryons. Those two systematics account for the apparent 6.9σ shape deviation from pure pISO. The satellite term uses fixed external halo-occupation, isolation-leakage, $c(M)$, and miscentering inputs; baryon extent uses a calibrated template with three fitted shape parameters. Together they permit full BTFR amplitude with no stellar-mass shift: $\delta = +0.02$ dex is a fitted consistency check (supplementary code). Budget-boundary truncation and orbital-time saturation fronts are excluded by the same data.

Stricter isolation of the same survey yields flat circular velocities to $\sim$1 Mpc for both early and late types [24]. The shape agrees with pure Φ_b^2; the amplitude is $+0.14$ dex high, within the SPS-versus-kinematic stellar-mass band also shared with MOND (supplementary code). With systematics terms, the joint fit gives $\chi^2 = 97/57$, retaining template imperfections. On the same stacks, a Navarro–Frenk–White (NFW) halo [50] with standard $c(M)$ gives $\chi^2 = 368/56$. Freeing concentration improves it to $\chi^2 = 77.6/52$, but requires $c \approx 2$ at $M_{200} \sim 3 \times 10^{12}\, M_\odot$, $\sim 5\sigma$ below $c(M)$ and outside ΛCDM's mass–concentration relation. Independently, the stated reference Ω_m fixes a halo mass function with $\sim$12% of B21's isolated lenses in group-mass halos, $\log M \sim 12.4$–13.1. Their NFW contribution reproduces mid-radius contamination at $\chi^2 = 100/60$, with fixed external inputs and zero parameters fitted to the stacks (supplementary code).

C.2 Environmental regimes

Spirals, ellipticals, and dwarf spheroidals follow the same radial acceleration relation [27, 21]; elliptical stars are collisionless despite dispersion support, with mean free path $\sim 10^{17}$ pc. Resolved ultra-faint dwarfs instead show excess dark matter in 2025 MUSE-Faint data compared with EDGE ΛCDM simulations [11]. The survey's tidal audit calls these systems weakly processed, placing the equilibrium profile's low-mass boundary at $M_b \lesssim 10^6\, M_\odot$. Tidal depletion has a separate onset: $\rho_{1/2} \lesssim 10\, \bar{\rho}_{\rm MW}(r_{\rm peri})$ and $N_{\rm peri} \gtrsim 2$–3 passages predict under-dark systems. Across Milky Way satellites, 0/4 meeting this condition are over-dark, versus 14/20 outside it (supplementary code).

Clusters contain collisional ICM gas: $T \sim 10^8$ K and Coulomb mean free path ~ 23 kpc $\ll R_{\rm vir} \sim 1.5$ Mpc [60]. Its Φ_b^2 is negligible, leaving primordial CDM from T3. The measured cluster acceleration relation is parallel to the galaxy relation at $\sim$17$\times$ its scale [28]. The primordial-NFW branch gives $\sim$5–9$\times$, undershooting the amplitude, and local slope 0.67–0.74, 4–6σ above the measured $0.51^{+0.04}_{-0.05}$. This single-sample shape strain is shared with any NFW-based expectation (supplementary code).

Groups interpolate with collisionless baryon fraction. At $r_{500}/2$, $M_{500} \sim 10^{13}$–$10^{14}\, M_\odot$ groups lie 0.97–1.09$\times$ above the galaxy relation, decreasing monotonically with hot-gas

fraction (supplementary code). Three limits qualify this band. Gas fractions from Sun et al. [79], Lovisari et al. [80], and Akino et al. [81], with stellar fractions from Gonzalez et al. [82] and Kravtsov et al. [83], imply collisionless fractions as low as $\sim$0.13: below the grid's 0.3 floor, although the offset saturates inside the band there (supplementary code). Globular-cluster (GC) kinematics in group-central galaxies [39] probe 0.21–0.23 r_{500}, below the grid's 0.35 r_{500} validity floor, and show enhanced central accelerations toward the cluster relation. Inside $\sim$0.3 r_{500}, the point-mass baryon approximation lowers the prediction to 0.92–0.97. Separately, cluster inner slopes average $\langle\beta_{\rm DM}\rangle = 0.50 \pm 0.10\,(\text{random})\,^{+0.14}_{-0.13}\,(\text{sys})$ [20], shallower than uncored NFW (β=1, $\sim 3\sigma$ with combined errors). Cored NFW fits equally well, but the ICM-collisional branch supplies no core scale; this measurement remains a gap.

Bullet Cluster. Primordial bookkeeping dominates a merging cluster's lensing mass ($1-f \approx 0.85$). It crosses ballistically like CDM, while shocked ICM stops and the small Φ_b^2 share follows the galaxies. The 8σ gas–lensing offset [16] is consistent with this accounting but cannot distinguish it from ΛCDM. It excludes only the already-excluded all-baryon-attached reading.

Satellites and early galaxies. Below atomic cooling, primordial bookkeeping clusters hierarchically with uncooled gas, giving the same subhalo population as ΛCDM. A Milky Way–mass halo contains 63 baryonic subhalos above the cooling threshold against $\sim$60 observed (supplementary code: `satellite_luminosity_function.py`). ΛCDM with baryonic physics reaches the same census; the distinctive prediction is the collisionless-hosted profile (Table 11). For satellite planes, Φ_b^2 follows baryonic filament collapse. In 10^4 Monte Carlo realizations (`satellite_planes.py`), 68% of MW-mass halos have planes as thin as observed, versus 11% for a ΛCDM baseline with 20% filamentary accretion. The 60/30/10% tidal/filamentary/isotropic mix is a modeling input: the anisotropy direction is framework content, the rate is model-dependent. Primordial clustering also preserves ΛCDM's assembly hierarchy at $z > 9$. The theory therefore offers no distinctive reduction of the JWST excess, including RUBIES z=7.3 [14] at $>100\times$ simulations.

C.3 Local clocks and reconstruction

H_0 and local reads. Boosting $H(z)$ at $z < 2$ shortens the distance to recombination. Preserving θ^* gives the common-background central value $H_0 \approx 65.31$ km/s/Mpc. Its physical uncertainty remains to be evaluated. $H_{\rm local}$ is an observer coordinate. The separate reference-background profile estimates use the scalar prescription $d\chi/dz = c/[H_{\rm bg}E(z)R_{\rm ms}(\chi)]$ and $D_L = (1+z)\chi$, with size, depth, observer offset and sightline specified (supplementary code). Self-consistent size–depth pairs give $H_{\rm local} \approx$ 67.6–69.1, with $\pm$0.4–0.5 from depth uncertainty at fixed shape. Profile shape contributes more, and small-scale density statistics remain a sensitivity. The observer's $\sim$30 Mpc offset gives 2–3 km/s/Mpc peak-to-trough directionality. The same structures source comparable residual dipoles in standard bulk-flow analyses, leaving the two interpretations degenerate in current data.

With conserved photon number, a central observer at rest in a supplied spherical metric with areal radius C measures $D_L = (1+z)^2 C_e$. The scalar prescription represents that same read only when its redshift law agrees and $(1+z)C_e = \chi$; a regional clock profile alone does not impose these two conditions. Conversely, on a monotone regular branch, the completed read defines $\chi_L = (1+z)C_e$ and $R_{\rm rec} = c[H_{\rm bg}E(z)]^{-1}\,dz/d\chi_L$. These reconstruct its scalar distance law, but identifying $R_{\rm rec}$ with the supplied density response remains a physical condition (supplementary code).

Within the fixed-width lognormal closure, let Y be the fixed positive, mean-one lognormal density factor. The regional clock average and its reference ratio are

$$G(\Delta) = \mathbb{E}\left[\sqrt{1+\tfrac{3}{2}[1-(1+\Delta)Y]_+}\right], \qquad R_{\rm ms}(\Delta) = \frac{G(\Delta)}{G(0)}. \tag{24}$$

For nonzero width, increasing Δ strictly lowers this expectation: denser regions have less idle substructure. Thus $1/G(0) < R_{\rm ms}(\Delta) < 1$ for finite $\Delta > 0$. Holding $R_{\rm ms} = 1$ there is a one-sided approximation to this same closure, not its exact evaluation (supplementary code). The actual distribution and its width conditioned on environment remain modeling inputs.

The reconstructed Manticore-Local density field gives a smaller straight-ray path-excess proxy: +2.7%, with a 68% interval of 2.5–2.8% across 80 posterior realizations [73]. On that reconstruction's reference background this corresponds to a proxy $H_{\rm local} = 67.4$, with a 0.80 km/s/Mpc dipole (supplementary code). This statistic averages $R_{\rm ms} - 1$ from the observer along each ray, then over path lengths in the nominal 100–280 Mpc window; it uses the one-sided $R_{\rm ms} = 1$ approximation in overdense regions and is not a sample-weighted distance-modulus fit. Every realization falls below the profile values, the falsification row's +3% monopole floor, and the dipole level used in the bulk-flow-degeneracy argument. Reconstruction scatter therefore cannot explain the shortfall. Depth remains uncertain: the field uses a ΛCDM prior and one galaxy survey, and this reconstruction class under-resolves void interiors, with bias sign unbounded here. The resulting $H_{\rm local}$ lies between the θ^*-anchored ∼65 background and TRGB/JAGB (tip of the red giant branch/J-region asymptotic giant branch) values near ∼70. Interpreting SH0ES's Cepheid value 73 [15] as an unmodeled calibration difference is increasingly strained by the 2026 method-robust local-ladder consensus (Table 10). A direct void-depth measurement decides.

A CosmicFlows-4 field analysis limits a local void to $\lesssim$70 Mpc [34], well below the Keenan–Barger–Cowie (KBC) fiducial volume. Independent CF4 analyses find bulk flow ∼3.7–4.6σ above ΛCDM at 150–200 h^{-1} Mpc, without a significance claim beyond 200 h^{-1} Mpc [12]. The theory requires a +3–6% boost, compared with SH0ES's +12%. Its clock-path effect dilutes as $\Delta\chi/d$, versus $(R/d)^3$ for gravitational outflow. At KBC contrast $\delta \approx -0.46$, also preferred by CF4's velocity-amplitude proxy, capped profiles give +3.5–4.8% over the SH0ES calibration volume; the reconstructed field gives +2.7%. A shallow $\delta = -0.15$ void at 70 Mpc gives only +1.2%. The binding datum is depth at $\lesssim$70 Mpc, fixed rather than measured in the CF4 fit (supplementary code).

The $\delta \approx -0.46$ count contrast is in redshift space. Its distortion correction gives matter depth near −0.3, also reproduced by the framework's apparent clock outflow. The +3–6%

band survives both self-consistent geometries: a large count-fitted void at matter depth, or a smaller, deeper CF4-capped void. Pairing the small radius with the large void's shallow depth fails (supplementary code). Gravitational-outflow fits to CF4 also prefer ∼60–70 Mpc [34], but their $(R/d)^3$ scaling differs from the clock's $\Delta\chi/d$. Beyond the void, the clock term saturates at $z_0 \approx +0.2\%$ in the capped geometry, consistent with the 2026 BAO-frame result $-0.1 \pm 0.7\%$ [111]; that result disfavors the $+0.84\%$ of number-count-fitted outflow models. Both profile branches retain $H_{\rm local} \approx 67$–69, while the field reconstruction sits at the lower end.

C.4 The averaging and continuum boundary

Void-clock geometry. Effective $H(z)$ volume-averages local clocks. Conditional on a manifold continuum, CSG and the substrate tick supply the cosmological rest-frame slicing (Section 4.2). In the measured event web, tick levels are causal antichains and differ from order depth by Planck-scale scatter (supplementary code); the microscopic atlas remains open (Appendix O). Green–Wald [58] bounds stress-energy from averaged small metric perturbations. The void clock instead reweights local proper time against background expansion per tick, so it lies outside those assumptions while retaining the averaging question. That exemption alone supplies no acceleration term in Einstein's equations. The geometric response comes from activation flux: idle capacity drives excess address growth, sourcing $\rho_{\rm void} = \nabla^2[(N_c/(N_c{+}1))\,\delta_v]/(4\pi G)$. Its $N_c/(N_c{+}1)$ coefficient is the active budget fraction (A2+A3). On a supplied Einstein–de Sitter background, the finite-resolution coupled scalar model below and its reconstructed total stress satisfy the Hamiltonian and momentum constraints and ADM evolution at linear order in the zero-slip metric ansatz; its clipped threshold has the corresponding weak jump balance. Exact dust modes, refinement controls and quadratic amplitude scaling check the implementation of the smooth linear limit (supplementary code). Since mass is demand, underdensity δ_v is idle share relative to mean load. The strict Jensen volume–distance gap for non-degenerate density PDFs changes shape by $\leq$0.3% after hybrid η_0 calibration, giving $|\Delta H_0| \leq 0.18$ km/s/Mpc, an order below the hybrid reference implementation's stated errors (supplementary code). The fixed-background ΛCDM acoustic diagnostic at $H_0 = 65$ gives $\ell_1 = 222$, 3.9σ off. The θ^*-anchored common CAMB/clock state gives official-covariance $\chi^2 \approx 15.5$ and $S_8 \approx 0.835$ (Section 4.7). The substrate uses CSG order vocabulary, with activation required across tested growth families and intrinsic neighbour/refinement law still open. Quantum mechanics follows $A2{+}A4 \to \mathbb{C} \to$ Born rule independently of growth (Section 1, Appendix I.1), at the semiclassical architecture of QFT on curved spacetime.

A coupled continuum realization. Subdivision is paced by void time; resident load and the volume carrying it must evolve together. A finite weak-scalar realization makes this distinction explicit on a supplied Einstein–de Sitter background ($c = 1$). Let δ be the density contrast on the cosmological time slices, $\theta = \nabla\cdot\dot{\mathbf{x}}$ the coordinate velocity divergence, and $\Phi = \Psi$ the metric potential. The resident mass-stock contrast in a fixed coordinate region is $\xi = \delta - 3\Phi$, since its proper volume is $a^3(1-3\Phi)$ to first order. This stock is resident demand, not A2's conserved capacity. On a finite periodic grid,

remove the spatial mean with P, invert $-\nabla^2$ only on mean-zero fields, and define $A = 4\pi G\bar{\rho}a^2(-\nabla^2)^{-1}$ and $J = 3Ha^2A(-\nabla^2)^{-1}$. The clipped constitutive response and dust equations then form one initial-value problem:

$$\begin{aligned} &\Phi = -A\delta - J\theta + bP[-\delta]_+, \qquad b = \frac{N_c}{N_c+1}, \\ &\dot{\xi} = -\theta, \qquad \dot{\theta} = -2H\theta - a^{-2}\nabla^2\Phi, \qquad \xi = \delta - 3\Phi. \end{aligned} \tag{25}$$

Here $[x]_+ = \max(x, 0)$. On each density branch the response Jacobian restricted to mean-zero fields is $I + 3A + 3bPDP$, with $D_{ii} = 1$ in underdense cells and zero otherwise. It is bounded below by I; the associated strictly convex piecewise-quadratic functional therefore gives a unique density and potential from (ξ, θ). The $J\theta$ term retains the full comoving matter constraint: omitting it would generate a spurious effective source even when $b = 0$. All linear effective source moments follow from the same metric. Writing $Q_X = \dot{\Phi} + H\Phi - A\theta$, its pressure obeys $4\pi G p_X = \dot{Q}_X + 3HQ_X$; at a clipping transition where $\dot{\Phi}$ jumps, this includes the distributional pressure needed to balance jumps in energy and momentum. The finite-grid calculation verifies the dust limit, source conservation, time refinement and moving-resident null reads on this same evolving geometry (supplementary code). The same ray bundle gives angular-diameter and luminosity distances, including observer aberration and assuming conserved photon number; a stated distance-modulus window then defines its Hubble intercept (supplementary code). On a fixed underdense branch, high-frequency growth scales as $\sqrt{b/(1+3b)}\,k/a$ for comoving wavenumber k; finite-grid uniqueness therefore does not establish a continuum limit. That dust branch presupposes nonzero resident density. A fully idle region has no resident velocity field to perturb; its intrinsic constant idle contribution has zero spatial gradient in its interior, while geometry sourced by its surroundings remains (Appendix O.1). This is a conditional finite-resolution construction: its clipped elliptic response and fixed mean expansion do not establish a causal continuum limit, select a physical length scale, or calibrate H_{local}. Exact photon integration in a chosen weak-metric completion does not upgrade the linear matter equations to a nonlinear Einstein solution.

D Detailed tests and measurement boundaries

D.1 Tensions

Table 10: *Strained comparisons and current status. Most observational strains at or above 3σ involve measurement disagreement: CMB/BAO (n_s), surveys (S_8), bottle/beam (τ_n), lattice determinations (T_{QCD}), or UDG tracers. The UDG GC measurement itself can trigger Table 11. Exceptions to this pattern are the DF44 class, the NFW cluster slope against the single CLASH sample, and the deep-RAR offset in SPARC. The table preserves these as direct exposures.*

Quantity	Framework	Observed	Status
A_s ($\times 10^{-9}$)	2.10	2.10 ± 0.03	0.1σ (exists $\neq$ computes: 30/29)
Δm^2_{21}	7.55×10^{-5}	7.48–7.53×10^{-5} †	0.2–0.7σ (EM-blind); both post-JUNO global fits lie below the prediction within their quoted uncertainties; JUNO 6-year (± 0.04) genuinely decisive
θ_{13}	8.59°	$8.62 \pm 0.11°$ (NuFIT 6.1)	0.3σ (with EM-blind factor; 2.7σ without)
τ_n	878.22 s ($O(\alpha_{EM})$)	$877.83^{+0.30}_{-0.28}$ s (UCNτ 2025)	1.3–1.4σ with $O(\alpha_{EM})$-corrected $g_A = (65/51)(1+\alpha_{EM}/5) =$ 1.2764 (Appendix D.4); bare $g_A = 65/51$ gives 880.3 s (8.2–8.8σ). Conditional on CKM-unitarity V_{ud}; Nab decides
m_W	80.368 GeV	80.3625 ± 0.0077	0.7σ (2026 LHC–TeV combination; measurement set compatible at 97% once CDF-II [17] is excluded)
m_H	125.15 GeV	125.13 ± 0.11	0.2σ; $+\alpha_{EM}$ seam correction follows A4's record requirement (Section 3.1); m_t-free ratio $+0.15\sigma$; the three-way row remains the falsification test

Tensions, continued

Quantity	Framework	Observed	Status
n_s, ω_b	0.9649, 0.0223	CMB+DESI combinations: n_s 0.969–0.977; ω_b 0.0226 ± 0.0001	Adding DESI DR2 BAO to CMB gives 2.8–3.5σ tension; CMB-SPA+DESI gives 0.9737 ± 0.0025, 3.5σ. CMB alone remains lower: Planck PR4 0.9638 ± 0.0040, 0.3σ; SPT-3G 0.948 ± 0.012. DESI BAO is itself 2.8σ from CMB within ΛCDM. This tilt shift is conditional on that joint model: the framework's own $H(z)$ gives official-covariance BAO $\chi^2 \approx 15.5$ on 11 points with the stated n_s (Section 4.7). The exact $1 - 2/57$ remains a forward risk. Joint A_s fixes N to 57.0 ± 0.4 and n_s to ± 0.0003 under the remaining-capacity pivot reading; the imported per-mode convention centers $\sim$0.001 lower (Section 4.3). No parameter escape
$\Lambda^{(5)}_{\rm QCD}$	204 MeV	210 ± 14	0.4σ ($\alpha_s = 2/17$ at 0.4σ; 4-loop MS-bar; exponential sensitivity amplifies the α_s difference 7$\times$)
$T_{\rm QCD}$	155.0 MeV	156.5(15)–158.0(6) (lattice) [85, 52]	1–5σ depending on which lattice determination is compared — a lattice-versus-lattice spread across groups and methods; named in the Section 5.1 exceptions list
BAO χ^2 (11 pts)	15.5 (common state, official covariance)	DESI DR2	Same-state diagonal diagnostic 14.6; all 13 points give 15.8 under official covariance. Refitted ΛCDM reference: $\sim$9.6 on 11 points. No joint fit or model-evidence conclusion (Section 4.7)

Tensions, continued

Quantity	Framework	Observed	Status
$w(z)$ shape	Common-state crossing $z \approx 0.72$; reference chain 0.73 ± 0.05; reference CPL $(-0.70, -0.74)$	DESI CPL crossing $z \approx 0.4$–0.5; 2025–26 non-parametric $f_{\rm DE} > 0$ at $\sim 2\sigma$ [66], crossing $z \approx 0.6$–0.8 [112]	The reference CPL comparison is consistent at the distance level with DESI DR2's 2.8–4.2σ departure from ΛCDM; within $\sim 1\sigma$ of Union3/DES-SN5YR CPL fits and $\sim 2.5\sigma$ from Pantheon+'s w_0 marginal. Non-parametric $\rho_{\rm DE}$ peaks bracket the zero-parameter crossing; CPL places it lower. Falsification tests existence and the w_0 bound, not redshift (Section 4.7). Reanalyses in 2024–25 lower evolving-w preference below $\sim 2\sigma$ after supernova-systematic correction [113]; under that reading, the prediction awaits DESI 5-year
S_8	0.835 (conditional common state)	Combined CMB $0.836^{+0.012}_{-0.013}$; DES Y6 0.789 ± 0.012	Quoted-data errors against the conditional central value: Combined CMB 0.1σ; KiDS Legacy 1.3σ; Planck PR4 1.1σ; DES Y6 3.8σ. Theory covariance and a joint likelihood remain uncomputed (Section 4.7)
H_0	65.31 (conditional background central value)	SH0ES 73.0 ± 1.0; TDCOSMO $71.6^{+3.9}_{-3.3}$ [75]	The separate KBC profile calculation gives $H_{\rm local} \approx 67$–69, between background and TRGB; a SH0ES calibration gap is assumed (Section 4.7). Time delays measure background here because the path term saturates at $\approx +0.2\%$: their central excess is 1.9 quoted lower-side data errors above the common value, before theory uncertainty; the local boost does not explain it. SH0ES/Planck tension in ΛCDM is $\sim 5\sigma$. The 2026 117-author ladder consensus gives $H_{\rm local} = 73.50 \pm 0.81$ [91]; CCHP is absent, with TRGB/JAGB at 68–70. If this result stands, its $+13\%$ boost exceeds the clock-path $+3$–6% at every measured depth. The CF4-successor depth measurement tests this exposed local-sector boundary

Tensions, continued

Quantity	Framework	Observed	Status
GGL profile	pISO: slope -1, $q = 0$	B21 stacks: $q = -0.22 \pm 0.03$	Decomposes into named systematics (satellite term + baryon extent); with them, zero shift needed; strict-isolation sample flat to $\sim$1 Mpc [24]; locked-$c(M)$ NFW fails (χ^2 368/56)
RAR deep regime	$\sqrt{a_0\, g_{\rm bar}}$ (deep limit, convention-matched)	SPARC, $g_{\rm bar} \ll a_0$	Mean $+0.04$ dex over-prediction; 5.9–7.5σ by galaxy-level bootstrap, 0.04–0.11 dex across the full acceleration range (Section 4.7) — against one converged sample: asymptotic agreement, not exact
Cluster lensing	Primordial CDM (ICM collisional)	NFW arcs	ICM gas (collisional) contributes negligible Φ_b^2; cluster DM is primordial CDM from T3. NFW profile at standard concentrations. Against the CLASH cluster acceleration relation the same branch undershoots the $\sim$17$\times$ amplitude (5–9$\times$) and runs 4–6σ steep on slope (Section 4.6) — a strain shared with any NFW-based expectation, against a single-sample determination

Tensions, continued

Quantity	Framework	Observed	Status
DF44 σ	Φ_b^2: $\sim$12 km/s (Jeans)	33 ± 3 km/s	Open falsification class. Kinematics changed from 47 to 33 in van Dokkum 2019 [76]. GC counts differ fourfold: 21^{+7}_{-9} in Saifollahi 2021 [77], implying a normal $\sim 10^{11}\,M_\odot$ dwarf halo, versus 78.3 ± 3.7 in a spatially extended system, from ultra-deep HST white-light imaging [94]. The latter gives $\log(M_{\rm vir}/M_\odot) = 11.6 \pm 0.3$, consistent with cored kinematics and GC richness for its stellar mass. BTFR-normalized Φ_b^2 gives $\sigma_{\rm lw} \approx 12$ km/s (supplementary code), matching GC-poor PUDG R16 in Perseus: $R_e = 4.2$ kpc, $\sigma = 12 \pm 3$, richness 1 [93]. The matching structural twin is GC-poor; DF44's GC richness marks the anomaly. DF44 is a Coma first infaller, $\leq$2 Gyr [78], with σ rising from 26 to 41 km/s, indicating tidal heating. Opposite, DM-deficient cases DF2, DF4, DF9, FCC 224/240 agree with stars alone [35, 36]. Over-dark first infallers and under-dark tidal debris correlate with interaction history; that extension is outside the theory's claims
Galaxies $z > 9$	CDM hierarchy inherited (the primordial branch clusters first)	JWST	No distinctive reduction of the excess; a tension shared with ΛCDM, at the measurement frontier

†Post-JUNO global fits: Bari [42] 7.48 ± 0.10 (0.7σ); NuFIT 6.1 [9] $7.530^{+0.096}_{-0.097}$ (0.2σ). Pre-JUNO NuFIT 6.0: 7.49 ± 0.19 (0.3σ); JUNO 59-day [8]: 7.50 ± 0.12 (0.4σ). B_c^* (2.3σ low; Section D.3) scores against the V-P splitting family's value but is not a mass-table entry. The two remaining members of Section 5.1's exceptions list live in their own instruments: θ_{23} (2.1–2.3σ against the NuFIT 6.1 global minimum; Section 3.2) and the birefringence null ($\sim 4.8\sigma$ joint, 3.5σ after dust-EB robustness; the falsification table's falsification condition closest to triggering).

D.2 Falsification

Table 11 gives each prediction, its test, and its falsifying observation; Fig. 19 shows the timeline. Seventeen rows are distinctive forecasts, ten are structural nulls shared with

standard expectations, and two have conditional branches: $\sum m_\nu$ uses the positive-count completion, and J_{CP} uses the composed-route condition. Only failure of a shared null discriminates.

Table 11: *Decisive tests: 29 falsification rows, plus one conditional row. Notes in the text below.*

Test	Prediction	Timeline	Falsifying observation
$a_0(z) = cH(z)/\Sigma$	MOND scale evolves	SKA ∼2030; ELT ∼2031	Constant a_0
$\sin^2 2\theta_{12} = 6/7$	Resolution-floor mixing	JUNO 6-yr (∼2031)	θ_{12} outside 33.6°–34.2° (3σ at JUNO's 6-year $\pm 0.10°$)
$r = 4/N^2 = 1.23 \times 10^{-3}$	Tensor-to-scalar**	LiteBIRD ∼2036	$r > 4 \times 10^{-3}$ (3σ above the prediction at LiteBIRD's $\sigma(r) \sim 10^{-3}$; a non-detection does not falsify, note **)
$r/(1-n_s)^2 = 1$	Consistency relation*	LiteBIRD ∼2036	Identity violated by $> 3\sigma$
n_s–A_s joint	fw n_s pinned to 0.9646–0.9652 (± 0.0003; no free N)	SO / CMB-SPA ∼2027–28	n_s converging above ∼0.972 (3σ at $\sigma(n_s) \sim 0.0025$; ACT-like 0.9752 needs N=81, forcing A_s $\geq 42\sigma$ high — no escape)
$\Delta m^2_{21} = 7.55 \times 10^{-5}$	EM-blind (structural: 7.87)	JUNO 6-yr (∼2031)	Outside 7.43–7.67 (3σ at JUNO's 6-year ± 0.04); at 7.87: EM-blind factor falsified
$\sum m_\nu = 65.2$ meV (conditional)	NH, positive-count minimum, EM-blind gap	DESI/Euclid	$\sum m_\nu > 0.1$ eV or cosmological bound <50 meV
No $0\nu\beta\beta$	Dirac neutrinos	nEXO/ LEGEND	Any $0\nu\beta\beta$ detection
$J_{\mathrm{CP}}^{\mathrm{PMNS}} = 0$††	No leptonic CP violation	DUNE ∼2032	$J_{\mathrm{CP}} \neq 0$ at 3σ
Normal ordering	A3 sign selection ($H_{11} < 0$)	JUNO ∼2030 / DUNE	Inverted ordering at $>3\sigma$
$d_n = 0$	No strong-CP neutron EDM (the P-even axioms fix $\theta_{\mathrm{eff}} = 0$ directly; Vafa–Witten [47] then forbids its non-perturbative restoration by spontaneous P/CP breaking; the CKM radiative floor $\sim 10^{-32}\, e$ cm is far below reach)	n2EDM ∼2030	Nonzero d_n at $>5\sigma$
$d_e = 0$	No electron EDM at tree level (CKM radiative floor $\sim 10^{-44}\, e$ cm, far below any experiment)	JILA HfF^+ (current: 4.1×10^{-30} [38]); ACME III ∼2027 (projected $\sim 0.3 \times 10^{-30}$)	Nonzero d_e at $>5\sigma$
Birefringence $= 0$	No P-odd fundamental field‡	SO ∼2028–31; LiteBIRD ∼2036	Confirmed detection at $>5\sigma$ in a clean-β determination (dust systematic modeled)
$d_L^{\mathrm{GW}}/d_L^{\mathrm{EM}} = 1$; nondispersive GW propagation	Same substrate (d=0)	LIGO/ET	Propagation test: energy-dependent v_{GW}

Table 11: continued

Test	Prediction	Timeline	Falsifying observation
$\alpha_{\rm EM}$, m_p/m_e constant	Integer arithmetic	Quasar spectra	Confirmed variation
$R(K) = 1$; $R(D^*)$ at SM value	Lepton universality	Belle II	Confirmed violation ($R(D^{(*)})$ now 3.8σ high, but lattice SM spread covers it — untriggered)
Φ_b^2 cores	pISO lensing shape (flat v_c in strict-isolation stacks)	KiDS (Mistele: flat to $\sim$1 Mpc) / Rubin	NFW cusp confirmed at $>5\sigma$; or v_c decline/steepening in strict-isolation stacks at $>5\sigma$
CMB anomalies	K=1 initial uniformity	Planck/ LiteBIRD	Hemispherical asymmetry $>5\sigma$ (a scale-dependent $\sim 3\sigma$ asymmetry persists in PR4 — below the row's threshold)
Isocurvature $= 0$	Adiabatic-only: one admission per e-fold, one shared δN (Sections 4.4, 3.7)	Planck (current) / SO / LiteBIRD	Confirmed primordial isocurvature (any species mode, including compensated) at $>5\sigma$
$w(z)$ phantom crossing	Reference CPL $\approx (-0.70, -0.74)$; common-state crossing $z \approx 0.72$	DESI 5-year $\sim$2027	No crossing at $>3\sigma$; or $w_0 < -0.85$
H_0 structure	Reference background test 65 ± 1.5 (θ^*); common central 65.31; $(H_{\rm local} - H_0)/H_0 = +3$–$6\%$ from local void depth	Standard sirens / inverse-distance ladder; CF4-successor depth measurement of the $\lesssim$70 Mpc volume	Background-only H_0 (beyond the local volume) outside the reference 65 ± 1.5 model band at $>3\sigma$; or measured local depth $\delta \gtrsim -0.2$ while $H_{\rm local}$ exceeds the concurrently measured background by $\geq +3\%$ — the path effect cannot deliver that boost at that depth
UDG falsification class	Undisturbed DF44-class UDGs sit at plain Φ_b^2 (Jeans $\sigma_{\rm lw} \approx 12$ km/s; PUDG R16: 12 ± 3)	Deep IFU dwarf spectroscopy (Coma/ Perseus, ongoing)	An undisturbed UDG (no rising outer σ profile, no recent-infall signature) with σ above the BTFR-normalized Φ_b^2 Jeans prediction at $>3\sigma$
ISW void/wall asymmetry	$\sim$20% modulation	SO $\times$ DESI $\sim$2028	No environment dependence at $>3\sigma$
$O(\alpha_{\rm EM})$ residual N=14	$K^\pm$ correction 52 keV	Kaonic atoms / Penning	$K^\pm$ mass at ± 1.6 keV not at N=14
m_H/m_W (m_t-free)	1.5573 ($+\alpha_{\rm EM}$, derived) vs 1.5546 (bare)	HL-LHC (m_H to 10–20 MeV)	ratio at $>3\sigma$ from $+\alpha_{\rm EM}$ value excludes the derived form
Group-scale relation	Groups at $r_{500}/2$ sit 0.97–1.09$\times$ the galaxy relation, offset decreasing with hot-gas fraction (Section 4.6)	Group weak-lensing / dynamics stacks (eRASS/ SDSS-class), $\sim$2027–29	Offset $>1.2\times$ or f-independence at $>3\sigma$; or offset below the predicted band at matched gas fraction by $>3\sigma$

Table 11: continued

Test	Prediction	Timeline	Falsifying observation
DM non-detection	DM is bookkeeping bE, not a particle (Section 4.5): no recoils, no annihilation products, no collider production	LZ/XENONnT full exposure ∼2028; CTA; HL-LHC	Any confirmed particle-DM detection: a direct-detection recoil spectrum, an annihilation line, or a collider discovery with matching relic abundance; the same class includes a confirmed dark-matter–capture heating signature in old, cold neutron stars (no particle → no capture)
No collapse noise	Quantum mechanics is exact (T1–T2): no objective-collapse dynamics, hence no anomalous CSL/GRW heating or diffusion	Levitated-sensor and space-based tests (LISA Pathfinder bounds; successors ∼2030s)	A confirmed objective-collapse signature (anomalous position diffusion or heating matching a collapse model) falsifies T1–T2
Flat-band pairing geometric	Superfluid stiffness in spin-degenerate flat-band superconductors exceeds the single-band Fermi-liquid (no-geometry) prediction (K=1, Sections 2.6, 4.8); MATBG's measured stiffness runs an order of magnitude above its conventional-only estimate [25], with the geometric/conventional split not yet experimentally resolved — unresolved likewise in the twisted-trilayer companion measurement (nodal gap, phase-fluctuation-limited T_c), whose stiffness is analyzed through an interaction-renormalized band picture without an isolated metric term [26]	Twisted-TMD / next-generation moiré stiffness measurements ∼2027–30	A clean flat-band superconductor whose measured superfluid stiffness matches its own single-band Fermi-liquid prediction — no excess beyond the conventional estimate at $>3\sigma$
$\lvert V_{cb}\rvert = 0.04262$	Conditional†	Belle II / lattice	Exclusive prevails: 6.2σ (PDG 2026 errors) to 8.7σ (lattice-only) exclusion

Notes to Table 11. $^{*}r/(1-n_s)^2 = 1$ is algebraically equivalent to $r = 4/N^2$, not an independent test. **The predicted $r = 1.23 \times 10^{-3}$ is 1/3 of Starobinsky R^2. At LiteBIRD's $\sigma(r) \sim 10^{-3}$ and ∼2036 launch, it is a $\sim 1\sigma$ signal; discrimination is marginal and sensitivity-dependent. CMB-S4 was canceled in 2025, leaving LiteBIRD as the sole listed r test. †The V_{cb} row agrees with inclusive data at 1.2σ (Table 9) and is excluded if exclusive data prevail. For a_0, the near-term test is evolution versus constancy: MOND holds it constant, the

framework has $a_0 \propto H(z)$, and ΛCDM's expansion gives similar apparent scaling without a fundamental a_0 (Section 4.7).

‡Birefringence is the nearest falsification threshold. ACT DR6 alone gives $\beta = 0.215^\circ \pm 0.074^\circ$ (2.9σ) after marginalizing dust EB leakage [88]. Joint ACT+Planck PR4 gives $\beta = 0.277^\circ \pm 0.057^\circ$ at 4.8σ, falling to 3.5σ under its dust-EB robustness test [87]. Unresolved systematics in both datasets prevent strong cosmological conclusions. No clean-β result exceeds 5σ; Simons Observatory (∼2028–31) will decide.

The tightest ΛCDM-conditioned $\sum m_\nu$ bound is 52 meV (ACT DR6+DESI DR2, 2026), versus earlier 53–64 meV. It lies 2 meV above the <50 meV line and excludes the conditional 65.2 meV sum under strict adiabatic ΛCDM. The limit relaxes to <163 meV in w_0w_aCDM or <92 meV with an NH-informed prior. One joint analysis gives 98^{+16}_{-37} meV, within one lower-side quoted uncertainty of this branch; another 2026 combination prefers positive mass at 2.7σ. These model and prior dependencies must accompany the comparison; a generic w_0w_aCDM limit is not a likelihood calculation for this theory's specific $w(z)$. The row retains >0.1 eV and <50 meV as coarse sufficient exclusions, not the complete rejection boundary for 65.2 meV. At the retained squared gaps, 58.6 meV is the reference normal-ordering floor obtained by taking $m_1 \to 0$; it is not the positive branch's sum. A sufficiently precise bound between that floor and 65.2 meV can reject the positive-count completion without rejecting every normal-ordering spectrum. The physical prediction remains conditional on the common-unit counts and their mode admissibility (Section 3.7).

For $J_{\rm CP}$, T2K+NOνA's normal-ordering 3σ interval includes $\delta = \pi$, while SK+T2K disfavors $J_{\rm CP} = 0$ at 1.9σ: net ∼1.5–2σ against, pending DUNE. ††The zero is conditional on the sequential route being its primitive legs' T1 composition. That graph is a tree, giving $J = 0$ exactly (supplementary code). An unmatched loop phase is the alternative. Nonzero DUNE CP would falsify this row and select that branch, changing all three mixing angles jointly; at the reference control phase, θ_{13} nearly doubles. The three angle matches would therefore also require reconsideration. Loop deviations 0 or π give $J = 0$ and conserve CP.

The UDG row's hardest present case is NGC5846_UDG1: GC-rich, flat dispersion, and no tidal features. Its $\sigma_{\rm GC} = 29.8^{+6.4}_{-4.9}$ km/s exceeds a tracer-matched Jeans prediction of 9.4–10.5 km/s by 3.9–4.6σ. The ∼20 measured clusters are more central than starlight, at median $R \approx 0.5\,R_{\rm eff}$, where cored Φ_b^2 predicts falling dispersion. Stellar-light dispersion 17 ± 2 is only 2.2σ above its own tracer prediction. This unexplained split carries a mass-segregation caveat and lacks an independently fitted GC density profile. A larger GC sample or resolved stellar dispersion near ∼3 kpc decides; confirmation of the GC value in an undisturbed system triggers the row.

D.3 The strangeness step and untested predictions

A flavor substitution swaps one register in a computed composite. Its price depends on whether the composite closes at a junction or along a string (Table 3). The junction rule has thirteen independent baryon-pair tests; Table 12 lists the unmeasured predictions.

The strangeness step. Within a multiplet, the junction step $= (2N_c^2{-}1 - \text{blocked}) \times (2N_c^2{-}1 + \text{overflow})$: heavy quarks block vertex modes and saturation overflows into the gate. For two charm quarks, $N_c^2{-}1 = 8$ of 17 modes are blocked, leaving 9. Overflow reaches $\Sigma = 6$ computational gate channels; existence does not participate. Thus $9 \times 23 = 207$ and $\Omega_{cc} = \Xi_{cc} + N_c^2 \times 23 = 7088 + 207 = 7295\,m_e = 3728$ MeV. LHCb measures 3725.9 ± 1.3 [63], 1.4σ from unrounded 3727.7. Its splitting $\Omega_{cc} - \Xi_{cc} = 104.3 \pm 1.4$ MeV is 1.1σ from $207\,m_e = 105.8$ MeV. Independent lattice $\Omega_{cc}(1/2^+) = 3712(11)(12)$ MeV [65] gives 90 ± 16 MeV, 0.9σ away, alongside the dozen-plus lighter comparisons (supplementary code). Four baryon families follow the rule (Figs. 27, 28):

- $17^2 = 289$ at zero heavy quarks (decuplet, 0.04–0.29%)

- base values 229–248 at one heavy quark, by spin and step order ($238 = 14 \times 17$ is the J=3/2 base); residuals 0.02–0.85%
- $9 \times 23 = 207$ at two (1.1σ)

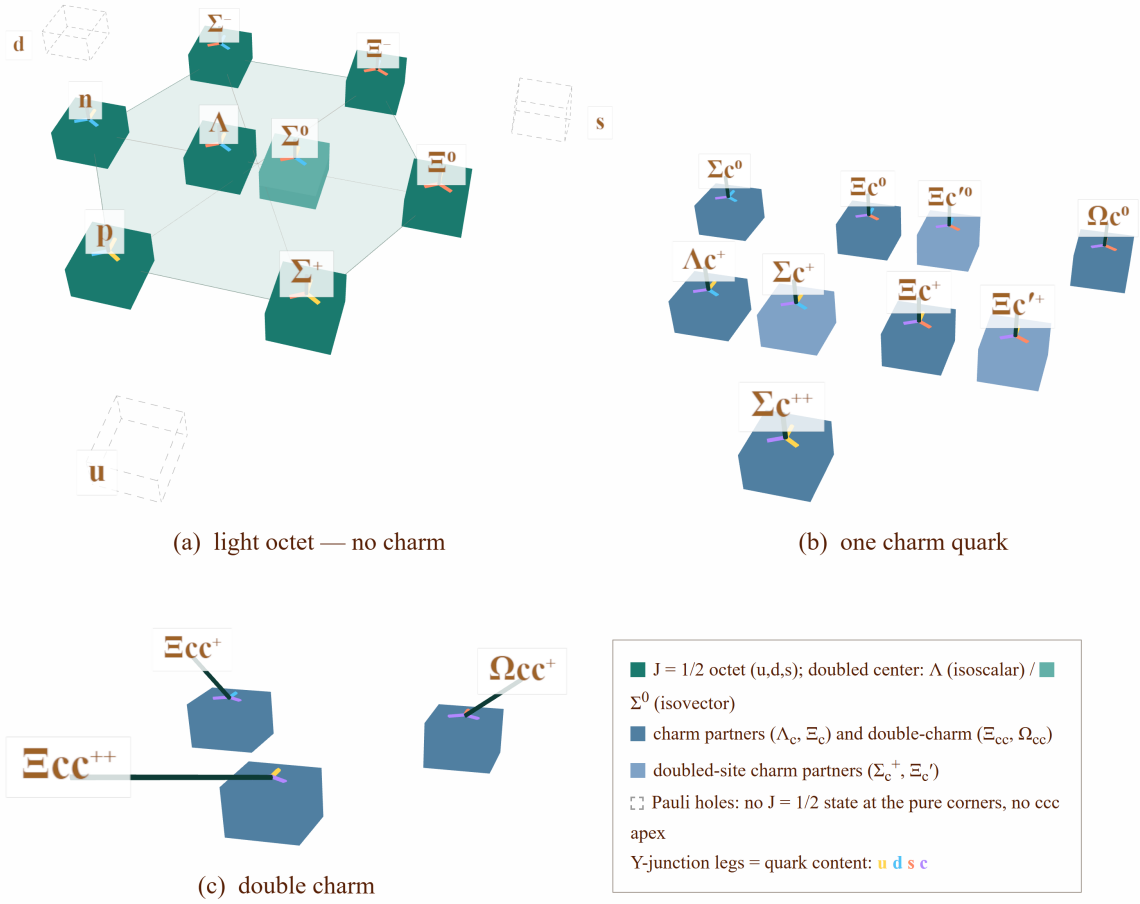


Figure 27: *The J=1/2 composition lattice and its charm floors. (a) Three-quark flavor words tile a triangular lattice of ten sites; Pauli exclusion removes the three pure corners from the J=1/2 octet (three identical flavors force symmetric spin), leaving eight states over seven sites with a doubled center (Λ/Σ^0: isoscalar versus isovector light pair). The strangeness ladder therefore cannot reach the sss corner within the octet: the $\Omega = \Xi + F$ row crosses multiplets by construction (the decuplet occupies all ten sites, Fig. 28), and its observed price equals the meson-sector string price raw ($\Omega - \Xi = 354.17$ MeV; $K - \pi = 354.11$ MeV; 0.016%). No first-order SU(3) relation connects the two sides — the octet, decuplet, and pseudoscalar sectors carry separate breaking parameters, and the meson relations hold in mass-squared — so the equality is the row's own content, not an inherited symmetry identity. The Λ-anchor family ($F/2$, Table 3) sits at the lattice center, where all three flavors are distinct; the full string price F appears at the purity corner. (b) One charm quark (rows 64–65, 67–68, 70 of Table 9): three sites are doubled, repeating the center's device (Λ_c/Σ_c^+, Ξ_c/Ξ_c'). (c) Two charm quarks: Ξ_{cc} (both charge states, row 72) and Ω_{cc} (the strangeness step, this section). All three panels share one camera and one scale, so box sizes are directly comparable; the lattice orientation (u/d/s corners) is that of panel (a).*

All thirteen independent baryon pairs agree within 1.5% (supplementary code). The tightest strains are $\Xi^* \to \Omega$ (1.4%, 3.0σ) and $\Sigma_b^{*-} \to \Xi_b^{*-}$ (0.85%, 3.1σ). Their u-channels lie below 1σ at the same bases; d-channels carry the strain. The zero-parameter isospin model reproduces both channel differences at 0.2σ (supplementary package,

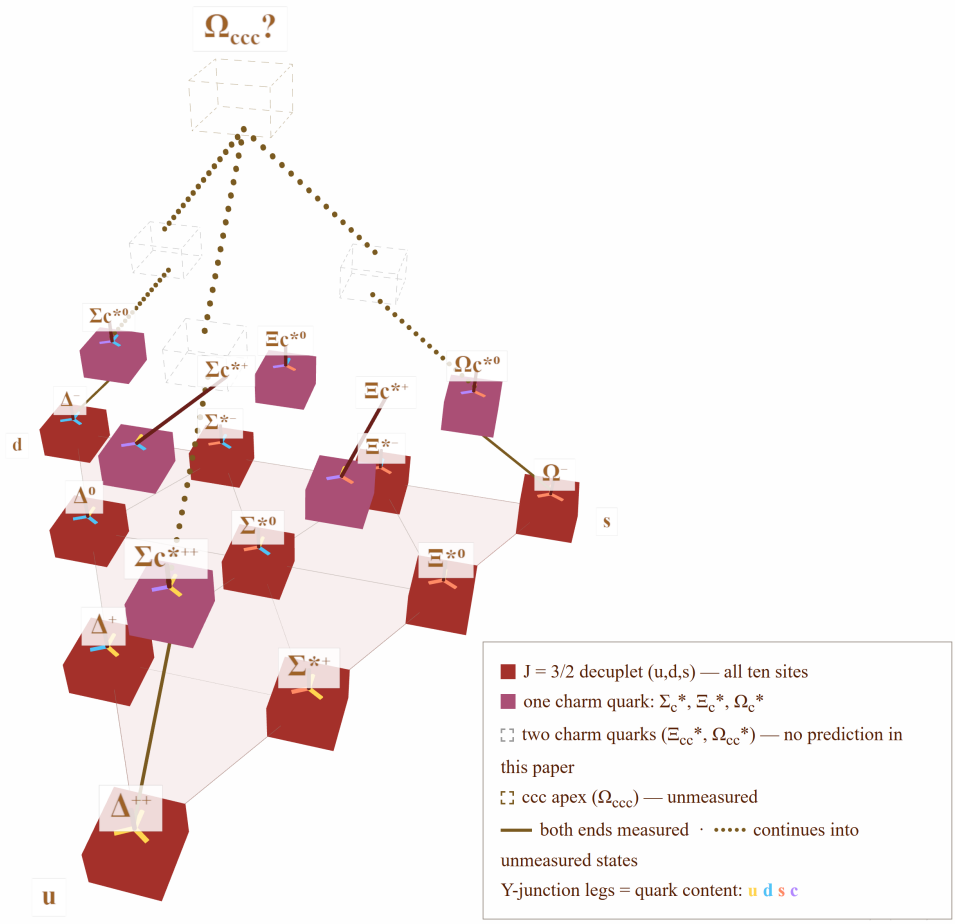


Figure 28: *The J=3/2 tower. The decuplet occupies all ten sites of the light lattice, including the pure corners forbidden to the octet. Charm count is drawn as height: each floor replaces one light quark with c (rows 66/69/71 at one charm) — each riser is one register substitution, priced by the swap-price table (Table 3) — and the tower ends at the pure-charm apex ccc (Ω_{ccc}) — the corner argument that removes sss from the octet removes the ccc apex from the J=1/2 cone entirely, so this apex belongs to the J=3/2 family alone. Solid connectors join states measured at both ends; dotted connectors continue into states with no prediction in this paper (dashed boxes: the J=3/2 double-charm pair Ξ^*_{cc}/Ω^*_{cc}, and the unmeasured apex).*

`zero_dial_isospin`); the charm analog is unresolved. This composes strangeness steps with isospin differences, without licensing absolute baryon corrections. Decuplet steps are $289+8 = 297$ for a new-channel surcharge N_c^2-1, then 289, then $289-13 = 276$ for existing-s reduction $N_c^2+N_c+1$. Charmed sextet steps $\Xi'_c \leftrightarrow \Sigma_c$ and $\Omega_c \leftrightarrow \Xi'_c$ follow the same blocked/overflow pattern at bases 243 and 229, with 0.02–0.62% residuals. The untested $\Omega_{bc} - \Xi_{bc} = 207\, m_e = 106$ MeV equals the $\Omega_{cc} - \Xi_{cc}$ step because bc and cc have identical D=1 color content (Table 12). Separately, the V-P family's B_c^* splitting $117\, m_e$ gives 6334 MeV (Appendix K.1), 2.3σ below ATLAS's 6339.0 [10]; this is that family's one strained cell (Section 5.1).

Quantity	Predicted	Experiment	Timeline	Status
Ξ_{bc} (bcu)	$\Lambda_b + \Lambda_c - p =$ 6968 MeV	LHCb	Run 3–4	Unmeasured
Ω_{bc} (bcs)	$\Xi_{bc} + 106$ MeV (strangeness step)	LHCb	Run 3–4	Unmeasured
Ω_b^* (ssb, J=3/2)	$\Omega_b + 42\, m_e =$ 6067.3 ± 1.7 MeV	LHCb	Run 3–4	Unmeasured
$\Xi_b^{\prime 0}$ (bsu, J=1/2)	$\Xi_b^{*0} - 40\, m_e =$ 5931.9 ± 1.0 MeV	LHCb	Run 3–4	Unmeasured
g_T	$50/51 = 0.980$	JLAB 12 GeV / EIC	∼2030	0.989 ± 0.033 (0.3σ)
$\Delta\Sigma_p/\Delta\Sigma_n$	$56/65 = 0.862$	EIC	∼2032	Unmeasured
r (tensor-to-scalar)	1.23×10^{-3}	LiteBIRD	∼2036	Bounds cover ($r < 0.032$)
$\sum m_\nu$	65.2 meV (conditional)	DESI / Euclid	∼2028	Positive-count branch; model-dependent bounds (Table 11 note)
κ_λ (Higgs self)	$= 1$	HL-LHC	∼2035	Bounds cover (−0.4–6.3)
T_{bb} ($\bar{b}\bar{b}ud$)	Does not bind	Lattice / LHCb	Ongoing	Lattice predicts binding
$\mu(\Sigma^*, \Xi^*)$ (J=3/2)	Five values (caption), +3.36 to −2.64 μ_N	Lattice QCD	—	Unmeasured
$\mu_{\Delta^{++}}$	$Q \times N_{\text{spin}} = +4\,\mu_N$	γp / lattice	Ongoing	Band 3.7–7.5 covers

Table 12: *Forward predictions — untested or not yet decisively tested (the status column carries any first data) — each exercising a different sector of the theory. The moment rows test the count-class grammar of Section 3.5; the voice sums are Σ^{*+} +3.36, Σ^{*0} +0.36, Σ^{*-} −2.64, Ξ^{*0} +0.72, Ξ^{*-} −2.28 μ_N. g_T and $\Delta\Sigma_p/\Delta\Sigma_n$ test the resonant-channel counting (Operation 3). r tests the inflationary $\varepsilon_1 = 1/(4N^2)$. $\sum m_\nu$ jointly tests the EM-blind gap scale and the conditional positive-count completion; the gap measurements alone do not select its absolute spectrum. $\kappa_\lambda = 1$ tests that m_H is geometric (SO(2)), not a potential minimum. T_{bb} tests the topology theorem's minimum-cost reading (Section 6). The Ω_b^* and $\Xi_b^{\prime 0}$ uncertainties are propagated from their measured anchor masses, not fitted.*

D.4 Residual structure by operation

$O(\alpha_{\text{EM}})$ **residuals.** Integer-formula residuals carry $O(\alpha_{\text{EM}})$ structure by operation type. Junction baryons use $N = 21 = N_c(\Sigma{+}1)$, matching the proton to about 6 eV and inherited by the neutron. The single addressed meson, $K^\pm$, is consistent with gluon-blind $N = 14 = N_{\text{spin}}(\Sigma{+}1)$: strings have no junction vertex. Elementary leptons use $N = 5/2$, matching the muon at 0.2 ppm, an $O(\alpha_{\text{EM}}^2)$ residual like the $1/\alpha_{\text{EM}}$ and proton $(1 - \alpha_{\text{EM}}/2)$ completions. The fourteen width formulas assign $N = 0$. Each selected rational is an identification charged in Appendix M, Rule 3; these ppm matches are not choice-free derivations.

Stocks (Operation 1, masses and bindings — receiptless mode counts):

Quantity	$O(\alpha_{\rm EM})$ form	N (kernel reading)	Match
m_p/m_e	$1836 + 21\alpha_{\rm EM}(1-\alpha_{\rm EM}/2)$	$21 = N_c(\Sigma{+}1)$: transparent cells; the $1/2$ is a gate-screening coefficient, its own type	6 eV
m_n/m_e	$m_p/m_e + (5/2)(82/81)$	21: junction correction inherited via m_p	57 eV
$m_n - m_p$	$(5/2)(82/81)\, m_e$, no $O(\alpha_{\rm EM})$	$N{=}0$: A4 origin, not EM	63 eV
B_d	$(13/3)(1 + 19\alpha_{\rm EM}/30)$	$19/30$ = vertex+spin/ total	3.5 eV
m_μ/m_e	$206.75 + 5\alpha_{\rm EM}/2$	$5/2 = (N_c{+}N_{\rm spin})/N_{\rm spin}$: internal/ spin	0.2 ppm

Ratios (Operation 3, charges — additive channel counts): $g_A = (65/51)(1 + \alpha_{\rm EM}/(N_c{+}N_{\rm spin})) = 1.2764$, versus PERKEO III's 1.27641 ± 0.00056 at 0.07σ [13]. Here $N = 1/(N_c{+}N_{\rm spin}) = 1/5$ reads per-channel EM resolution of the denominator. With CKM-unitarity $V_{ud} = 0.97436$, this gives $\tau_n = 878.22$ s, 1.3–1.4σ from UCNτ; bare g_A gives 880.3 s, 8.2–8.8σ (Section 3.6). Unitarity is exact, with derived V_{us} and V_{ub} (Table 4). Direct superallowed-β V_{ud} of 0.97367, the 2–3σ Cabibbo-angle anomaly, instead gives $\sim$879.5 s.

Couplings (Operation 5, shell-to-shell — magnetic moments): $\mu_p/\mu_N = (39/14)(1 + 6\alpha_{\rm EM}/17)$, $N = \Sigma/(2N_c^2{-}1)$: gate-to-source coupling ratio (15 ppm). $\mu_n/\mu_N = -(21/11)(1 + 2\alpha_{\rm EM}/7)$, $N = N_{\rm spin}/(\Sigma{+}1)$: spin-to-channel ratio (15 ppm). Ratio μ_p/μ_n: the two corrections conspire to give -1.45990 (obs -1.45990, 0.3 ppm; bare $-143/98 = -1.45918$, 489 ppm — 1943$\times$ improvement).

Drains (Operation 4, widths — exit-path mode ratios): $N = 0$ across all 14 width formulas. The reference width data give the combined fit $N = -0.3 \pm 0.3$, excluding the mass-sector $N{=}21$ at $\sim 70\sigma$, or 9.9σ by π^0 alone. The PDG 2026 χ_{c0} comparison is given in Section 3.4. Widths are *receipted flows*: each exit emits a zero-cost photon receipt. The ledger does not bill its own transactions, the same no-self-reference rule that forbids the graviton (Section 4.8).

The drain correction depends on the depth each operation reads. Composite-anchor stocks scale with transparent cells; binding and lepton stocks use ratio-form N; charge ratios use $1/\text{channels}$; shell couplings use the bridge ratio. Drains carry no correction: the electromagnetic field cannot correct itself. Seven data points span four operation types, with the shared drain $N{=}0$ giving eight measured values. The operation-to-type rule is structural; the individual N values remain identifications. Code enumeration finds no non-tautological rule family covering more than four values (supplementary code). The choice-free distinguishable-parton count $\times\,(\Sigma{+}1)$ covers only verified composite stocks 21 and 14. Kernel expressibility alone contains no assignment information: seven of eight are exact kernel ratios, so every permutation scores. The pion residual $N = 18.15 \pm 0.05$ has no simple kernel form at that precision. One structured candidate uses the chiral transparent cell (Section 2.7): the EM read traverses its occupancy atop string 14, giving $N = 14 + P$ with $P = (N_c{+}N_{\rm spin})^2/\Sigma = 25/6$. This is the parked $d^2 = 4$ plus one read-citation unit over six computing channels; existence is the cited mark, not a computing

channel (Section 2.5). It gives $N_\pi = 18.17$ ($+0.4\sigma$), applies only to string-priced chiral states, and predicts $K^\pm$ at $N \approx 18.2$. The η leading read is shell-wide.

The measured $K^\pm$ mass permits $N = 14$ with ± 4.3 uncertainty from ± 16 keV. Its $N{=}14 = N_{\text{spin}}(\Sigma{+}1)$ reading follows meson topology: no junction, so the photon traverses spin-transparent rather than baryon vertex-transparent $N{=}21$ modes (Appendix I.7). Only $K^\pm$ tests this reading. Other precision meson anchors, $\pi^\pm$, η, $D^\pm$, $B^\pm_{(s)}$, quarkonium, and ϕ, remain outside its scope. The ϕ central $N = 5$ has ± 5.1 uncertainty, making 0 through 9 indistinguishable. Other residuals span tens to over a thousand α_{EM} units, reaching ~ -1200 for $\Upsilon(1S)$. Baryon anchor-plus-offset rows such as $\Lambda = p + F/2$ are likewise swap-price results, with residuals of tens of α_{EM} units. The exact-tier $N{=}21$ claim applies to the proton and is inherited by the neutron. Improving $K^\pm$ precision to ± 1.6 keV would pin $N{=}14$ at 2σ and separate universal-string from parked-cell readings at $\sim 10\sigma$.

E Field-theory and transport constructions

E.1 Reads, records, and energy

Light is the substrate's query. A photon is how one event knows about another. It carries zero internal state ($d{=}0$): it is an origin coordinate being read, propagated freely at the information speed c (A1). A photon from address X to address Y is the search for that route. The destination Y is certified where the search first completes, at the read (Section 4.5). Between emission and absorption, the search explores all paths simultaneously (the path integral, from A1+A2), and laziness selects the cheapest.

The search is the wave; its completion is a localized event. A standing wave carries the origin written by its last resolved event. At the next resolution, the destination reads that origin. The minimum-cost origin-to-destination relation enters the ledger as a localized event, photonic when the operation is a read.

The address stays put and spends nothing; A2 holds its slot whether visited or not (Section 4.5). What travels is its citation: "I came from here." This one-bit pointer is the smallest transferable fact, the same for radio and gamma rays. The photon is its cheapest carrier: ledger content alone, with no register state ($d{=}0$) to refract. Propagation is free. Its energy, $E = \hbar\omega$, is the citation rate: per-tick demand at the arrival clock, delivered to the absorber. Reference upkeep is billed to the holder at $1/3$ bE per pointer, independently of payload. Read from afar, the citation looks like a line back to its origin. A resident record whose registers have all frozen is finished: written once, billed once, never clocked again. The substrate's stable objects are these completed entries (Section 2.5).

E.2 Closed flux loops

A2 permits flux to depart only where a register supplies a source (Section 1). The seven channels contain no magnetic charge, so sourceless flux closes. A monopole would be an

exit without an entry; the grammar supplies no such unpaired exit. Departure writes an address and arrival reads the same address. Turning back writes no second citation, costs nothing, and leaves the loop unresolved until a read puts the books in order.

E.3 The local ledger

Storage. The substrate keeps a current-state ledger (A2) — each pattern's latest record, nothing older kept; an update overwrites its entry, the superseded content disperses outward through the environment's relations at read speed, and erasing a record is billed, never free: the second law is the dispersal accumulating.

Consistency. Consistency is distributed — every event announces its origin on everything that leaves it, agreement spreads at the speed of the read, and when a local transaction cannot balance, the machine emits the minimum correction automatically (the neutrino): no global lock, no central clock, and the books still balance exactly.

Identity. Identity is content-addressed — no event carries a serial number; two histories agreeing on every relation are one history, stored once (Section 4.2), and identical particles interfere exactly where the ledger holds one entry for their swapped histories (scoped in Appendix E): deduplication, visible in every interference experiment.

The minimum correction described above carries its own guarantee and is invisible to the read; what is guaranteed needs no verification.

E.4 Field-theory constructions

The field-theory bridge. The derived skeleton comprises a local ternary occupation basis, antiparticle relabeling $W^+ \leftrightarrow W^-$ (A2), classical past-lightcone microcausality at CSG tier (A1), and channel-resonance interaction selection rules. For source or sink pairs with at least one relationally symmetric end, histories agreeing in every relational fact are summed once. This derives the identical-particle quotient from the relational substrate (Section 4.2), where Leinaas and Myrheim take it as a postulate [92]; the result is verified on toy causal sets (supplementary code). Interference survives exactly where the web has no which-path record, matching delayed-choice and quantum-eraser decoherence. A fringe maps where a record can next appear: bright where unresolved histories reinforce, dark where they cancel.

The spatial exchange phase is conditional on the manifoldlike continuum (Appendix O). At $D \geq 3$, π_1 of identical-particle configuration space is S_n [92], whose one-dimensional characters are exactly $\{+1, -1\}$. A discrete graph's free fundamental group cannot supply this $\mathbb{Z}_2$ (supplementary code). A3 selects antisymmetry for finite discrete registers. The constitutive step identifies no exchange-relevant classical surplus beyond A4's outcome count: an unused which-way identity marker does no derivational work, so A3 forbids it (Section 4.2). This is a structural consequence, not an identification. The topology $\pi_1 = S_n$ chooses its character globally, including asymmetric endpoints lacking a compensating relabeling. The exchange postulate is therefore unconditional

within the assumed continuum; its microscopic atlas remains open. Appendix I.1 separately derives total antisymmetric sector selection and excludes parastatistics, with its own import and identification. Operator microcausality is ceded to standard QFT on the derived field content, like vertex Lorentz structure below.

Two-read response. A photonic read's origin-to-destination line selects projectors on the gate doublet. Conditional on the manifoldlike continuum, that line is the unique frame-local axis. Projection follows because the isotropic cell provides no axis, the register rides the wave, and the read collapses it along its sole bearing. The imported representation-theory step, at T1's tier, is that the unique equivariant complex-doublet coupling to spatial rotations is fundamental. This applies to photon-mediated $d{=}0$ doublet reads; W-mediated vertices remain outside its domain (Appendix I.6).

Conditional on this carrier typing, the complex doublet forces $P(\text{same}) = \cos^2(\theta/2)$. Response factors through the written state because the standing wave is the only inter-read channel. Isotropy makes axis assignment equivariant; the stabilizer selects the $\mathbf{n} \cdot \boldsymbol{\sigma}$ eigenbasis without a continuity premise. A3 selects a faithful, definite write because degradation needs extra billed participation. Projector overlap then gives the law, with the double cover as corollary, exactly at the conditional continuum tier (Section 4.2; supplementary code). The separate internal SO(2) phase supplies spin-statistics and seam monodromies (Appendix I.1).

Bridge boundary. Continuum-EFT vertex strength and renormalizability differ in kind from substrate selection rules and coupling values (Section 3.3). A lazy neighbour-only substrate gives a local derivative expansion dominated by its lowest operators, but locality alone selects no spinor representation. The first-order spinor carrier remains open: a gate SU(2) doublet on the scalar transport cone gives two scalar copies, not a Clifford/Dirac principal symbol. With a spinor action and manifold limit, local Lorentz invariance, gauge structure, and locality would constrain vertices through standard uniqueness theorems. The scalar walk does not supply those premises. Likewise, the 29-state content count is no gauge projection; singlet structure comes from composite topology (Appendix I.7).

A 1+1 two-component fixed-cell existence witness is one-cell local, exactly unitary, locally $U(1)$-covariant, and Dirac-dispersive: $\cos\omega = \cos k \cos\theta$. It is not yet transistor dynamics, with no link dynamics or non-Abelian content. The coin toll θ is identified as mass per tick: rest phase equals θ, the $\theta{=}0$ limit travels at c, and maximum group speed is $\cos\theta$ (supplementary code). The smallest native reversible census has eight equal-cost rules in six inverse-time classes. Checkerboard space/time and register recodings connect them into two orbits of four, distinguished by the previous-slice coefficient, their sole recoding invariant. These recodings use only A2's unobserved swap of nonzero labels. Unitarity fixes the witness coefficient to -1 and selects one orbit; the shadow pair, including the mass-dial sign, is itself one orbit. One unitary census dynamics survives up to relabeling (supplementary code).

Every A2-reversible rule bijects prior-slice values, preserving the uniform ensemble and equal equilibrium bills. The billed overwrite fraction is exactly 2/3 for every reversible rule; equilibrium kicks also cost equally in expectation, so A3 cannot discrim-

inate on any stationary ensemble or its linear response (supplementary code). The selected orbit's four representatives are recodings, not distinct physics. On a matched three-cell, three-tick grid, linear sum and sample blocks reproduce their representative exactly. Plurality blocking has no closed rule because the blocked future is not a function of the blocked state (exact code certificates; supplementary code). Eliminating one witness component gives the census recurrence with fixed coefficients: resident 0 because both components move, previous slice -1 by unitarity and reflection with a real coin, and neighbour $\cos\theta$. This assignment lies outside the eight-rule census because its dependency criterion requires an active resident term (supplementary code). Among eighteen reversible reflection-symmetric ternary rules, only the shadow pair transports arbitrary waveforms one cell per tick. Initial-state cost does not select it: cheapest for a single-cell seed, dearest for generic seeds because transport retains everything (supplementary code). On the 3+1 six-neighbour lattice, one amplitude per direction and octahedral covariance give three coin phases through $6 = A_{1g} \oplus E_g \oplus T_{1u}$. The isotropic massless walk has a linear scalar cone. Its rest phase is half the k=0 branch gap; massive curvature is $(1/D)\, m \cot m$, reducing to $\theta \cot\theta$ at D=1 (supplementary code).

No two-component spinor cone exists on that lattice: every two-dimensional projective octahedral representation violates opposite-pair unitarity. The 3+1 witness is scalar, and internal $N_{\text{spin}} = 2$ adds only two scalar copies. On the actual exact-six medium, even the scalar relativistic cone is conditional on homogenization; first-order spinor/Clifford coupling remains open.

The walk's t-tick reach is its t-hop ball. The inscribed sphere caps a direction-independent front at $1/\sqrt{D}$ of the hop rate; the walk saturates this in one, two, and three dimensions, including the 1+1 witness's D=1 speed c. One cell per tick remains exact as path length (supplementary code). A variable-degree contact graph is only a localization control: its degree disorder made transport sub-ballistic, but this does not transfer to literal exact-six address graphs. Angular and topological disorder may still renormalize, anisotropize, trap, or localize finite-frequency content. Isotropy and ballistic propagation require scaling-limit proofs.

For a fixed undirected exact-six graph, the centered massive cubic coin has the frame-free scattering form

$$W_\theta = e^{i\theta} P_{\mathbf{1}} - e^{-i\theta}(I - P_{\mathbf{1}}), \qquad P_{\mathbf{1}} = |\mathbf{1}\rangle\langle\mathbf{1}|/6. \tag{26}$$

It is the generalized Grover reflection at $\theta = 0$, commutes with every port relabeling, and has scalar–vector phase gap 2θ, hence rest mass $|\theta|$. If L_{norm} is the normalized graph Laplacian, every inherited walk phase obeys the exact relation

$$\cos\omega = \cos\theta\,(1 - \lambda_{\text{norm}}). \tag{27}$$

Thus the fixed-graph walk is an exact functional-calculus object. If a spacing-ϵ graph family satisfies $\epsilon^{-2} L_{\text{norm}} \to H = -\nabla \cdot A_{\text{eff}} \nabla$ in the Mosco/strong-resolvent sense, $\theta = m\epsilon$, $n\epsilon \to t$, and the source and read profiles are low-energy prepared, its endpoint compression converges to

$$dU^n d^* \longrightarrow \cos(tE) + imE^{-1}\sin(tE), \qquad E = \sqrt{m^2 + 2H}. \tag{28}$$

Here d averages six endpoint port amplitudes. Stationarity and ergodicity make A_{eff} deterministic; rotational covariance gives αI. A separate nontrapping estimate must establish $\alpha > 0$. Only then does the scalar Klein–Gordon dispersion $\Omega^2 = m^2 + 2\alpha|q|^2$ follow. Selected finite exact-six graphs have positive, nearly isotropic low-twist tensors. They do not establish infinite-volume correctors, Poincaré bounds, nondegeneracy, or homogenization.

A conditional local spinor carrier. A supplied response on six reciprocal ports admits a four-component first-order carrier. In a Cartesian control frame, let $\sigma_p = (\sigma_x, -\sigma_x, \sigma_y, -\sigma_y, \sigma_z, -\sigma_z)$, $J_0 = (I_2, \dots, I_2)^\dagger/\sqrt{6}$, $J_1 = (\sigma_1, \dots, \sigma_6)^\dagger/\sqrt{6}$, $V = (J_0, J_1)$ and $P = VV^\dagger$. These two isometric doublets are orthogonal. Let S_0 exchange opposite ports, and let the one-grain Fourier hop satisfy $S(\mathbf{k})_{\bar{p},p} = e^{i\mathbf{k}\cdot\mathbf{d}_p} I_2$. The complete local response and update are

$$C_* = S_0(2P - I_{12}), \qquad M_* = 2J_0J_0^\dagger - I_{12}, \qquad U_\vartheta(\mathbf{k}) = S(\mathbf{k})C_* e^{-i\vartheta M_*}. \tag{29}$$

Both C_* and M_* are Hermitian involutions. At $\mathbf{k} = \vartheta = 0$, $U = 2P - I_{12}$ has four eigenvalues $+1$ and eight -1. With $\mathbf{k} = \epsilon\mathbf{p}$ and $\vartheta = \epsilon m$, the isolated four-dimensional sector has first-order generator

$$H_{\mathrm{loc}} = m\,\tau_z \otimes I_2 + \frac{1}{3}\tau_x \otimes (\boldsymbol{\sigma}\cdot\mathbf{p}), \qquad H_{\mathrm{loc}}^2 = \left(m^2 + \frac{|\mathbf{p}|^2}{9}\right) I_4. \tag{30}$$

For $e^{-4i\vartheta} \neq 1$, this full three-direction update family has no nonzero proper momentum-independent local invariant subspace. Replacing J_1 by $(\sigma_x, -\sigma_x, \sigma_y, -\sigma_y, 0, 0)^\dagger/2$ preserves a normal-spin line across all six travel directions. The full $D{=}3$ response is a scalar opposite-port reversal plus a correction through two shared coherent doublet reads; their four complex outputs are minimal in this linear factorization class, whose rank bound does not set the physical work price. Here τ_x, τ_z act between the two read doublets; spin remains in the carried wave. The full twelve-component update retains its complement, and the low-frequency subspace is dressed at finite momentum. The coefficient $1/3$ is in this graph's hop units. This construction supplies a conditional Clifford principal symbol, rather than a selected substrate law: the local response, physical speed normalization, frame transitions and their spin connection, and the action-to-demand map remain to be derived. At zero load, additional zero-phase branches elsewhere in the Brillouin zone require physical mode selection before a unique particle spectrum follows (supplementary code).

A local chiral read. On this supplied carrier, let $\Xi = \mathrm{diag}(\sigma_p)$ and $P_L = (I_{12} - \Xi)/2$, $P_R = (I_{12} + \Xi)/2$. The identities $\Xi J_0 = J_1$ and $\Xi J_1 = J_0$ give $V^\dagger \Xi V = \tau_x \otimes I_2 = \gamma^5$ for the principal Clifford symbol above. The complete massless update commutes with Ξ; each chiral sector retains two principal and four complementary amplitudes. Proper cubic rotations preserve this read, while opposite-port parity exchanges its signs. It uses the native spin line correlated with each travel port: a port-independent operator $I_6 \otimes g$ has zero off-diagonal blocks between J_0, J_1 and cannot implement this read. For real λ, the specified local response $P_R + e^{-i\lambda}P_L$ is unitary on all twelve amplitudes and acts identically on P_R. It introduces no extra physical tick. This identifies Lorentz chirality at the

principal-symbol level; it does not establish exact lattice Lorentz symmetry. Chirality is distinct from momentum-dependent helicity, and the massive response generically mixes its two sectors.

An independent internal weak action requires distinguishable roles in addition to a Weyl spin doublet: the matrices commuting with all three spin Pauli matrices on that doublet are scalar. On two supplied roles with identical massless propagation, the generators $P_L \otimes \sigma_a/2$ constructively realize a left-only SU(2), commuting with principal Lorentz generators. This has twenty-four full amplitudes and eight principal amplitudes, equivalently two copies of the retained carrier. Role multiplicity is not another cell channel, but neither can it be erased from the representation or its eventual work accounting. The physical role pairing, active handedness, gauge response and compatible mass generation remain to be selected; identical massless propagation here does not identify the native D=2 neutrino and D=3 electron responses (supplementary code).

A controlled full-wave limit. The supplied Cartesian carrier of Eq. (29) admits a many-tick continuum limit with all twelve amplitudes retained. Write $Q = I_{12} - VV^\dagger$ and $D_{\mathbf{p}} = \mathrm{diag}(\mathbf{d}_p \cdot \mathbf{p}) \otimes I_2$. For bounded Lipschitz, spatially homogeneous controls $m(t), g(t)$, use the complete tick

$$U_\epsilon(t) = S(\epsilon\mathbf{p}) C_* e^{-i\epsilon m(t) M_*} e^{-i\epsilon g(t) P_L}, \qquad H(t) = m(t)\beta + \frac{1}{3}\boldsymbol{\alpha}\cdot\mathbf{p} + g(t)\frac{I_4 - \gamma^5}{2},$$

where $\beta = \tau_z \otimes I_2$ and $\alpha_i = \tau_x \otimes \sigma_i$. The parameter ϵ indexes long wavelengths, weak per-tick phases and macroscopic time $t = \epsilon n$; each native update remains one complete tick. At zero arguments, $U_0 = 2VV^\dagger - I_{12}$ has an exact complementary gap of two. The perpendicular corrector and its norm are

$$X = -\frac{i}{2} Q D_{\mathbf{p}} V, \qquad X^\dagger X = \frac{|\mathbf{p}|^2}{18} I_4, \qquad x := \|X\| = \frac{|\mathbf{p}|}{3\sqrt{2}}. \tag{31}$$

Both control operators preserve $\operatorname{ran} V$, so X is independent of m, g. Let $m_* = \sup|m|$, $g_* = \sup|g|$, $a = \|\mathbf{p}\|_\infty + m_* + g_*$, $h = |\mathbf{p}|/3 + m_* + g_*$, $B = (a^2 + h^2)/2 + (a + h)x$, and $L = \mathrm{Lip}(m) + \mathrm{Lip}(g)$. If $G(t, 0)$ solves $\partial_t G = -iH(t)G$, $G(0, 0) = I_4$, then for $\epsilon > 0$ and integers $n \geq 1$, in operator norm,

$$\|U_\epsilon((n-1)\epsilon) \cdots U_\epsilon(0)V - VG(n\epsilon, 0)\| \leq \epsilon\big[2x + n\epsilon(B + L/2)\big]. \tag{32}$$

Rounding a specified macroscopic time down to $n\epsilon$ adds at most ϵh.

To obtain the bound, set $A(t) = \partial_\epsilon U_\epsilon(t)|_0$. The identity $(I - U_0)X = A(t)V + iVH(t)$ cancels the first-order defect of $V + \epsilon X$. The second derivatives of the full tick and $e^{-i\epsilon H(t)}$ have norms bounded by a^2 and h^2, giving a one-step defect at most $\epsilon^2 B$. Exact unitarity telescopes that defect through all ticks; restoring the actual initial V contributes $2\epsilon x$. The chronological comparison contributes at most $n\epsilon^2 L/2$. The corrector is a proof device: propagation starts from V, with no projection, renormalization or discarded complementary wave.

The estimate is uniform on compact momentum sets and bounded times, and is unchanged by tensoring an external reference. Parseval's identity gives the full-state L^2

limit for band-limited envelopes. For a general fixed L^2 envelope, prepare its Fourier restriction to the rescaled Brillouin zone with the truncated norm retained; compact cutoffs and unitarity then give strong convergence, without a uniform rate for arbitrary high-frequency input. This is the continuum limit of the specified response and prepared four-component sector. Physical response and speed calibration, spatially varying connections and changing-atlas homogenization retain their separate requirements. Other native spectral branches remain in the complete update (supplementary code).

Local transport speed bounds. For a momentum-independent local unitary followed by one orthogonal-port unit hop, let V embed a complete isolated same-eigenphase sector of dimension q. The compressed port effects obey $E_p = V^\dagger \Pi_p V \geq 0$ and $\sum_p E_p = I_q$. On six Cartesian ports the velocity matrices are $A_i = E_{+i} - E_{-i}$, so $\sum_i \|A_i\|_1 \leq q$, where $\|\cdot\|_1$ is the trace norm. An isotropic Clifford symbol $A_i = v\alpha_i$, $\alpha_i^2 = I_q$, therefore has $|v| \leq 1/3$. At $v = 1/3$, equality fixes $E_{\pm i} = (I_q \pm \alpha_i)/6$; for $q = 4$ their six rank-two supports require at least twelve orthogonal port coordinates. The carrier above attains this bound, independently of its separated complementary phases.

A direct coherent-neighbour update has a different bound. For an exactly unitary finite stencil $U(\mathbf{k}) = \sum_d T_d e^{i\mathbf{k}\cdot d}$ with unit-length displacements and no zero-displacement term, set $C_i = -iU(0)^\dagger \partial_i U(0)$. Differentiating unitarity and using $\Delta U(0) = -U(0)$ gives $\sum_i C_i^2 = I$. Compression to an isolated same-eigenphase sector with an isotropic Clifford principal symbol yields $|v| \leq 1/\sqrt{3}$. A supplied four-component six-axial-neighbour stencil attains this bound, with massless eigenphases $\cos\omega = (\cos k_x + \cos k_y + \cos k_z)/3$. A local mass turn gives the principal square $(m^2 + |\mathbf{p}|^2/3) I_4$. At zero mass its only zero-phase point is $\mathbf{k} = 0$; the retained π-phase partner lies at (π, π, π), modulo 2π. This stencil is isotropic at first order but lacks full cubic covariance under a fixed onsite unitary. These are bounds and witnesses for specified transport classes in unit-hop, unit-tick coordinates; the physical speed and atlas identifications remain open (supplementary code).

An eight-component direct stencil admits a projective onsite unitary action of all 24 proper cubic rotations and an invariant mass involution M. Five mutually anticommuting Hermitian stencil involutions, together with M anticommuting with all five, require at least eight components; the supplied construction attains this dimension. Its exact phases satisfy $\cos\omega = \cos m\,(\cos k_x + \cos k_y + \cos k_z)/3$. Its joint small-mass/momentum principal carrier comprises two Dirac copies; the complete momentum-and-mass family has no proper momentum-independent invariant subspace. On the same Cartesian lattice with constant m, let J isometrically embed the $M = +1$ eigenspace, let d average the scalar walk's six endpoint ports, and denote that walk at $\theta = -m$ by U^{G}_{-m}. The compressed propagators obey $J^\dagger U_m(\mathbf{k})^n J = [d U^{\mathrm{G}}_{-m}(\mathbf{k})^n d^\dagger] I_4$ for every integer n. The common matrix kernel has seeds $F_0 = I_4$ and $F_1 = e^{-im}(\sum_i \cos k_i) I_4/3$ and obeys the same second-order recurrence. This is an equality of the specified source/read amplitudes; the compressed evolution is not itself a unitary closed sector. The scalar walk's propagating plane has trivial axial little-group action, while its transverse cycle modes have flat phases. Thus the common endpoint kernel leaves the physical photon response and growing-atlas identification open (supplementary code).

For these fixed updates, the common endpoint moments also determine the complete

source-generated cyclic wave up to unitary equivalence. The Gram matrix of any finite sum $\sum_n U^n J f_n$ contains only F_{m-n}, so equality of all integer moments gives an isometric correspondence that intertwines the updates and source maps. Matched endpoint-supported read projectors $JP_rJ^\dagger$, together with their no-read complement, therefore give identical finite outcome histories, survival probabilities and external-reference correlations. Read times and projectors may vary; the propagation law remains fixed. This statement specifies an instrument class without selecting the physical photon detector or equating the complete carrier spaces (supplementary code).

Neighbor-link consistency. Attach reciprocal scalar phases, or unitary links on a separate tensor factor, to the eight-component direct stencil while holding its six coefficients A_d fixed. For orthogonal signed axial directions u, v, the two routes between opposite square corners contribute $C_{uv} \otimes (Q_1 - Q_2)$ to the off-diagonal block of $U^\dagger U$, where $C_{uv} = A_u^\dagger A_{-v} = -A_v^\dagger A_{-u}$ and $\|C_{uv}\|_F^2 = 2/9$. Hence exact unitarity requires and, with the corresponding output identities, is guaranteed by equal ordered transports $Q_1 = Q_2$ around every elementary square. Nontrivial global winding remains possible. Single-site inputs do not expose a failed cancellation; coherent inputs at different sites do. This restriction concerns the fixed direct coefficients and external links, not an arbitrary spin connection. A retained orthogonal-port hop instead remains unitary for arbitrary reciprocal unitary links, followed or preceded by any complete local unitary, with no intermediate physical tick or projection (supplementary code).

On any supplied fixed reciprocal graph without isolated vertices, let S_L be that linked port hop and d the normalized uniform-star read. Then $S_L^\dagger = S_L$, $S_L^2 = I$ and $T_L = dS_Ld^\dagger$ is a Hermitian contraction, irrespective of loop holonomy. With $P = d^\dagger d$ and constant θ, the complete update $U = S_L[e^{i\theta}P - e^{-i\theta}(I - P)]$ has endpoint history $F_0 = I$, $F_1 = e^{i\theta}T_L$ and $F_{n+2} = 2\cos\theta\, T_L F_{n+1} - F_n$. This extends the scalar endpoint recurrence to prescribed reciprocal connections without a Fourier or flatness assumption. Spatially varying centered mass phases generally prevent this closed vertex recurrence. They still admit an exact complete-wave description: for vertex phases A, their arc lift $\widetilde{A}$, $\widetilde{B} = -\widetilde{A}^\dagger$, $f = d\psi$ and $h = (I - P)\psi$,

$$f' = T_L A f + dS_L \widetilde{B} h, \qquad h' = (I - P)S_L d^\dagger A f + (I - P)S_L \widetilde{B} h. \tag{33}$$

Both fields update in the same tick and reconstruct $\psi' = d^\dagger f' + h'$. The retained port complement is not an added scalar endpoint state. The physical connection, photon read and continuum identification remain unspecified (supplementary code).

The twelve-port Clifford carrier likewise admits arbitrary reciprocal unitary links on a separate q-dimensional external fiber while retaining all $12q$ amplitudes. For a supplied smooth weak-link scaling, $X = \epsilon x$, $L_{x+e_i,x} = I_q - i\epsilon \mathcal{A}_i(X) + O(\epsilon^2)$, $\mathbf{k} = \epsilon \mathbf{p}$, and a mass turn of phase ϵm with constant m, differentiation of the complete update before compression gives

$$H_{\text{eff}} = m\beta \otimes I_q + \frac{1}{3}\sum_i \alpha_i \otimes \Pi_i, \qquad \Pi_i = p_i I_q - \mathcal{A}_i, \quad \alpha_i = \tau_x \otimes \sigma_i, \quad \beta = \tau_z \otimes I_2. \tag{34}$$

Here the hop convention $e^{i\mathbf{k}\cdot d}$ uses envelopes $e^{-i\mathbf{p}\cdot X}$ and $p_i = +i\partial_i$. The square retains the terms $\sum_{i<j} \alpha_i\alpha_j \otimes [\Pi_i, \Pi_j]/9$ when covariant momenta do not commute. This is a principal

expansion with the full complement retained, not a continuum convergence theorem or a physical identification of the external connection (supplementary code).

A moving native plane. Native dimension need not fix spatial orientation. In the same Cartesian six-port control, choose a unit normal $\mathbf{n}$ and project each travel direction into its tangent plane:

$$\mathbf{t}_p = (I_3 - \mathbf{n}\mathbf{n}^\top)\mathbf{d}_p, \qquad B_\mathbf{n} = \tfrac{1}{2}(\boldsymbol{\sigma}\cdot\mathbf{t}_1, \ldots, \boldsymbol{\sigma}\cdot\mathbf{t}_6), \qquad A_\mathbf{n} = I_{12} + (e^{-i\vartheta} - 1)B_\mathbf{n}^\dagger B_\mathbf{n}. \tag{35}$$

The tight-frame identity $\sum_p \mathbf{d}_p\mathbf{d}_p^\top = 2I_3$ gives $B_\mathbf{n}B_\mathbf{n}^\dagger = I_2$, so $A_\mathbf{n}$ is unitary for real ϑ. Tangential Pauli products commute with $\boldsymbol{\sigma}\cdot\mathbf{n}$. Each normal eigenspace therefore retains six travel amplitudes, with one combination receiving the phase $e^{-i\vartheta}$ and five combinations unchanged by $A_\mathbf{n}$. At $\mathbf{n} = \hat{\mathbf{z}}$ the native read has zero normal-port entries, while those ports still carry traveling amplitudes. This is a two-dimensional native response in three-dimensional travel, not a restriction of spatial support to a flat sheet. The plane is unchanged by $\mathbf{n} \mapsto -\mathbf{n}$; it does not select which normal eigenspace the wave occupies.

For changing normals, a unitary native link on each actual neighbor hop can preserve the carried line by satisfying $q_{\mathbf{n}_{y,t+1}}L_{yx,t} = L_{yx,t}q_{\mathbf{n}_{x,t}}$, where $q_\mathbf{n} = (I_2 + \boldsymbol{\sigma}\cdot\mathbf{n})/2$. With the same bijective six-port routing, the full turn and linked hop then preserve norm and that moving line without intermediate projections or extra physical sub-ticks. This construction selects neither the orientation history nor its physical connection, occupied modes or mass read. A common unitary on travel and native state that acts identically on flavor leaves the unconditioned reduced flavor state unchanged; random plane orientation alone does not supply the relative mode evolution used in Appendix J.3.

A photon completes a look-back read; it requires no reversible state injection when an address activates. For source X, read Y, and realized timestamped causal interval $\mathfrak{C} = J^+(X) \cap J^-(Y)$, its amplitude is

$$K_\mathfrak{C}(Y, X) = \sum_{\substack{\gamma: X \rightsquigarrow Y \\ \text{activation-respecting}}} \prod_{p\in\gamma} W_p. \tag{36}$$

Routes use relations only after activation. Later activation, or activation outside $\mathfrak{C}$, leaves a completed read unchanged. Fixed adjacency reduces this kernel to the exact Laplacian walk. Changing adjacency still requires a time-ordered causal-history or spacetime-homogenization theorem. RG running remains ceded downstream (Section 5.3).

E.5 Finite-tick thermal limit

The accelerated-register calculation in Section 4.2 uses the chirp $f(\tau) = \exp[-(\delta - i\Omega)e^{-\tau}]$, with $\Omega, \delta > 0$. Here τ is proper time in units c/a, $\nu = \omega c/a$, and ϵ is the proper sampling interval in the same units. The continuum amplitude and uniformly sampled amplitude are

$$F(\nu) = \Gamma(-i\nu)(\delta - i\Omega)^{i\nu}, \qquad D_\epsilon(\nu) = \lim_{\eta\downarrow 0} \epsilon \sum_{n\in\mathbb{Z}} e^{(i\nu-\eta)n\epsilon} f(n\epsilon).$$

The auxiliary η defines the constant late-time tail by Abel summation; δ regulates the early-time chirp. They are distinct limits. Set $L = 2\pi/\epsilon$, $b = \arctan(\delta/\Omega)$ and $C(s) = \sqrt{2\pi/[s(1-e^{-2\pi s})]}$ for $s > 0$.

Proposition 1 (Controlled uniform-tick sampling)**.** *For $\nu \notin L\mathbb{Z}$,*

$$D_\epsilon(\nu) = \sum_{k\in\mathbb{Z}} F(\nu + kL).$$

For $0 < |\nu| < L$, the absolute sampling error is bounded by

$$|D_\epsilon(\nu) - F(\nu)| \le B_\epsilon(\nu) := \frac{C(L+\nu)e^{-b(L+\nu)}}{1-e^{-bL}} + \frac{C(L-\nu)e^{-(\pi-b)(L-\nu)}}{1-e^{-(\pi-b)L}}. \tag{37}$$

At fixed $\delta, \Omega > 0$, convergence is exponential in $1/\epsilon$ on every fixed compact frequency band away from zero. At any fixed $\epsilon, \delta > 0$, the sampled probability ratio cannot equal $e^{-c_*\nu}$ with constant $c_* > 0$ throughout any open frequency band where the ratio is defined.

Proof. For $\eta > 0$, $e^{-\eta\tau}f(\tau)$ is Schwartz, with Fourier transform $\Gamma(\eta - i\nu)(\delta - i\Omega)^{-\eta+i\nu}$. Poisson summation and then $\eta \downarrow 0$ give the alias series; the Gamma decay makes this limit locally uniform away from $L\mathbb{Z}$. The exact identity $|\Gamma(is)|^2 = \pi/[s\sinh(\pi s)]$ [97, DLMF 5.4.3] gives $|F(s)| = C(s)e^{-bs}$ and $|F(-s)| = C(s)e^{-(\pi-b)s}$. Since C decreases, the two alias tails are bounded by the geometric sums in Eq. 37.

For the last assertion, $G(\nu) = |D_\epsilon(\nu)|^2$ is L-periodic and real analytic between its poles. If $G(-\nu) = e^{-c_*\nu}G(\nu)$ holds on an open subinterval of $(0, L)$, real analyticity extends it throughout that interval. Replacing ν by $L - \nu$ and using periodicity gives $G(\nu) = e^{-c_*L}G(\nu)$, a contradiction: G is not identically zero, since D_ϵ has the nonzero constant-tail pole at zero. On an interval $(mL, (m+1)L)$, write $\nu = s + mL$. The same argument gives $G(s) = e^{-c_*(2m+1)L}G(s)$, impossible for integer m and $c_* > 0$. □

For $0 < \nu < L$, put $e_\pm = B_\epsilon(\pm\nu)/|F(\pm\nu)|$. Whenever both are below one, the triangle inequality gives the explicit probability-ratio bounds

$$\left(\frac{1-e_-}{1+e_+}\right)^2 \le \frac{|D_\epsilon(-\nu)/D_\epsilon(\nu)|^2}{e^{-4\nu\arctan(\Omega/\delta)}} \le \left(\frac{1+e_-}{1-e_+}\right)^2.$$

Thus the continuum limit followed by $\delta \downarrow 0$ yields $e^{-2\pi\nu}$ and $k_BT = \hbar a/(2\pi c)$. A sufficient joint limit on fixed compact nonzero bands is $\delta/\Omega = \sqrt{\epsilon}$ at fixed Ω. The error bound is nonuniform when $\delta \downarrow 0$ at fixed ϵ; the Nyquist band $|\nu| < \pi/\epsilon$ alone supplies no thermal-accuracy guarantee. These statements concern the specified uniform proper-tick instrument. Its realization as the substrate's microscopic read law remains part of the instrument and continuum construction in Appendix O. Independent time-domain and alias-domain calculations verify the identities and bounds (supplementary code).

F The compute matrix

F.1 Registers and particle patterns

Each sphere has seven *channels*: internal registers that particles excite in different combinations. Three carry color, the strong force's three-valued charge; two carry spin, written W^+ and W^- below. The weak interaction reads this pair and owns no separate channels (Appendix F). One channel carries electric charge. The seventh, existence, marks local occupancy for later reference. Any occupied channel implies existence, so the mark costs nothing to hold (Section 4.5). The pattern repeats across three generations, progressively heavier copies of the particle families.

The selected channel patterns are (Fig. 30):

- **Neutrino**: only W^+ and existence resonate — no color, no charge, invisible to the drain. The faintest wave.
- **Electron**: opens the gate fully (W^+, W^-) and couples to the drain (Q) but never touches the source — no color, no confinement.
- **Up and down quarks**: couple to all three shells — color-locked in the source, spin in the gate, charge in the drain. The down quark's one extra gate channel (W^-) over the up quark gives the mass ratio 13/6. Its arctangent matches the measured CKM angle γ; this is the structural gate angle, not yet the rephasing-invariant CP carrier J, whose time-ordered mechanism remains open (Section 3.2).
- **Top quark**: engages all seven channels, the maximum channel occupancy. The bottom quark has the same binary occupancy; their charge values distinguish them in the matrix below.
- **Photon (the read operation)**: not a pattern the substrate maintains but the query itself — excites no channels (d=0), zero internal state, though it interacts with charged particles (α_{EM}). In dense matter, these encounters are frequent (opacity); in vacuum, the photon travels unimpeded.

Under the independent ternary-alphabet convention, each channel inscription is empty, particle, or antiparticle, giving raw register capacity $3^7 = 2187$. Antiparticles are A2's mirror relabeling, $W^+ \leftrightarrow W^-$, reversing each computing channel's sense. Existence has no sense to reverse, so its unsigned catalogue contains $2 \cdot 3^6 = 1458$ inscriptions before occupancy constraints. At a marked address ($E = 1$), the six computing channels admit $3^6 = 729$ inscriptions. The implication that any engaged channel carries E leaves $1 + 3^6 = 730$ inscriptions in the full catalogue: the unmarked vacuum plus all marked inscriptions, including a marked but noncomputing one. This implication does not decide whether the two empty descriptions are physically distinct; the stronger convention $E = \text{OR}(\text{computing occupancies})$ would instead retain 729. These are inscription counts under the stated conditions, before further compatibility or spectrum selection. The gate's $3^2 = 9$ occupancy inscriptions are distinct from the dimension $N_{\text{spin}} = 2$ of its coherent spin role. The counting rules price a content space of 29 states, including vacuum: an accounting over independent shells, rather than a row census (Section 4.8). Mirror relabeling changes gate occupancy without changing a count. Since mass formulas price counts (Rule 1, Appendix I.6), particles and antiparticles have equal mass. The extra gate

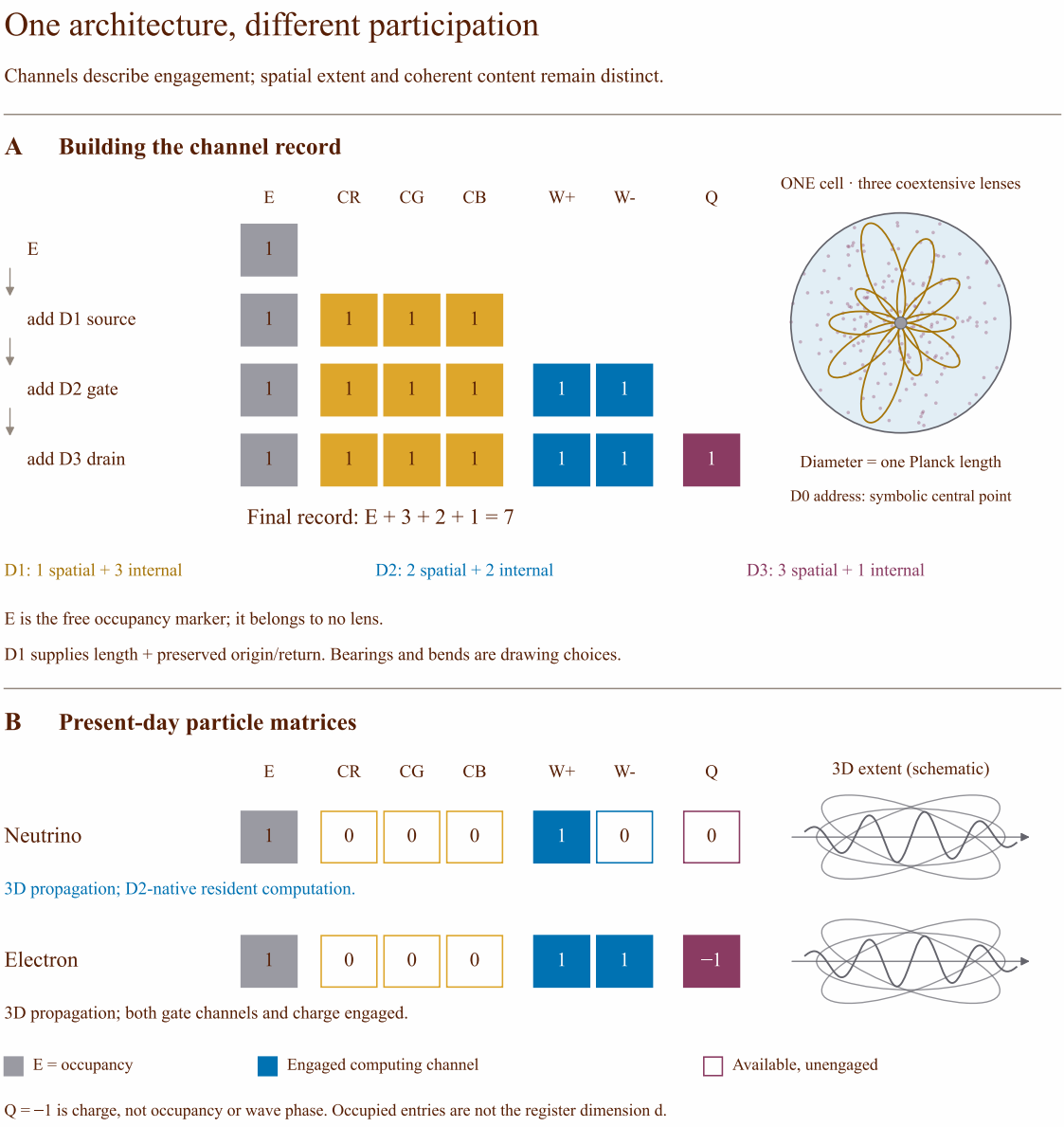


Figure 29: *Dimensional architecture and participation. (A) The free occupancy marker E is separate from the three completely coextensive lenses and their spatial/internal budgets. The sphere is a qualitative projection; textures are not calibrated physical densities. (B) The two present-day records share three-dimensional spatial extent; the equal-sized wave icons do not compare wavelengths. $Q = -1$ denotes charge. Occupied entries do not count coherent register dimension. Panel C continues on p. 114.*

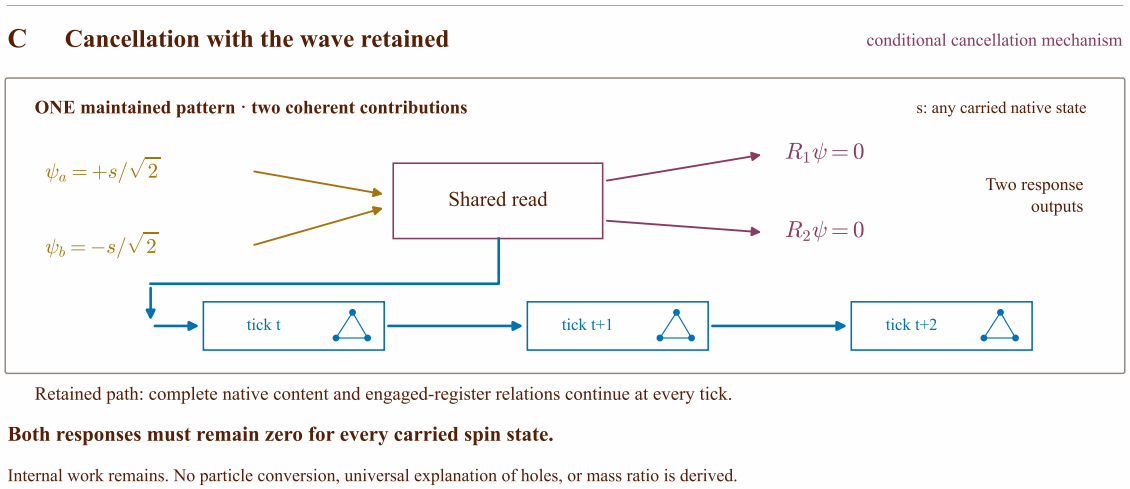


Figure 29: *Continued: conditional cancellation. A supplied shared read cancels two response outputs while retaining the complete native content. Both zeros must persist for every carried spin state under the local update. Linked-node icons represent maintained relations, not a state census. Zero response neither erases charge nor removes internal work. This conditional mechanism does not derive particle conversion, the existing matrix holes, or a mass ratio.*

channel separating d from u gives the same 13/6 ratio between $\bar{d}$ and $\bar{u}$ in the mirrored column.

F.2 Matrix entries and their reading

A full wave with fewer resident operands. A native computing dimension can be smaller than the representation carrying its orientation. On a supplied one-dimensional active domain, write $\psi_{\nu,az} = a_{az} v_{az}$ with $v_{az}^{\dagger} v_{az} = 1$. Every retained spatial amplitude a_{az} is independent; both displayed spin components share that amplitude through v_{az}. An intrinsic scalar update $[u_a]$, with $|u_a| = 1$, has one ordered relation, whereas a freely engaged electron doublet has four. A supplied local unitary transport $U_{az} v_{az} = v'_{az} u_a$, with normalized v'_{az}, preserves the entire allowed wave as orientation changes. The active domain and transport are premises; their construction retains the geometry label whenever it is independently coherent. The complete mass bill must also price orientation, charge and transfer work. Arbitrary independent two-component input requires the larger domain.

For a separate controlled comparison holding the native doublet fixed, let a label transverse samples, $z = 1, \ldots, n$ a retained third-coordinate label, and $\psi_{az} \in \mathbb{C}^2$ the complete native doublet. A resident suboperation independent of z has $U_v = \bigoplus_a (I_n \otimes A_a)$, whereas a supplied third-coordinate-sensitive suboperation has $U_e = \bigoplus_{a,z} B_{az}$, with A_a, B_{az} unitary. Both retain arbitrary three-dimensional wave information; neighboring-cell transfer remains part of the complete tick. For m transverse labels, the linear spans of these operator families have complex dimensions $4m$ and $4mn$. A freely engaged joint n-coordinate/doublet block would instead have $4n^2$ relations per transverse label. These

are distinct operation choices. Repeated coefficients do not remove their physical occurrences or select the local weights that convert them into demand (supplementary code).

Cancellation with the native state retained. For supplied response maps $R_{1,t}, R_{2,t}$, let V_t encode the complete native space isometrically, $V_t^\dagger V_t = I$. The identities

$$R_{1,t}V_t = R_{2,t}V_t = 0, \qquad U_t V_t = V_{t+1}A_t, \qquad A_t^\dagger A_t = I \tag{38}$$

express persistent cancellation while the native state continues turning. Their product preserves both response zeros and every native input through successive ticks. For two paths with response rows $R_W = [W_1, W_2]$ and $R_q = [q_1 I, q_2 I]$, $(q_1, q_2) \neq (0, 0)$, the common cancelling combination retains every native state precisely when $q_2 W_1 - q_1 W_2 = 0$. A single cancelled polarization is insufficient. The response maps must be physically identified before this condition explains a matrix hole; a vanishing output neither changes conserved charge nor removes the engaged block's internal work (supplementary code).

The compute matrix records how particles couple to the transistor's landscapes (Section 1.1): a row is a particle's coupling signature; a column is one landscape channel. A3 selects the minimum integer satisfying each constraint, so every kernel integer is the cheapest mode count forced by the axioms. Operation 1 derives the equal relation bill at K=1. The separate matrix-to-register convention assigns each occupied entry one A4 distinguishability unit. Discrete kernel-integer counts weight these units equally: disjoint units add, and an n-unit domain normalizes each to $1/n$. Continuous SO(2) seam measures (Section 3.1) and dynamically bE-weighted operations retain their separately disclosed types.

A particle's identity combines its full channel content with its generation index. Quarks spend the N_c^3 generation space as cell occupancy; leptons carry it as internal dimension. Identical binary signatures need not identify the same state: $e/\mu/\tau$ differ by generation, while b/t at generation-3 saturation differ in their drain values, $Q_b = -1/3$ and $Q_t = +2/3$. The 11×7 compute matrix displays resonance and generation, not every channel value.

Its columns are the cumulative boot record 1+3+2+1: a pre-shell existence marker, three D=1 source channels, two D=2 gate channels, and one D=3 drain channel. The E marker predates the spatial eras and belongs to no shell. This grouping follows from the overlaid architecture; it does not yet derive resident row occupations or their holes. The column meanings are C_R, C_G, C_B = color (source, D=1, K=r); W^+, W^- = weak isospin (gate, D=2, K=1); Q = electric charge (drain, D=3, K=$1/r$); E = local occupancy (the existence mark, A2 conservation). A resonant cell marks a mode excited by the wave. The photon is the read operation (d=0, zero internal state and compute cost).

Local occupancy. E is the local occupancy mark. Compatible maintained patterns whose computing-channel supports do not overlap share this mark: E combines by OR, while their channel values and carried states remain distinct. It is counted once in a local occupancy tally, regardless of how many compatible patterns are present. A shared occupancy mask is not itself a new particle identity or a mass bill. The matrix lists alternative patterns, so its E column records this property separately in each row.

Charge quantization distributes the drain's one charge state over $N_c(D{=}1) = 3$ color states: $Q(u) = +2/3$, $Q(d) = -1/3$. Colorless leptons have $Q(\nu) = 0$, $Q(e) = -1$ by A2 conservation. A1's serialized clock orders the gate's $N_{\text{spin}} = 2$ states: left-handed processing activates the $K{=}1$ doublet; right-handed processing leaves it passive as a singlet. Occupation of the negative-isospin W^- column licenses a fermion to source or absorb a W^- quantum. The boson's mass is separately priced as a seam row (Section 3.1).

One wave, two gate reads. The wave carries its spin information; the gate is the coextensive $D{=}2$ shell to which that part of the wave couples and through which it refracts. The mass read uses the wave's two spin states, giving $d^2 = 4$; the matrix records the $W^{\pm}$ occupancy of the corresponding gate channels. Section 3.5 calls them the $D{=}2$ spin channels (W^+, W^-). The seven-channel budget contains no separate weak-isospin slot: spin and weak occupancy are operation-specific reads of the same coupled wave and shell. Masses and moments read coherence count; matrix rows and the Gottfried integral read occupancy. No formula bills both roles at once.

Only the weak force reads handedness. Color does not directly act on this spin factor, and charge couples parity-evenly: Operation 5 reads the spin magnitude. At the weak vertex, the wave's chirality determines which fermions couple, with the left-handed doublet active and right-handed singlet passive.

Compatibility with Coleman–Mandula is structural. Alignment with processing refers to the particle's own worldline; a universal axis would contradict substrate isotropy. Spin projected along that direction is a rotational scalar, so a weak charge built from it commutes with every rotation generator. Conditional on one irreducible complex two-state gate factor, its projective state space is the Bloch sphere, with $SU(2) \to SO(3)$ double cover and $2\pi \to -\mathbb{1}$ (exact representation theory; supplementary code). Its scalar commutant is consistent with the separate roles. The theorem's Lorentz, S-matrix, and scattering premises have not been built here. The physical rotation-group action remains the conditional map of Appendix E: the read-axis carrier forces its argument; existence is the one named premise; the two-read law then follows.

The coextensive source, gate, and drain have widths $3, 2, 1$, ordered only by D (Fig. 1); the pre-shell E marker records existence without adding a fourth shell. Filling rises from 43% in $D{=}1$ color to 100% in existence, mapping to the force hierarchy: sparser shells have stronger couplings. The identical binary patterns of E and W^+ give rank 6. Every computing particle exists, so the structurally present "+1" existence channel is linearly dependent: exists $\neq$ computes in the matrix's algebra (Section 2.5).

Generations and Pauli exclusion. Pauli exclusion (A3+A4, Appendix I.1) applied to the color register gives $N_c(D{=}1) = 3$ generation slots. A2+A3 saturate them, with each generation occupying a more complex configuration. Under this progression rule, the third-generation top saturates every channel and strictly nested color chains cannot exceed $N_c = 3$ (supplementary code). No fourth, strictly more complex quark configuration remains. This exclusion is conditional on the progression grammar.

The lepton ceiling has the same conditional tier. A colorless row can borrow one unoccupied block as internal dimension, used by generation 2, and has one circulant closure: the Z_3 Koide relation, realized by the observed masses at 10^{-5}. The grammar supplies

Particle	Gen	E	C_R	C_G	C_B	W^+	W^-	Q	Mass formula	m/m_e
Colorless patterns										
ν_e	1	✓				✓		0	$m_i = \beta_\nu(1,2,10)_i$ (conditional)	$(1,2,10)/B$
e	1	✓				✓	✓	-1	m_e (anchor)	1
μ	2	✓				✓	✓	-1	$(d_\mu^2-14)/N_{\text{spin}}^2$, d_μ=29	206.8
τ	3	✓				✓	✓	-1	Koide(m_e, m_μ)	3477
Down-type configurations										
d	1	✓	✓			✓	✓	$-1/3$	$m_u + 5m_e$	9.3
s	2	✓	✓	✓		✓	✓	$-1/3$	$m_d(9/2)^2$	188
b	3	✓	✓	✓	✓	✓	✓	$-1/3$	$m_s(27/4)^2$	8567
Up-type configurations										
u	1	✓	✓			✓		$+2/3$	$30/7$	4.3
c	2	✓	✓	✓	✓	✓		$+2/3$	$m_u(73/3)^2$	2537
t	3	✓	✓	✓	✓	✓	✓	$+2/3$	$m_c(35/3)^2$	345398

Table 13: *The compute matrix (10 resident particle patterns, 49 resonant cells), grouped by family with columns in boot order: existence, source, gate, drain. The Q column shows charge values; each nonzero value marks one resonant entry. The ν_e row represents the shared resident register: its three flavors are weak-endpoint reads of the same pattern, and its three masses are positive square roots of that register's mass-squared eigenvalues; $e/\mu/\tau$ share a binary signature and differ by generation; b/t share both their generation-3 index and binary signature, but differ in drain charge (the identity clause above). Thus 10 resident patterns encode 12 operational fermion labels (3×4 per generation). Quark-row mass formulas are the internal generation ladder (scaffolding, not comparison rows; Section 3.1). Antiparticle rows are not tabulated; each follows by A2's $W^+ \leftrightarrow W^-$ exchange (Section 3.7). In the one-register typing, a weak conversion resolves flavor against the charged-lepton endpoint basis; that instrument measures the propagation state and cannot be the mechanism that mixes it (the endpoint effects are the three charged-lepton-diagonal projectors, unique up to relabeling; supplementary code). The mass column uses the conditional positive-count spectrum of Section 3.7: $\mathcal{M}_*^2$ has eigenvalues $\beta_\nu^2(1,4,100)$. Its gap matrix has eigenvalues $m_\nu^2(0,1,33)$, with $\beta_\nu = m_\nu/\sqrt{3}$; the displayed mixing generator's eigenvalue magnitudes are not masses. The integers are propagation-mode work counts, not three different flavor masses or extra occupied cells.*

no fourth-lepton distinctness mechanism (supplementary code). The $D = 1$ generation space contains $N_c^3 = 27$ configurations, the ternary register's color sector (Section 4.8). Quarks realize Pauli distinctness through cell occupancy; leptons, barred from color cells, carry it internally. The composition lattice of Fig. 27 and the lepton d-progression are these two uses of the same space.

Quarks: the Pauli progression is visible as color-shell excitations. The down-type staircase is clean (d: 1, s: 2, b: 3 active colors); the up-type runs ahead (u: 1, c: 3, t: 3) — charm saturates color early, the counterpart of its closed W^- gate. Down-type (W^- active): generation ratio N_c^2/N_{spin} at the first step ($m_s/m_d = (9/2)^2$); second step N_c^3/N_{spin}^2 ($m_b/m_s = (27/4)^2$). Unsaturated up-type (u, c: W^- absent; the top row is generation-3-saturated and occupies W^-): gauge redundancies subtracted explicitly. $m_c/m_u = ((N_c^4 - (N_c^2-1))/N_c)^2 = (73/3)^2$.

Leptons: all three generations share one channel signature. Pauli distinctness acts through internal dimension d, conditional on faithful complete-content association: a carried association distinguishes every possible D=1 partner content, all $N_c^3 = 27$. Color

symmetry alone permits 10- or 6-label response quotients (supplementary code). The faithful rule makes 27 the collision-free A3 minimum and gives $d_\mu = 27 + N_{\text{spin}} = 29$. At K=1, its gross bill is $d^2 = 841$ coherences. The family coordinate indexes the complete resident matter-lepton block, whose transparent ground is $5_\nu + 3_e + 3_\mu + 3_\tau = 14$ A4 units (Fig. 30). Rule 4 debits that ground once; the electron anchor has no such coordinate. Division by $N_{\text{spin}}^2 = 4$ spin relations gives $m_\mu/m_e = (29^2 - 14)/4 = 206.75$ (obs 206.768, 0.009%). Koide closes gen-3 through Z_3 symmetry at K=1, with family phase $\delta = 2/9$ selected conditionally by the minimum directed-route read (Appendix I.5). The progression is $d_e = N_{\text{spin}} = 2$, $d_\mu = 29$, $d_\tau = 118 = N_c^4 + N_c^3 + N_c^2 + 1$.

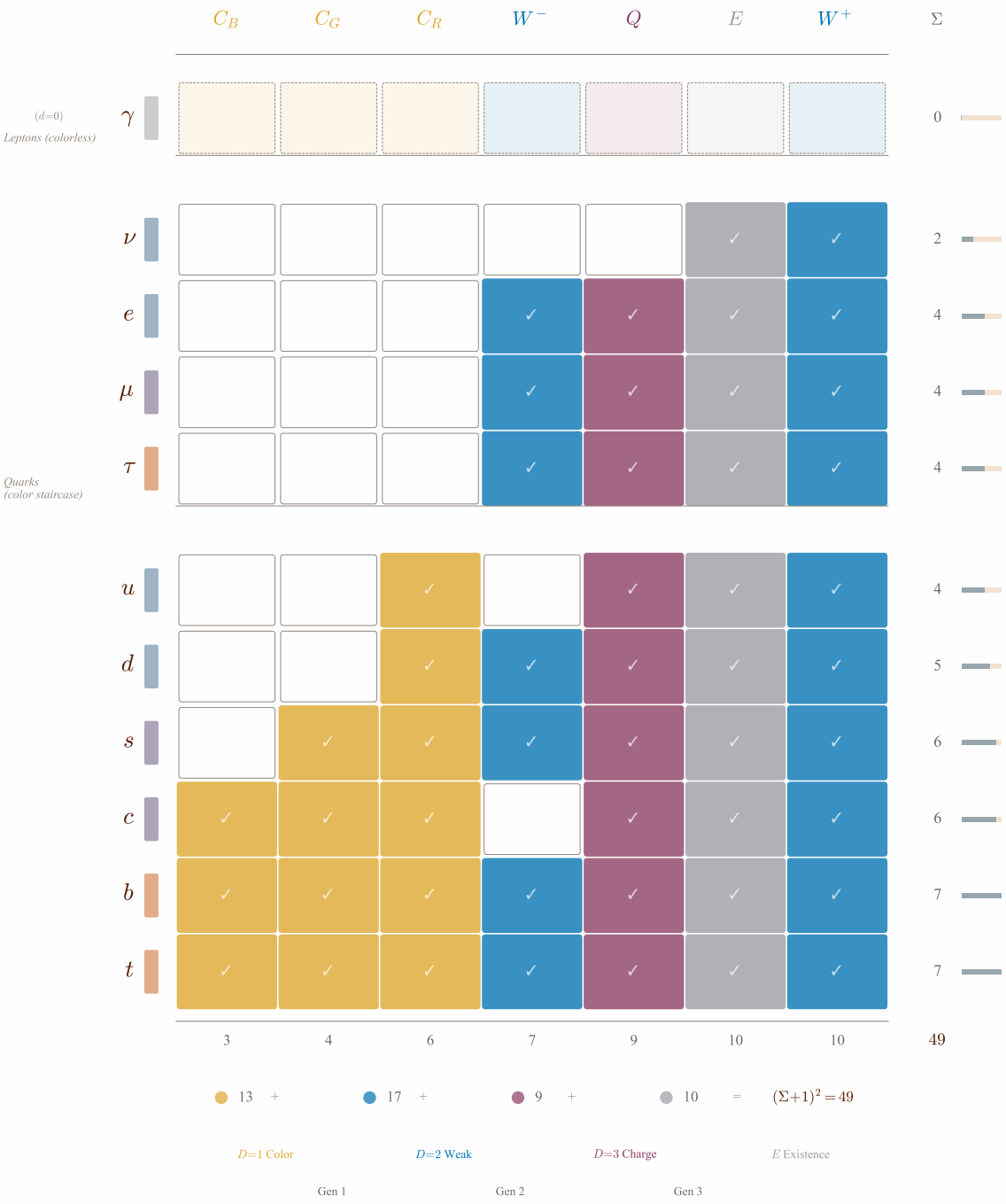


Figure 30: *The compute matrix visualized with columns reordered by ascending coupling count. Resonant cells* $= 49 = (\Sigma{+}1)^2$ *(figure); transparent cells* $= 28 = N_c^3{+}1$ *(ground). In the resident matter-lepton block the holes themselves give* $5_\nu + 3_e + 3_\mu + 3_\tau = 14$ *transparent cells; the photon is excluded because it is the d=0 read, not a resident. The same block contains 14 resonant cells, a figure–ground equality unique to* $N_c = 3$ *on the paper's ladder. The* W^- *column is the sole column that breaks monotonicity — the geometric origin of parity violation. Given the row and column sums plus a single ordering bit (the* W^- *column packed as the costliest — a convention of the reconstruction, not a price: the mirror register of Appendix F.1 weighs the same), heaviest-first greedy packing reproduces this interior uniquely (supplementary code). The drawing's layout — cell order, visible holes, adjacency — is a visualization convention: it defines no physical routes, punctures, or holonomy.*

G Theorems T1–T3

T1: The cost principle (A1–A4). Sequential computation with additive cost gives the path integral. On the coherence space, bE share is the norm or probability share, distinct from per-tick demand. A2 therefore conserves transition probabilities and forces norm preservation (Wigner), hence unitarity: $e^{iS/\hbar}$, not $e^{-S/\hbar}$. A1's forward ordering excludes the antiunitary branch between ticks. The classical limit is least action.

Norm preservation and invertibility govern the resident state in one local update. Every resolution also overwrites a record and disperses its former content outward at read speed. A recurring register therefore returns within a strictly larger dispersal; the combined state never recurs (supplementary code). The billed overwrite makes the arrow fundamental at the bookkeeping tier (Section 4.8); statistical entropy growth follows from it.

T2: Born rule (A2–A4). On a fixed finite complete unresolved-read block, ledger completion derives the Hermitian norm, unitary read evolution, and $\Pr(P|\psi) = \|P\psi\|^2/\|\psi\|^2$. Balanced completion of an equal-role closed pair is unique up to an output $U(1)$; A4's serialized refinement gives the quotient rule (supplementary code). This result does not address growth of the activated state or address graph. Gleason's theorem checks consistency and uniqueness of the resulting measure in the derived $\mathbb{C}$ Hilbert space (dim $= \Sigma{+}1 = 7 \geq 3$); ledger completion supplies its operational meaning.

T3: Localization (A2+A3+T2): locked bE contributes to the clock-rate gradient at its source's position; the equilibrium DM profile is Φ_b^2 (Section 4).

H $D^{\mathrm{s}}_{\max}$ proofs

$D^{\mathrm{s}}_{\max} = 3$ is fixed by four conditions (plus one redundant) — one stability theorem and three consistency identities on the structure built at $N_c = 3$, the framework's own closure rather than independent external selection. First, stable bound orbits under $K = r^{2-D}$ require $D = 3$ [40] (for $D \geq 4$ the potential is too steep; for $D \leq 2$ there is no attractive inverse-distance force).

Second, gauge–channel consistency: each D-shell has mode count $2N_c(D)^2 - 1$ (color states plus generators); equating the total channel count $\Sigma{+}1$ with the $D{=}2$ boundary shell's gauge dimension gives $3D^{\mathrm{s}}_{\max}(D^{\mathrm{s}}_{\max} - 3) = 0$. Mode-matching (the $D{=}1$ confining shell accommodating the full channel count) gives the same polynomial $N_c(N_c{-}3) = 0$ — algebraically equivalent, different physical content.

Third, gauge-dimension closure equates $\sum_{D=1}^{3}(2N_c(D)^2{-}1) = 17{+}7{+}1$ with $(N_c{+}N_{\mathrm{spin}})^2$. It gives $(N_c{-}3)(N_c{+}1)(N_c{+}2) = 0$ and unique positive root $N_c = 3$, with the gate held at universal $N_{\mathrm{spin}} = 2$ by Section 2.6's A3 selection. Under the bare spending-rule generalization, closure also admits $N_c = 1$. Its uniqueness is therefore conditional on gate typing; the first two constraints select $D^{\mathrm{s}}_{\max}$ unconditionally. Self-consistency requires total gauge dimension to equal a kernel integer.

The filling fraction $\phi = (\Sigma{+}1)/(2\Sigma{-}1)$ provides a further comparison. It falls in the measured random-close-packing window 0.634–0.640 only at $N_c = 3$, with $\phi = 7/11 = 0.636$. At $N_c = 2$, $\phi = 0.80$ is unphysical; at $N_c = 4$, $\phi = 0.58$ is below RCP.

A fourth: the transparent-cell total equals the triangular number of the channel count, $T(\Sigma{+}1) = 28 = N_c^3{+}1$, requiring $N_c(N_c{-}3)(N_c^2{-}3N_c{-}2) = 0$ — unique positive *integer* root $N_c = 3$ (the quadratic factor's positive root, $(3{+}\sqrt{17})/2$, is irrational).

Contact count and channel sum. Their equality restates the same constraint. The spending rule $N_c(D) = D^{\mathrm{s}}_{\max}{+}1{-}D$ gives $\Sigma = D^{\mathrm{s}}_{\max}(D^{\mathrm{s}}_{\max}{+}1)/2$, while the isostatic count is $z = 2D^{\mathrm{s}}_{\max}$. Thus $\Sigma - z = N_c(N_c{-}3)/2$ agrees exactly with the selector's zero set (Fig. 31). Unlike $\Sigma - 6 = (N_c{-}3)(N_c{+}4)/2$, it adds no spurious root at -4. Only counts agree: a channel cannot be assigned to a direction through this equality. A union of antipodal pairs has even size, whereas the blocks 3 and 1 are odd (supplementary code).

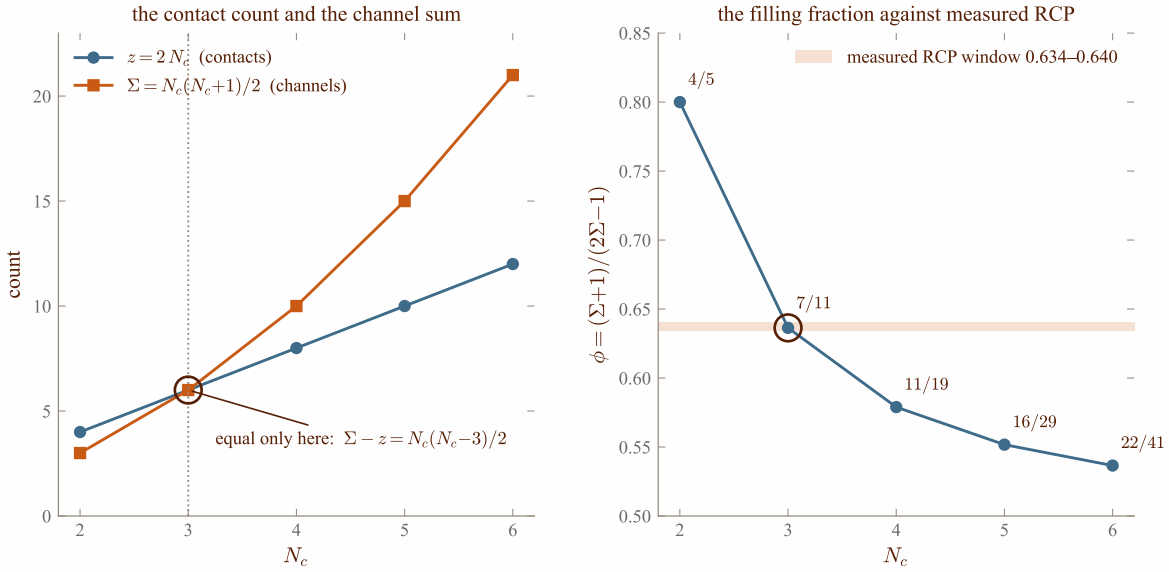


Figure 31: *Two of the $N_c = 3$ selections, drawn. Left: the isostatic contact count $z = 2N_c$ against the channel sum $\Sigma = N_c(N_c{+}1)/2$ — equal exactly on the selector's zero set, $\Sigma - z = N_c(N_c{-}3)/2$, with no extraneous root. The agreement is of counts only (this appendix): no channel is identified with any direction. Right: the filling fraction $(\Sigma{+}1)/(2\Sigma{-}1)$ enters the measured random-close-packing window (0.634–0.640) only at $N_c = 3$; $N_c = 2$ gives 4/5 (unphysical), $N_c = 4$ gives 11/19, below RCP.*

The channel structure gives ten identities with unique positive integer root $N_c = 3$. Their independence means distinct, nonproportional factored polynomials: matrix cross-section enumeration gives twelve, with all further cross-sections duplicating them. Two admit extra roots at $N_c = 2$ or $N_c = 1$ and are excluded (supplementary code). Twelve numerical comparisons re-evaluate existing prediction rows across N_c and add no independent weight; Section 3.1 gives the structural comparison. Examples are lepton resonant/transparent symmetry $14 = 14$ from $(N_c{-}3)(N_c^2{+}N_c{+}2) = 0$; the Fermi–Dirac factor $7/8 = (\Sigma{+}1)/(N_c^2{-}1)$ from $(3N_c{+}5)(N_c{-}3) = 0$; and the transparent-cell identity $21 = N_c(\Sigma{+}1)$ from $N_c(N_c{-}3) = 0$.

Experimentally: the quark mass ratio $m_d/m_u = 13/6 = (N_c^2+N_c+1)/N_c!$ is independently confirmed by both the PDG mass ratio (2.165, 0.02σ) and, numerically, the CKM phase $\gamma = \arctan(13/6) = 65.2^\circ$ (0.4σ; the rephasing-invariant carrier is separately open, Section 3.2); across N_c the phase sits at 0.4σ at $N_c = 3$, against 2.8σ at $N_c = 2$ and 9.0σ at $N_c = 4$ (the cross-N_c pulls follow the printed $N_c!$ typing of the denominator; the value 6's other kernel readings are degenerate at the selector).

I Derivation details

I.1 $d^2 = n^2 \to \mathbb{C}$

At D=2, K=1 gives SO(2) and continuous phases between $N_{\text{spin}} = 2$ spin states. Each of the engaged register's n physically labeled components refracts against the entire same-instant superposition. Its relation space has n maintenance updates and two real, oriented components per cross-relation: $n + 2\binom{n}{2} = n^2$. A Hermitian-conjugate pair therefore carries two real operator bills, rather than two freely specifiable complex numbers. Trace normalization removes one affine state parameter without deleting an update. A4 distinguishes endpoint labels; K=1 assigns equal cost; A3 permits no excess. The orientations coexist rather than following one another in time.

The real count $n(n+1)/2$ is too small, and A3 excludes the quaternionic excess $n(2n-1)$. Thus $d^2 = n^2$ selects $\mathbb{C}$, confirmed at $N_{\text{spin}} = 2$, $N_c = 3$, and $\Sigma+1 = 7$. Solèr's theorem [43] restricts the infinite-dimensional limit to $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$; the count selects $\mathbb{C}$ within that list. Gleason's theorem [44] then confirms Born-measure uniqueness at $\dim = 7 \geq 3$. Appendix G derives its operational rule on finite complete read blocks.

For spin-statistics, K=1 makes the $N_{\text{spin}} = 2$ states cost-degenerate, A4 distinguishes exchange labels, and A3 selects the antisymmetric fermion sector, cost $1 < 9$.[4] Gauge bosons have continuous $d^2 = 2\pi$ from SO(2), satisfying A4 without exchange constraints (Bose). This discrete/continuous D=2 split is the spin-statistics connection.

The scalar seam's π (Section 3.1) is the circle's unoriented half-turn, conditional on two typings: a crossing bills distinguishable phase-domain measure (the continuous seam convention, Appendix F), and a neutral read is conjugation-blind. Charged crossings distinguish conjugation because A2's $W^+ \leftrightarrow W^-$ relabeling exchanges emitted identities and reflects phase. They therefore bill 2π; neutral crossings bill its half-turn quotient π. Conjugation is an involution, leaving no intermediate seam measure in this inventory. Doubling the gate's half-turn monodromy by the same oriented cross-relation pair also gives the Unruh 2π (Section 4.2; supplementary code).

Statistics from the ledger (scoped). The two exchange characters of an identical pair are imported representation theory at the Gleason tier. A4 and A3 select among them. One priced identification bills the joint state in the registers' ordered-pair coherence basis, a reading suggested but not forced by the Cooper-pair construction (Section 4.8).

[4] For two particles the antisymmetric sector is cheaper at every register dimension, so this clause alone does not discriminate. Three or more identical constituents supply the distinction described below.

For three or more identical constituents, the A3 minimum discriminates sectors. Three identical quarks have joint register $(\mathbb{C}^3 \otimes \mathbb{C}^2)^{\otimes 3}$, decomposing into totally symmetric (56), two mixed-symmetry (70 each), and totally antisymmetric (20) sectors. The cheapest is the antisymmetric 20, excluding parastatistics. Within it, confinement's singlet condition (Appendix I.7) leaves the observed Δ^{++}'s four states, color-antisymmetric $\times$ spin-aligned. A3 selects the sector; confinement restricts it.

The rule applies to fundamental discrete registers, present only at $D^{\mathrm{s}}_{\max} = 3$ (Appendix H). For $D \geq 3$, exchange paths with equal endpoints are homotopic: this standard topological fact permits endpoint counting. Emergent D_{eff}=2 collective statistics lies outside the rule. Joint register dimension must also be at least the constituent count; below it, resolution cedes to the continuum register. Constituents within a composite obey this rule, while whole composites exchange by constituent-count parity. The spatial exchange phase remains derived-scoped (Appendix E): the assumed manifoldlike continuum supplies the quotient. Neither the microscopic atlas nor an unconditional environment-independent postulate is claimed.

I.2 Shell-crossing exponent

The cross-shell exponent $N_c(b)/N_c(a)$ is uniquely forced by derived gauge symmetry in four steps: (i) within the source shell, SU($N_c(a)$) makes all $N_c(a)$ axes equivalent; the per-source-axis factor is $(d_i/d_j)^{1/N_c(a)}$; (ii) SU($N_c(b)$) symmetry of the destination forces all $N_c(b)$ axes to participate equally; (iii) A4 requires the amplitude to reach each axis; A3 selects the minimum: exactly one copy per axis; (iv) the tensor product over $N_c(b)$ axes gives $[(d_i/d_j)^{1/N_c(a)}]^{N_c(b)} = (d_i/d_j)^{N_c(b)/N_c(a)}$.

I.3 Anomaly cancellation at general N_c

This subsection varies SU(N_c) with the fixed SU(2)×U(1) matter assignment used in the anomaly literature. The theory's spending ladder instead scales every shell (Section 1.1); both generalizations agree at $N_c = 3$. The A2-derived charges $Y_Q = 1/N_c$, $Y_L = -1$ cancel every perturbative gauge anomaly for any N_c. The global SU(2) (Witten) condition requires an even number of left-handed doublets, $N_c + 1$ per generation: four at $N_c = 3$, and even at every odd N_c. The cancellations are gravitational, $N_c \cdot 2/N_c + 2(-1) - (N_c{+}1) + (N_c{-}1) + 2 = 0$; SU($N_c$)^{2}U(1), $(2 - N_c{-}1 + N_c{-}1)/N_c = 0$; SU(2)^{2}U(1), $N_c/N_c - 1 = 0$; and U(1)^3, $2/N_c^2 + 6 - 6N_c^2/N_c^2 - 2/N_c^2 = 0$.

Conditional on these charges, exactly Z_6 of the product center acts identically on every complete matter role (exact census of all 36 center labels; supplementary code). The content-addressed identity law (Section 4.2) therefore gives the faithful global action $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/Z_6$. Local generator counts, couplings, and mass formulas retain their values; product notation elsewhere names the local algebra. An endpoint-dependent relation-phase construction is an existence witness for one surviving physical U(1). A2 norm conservation alone does not supply determinant-one.

I.4 Proton mass

The proton mass is derived in Section 2.3: $m_p/m_e = N_{\text{spin}}^2 \times N_c^3 \times (2N_c^2 - 1) = 4 \times 27 \times 17 = 1836$ (obs 1836.15, 0.008%). The neutron–proton mass difference: $(5m_e/2)(82/81) = 1.293\,\text{MeV}$ (obs 1.293, 0.006%). The charge radius $r_p = 4\hbar/(m_p c) = N_{\text{spin}}^2/m_p$ (obs $0.841 \pm 0.0004\,\text{fm}$, 0.04%). The neutron charge radius: $\langle r_n^2 \rangle = -r_p^2/\Sigma$ (obs $-0.1161 \pm 0.0022\,\text{fm}^2$; predicted -0.1180, 0.9σ). The pion charge radius: $r_\pi = (N_c^2{+}N_c{+}1)/(N_c^3{+}1) \times \hbar/(m_\pi c) = (13/28)/m_\pi = 0.657\,\text{fm}$ (obs 0.659 ± 0.004, 0.6σ): pair cost divided by full transparent total, times Compton wavelength.

I.5 Lepton mass

The gen-2 lepton must be Pauli-distinct within the $N_c^3 = 27$ generation space. Because it has no color, that space enters at $K{=}1$ through distinctness rather than confinement. Faithful complete-content association gives the collision-free minimum $d_\mu = N_c^3 + N_{\text{spin}} = 29$: generation space plus spin doublet, added because no color couples them. Symmetry alone permits 10- or 6-label quotients; the faithful rule excludes them (supplementary code). The sum is one internal dimension in $\mathbb{C}^{29}$ at $K{=}1$.

The same join-then-square operation appears in the color–spin 5-space: $(N_c{+}N_{\text{spin}})^2 = 25$, including $2N_cN_{\text{spin}} = 12$ cross-sector pairs. Three measured applications are the Λ moment's $24/25$ damping (Section 3.5), the Ω_c's $+25$ color–gate coherences (Appendix K.2), and the quarkonium normalizations $25/3$, $25/2$. The blockwise alternative $3^2{+}2^2 = 13$ is excluded at all three: 5.9σ for Λ, 14.8σ for Ω_c against PDG-2025, and 4.3σ for η_c. The tensor alternative $6^2 = 36$ is excluded at Ω_c and η_c (14.6σ; 5.4σ), and disfavored at Λ (1.95σ). A fourth application, the ϕ width $(N_c{+}N_{\text{spin}})^2/N_c$ (Section 3.4), excludes both with its 0.3% precision. Since $25 = 17{+}7{+}1$ also, the shared operation carries the evidence, rather than the value alone (supplementary code).

For the lepton read, the complete resident matter-lepton block has transparent ground H_L, with $|H_L| = 5_\nu + 3_e + 3_\mu + 3_\tau = 14$. Rule 4 gives $B_\ell = d_\ell^2 - \chi_\ell |H_L|$, where $\chi_e = 0$ for the anchor and $\chi_\mu = \chi_\tau = 1$ thereafter. This scalar debit uses equally weighted A4 units; it does not delete operators from $\mathbb{C}^{d_\ell}$. Hence $m_\mu/m_e = B_\mu/N_{\text{spin}}^2 = (841 - 14)/4 = 206.75$ (obs 206.768, 0.009%).

Koide supplies the τ derivation: the Z_3 circulant has $K{=}2/3$ and phase $\delta = 2/9 = N_{\text{spin}}/N_c^2$. The phase agrees at 0.01% with $\delta_{\text{fit}} = 0.22222 \pm 0.00001$, versus $2/9 = 0.22222$. The error propagates PDG-2024's $\sigma(m_\tau) = 0.09\,\text{MeV}$. Brannen first exhibited the circulant and $2/9$ as a fit [107] (Section 3.1). Koide's algebra is exact given the separately typed $T_K = 1/2$; Z_3 alone does not select the phase. In the directed-route construction, the transparent source is identified with a relative-family-displacement register whose ordered-pair space has nine relations. Assigning action $\hbar/9$ to each permitted forward hop, the required inverse-neighbor displacement takes $2 + 3k$ hops ($k \geq 0$); A3 selects $k = 0$ and hence $\delta = 2/9$. With U the unweighted forward shift, the phased two-hop operation is $W = e^{2i/9}U^2$, with $W^3 = e^{2i/3}I$. The normalized Hermitian root read $R = I + (W + W^\dagger)/\sqrt{2}$ then has $\operatorname{Tr} R = 3$ and $\operatorname{Tr}(R^2) = 6$; its three eigenvalues are

positive on this branch. This conditional selection uses the source-to-displacement identification and the specified route-action read; Z_3 symmetry and equal costs alone do not supply those identifications (supplementary code).

Inversion through the d-progression gives $d_\tau = 118$. Its decomposition $N_c^4+N_c^3+N_c^2+1$ is a consistency label, not a second mass derivation. The forward register construction yielding $d_\mu = 29$ does not extend to 118 (supplementary code).

Association through complete read histories. An association can share content only to the extent that its allowed later refractions and reads cannot distinguish it. On the source-inscription carrier $H = \mathbb{C}^{\{-1,0,1\}^3}$, the ten permutation orbits, or six orbits after simultaneous linear sense reversal, admit normalized coherent representatives

$$J|o\rangle = |o|^{-1/2} \sum_{x\in o} |x\rangle, \qquad J^\dagger J = I. \tag{39}$$

These are complete coherent descriptions of their prepared sectors when every allowed unitary update U_t and resolved/no-read projector $R_{t,a}$ obeys

$$U_t J = J u_t, \qquad R_{t,a} J = J r_{t,a}. \tag{40}$$

Products then preserve every finite read history, including arbitrary external-reference correlations. The initial wave must lie in ran J; permutation symmetry of the dynamics alone does not prepare it. Equal orbit probabilities can conceal phases that later symmetric refraction resolves. In particular, this is not an isometric compression of arbitrary H (supplementary code).

For arbitrary input and the entire operation algebra commuting with source-role permutations, the exact decomposition is

$$H \simeq \mathbb{C}^{10} \oplus (\mathbb{C}^8 \otimes \mathbb{C}^2) \oplus \mathbb{C}, \qquad \mathcal{A} \simeq M_{10}(\mathbb{C}) \oplus M_8(\mathbb{C}) \oplus \mathbb{C}. \tag{41}$$

Discarding inter-sector coherence and tracing the unread representation factors gives a block-diagonal density-state description on $10 + 8 + 1 = 19$ dimensions. Nineteen mutually orthogonal readable projections prove its minimum among deterministic quantum encodings of this full operation class, including all carried records. Requiring simultaneous linear sense-reversal invariance splits the response blocks into sizes $6, 4, 4, 4, 1$, retaining the same minimum. This channel preserves allowed history probabilities and conditional reference states when each outcome's Kraus operators belong to the stated algebra; it does not recover the full source wave or its discarded correlations. These operational dimensions are not native mass bills.

Allowing an ordered signed-role read changes the response algebra and supplies a sufficient condition for all 27 content distinctions. Take the recurring color-coordinate cycle $C|a, b, c\rangle = |b, c, a\rangle$ and projectors P_a reading the first role's value $a \in \{-1, 0, 1\}$. Three completed reads give

$$CP_cCP_bCP_a = |a, b, c\rangle\langle a, b, c|. \tag{42}$$

The distinctions grow as $3, 9, 27$; the complete source plus three ternary output records is a reversible 729-state map. Its coherent extension retains the wave and records, without cloning an unknown state. A relative realization supplies an ordered comparison frame with six role permutations. Holding the source fixed while cycling this frame reproduces the same relative-coordinate reads. Every read and the complete update commute with simultaneous source/frame relabeling: the 27 relative outcomes have rank six on the full 162-dimensional source/frame carrier, or rank one on its normalized 27-dimensional invariant input sector. The full source/frame/output map retains all 4374 amplitudes. This construction supplies a finite role-permutation context, not a continuous color action or an unpriced lepton register; the role order is not the placement of a column in Fig. 30.

For a fixed device implementing all six independently selectable orders on every source, six orthogonal context states are necessary and sufficient: the same all-distinct source gives six deterministic output words. An independently preserved spin doublet requires twelve joint context/spin states. An already accessible first-role phase can supply the ternary part, leaving two orientation alternatives. These are complete information resources, not mass tariffs.

A relative pattern can also become an autonomous resident. Write $y_i = x_{g(i)}$, with inverse $x(g, y)$, and supply a resident payload z and physical spin s. The complete involution

$$F|x, g, z, s\rangle = |x(g, z), g, y, s\rangle \tag{43}$$

exchanges relative content and payload, commutes with simultaneous finite role relabeling, and is identity on spin. On the normalized invariant input $J_{\mathrm{rel}}|y\rangle = 6^{-1/2}\sum_g |x(g, y), g\rangle$ with fixed resident z_0, it leaves $J_{\mathrm{rel}}|z_0\rangle$ and transfers $|y\rangle$ into the resident. A supplied resident cycle $C|a, b, c\rangle = |b, c, a\rangle$ then recurs without source access. For supplied additive diagonal energy $e_s(y) + e_r(z)$, the transfer conserves every input if and only if $e_s - e_r$ is constant; closed recurrence separately requires $e_r(Cy) = e_r(y)$. Each conserved additive charge obeys the analogous conditions. Positive orbit-constant spectra matched up to a constant satisfy both. Stationary cycle modes are its eigenvectors; their update phases alone do not select physical energies. The full carrier retains $27 \times 6 \times 27 \times 2$ amplitudes, and the prepared content/spin domain has dimension 54; neither is the native $2 + 27$ mass operand.

Thus faithful association follows from a specified separating read law rather than a label count alone. Applying this source-side criterion to d_μ still requires the colorless lepton's carried association to reproduce those responses when the source and other records cannot be read instead. The transfer establishes finite-symmetry feasibility; the physical source-to-lepton refraction, charge action and mass operand still require selection (supplementary code).

I.6 The matrix-reading rules

The matrix-reading rules specify how parts combine; the five operations specify what a read computes (Section 2.2). In the optical picture, simultaneous coupling multiplies state spaces: the coextensive shells engage at once, giving A4's $a \times b$ joint outcomes (Sec-

tion 1.1). This is the tensor product shared by a wave's polarization, mode, and frequency. Rule 2 assigns each generation staircase its ratio. Rule 3 includes an instrument in the system it reads. Rule 4 removes modes passed untouched. The four general rules follow; Appendix K.2 gives the fifth, the W^- mismatch rule, beside its composite applications.

1. *Multiply or add?* Coupled channels (color) form a tensor product (multiply); independent channels add. Excitation costs are additive corrections.
2. *Which generation formula?* Down-type (W^- active): clean generation ratios. Unsaturated up-type (u, c: W^- absent; t saturates every channel): explicit gauge subtraction. Leptons: internal-dimension counting — lepton generation transitions, like $t \to b$, toggle no occupied cell (identical patterns): the zero-motion class is exactly where distinctness is carried internally (the d-progression) or exhausted (gen-3 saturation), a shared signature, not a shared mechanism.
3. *Where does $+1$ appear?* Three mechanisms give near-unity corrections or additive $+1$ terms.
   (a) *Exists $\neq$ computes*: a frozen or EM-blind channel redistributes energy over $n-1$ active modes, giving $30/29$ in A_s and $47/49$ in Δm^2_{21} and θ_{13}.
   (b) *A4 observer participation*: an instrument in the beam becomes part of the optical system. A W-mediated vertex, with mediator $d \neq 0$, adds $+1$ to the measurement basis: $82/81$ in V_{us} and $m_n - m_p$, $28/27$ in V_{cb}. Gravity applies the same principle to a column census, $631 = 630+1$, with $630 = 7 \times 9 \times 10$ (Section 3.3), rather than a mode-space orbit. Photonic vertices ($d=0$) add nothing, so meson EM splittings remain flat kernel integers. Participation cancels in ratios measured at one vertex: g_A, g_T, and mixing angles.
   The participation space is N_c^3 generation configurations times the destination color pattern's permutation orbit, $\binom{N_c}{k}$. Partial rows ($k = 1, 2$) have orbit $N_c = 3$, hence $81 \to 82/81$ for V_{us} and $m_n - m_p$. Saturated rows ($k = N_c$) have orbit 1, hence $27 \to 28/27$ for V_{cb}, saturated at both endpoints. The identity $\binom{N_c}{N_c} = 1$ holds at every N_c without a collapse postulate. One priced identification resolves the destination pattern only up to color permutation, at the same tier as the junction step (Appendix I.7). The CKM-specific sequential assignment $V_{ub} = V_{us} V_{cb} \times 5/13$ inherits each factor once. The conditional PMNS route theorem does not automatically apply to this row, and $5/13$ is a structural branch-weight identification with an exact matrix census (supplementary code). Its comparison is -0.08σ, versus $+0.78\sigma$ for the over-collapsed alternative.
   (c) *Structural mode counts*: $13 = N_c^2 + N_c + 1$ counts the resonant $D=1$ block, and $28 = N_c^3 + 1$ the transparent total. They have independent physical origins as complete cross-sections; reading them as "12+1" or "27+1" corrections would conflate these types.
4. *What baseline is subtracted?* The complete resident matter-lepton block is $L = \{\nu, e, \mu, \tau\} \times \{C_B, C_G, C_R, W^-, Q, E, W^+\}$. The photon is excluded as the $d=0$ read. Its transparent ground is $|H_L| = 5_\nu + 3_e + 3_\mu + 3_\tau = 14$, independent of layout. Equality with its 14 resonant cells holds only at $N_c = 3$, through $(N_c - 3)(N_c^2 + N_c + 2) = 0$. Thus $N_{\text{spin}}(\Sigma + 1) = 2 \times 7 = 14$ is an identity of this matrix, not a separate seven-coherence mechanism. A family coordinate indexes

the full block, so Rule 4 debits its ground once: $d_\mu^2 - 14$ and, downstream, $d_\tau^2 - 14$. The electron anchor has no post-anchor coordinate and no subtraction. This scalar cost debit uses the gross bill's equal A4 units; it neither superselects a sector nor deletes 14 state operators. The τ checks the selected holes independently (supplementary code).

I.7 Baryon and meson topology

At $K{=}r$, A2 conducts color flux along A4-quantized lines terminating on SU(3)-invariant couplings. The complete invariant list is δ for quark–antiquark pairs and ϵ_{abc} for three quarks. Two like sources cannot cap one another: 3×3 ($= \bar{3} \oplus 6$) has no two-index invariant.

A3 selects the shortest legal network. A quark–antiquark pair forms a straight color-flux connection, called a *string* in the topology and QCD comparisons below; the term describes this connection within the composite wave. Interior nodes add length, and the δ coupling is position-degenerate along it. A color-singlet three-quark state forms a Steiner tree with one ϵ-node at the Fermat point. It is unique for every spatial configuration: interior if all angles are below 120°, at the obtuse vertex when one reaches 120°. In that branch the two edges meet at the node. A closed three-source cycle is excluded independently of geometry because quantized flux conservation cannot absorb three source insertions around it.

One interpretive step (selection cost $\sim$4 bits at the adversarial accounting): an A3-localized coupling is a computational element carrying a vertex factor; a position-degenerate coupling is not. The baryon/meson asymmetry of the mass grammar follows: the baryon's node reads the transistor's $D{=}1$ (source) shell modes ($2N_c^2{-}1 = 17$, "at the confining vertex," Appendix I.4; the transistor-shell census, not a matrix-block total); the meson, having no node, carries no vertex factor — no open-flavor meson formula in Table 9 contains 17.

A2 ties each edge's endpoint color labels, while source-shell SU(N_c) leaves their common label unresolved. Each edge therefore retains the full N_c-sector diagonal. A meson has one edge and color cost $N_c = 3$. A baryon's three edge registers coexist, giving Rule 1's tensor-product bill $N_c^3 = 27$, with the ϵ_{abc} node performing simultaneous closure. That contraction correlates labels without removing same-tick edge work; the vertex supplies its separate 17 factor. Both color factors follow without billing the three legs as independent particles.

Rule 1 also fixes when the vertex product dominates. Light quarks have negligible internal generation cost, so the junction leads, as in $4 \times 27 \times 17$. Heavy quarks carry dominant independent generation cost, giving anchor-plus-offset or heavy-led formulas (Appendix K). Six of fifteen heavy-baryon rows are products. Two, $\Sigma_c = 2 \times 7^4$ and $\Xi_c = 3 \times 23 \times 70$, lack cross-sector recurrence and remain structurally isolated fits (supplementary code).

I.8 Baryogenesis

Baryogenesis is conditional at the $2D \to 3D$ transition. Two ingredients follow from the axioms: baryon number appears there ($K{=}r$ confinement defines B, absent before the transition), and the one-tick transition (A1 + discrete D) gives maximal departure from equilibrium. The gate supplies an exact occupancy asymmetry: u and c lack W^-, while t is saturated. It also supplies the structural angle $\arctan(m_d/m_u) = \arctan(13/6)$.

The missing ingredient is a physical CP-odd carrier. Equal static gate phases are removable, so the angle alone cannot provide it. The open candidate is the non-separable phase accumulated through A2's monotonic, time-ordered sequence of complete source loops (Section 3.2). Conditional on that carrier, $N_c{+}N_{\text{spin}} = 5$ channels are active at the transition and the asymmetry occupies the $N_{\text{spin}} = 2$ gate channels. The exponent is therefore $(m_e/m_P)^{N_{\text{spin}}/(N_c+N_{\text{spin}})} = (m_e/m_P)^{2/5}$. The prefactor $N_c!/(N_c^2{+}N_{\text{spin}}) = 6/11$ counts color-singlet permutations among color–spin states.

I.9 QCD running

The one-loop β-function coefficient $33{-}2N_f$ steps through framework kernel integers at each flavor threshold: $33 = N_c(N_c^2{+}N_{\text{spin}})$ re-expresses the known QCD coefficient in framework variables. At successive thresholds ($N_f = 1, 2, \ldots, 6$), the coefficient takes values $31, 29, 27, 25, 23, 21$ — each independently a named kernel integer with its own physical origin. Each re-expresses a known QCD coefficient as a kernel integer: $31 = N_c^3{+}N_{\text{spin}}^2$, $29 = N_c^3{+}N_{\text{spin}}$, $27 = N_c^3$, $25 = (N_c{+}N_{\text{spin}})^2$, $23 = 2(\Sigma{+}1){+}N_c^2$, $21 = N_c(\Sigma{+}1)$.

The pattern extends to two loops: $153{-}19N_f$ decomposes as $153 = N_c^2(2N_c^2{-}1) = 9 \times 17$; each flavor subtracts $19 = (2N_c^2{-}1) + N_{\text{spin}}$ (vertex+spin). At successive thresholds: $N_f = 3 \to 96 = (N_c^2{-}1) \times N_c(N_c{+}1) = 8 \times 12$; $N_f = 4 \to 77 = (\Sigma{+}1)(N_c^2{+}N_{\text{spin}}) = 7 \times 11$ (the transistor's total mode count); $N_f = 5 \to 58 = N_{\text{spin}}(N_c^3{+}N_{\text{spin}}) = 2 \times 29$; $N_f = 6 \to 39 = N_c(N_c^2{+}N_c{+}1) = 3 \times 13$ (the same pion pair cost 13 that enters $\sqrt{\sigma}$ and B_d).

I.10 The Φ_b^2 profile

Position is a substrate mode of the $D{=}3$ drain, the same ontological type as spin and color because space is the substrate. At dispersive $K{=}1/r$, the relation rule counts simultaneous spatial cross-relations. This read excludes diagonal citations: an address does not reference itself, and component maintenance belongs to the resident mass bill. Weighting each distinct-address relation by $K{=}1/r$ gives $\sum_{i\neq j}\phi_i\phi_j = \Phi_b^2 - \sum_i \phi_i^2 = \Phi_b^2(1 - 1/N_{\text{eff}})$, with $N_{\text{eff}} \equiv \Phi_b^2/\sum_i \phi_i^2 \gg 1$ macroscopically. The isolated potential has $\Phi_b \to 0$ at infinity; overlapping-system cross terms remain the ambient-bookkeeping question of Section 4.6.

A3 cannot delete an occupied wave's simultaneous support without changing the wave. A2 conserves it until a distinguishing read rewrites it. Its distance-weighted mean-field demand produces gravity through the clock-rate gradient. In equilibrium, T3 local-

izes bookkeeping bE at baryonic positions and its source is Φ_b^2. At the linear tier and in the primordial branch, it responds instead to the total potential, including its own share (Section 4.5).

The baryon restriction follows from the EM-family read accumulating the sum. It skips the only EM-blind row, the neutrino with its two occupied cells (the 47/49 census, Appendix J). Elimination resolves an intermediate only when A4 requires it, as in mixing; A3 does not perform unforced coherence accumulation. No Φ_ν^2 term appears, and the cosmic neutrino background contributes only its standard background density under the supplied relic-population assumptions. When the positive-count branch is used, that density and its transfer calculation must use the conditional propagation masses of Section 3.7 (supplementary code). Requiring the enclosed-mass integral to reproduce the BTFR $v^4 = GM_b a_0$ fixes the normalization:

$$A = \frac{\sqrt{a_0}}{4\pi\, G^{5/2}\, M_b^{3/2}}\,, \tag{44}$$

so $A \propto M_b^{-3/2}$, with no per-galaxy freedom.

Finite radius and normalization. For a spherical enclosed-mass calculation, define

$$g_I(r) = g_b(r) + \frac{4\pi G A}{r^2}\int_0^r \Phi_b(s)^2 s^2\, ds, \qquad g_C(r) = g_b(r) + \frac{\sqrt{GM_b a_0}}{r}. \tag{45}$$

Here g_C replaces the dark contribution by its fixed-mass asymptote. The algebraic curve $g_R = g_b + \sqrt{a_0 g_b}$ additionally substitutes the local baryonic acceleration for GM_b/r^2; at finite disk radii these operations differ. An outer control $g_O = g_b + s(g_I - g_b)$ changes only the dark amplitude, using

$$s = \frac{\sqrt{GM_b a_0} - \langle v^2_{b,\mathrm{unsigned}}\rangle_{\mathrm{out}}}{\langle r(g_I - g_b)\rangle_{\mathrm{out}}}, \tag{46}$$

where the averages run over the last $\min(5, N)$ original curve points. The target uses unsigned baryonic components; g_b in the predictions retains their signed radial contributions.

On a common 139-galaxy, 2608-point SPARC domain, with $\Upsilon_{\mathrm{disk}} = 0.5$, $\Upsilon_{\mathrm{bulge}} = 0.7$, current baryonic mass and $a_0 = 1.10 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, the point mean of $\ell_X = \log_{10}(g_{\mathrm{obs}}/g_X)$ obeys

$$\langle \ell_O - \ell_R\rangle = \underbrace{\langle \ell_I - \ell_R\rangle}_{0.13166\,\mathrm{dex}} + \underbrace{\langle \ell_O - \ell_I\rangle}_{0.01554\,\mathrm{dex}} = 0.14720\,\mathrm{dex}. \tag{47}$$

The first term passes through g_C as $-0.08260 + 0.21426$ dex. These are paired implementation differences, not uncertainties: giving galaxies equal weight changes the two terms to 0.15670 and -0.01781 dex. The domain requires SPARC quality 1 or 2, inclination at least 30°, relative velocity error below 10%, at least three retained points with positive baryonic, observed and finite dark accelerations per galaxy, and a positive outer target.

The potential reconstruction uses unsigned baryonic rotation components, a rigid inner continuation and the outer boundary $|\Phi_b(r_{\max})| = GM_b/r_{\max}$; it is a spherical quadrature diagnostic, not an exact three-dimensional disk force. The outer control uses that same potential and current mass; the rotation-curve benchmark above uses a distinct implementation.

With the same quality, inclination and velocity-error cuts, a separate common-mask nuisance control retains 141 galaxies and 2686 points positive throughout $\Upsilon_{\rm disk} \in [0.15, 1.20]$ in steps of 0.01, holds bulge $\Upsilon = 0.7$, distances and inclinations fixed, and recomputes each model at every grid value. Minimizing each galaxy's absolute mean ℓ gives median $\Upsilon_{\rm disk} = (0.68, 0.34, 0.43)$ for $(g_I, g_R, \mathrm{MOND})$; minimizing velocity χ^2 with a $2\,\mathrm{km\,s^{-1}}$ error floor gives $(0.68, 0.36, 0.43)$. MOND retains its comparison value $a_0 = 1.20 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. In the mean-offset control, 21% of algebraic-curve fits reach the lower grid boundary, while 23% of integral fits reach the upper boundary. Thus nuisance adjustments depend on the profile and objective; these in-sample grid controls neither select a unique amplitude prescription nor derive the local endpoint response (supplementary code).

A separate SPARC comparison excludes a stars-only coherent share, $\rho = f\,\Phi_\star^2 + (1-f)\,\rho_{\rm prim}$. In 21 gas-rich dwarfs ($f_{\rm gas} \geq 0.8$), pooled χ^2/N worsens by 22%, from 13.3 to 16.3. A separate shape-only comparison of the ten most gas-rich dwarfs leaves 2/10 in the cored band, against 7/10 for the all-baryon sum (supplementary code). Their cores require gas amplitudes in Φ_b^2.

J The EM-blind and inflationary factors

J.1 The 47/49 EM-blind factor

The compute matrix has 49 occupied, equally weighted A4 units (Appendix F). Only the neutrino's two cells, W^+ and E, are EM-transparent because Q is empty. The matrix's total occupancy equals the 7×7 channel-pair count, $49 = (\Sigma{+}1)^2$, at $N_c{=}3$; the spaces themselves are not identified. All three neutrino flavors are EM-silent. The EM-blind factor depends on coupling order: whether a path includes an intermediate requiring EM resolution.

The $D{=}1$ color channels form an SU(3) triplet. Adjacent generation couplings W_{12} and W_{23} are first-order boundary crossings; $1 \leftrightarrow 3$ is second-order. Color containment forces this sequential route: gen-2 contains gen-1, and gen-3 contains gen-2, so the amplitude factors through the intermediate. SU(3) itself permits direct $1 \leftrightarrow 3$ generators; the routing follows containment rather than their absence or a cost selection.

A4 requires the intermediate generation to be distinguished. The photonic EM read responds to its occupied cells. If they answer, resolution is local and complete, with no factor. If all are EM-transparent, uniquely for the neutrino, resolution uses elimination against the readable register and carries density $47/49$. The receiving slot determines how the share enters: a Born-share slot takes $\sqrt{47/49}$ because T2 squares amplitudes; a

Hamiltonian slot takes 47/49 through T1 additive phase composition. The route projector contributes $T_{\text{route}} = 2/4$ (Section 3.2), numerically equal to the per-axis exponent T_F but separately typed. The resulting EM-blind coupling is:

$$W_{13} = T_{\text{route}} \times \frac{47}{49} = \frac{47}{98} \tag{48}$$

Adjacent couplings have no intermediate step — A4 resolves only the initial and final states, whose EM-blind factors are already absorbed into the mass eigenvalues via $\sqrt{47/49}$ in the mass formula. No additional census factor.

A quark intermediate answers the read locally at every step, so the CKM analogue V_{ub} receives no census factor. Its sequential route is separately forced by color containment, with gen-3 containing gen-2. Local resolution and route factorization are thus consistent branches of the same rule.

A response share and a cutoff. For a supplied, time-independent reciprocal response in Planck units, the complete generator $H = \left(\begin{smallmatrix} 0 & R^\dagger \\ R & 0 \end{smallmatrix}\right)$ gives $\ddot{x} = -R^\dagger R x$. If $R^\dagger R = p\lambda_0^2 I$ on the complete native block, the cutoff magnitude is $\sqrt{p}\,\lambda_0$ for every native state, with the orthogonal complement retained. This is a dynamical square-root mechanism with fixed time normalization. Restricting a later detector output while keeping H unchanged leaves its frequencies unchanged. These cutoff frequencies do not by themselves specify the positive resident work bill. The census share $p = 47/49$ does not by itself construct the required physical response R (supplementary code).

Evidence: $\sqrt{47/49}$ in the mass formula gives Δm^2_{21} at 0.2–0.7σ (post-JUNO global fits); 47/49 on W_{13} gives θ_{13} at 0.3σ (NuFIT 6.1) — two independent observables from one principle. Without the EM-blind factor, these would be 3.5–3.9σ (post-JUNO fits) and 2.7σ respectively (Fig. 32).

J.2 The 30/29 inflationary factor

During the D=2 phase, the transistor is a 2D disk — the D=3 shell does not exist. The Q channel receives energy (S_7 acts on *cost*, not identity: channels retain their shell locations, as in flat-band systems where energy degeneracy coexists with orbital geometry) but cannot spatially fluctuate. Of $N_c^3+N_{\text{spin}}+1 = 30$ existing substrate states, $N_c^3+N_{\text{spin}} = 29$ actively compute; A2 forces the frozen channel's energy to the active modes, enhancing the per-mode power weight by 30/29 (same principle as the 47/49 EM-blind factor, Section 2.5).

At current precision, (N=57, 30/29) and (N=58, no factor) are degenerate, with $58 = 49+N_c^2$ and joint χ^2 values 0.01 and 0.05. The structural 30/29 requirement selects the first pair; JUNO independently selects the related 47/49 factor.

The perturbation amplitude, including the 30/29 factor, is Eq. 13.

During D=2, K=1 imposes S_7 cost symmetry and equal energy for all $\Sigma+1 = 7$ channels. The complete cost-allocation block has dimension 7^7; the relation rule then gives $7^7 \times 7^7 = 7^{14}$ entries. Equal per-entry allocation follows separately from A2 and S_7. The

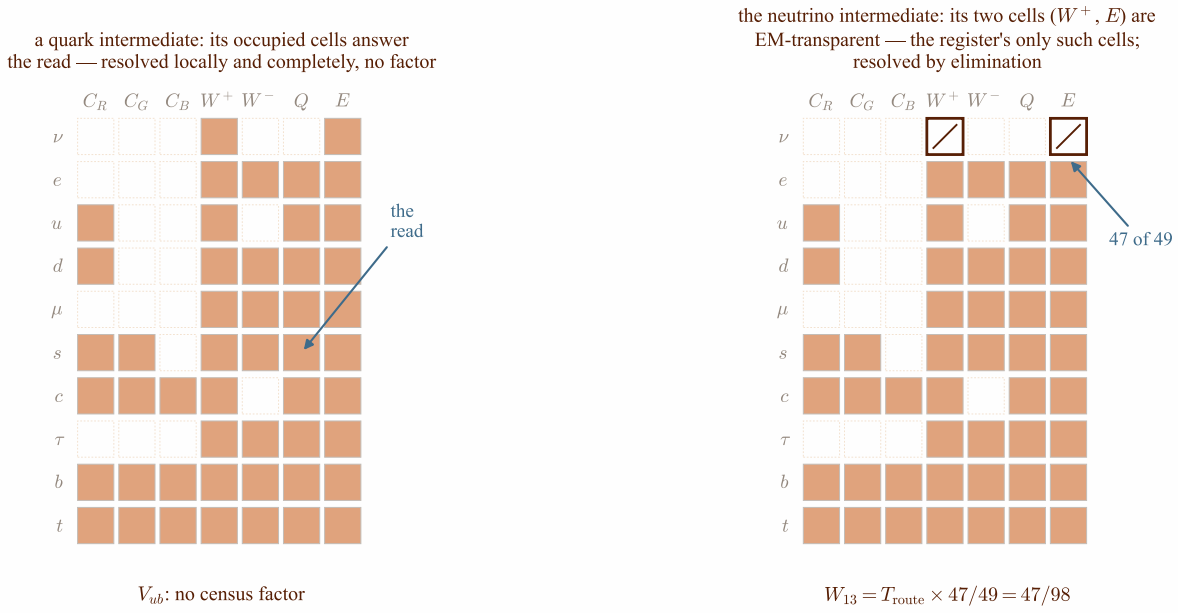


Figure 32: *The elimination read. Of the compute matrix's 49 occupied cells (Appendix F), the neutrino's two (W^+ and E) are EM-transparent (Q empty) — the register's only such cells. An intermediate whose occupied cells answer the read is resolved locally and completely, no factor (left: the quark case; the CKM cross-check — V_{ub} receives no census factor); an intermediate whose occupied cells are all EM-transparent — uniquely, the neutrino — is resolved relationally, by elimination against the readable register, and carries the readable density* $47/49$ *(right), giving* $W_{13} = T_{\text{route}} \times 47/49 = 47/98$. *The photon's all-transparent row carries no occupied cell and is not drawn; the display orders columns as source, gate, drain, existence, and the drawn grid defines no physical adjacency.*

seven-fold factor per channel is a structural identification of this era register, distinct from the ternary matter register (Section 4.8); squaring uses the complete-block relation rule. This is allocation, not a local particle-mass bill.

The resulting density is $\rho = m_P^4/(8\pi \times 7^{14})$: 7^{14} is the framework count, 8π the standard gravitational-coupling convention. Substitution into $A_s = \rho/(24\pi^2\varepsilon_1 \bar{m}_P^4)$ with $\varepsilon_1 = 1/(4N^2)$ gives exactly $4N^2/(3\pi \times 7^{14})$.

At the $D{=}2 \to D{=}3$ transition, the S_7 cost degeneracy lifts, leaving the three shells' gauge costs distinct (SU(3)×SU(2)×U(1)); the released compute capacity becomes available to the matter sector; the equal partition over $D^{\text{s}}_{\text{max}}{+}1 = 4$ channels supplies the initial shares for the temperature and expansion map in Section 4.4; the 30/29 reversal is an account identity conditional on the equal post-lift endpoint, and the photon is the contentless read of the resulting state, not the carrier that creates it.

J.3 PMNS Hamiltonian and resolution floor

The PMNS Hamiltonian in the charged-lepton-diagonal basis:

$$H = \mathbf{I} + R_1 \,|1\rangle\langle 1| + R_2\, W \tag{49}$$

Here $R_1 = -(m_d/m_u)^{1/3} = -(13/6)^{1/3}$, $R_2 = -(m_u/m_d)^{1/2} = -(6/13)^{1/2}$, $W_{12} = W_{23} = 1$, and $W_{13} = T_{\text{route}} \times 47/49 = 47/98$. The route-projector weight $T_{\text{route}} = 2/4$

(Section 3.2) multiplies the relational resolution density at the second-order intermediate (Section 2.5). All three neutrinos have Q=0.

Conditional on resident weak doublets sharing one generation coordinate, the diagonal up/down row loads are $(4,5)$, $(6,6)$, $(7,7)$. Their defect is exactly the generation-1 projector, fixing R_1's placement. A unique color-relabeling-symmetric linear occupation response fixes $W_{12} = W_{23}$. Both are exact conditional on the basis and response typings (supplementary code). Treating 12 and 23 as primitive covers and 13 as their composite remains structural routing (Appendix J); A4 resolves its intermediate relationally at the 49-cell register's 47/49 readable density.

A3 selects signs by minimizing $\lambda_{\min}$, uniquely $(-,-)$. Then $H_{11} = 1 + R_1 = 1 - (m_d/m_u)^{1/3} < 0$ places gen-1 lowest. Eigenvalue ordering is read as mass ordering; their magnitudes are not masses.

At the minimum splitting Δm^2_{21}, A4 imposes the resolution floor. The bare Hamiltonian gives $\theta_{12} = 32.2°$, disfavored by NuFIT 6.1 at 3.5–3.8σ (Section 3.2). The native active line has $d_\nu = 1$, distinct from the three-dimensional weak-endpoint carrier and from the propagation-mode work counts. Activation of the resolution floor only at the smallest splitting remains the structural identification used here. The maximum transition probability $\sin^2(2\theta)$ in the normalized lowest two-state block is A4's distinguishability measure and saturates at $\Sigma/(\Sigma{+}1) = 6/7$. At least one of $\Sigma{+}1 = 7$ states remains distinguishable, leaving at most $\Sigma = 6$ to mix.

Saturation is structural: below the floor, an unmixed remainder maintains a distinction no read can resolve, so A3 drives mixing to A4's boundary. The result is $\theta_{12} = 33.9°$. Transfer from count share to transition probability is exact conditional on slot typing. Equal, additive A4 units require a normalized k-unit share to receive $k/7$. Once $\sin^2(2\theta)$ is typed as a Born share, T2 assigns its squared amplitude that share (supplementary code). The slot typing and activation only at the minimum splitting remain structural identifications tested by JUNO.

The splittings governing θ_{23} and θ_{13} are 32× and 33× larger, so the Hamiltonian controls them without an override. CKM quarks have $d \gg 1$ and remain sharply distinguishable; no CKM angle is overridden.

One propagation operator. On the conditional positive-count and composed branches, let v_i be H's ordered eigenvectors, $\Pi_i = |v_i\rangle\langle v_i|$, and $Q = \mathbf{I} - \Pi_3$. Normalize the electron endpoint in this subspace as $a = Q|e\rangle / \|Q|e\rangle\|$. Choose the unit $b \perp a$ in Q so that $v_1 = \sqrt{q_0}\, a + \sqrt{1-q_0}\, b$, up to overall phase. The floor gives

$$\begin{aligned} q &= \frac{1 + 1/\sqrt{7}}{2}, \qquad v_1^* = \sqrt{q}\, a + \sqrt{1-q}\, b, \\ \Pi_1^* &= |v_1^*\rangle\langle v_1^*|, \qquad \mathcal{M}_*^2 = m_\nu^2[\tfrac{4}{3}\mathbf{I} - \Pi_1^* + 32\Pi_3]. \end{aligned} \tag{50}$$

The floor correction changes only the lowest two-state subspace. On the conditional positive-count branch, the scalar term also supplies the common squared-mass floor $\beta_\nu^2 = m_\nu^2/3$. The physical spectrum is $\beta_\nu^2(1, 4, 100)$; subtracting $\beta_\nu^2\mathbf{I}$ gives the gap matrix with

spectrum $m_\nu^2(0,1,33)$. The positive square root is $\mathcal{M}_* = \beta_\nu(2\mathbf{I} - \Pi_1^* + 8\Pi_3)$. Its eigenvectors give $\theta_{12} = 33.896°$, $\theta_{23} = 49.411°$, and $\theta_{13} = 8.593°$. The weak endpoint projectors stay fixed. In units $\hbar = c = 1$, the leading relativistic fixed-energy evolution $\exp[-i\mathcal{M}_*^2 L/(2E)]$ is unitary, with no flavor change at zero propagation distance; adding the common squared-mass floor gives only a common phase in this law. The eigenvectors and squared gaps are exactly unchanged, while exact finite-momentum energies $\sqrt{|\mathbf{p}|^2 + m_i^2}$ and rest phases depend on the positive absolute masses. A weak emission prepares the endpoint state once. Its continued intrinsic refraction carries the rest demand. On this operator branch the superposed propagation modes evolve coherently, so a later weak interaction can resolve a different flavor. The initial emission label needs no repeated read; the guaranteed correction remains present during free propagation without electromagnetic verification (supplementary code).

Event balance. For an isolated event in a stationary local background, the outgoing waves obey $\sum_i E_i = E_{\text{in}}$, $\sum_i \mathbf{p}_i = \mathbf{p}_{\text{in}}$ and $E_i^2 = m_i^2 + |\mathbf{p}_i|^2$. The rest-mass difference therefore also funds motion. A neutrino pair has invariant mass $W \geq m_i + m_j$; on the positive branch its six unordered thresholds are $\beta_\nu(2,3,4,11,12,20)$. For $2\beta_\nu \leq W < 3\beta_\nu$, only the lightest pair is allowed. Above the largest threshold all nine ordered pairs are kinematically open. Their probabilities and directions require the complete emission response, including the wave's spin and phase (supplementary code).

The prescription selects the closest lowest-state projector at the specified floor. Every candidate line in Q has representative $\sqrt{q}\,a + e^{i\varphi}\sqrt{1-q}\,b$. Its overlap with v_1 is bounded by $\sqrt{qq_0} + \sqrt{(1-q)(1-q_0)}$, uniquely attained at $\varphi = 0$ here. The other floor octant is farther from the original state. This minimum-change prescription is an additional conditional choice, not a derived microscopic A3 action; the floor and route identifications are inherited. It preserves the real composed branch and does not select the open leptonic CP branch. The $6/7$ coefficient belongs to the normalized solar block: in full electron survival its coefficient is $c_{13}^4(6/7) = 0.81930$, alongside the two larger-splitting terms, where $c_{13} = \cos\theta_{13}$.

Mixing and mass. The PMNS Hamiltonian gives Operation 2 mixing, while Operation 1 supplies $\Delta m_{32}^2/\Delta m_{21}^2 = 32$. Its eigenvalue ordering is physical; magnitudes are not masses (Appendix J.3). On the conditional positive-count and composed branches, the floor-completed $\mathcal{M}_*^2 = m_\nu^2[\frac{4}{3}\mathbf{I} - \Pi_1^* + N_{\text{spin}}^2(N_c^2-1)\Pi_3(H)]$ realizes both the supplied positive spectrum and the final angles. Appendix J.3 constructs Π_1^* by a specified minimum-change prescription. Its eigenvalues are $\beta_\nu^2(1,4,100)$, and its positive square root has masses $\beta_\nu(1,2,10)$, with $\beta_\nu = m_\nu/\sqrt{3}$. The corresponding gap matrix $\mathcal{M}_*^2 - \beta_\nu^2\mathbf{I}$ has eigenvalues $m_\nu^2(0,1,33)$; its zero is a reference for squared gaps, not a massless state. Its own eigenvectors give all three final mixing angles (supplementary code). The fixed-unit integer completion, its physical mode admissibility and the absolute gap scale remain separately typed. The apparent shared $1/N_c$ exponent in R_1 and m_ν does not unify them: mixing's $1/3$ is $(N_c(2)/N_c(1)) \times T_F = (2/3)(1/2)$, equal to $1/N_c$ only at $N_c = 3$ on this theory's ladder (Section 1.1). Appendix I.3's fixed-gate comparison would equate them identically; the ladder comparison used here does not (supplementary code).

K Extended predictions

K.1 Meson pair rule and V-P splittings

Flavored pseudoscalars obey $m/m_e = N_c \times (\text{spin-channel}) \times (\text{pair factor})$. The antiquark is conjugate, leaving $N_c = 3$ singlet configurations. Spin-channel cost is $\Sigma{+}1 = 7$ for the pion and $N_{\text{spin}}(\Sigma{+}1) = 14$ for the remaining family. Pair factors are gen-1×gen-1 = 13, gen-1×gen-2 (down) = 23, gen-2 (charm)×gen-1 = 87, and gen-3×gen-1 = 246.

Rule 1 combines unsaturated pairs additively: 13 counts $D{=}1$ quark cells and $23 = 13{+}10$ spans quarks in $D{=}1{+}D{=}2$. A color-saturated letter, charm or bottom (Appendix F), couples color as a tensor: $87 = N_c \times 29$, $246 = N_c \times 82$. Their kernel scales are $N_c^3{+}N_{\text{spin}}$ and $N_c^4{+}1$. Reading the latter's $+1$ as the 82/81 participation unit is a disclosed interpretation (Appendix I.6), not a derivation here. The spans are the matrix cross-sections of Table 2 (supplementary code).

Charm-vector V-P splittings add $+\pi + (\Sigma{+}1)$, giving $D^* = D + \pi + 7$. Strange partners use $+N_c^2$: it replaces charm's $+(\Sigma{+}1)$ in $D_s^* = D_s + \pi + 9$ and adds to the bottom splitting in $B_s^* = B_s + 87 + 9$. Bottom vectors add $+87 = N_c \times 29$. The dual-heavy $b\bar{c}$ vector is outside this rule; its comparison value $117\, m_e$ is row-level (Section D.3).

K.2 Rule 5 (the W^- mismatch rule)

Rule 5 complements the four general rules in Appendix I.6.

In composites where heavy quarks have different W^- status and neither a flavor-step base nor spin pairing absorbs the mismatch:

Case A (W^--active minority or tied): correction = N_c(active quark). Appendage is independent; costs 1 per color at $K{=}1$. B_c: $+3$; Ξ_c^*: $+2$.

Case B (W^--active majority): correction = $(N_{c,\text{absent}} + N_{\text{spin}})^2$. Deficit creates coherences between color and the gate shell; $d = N_c{+}N_{\text{spin}} = 5$; cost = $d^2 = 25$. Ω_c: $+25$.

The rule correctly classifies the nine measured composites with heavy W^- mismatch. The Ω_{cc}/Ω_c mirror pair illustrates the asymmetry: an extra channel can be shared; a missing channel cannot be created. The Ω_{cc} mass is derived via the additive strangeness grammar (Section D.3); the W^- mismatch classification itself stands on the nine measured composites.

K.3 Extended widths

The η' width is $\Gamma_{\eta'} = 150/(17 \times 23)\, m_e = 196.0\,\text{keV}$, against the current PDG fit $188 \pm 6\,\text{keV}$ (1.3σ). Its comparison is fit-sensitive: earlier PDG fits gave 194–196 ± 9, and the direct COSY-11 measurement is higher. The decuplet ratio is $\Sigma^*/\Xi^* = (N_c^2{-}1)N_c^2/(N_{\text{spin}}N_c^2) = 4 = N_c{+}1$ (obs 3.96, 1%). The $\rho/J/\psi$ cross-ratio spans three orders of magnitude: $\Gamma_\rho/\Gamma_{J/\psi} = 17^2 \times 11/2 = 1589.5$ (obs 1592, 0.14%).

K.4 Charmonium and bottomonium excitations

The P-wave cost $899 = 29 \times 31$ appears identically in charmonium (χ_{c2}) and bottomonium (χ_{b0}) — different quark families, same structural constant. Both charm and bottom saturate the color columns; bottom additionally occupies the W^- channel. Interpreting the shared 899 offset as a generation-saturation boundary, with gen-3's floor at gen-2's ceiling, requires a source/gate engagement rule for the different J states. The strangeness cost $F = 693$ recurs across light baryons (Λ, Ξ, Ω) and bottom baryons (Λ_b, Ξ_b). The η' singlet premium $11 \times 73 = 803$: the flavor singlet spans all $N_c^2{+}N_{\text{spin}} = 11$ color-spin coherences through the gauge-physical amplitude $73 = N_c^4{-}(N_c^2{-}1)$.

Offset decompositions. Representative mass-table offsets decompose into the kernel integers of Table 2. Radial selections retain the structurally constrained fit tier of Section 5.1. The non-radial offsets 397 and 384 are open objects of a different kind: a strangeness step and a junction–isospin composite, each the difference of two priced rows. A difference is not a mode count, so its factorization carries no constraint. The h rows are centroids of their χ rows (Table 9). Examples are: $575 = 25 \times 23 = (N_c{+}N_{\text{spin}})^2 \times (2(\Sigma{+}1){+}N_c^2)$ $(\Delta{-}p)$; $621 = 27 \times 23 = N_c^3 \times (2(\Sigma{+}1){+}N_c^2)$ (χ_{c0}); $810 = N_c^4(N_c^2{+}1)$ (χ_{c1}); $1092 = 4 \times 273 = N_{\text{spin}}^2 \times m_{\pi^\pm}$ $(\Upsilon(2S){-}\Upsilon(1S)$: four pion masses); $651 = N_c(\Sigma{+}1)(N_c^3{+}N_{\text{spin}}^2)$ $(\Upsilon(3S){-}\Upsilon(2S))$; $438 = N_{\text{spin}}N_c \times 73$ $(\Upsilon(4S){-}\Upsilon(3S))$; $598 = N_{\text{spin}} \times 13 \times 23$ $(\Upsilon(5S){-}\Upsilon(4S))$; $171 = N_c^2(2N_c^2{-}1{+}N_{\text{spin}})$ $(\psi(3770){-}\psi(2S))$; $95 = (N_c{+}N_{\text{spin}})(2N_c^2{-}1{+}N_{\text{spin}})$ $(\psi(2S){-}\eta_c(2S))$.

P-wave shell-boundary structure. The three charmonium P-wave offsets from J/ψ factorize as source × gate products at the $D{=}1/D{=}2$ boundary: $J{=}0$: $N_c^3 \times (2(\Sigma{+}1){+}N_c^2) = 27 \times 23 = 621$; $J{=}1$: $N_c^4 \times (N_c^2{+}1) = 81 \times 10 = 810$; $J{=}2$: $(N_c^3{+}N_{\text{spin}})(N_c^3{+}N_{\text{spin}}^2) = 29 \times 31 = 899$. All six factors are kernel integers. The source-factor sum $27{+}81{+}29 = 137 = \lfloor 1/\alpha_{\text{EM}} \rfloor$; the gate-factor sum $23{+}10{+}31 = 64 = 1/\kappa_2$. Both identities hold via $(N_c{-}3) = 0$: for general N_c the differences factor as $(N_c{-}3)(N_c^3{+}5N_c^2{+}15N_c{+}45)$ (source) and $(N_c{-}3)(N_c^2{+}6N_c{+}19)$ (gate). Three independent sectors — charmonium P-wave excitations, the electromagnetic coupling, and the QCD phase curvature — are linked by the same algebraic root. These are structural decompositions of the printed offsets; a common microscopic rule selecting the participating source/gate relations for each J remains to be derived.

L Chiral structure derivation

The vacuum's sense-paired ground state admits flavored re-pairings that route through transparent cells of the compute matrix (tracelessness parks the mismatched sense in a hole — machine-verified on the canonical matrix), making flavored rotations ledger-cancelling valley modes. The flavor-singlet rotation is blocked: the W^+ and E columns contain zero transparent cells in any massive row, and the Q column only the neutrino's single cell, against a parking requirement of at least three per column. The unparkable

ledger sits in channels whose only transparency outside that lone cell is the photon row — the $\eta' \to \gamma\gamma$ anomaly channel: the singlet's toll, paid to the read sector.

With rotation cost quadratic in angle (T1) and stocks billed once per read (A2), the valley masses obey $m_{\mathrm{PS}}^2 = B\,(m_{q_1}+m_{q_2})$ — the GMOR relation at leading order, a consequence of A3's own linear cost form. The framework's stocks give $m_K^2/m_\pi^2 = 14.2$ (obs 13.1–13.6), and one-loop composition with the derived $f_\pi = 180\,m_e$ closes the K and η_8 deviations at a single conventional scale $0.78 \pm 0.06\,\mathrm{GeV}$ with zero fitted constants (mass sector only: f_K/f_π requires the standard analytic term $L_5 \approx 1.2 \times 10^{-3}$).

The leading-order mechanism is derived: A3 pairing, code-confirmed hole routing, T1 quadratic rotation cost, and A2 stocks billed once. Its scales remain flows (Section 5.3). The condensate scale B is a per-channel consistency band whose chiral-limit reading matches the lattice condensate; neither B nor the loop scale is a framework integer. Per-channel vector masses are excluded within the analysis, and quadratic-in-stock alternatives fail by $16\times$.

The blockage is overdetermined: two columns have no supply and a third has too little, even with lepton rows as absorbers. Falsification requires a revised matrix with at least three same-column massive-row holes in each of W^+, Q, and E (supplementary code).

M Description length and forcing tiers: the full audit

Scope of the overview accounting.

The overview accounting favors the Standard Model at its adversarial floor; the two ledgers below give the accounting conventions and ranges. Both sides exclude the five lattice quark-mass labels (m_u, m_s, m_c, m_b, m_t); m_d enters only through m_d/m_u because the theory prices no lone-quark observable (Section 3.1). The comparison is asymmetric: the same $\sim$164 bits also carry cosmology, the dark-matter budget and profile, and the derivations of quantum mechanics and general relativity, beyond the Standard Model parameters' scope. The cosmological values are in Sections 4.5 and 4.7; Section 4.1 gives the clock-channel reading.

The compressed statements of Section 5.1 are priced here in full.

The parameter ledger. This ledger prices the theories' dials on the 23 SM+ΛCDM parameters the theory independently predicts. It excludes five lattice quark-mass labels from both sides because the framework computes no lone-quark observables (Section 3.1). The two-part code costs $\sim$164 bits versus $\sim$344 for the Standard Model, with its 19 continuous free parameters: $2.1\times$ raw compression. Charging $\sim$148 bits of assembly selection reduces this to $\sim 1.10\times$. The code enumerates $\sim$64 bits for anchors and $\sim$82 for the anchor-plus-offset family. The latter charges its three grammar-open offsets at the uncovered-row rate 61/7 bits each; a minimal decodable window pointer gives $\sim$64 instead. The radial family contributes its measured $\sim$24 bits.

The harshest constructible accounting uses a literal pointer across the grammar's offset range and falls below parity, $\sim$0.72–0.79$\times$: the Standard Model code is shorter. This is the conservative reading because selections informed by the comparison data incur their full pointer cost. Only predictions fixed before their data receive blind credit. That charge applies directly to offset and uncovered-anchor families. The largest remaining penalty prices the kernel integers as free vocabulary selections, although they are code-confirmed consequences of $N_c = 3$. The accountings therefore bracket the comparison: the freedom-priced floor, $\sim$1.10$\times$ with specified grammar selections charged, and optimistic raw 2.1$\times$.

The same $\sim$164 bits also carry S_8, $w(z)$, BAO, H_0, the dark-matter budget, and BBN; the Standard Model side prices none of that cosmological content.

The observable ledger. This second ledger prices the same 69 hadron and lepton masses. It restores the four quark-mass inputs required by lattice QCD; the fifth label, the top's, prices no hadron in the set. All five remain excluded from the parameter ledger. Here the framework costs $\sim$163–429 bits, raw to adversarial floor, against $\sim$351–400 for the Standard Model. It leads centrally, with ratios $\sim$0.82$\times$ at the adversarial floor and $\sim$0.79$\times$ in the harshest constructible accounting. That floor assigns zero cost to lattice machinery and grants full precision parity even for channels such as Ω_c without a published blind lattice prediction at any precision (supplementary code).

Selection costs. Of the kernel integers, eight follow from matrix arithmetic, five from Rule 1, and eight each involve one interpretive identification. Pricing all as free choices produces the largest adversarial penalty. The core grammar decisions—$d \leftrightarrow d^2$, vertex 13 versus 17, junction placement, and addition versus multiplication—cost $\sim$1–2 bits: every alternative is excluded by an anchor at 0.15%. The lepton baseline's complete-family read is the priced interpretation; its count and negative sign follow from the matter-lepton ground and Rule 4. The anchor-plus-offset family's choice among offsets is a separate family-level charge, included with vocabulary-pointer and uncovered-anchor costs, not in the $\sim$1–2-bit figure. Those residual freedoms drive the worst parameter-ledger case below parity.

Five experimental splits add one further class: V_{cb} inclusive/exclusive; τ_n bottle–beam and V_{ud} unitarity–direct; m_W with CDF-II excluded; the H_0 ladder; and the T_{QCD} lattice spread. Structures fix the comparison sides (Section 6). T_{QCD} chooses no side; τ_n uses the harsher quoted datum; the m_W combination supplies its own compatibility criterion; and the H_{local} band cannot reach the opposite side. The additional charge is the V_{cb} route typing, $\sim$1–2 bits. Resolving all splits against these sides would still place the corresponding rows in tension simultaneously.

At greater precision, structural residuals grow while Standard Model residuals vanish by construction. The proton residual has structure, $0.15\, m_e = 21\, \alpha_{\mathrm{EM}}(1-\alpha_{\mathrm{EM}}/2)\, m_e$ (Appendix D.4). If this $O(\alpha_{\mathrm{EM}})$ pattern extends, residual bits approach zero at the cost of one N per particle.

Forcing tiers. Ground-state masses (proton, leptons, mesons) are *derived* at their stated local tiers. Gauge couplings retain the identifications below; the neutral-wave counts underlying $\sin^2\theta_W$ remain open. Each mass window of 0.15% contains exactly one grammar-

licensed integer: 1836, 273, and 966 are unique among 430–560 licensed expressions (supplementary code). Their per-family assignment cost, ∼64 anchor bits, remains charged in selection-adjusted minimum description length (MDL).

Orbital excitations (χ_c, χ_b P-waves) are *structurally constrained*: the charmonium source × gate factorization has source sum $= 137 = \lfloor 1/\alpha_{\mathrm{EM}} \rfloor$ and gate sum $= 64 = 1/\kappa_2$ (Appendix K.4). These identities do not yet select the J-dependent local operation. Radial excitations ($\psi(2S)$, $\Upsilon(nS)$) are *structurally constrained fits*. Their offsets decompose into kernel integers, but the $K{=}r$ Airy spectrum lacks protected rational structure (Appendix N); no kernel-integer derivation is established. The measured ∼24-bit residual belongs to the present fit grammar; neither the absence of hidden symmetry nor transcendence alone proves a universal fit floor.

The weak-angle boundary. In the structured tier, $\sin^2\theta_W = 117/506$ ranks first in combined simplicity and closeness among 212 kernel-integer fractions within 3σ; the only simpler candidate lies at 2.4σ. Its scheme identification is unforced. The three $\sin^2\theta_W$ conventions differ by ∼1%, and no framework rule chooses one. Likewise, nothing pins MS-bar at M_Z as the scheme and scale for $2/17$ (Section 3.3). The Thomson-limit referent of $\alpha_{\mathrm{EM}}(0)$ is convention-free.

The operation is typed but remains unevaluated. Mixing the two shell couplings is a gate/drain cross-shell refraction (Operation 2, Section 2.2), with both lenses engaged at once. Its $D{=}2 \to D{=}3$ amplitude exponent $N_c(3)/N_c(2) = 1/2$ makes the observable a bare ratio of joint mode counts through T2. Deriving which two counts the neutral wave presents remains open; the stated factorization does not supply that ratio. The convention-free pole-mass referent $1 - m_W^2/m_Z^2$ is 0.22333. The fraction matches the convention-bound MS-bar value, reached through the ceded radiative machinery (Section 5.3).

Grammar boundaries. Code perturbations of $d \leftrightarrow d^2$, vertex 13 for 17, junction factors on mesons, and additive baryon composition each fail at least one anchor's 0.15% window. Proton, pion, kaon, and η are unique 2–4-factor vocabulary products in their windows. The η enters at 0.11%; proton and kaon uniqueness requires the licensed small factors 2–6. The shared pion–kaon spin factor, doubled for the kaon, has exactly one solution.

The lepton subtraction has one priced structural identification: a post-anchor family coordinate reads the complete resident matter-lepton block. Rule 4 debits its $5{+}3{+}3{+}3 = 14$ transparent no-ops once; the electron anchor carries no coordinate. Four measured applications of join-then-square support the gross d^2 bill (Λ moment, Ω_c, quarkonium normalizations, ϕ width; Appendix I.5). The matrix complement supplies the subtraction. Rule 1 separately fixes multiplicative color-carrying anchors through coupled-register tensor products: every hadron anchor is a pure product.

Grammar-free product searches at $N_c = 4$ still find isolated accidental matches, including an η closer than at $N_c = 3$. The structured failure is that no $N_c = 4$ grammar assignment reaches the pion, kaon, or muon.

N The Airy transcendence conjecture

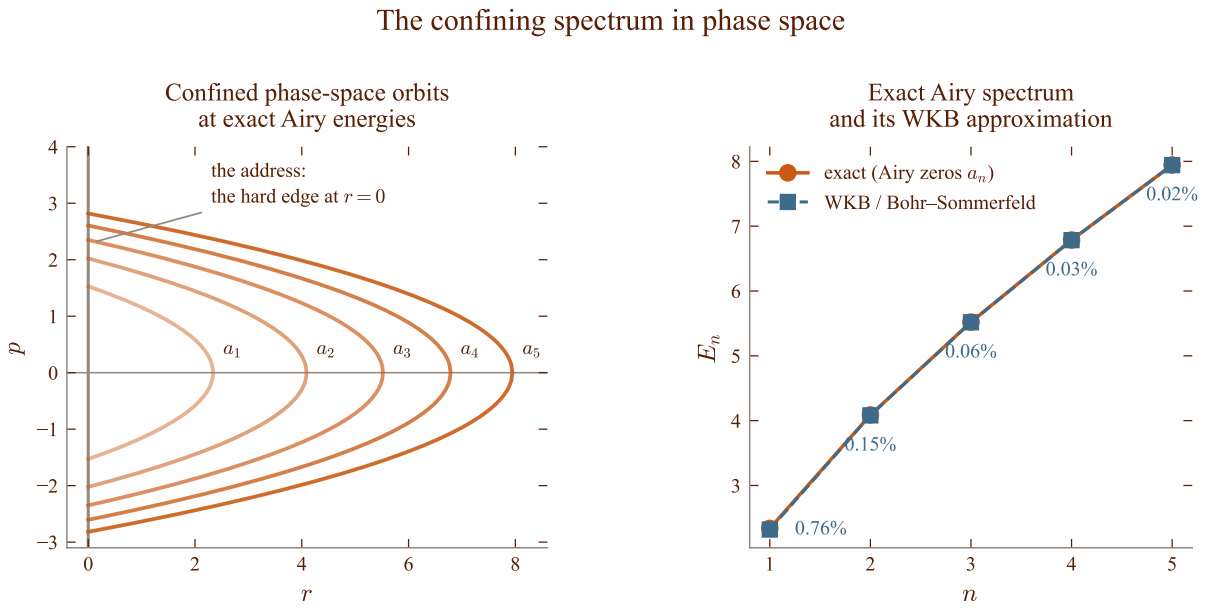


Figure 33: *The confining spectrum in phase space. In dimensionless units, $p^2 + r = E$, $r \geq 0$, has a soft turn at $r = E$ and an elastic momentum reversal at the fixed address $r = 0$. These curves describe radial position and momentum; their lens shapes are not spatial return paths. Left: five classical curves evaluated at exact Airy energies. Right: the Wentzel–Kramers–Brillouin (WKB) approximation lies below those energies, with errors falling from* 0.76% *at $n = 1$ to* 0.02% *at $n = 5$. The action area is $\oint p\,dr = 4E^{3/2}/3$; setting it to $\pi(4n-1)/2$ gives the WKB levels. An oscillator has elliptical phase-space curves (circular after rescaling); its quadratic potential belongs to a different kernel. The exact Airy levels are obtained from the boundary problem.*

The source has one preserved address and a family of radial levels. For the effective scalar equation already used here, take fixed $A, B > 0$ and measure $r \geq 0$ from that address. Set $x = \beta r$, with $\beta = (B/A)^{1/3}$ and $\epsilon = \beta E/B$. Then

$$-Au'' + Bru = Eu \quad \Longleftrightarrow \quad H\varphi = \epsilon\varphi, \qquad H = -\frac{d^2}{dx^2} + x, \quad x \geq 0. \tag{51}$$

Every mode has the same boundary $\varphi(0) = 0$ and decays as $x \to \infty$. Decay selects $\varphi_\epsilon(x) \propto \mathrm{Ai}(x - \epsilon)$; the boundary selects $\mathrm{Ai}(-\epsilon) = 0$, hence $\epsilon = a_n$. The operator and address stay fixed while a_n labels the radial level. The argument $t = x - \epsilon$ used to write the standard Airy equation absorbs the spectral parameter; it is neither a moving address nor time. Evolution belongs to the wave around the frozen D=0 point.

The common-node result below identifies the one address shared by these profiles (Proposition 16). The arithmetic question asks instead which spectral levels of this same operator are algebraic. It does not determine the number of addresses. The source's linear kernel lacks the oscillator or Coulomb protection discussed in Section 5.3; its stronger number-theoretic statement remains open.

Conjecture. Every zero of the Airy function Ai is transcendental.

Status. The reduction is exact. For $a > 0$, DLMF 9.6.6 reads $\mathrm{Ai}(-a) = \frac{1}{3}\sqrt{a}\,\big(J_{1/3}(\zeta) + J_{-1/3}(\zeta)\big)$ at $\zeta = (2/3)a^{3/2}$; the prefactor never vanishes, so $\mathrm{Ai}(-a) = 0$ iff $J_{1/3}(\zeta) + J_{-1/3}(\zeta) = 0$. Every zero of Ai is negative real, all functions are read on the positive real axis with one fixed branch of $z^{1/3}$, and a is algebraic iff ζ is. For a *single* Bessel function of rational order $\nu \notin \frac{1}{2} + \mathbb{Z}$ the corresponding statement is Siegel's [68] — there the level is 0, which is algebraic, and the E-function machinery applies at once. For the Airy sum it does not. Writing $f_\pm$ for the E-normalized solutions ($f_\pm(0) = 1$; these are Siegel's own K_λ at $\lambda = \pm\frac{1}{3}$), the zero condition pins the ratio $(\zeta/2)^{2/3} f_+(\zeta)/f_-(\zeta)$ to the level $-\Gamma(4/3)/\Gamma(2/3) = -\sqrt{3}\,\Gamma(1/3)^2/(6\pi)$, transcendental by Nesterenko's theorem [69] whose equality to the ratio is not decided by the E-value theorem used here.

At most one zero of Ai *is algebraic.* The proof below extends this bound to Ai and Bi together, their derivatives, and the algebraic pencil $c_1\mathrm{Ai} + c_2\mathrm{Bi}$ (Proposition 25). Two algebraic zeros give conditions at one level up to sign. Cross-multiplication removes that level, leaving a homogeneous Bessel E-function relation. The refined Siegel–Shidlovskii theorem [70] lifts it to a functional identity, which the differential-Galois calculation excludes.

A single zero instead places $f_+(\zeta)/f_-(\zeta)$ on $\overline{\mathbb{Q}}^\times \cdot \Gamma(1/3)^2/\pi$. This joins the E-function and period theories without eliminating the level; the argument therefore stops at one. The counting sector requires only Bertrand's theorem and survives either resolution of the conjecture. (Supplementary code.)

The proof first reduces the zeros to normalized Bessel values, then combines functional independence with arithmetic transfer to exclude two algebraic zeros. Later extensions treat finite boundaries, the common source address, characteristic two, and derivative families. No scored prediction depends on these arithmetic results; Bertrand's theorem supplies the physical demarcation.

N.1 Introduction

Siegel proved in 1929 that all nonzero zeros of the Bessel function J_ν are transcendental for rational $\nu \notin \frac{1}{2} + \mathbb{Z}$ [68]; see also [95]. The zeros of the Airy function

$$\mathrm{Ai}(-x) = \frac{\sqrt{x}}{3}\Big(J_{1/3}(\zeta) + J_{-1/3}(\zeta)\Big), \qquad \zeta = \tfrac{2}{3}x^{3/2}$$

(see [97, 9.6.6]) lie outside Siegel's theorem for a structural reason. No prior treatment of their transcendence is identified here. In the basis $(J_{1/3}, J_{-1/3})$ displayed above the two coefficients of Ai are equal; but Siegel–Shidlovskii theory does not run in that basis — it runs on E-functions, and passing to them absorbs the Γ-factors and the branch factors $z^{\pm 1/3}$ into the basis vectors. In the E-function normalization

$$f_+(z) = {}_0F_1\big(;\tfrac{4}{3};-\tfrac{z^2}{4}\big), \qquad f_-(z) = {}_0F_1\big(;\tfrac{2}{3};-\tfrac{z^2}{4}\big), \qquad J_{\pm 1/3}(z) = \frac{(z/2)^{\pm 1/3}}{\Gamma(1 \pm \frac{1}{3})}\, f_\pm(z),$$

the condition $\mathrm{Ai}(-x) = 0$ reads, in cleared form,

$$(\zeta/2)^{2/3} f_+(\zeta) + \gamma_* f_-(\zeta) = 0, \qquad \gamma_* := \frac{\Gamma(4/3)}{\Gamma(2/3)} = \frac{\sqrt{3}\,\Gamma(1/3)^2}{6\pi}; \tag{52}$$

equivalently — division being legitimate at such points because f_- does not vanish there (Lemma 6) — the ratio $h(\zeta) := (\zeta/2)^{2/3} f_+(\zeta)/f_-(\zeta)$ is pinned to the level $-\gamma_*$. A zero of a single Bessel function is the level-0 case of such a condition, and 0 is algebraic: the Siegel–Shidlovskii machinery, which controls all *homogeneous* algebraic relations among Bessel E-function values at algebraic points, applies at once. The Airy condition sits at the level γ_* — an algebraic multiple of $\Gamma(\frac{1}{3})^2/\pi$, a constant of the same $\Gamma(\frac{1}{3})$–π arithmetic that governs the periods of the equianharmonic ($j = 0$) CM curve, and transcendental by Nesterenko's theorem [69]. The proof uses only $\gamma_* \neq 0$. Each machinery is complete in its own domain; no theorem coupling them is identified or used here.

Two algebraic zeros would sit on the same level up to sign. Their ratio eliminates γ_*, giving a homogeneous polynomial relation with algebraic coefficients among f_+, f_- and their derivatives at two algebraic points. Beukers' refined Siegel–Shidlovskii theorem [70] lifts it to a functional identity. Differential Galois theory excludes that identity, giving:

Theorem 2. *At most one zero of* Ai *is an algebraic number.*

Corollary 3. *At most one algebraic number occurs among the negative real zeros of* Ai *and* Bi *together.*

All Ai zeros are negative real. Corollary 3 therefore contains Theorem 2, and Proposition 25 contains both. The infinitely many complex zeros of Bi [97, §9.9] remain outside this joint bound; Theorem 13 below covers each fixed Airy solution individually throughout $\mathbb{C}$.

Several comments place the statement.

(1) Bornemann's functional-independence theorem for the Airy equation [72] uses Shidlovskii's lemma and, in Singer's appendix, differential Galois theory. It concerns functions and supplies no immediate arithmetic consequence for values, including algebraic or linear independence over $\mathbb{Q}$ at nonzero algebraic arguments: Ai is not an E-function. That work does not raise the zeros question, and no earlier treatment is identified here.

(2) Granting André's conjecture on arithmetic Nilsson–Gevrey series [98, Conj. 4.5.1], at most finitely many zeros of Ai are algebraic; Theorem 2 is sharper and unconditional.

(3) The statement shape has precedent: Murty proved that all zeros of the digamma function are irrational with at most one possible exception [101], by Baker's theory.

(4) Lorch–Muldoon [100] proved transcendence of the zeros of $\frac{d^n}{dx^n}\, x^{-\mu} J_\nu(x)$ (ν rational, μ algebraic) away from explicit algebraic exceptions — results on the single-equation side of the divide that the Airy combination crosses.

(5) The one-exception form is intrinsic to the method: a *single* algebraic zero is, by (52), a relation between Bessel values and the line $\overline{\mathbb{Q}} \cdot \Gamma(\frac{1}{3})^2/\pi$ — or, for the derivative families of Proposition 25, the *reciprocal* line $\overline{\mathbb{Q}} \cdot \pi/\Gamma(\frac{1}{3})^2$, since $\Lambda = 4\pi/(3\sqrt{3}\,\Gamma(\frac{1}{3})^2)$ — invisible to E-function methods by construction (Remark 12).

Throughout, $\overline{\mathbb{Q}} \subset \mathbb{C}$ is the field of algebraic numbers. In the negative-real-zero reductions, Bessel evaluation points are positive and fractional powers are principal. The functional calculation and arithmetic transfer state their larger complex domains explicitly.

N.2 Reduction, Wronskian, nonvanishing

Lemma 4. *Let $a > 0$ and $\zeta = \frac{2}{3}a^{3/2}$. Then*

$$\mathrm{Ai}(-a) = 0 \iff (\zeta/2)^{2/3} f_+(\zeta) + \gamma_* f_-(\zeta) = 0,$$

$$\mathrm{Bi}(-a) = 0 \iff (\zeta/2)^{2/3} f_+(\zeta) - \gamma_* f_-(\zeta) = 0.$$

Moreover a is algebraic if and only if ζ is algebraic, and $a \mapsto \zeta$ is strictly increasing on $a > 0$.

Proof. Insert the normalizations of f_+, f_- into the connection formulae $\mathrm{Ai}(-x) = \frac{\sqrt{x}}{3}(J_{1/3} + J_{-1/3})(\zeta)$ and $\mathrm{Bi}(-x) = \sqrt{x/3}\,(J_{-1/3} - J_{1/3})(\zeta)$ [97, 9.6.6, 9.6.8] (note the two prefactors differ) and multiply by $\Gamma(\frac{4}{3})(\zeta/2)^{1/3}$; the outer prefactors multiply a vanishing bracket and drop out, and $\Gamma(\frac{4}{3})/\Gamma(\frac{2}{3}) = \gamma_*$ by the reflection formula. The maps $a \mapsto \zeta = \frac{2}{3}a^{3/2}$ and $\zeta \mapsto a = (\frac{3}{2}\zeta)^{2/3}$ preserve algebraicity in both directions. □

Lemma 5. $z\big(f_+ f_-' - f_+' f_-\big) = \frac{2}{3}\big(f_+ f_- - 1\big)$ *identically.*

Proof. From $W\{J_{1/3}, J_{-1/3}\}(z) = -2\sin(\pi/3)/(\pi z)$ [97, 10.5.1], rewritten through the normalizations, using $\Gamma(\frac{4}{3})\Gamma(\frac{2}{3}) = 2\pi/(3\sqrt{3})$. □

Lemma 6. *If ζ satisfies either equation of Lemma 4, then $f_+(\zeta) \neq 0$ and $f_-(\zeta) \neq 0$.*

Proof. If $f_-(\zeta) = 0$, the cleared relation forces $f_+(\zeta) = 0$ as well ($\gamma_* \neq 0$, $\zeta \neq 0$), and Lemma 5 then reads $0 = -\frac{2}{3}$. Symmetrically for f_+. □

Note that f_+ and f_- satisfy *different* second-order equations, $zy'' + \frac{5}{3}y' + zy = 0$ and $zy'' + \frac{1}{3}y' + zy = 0$ respectively; they are not a fundamental pair of a common equation, and Lemma 5 is a relation between solutions of two distinct operators.

N.3 Functional independence at two scaled arguments

Fix $\lambda \in \mathbb{C}^\times$ with $\lambda^2 \neq 1$. Consider the eight functions

$$\begin{aligned} u_1 &= f_+(z), & u_2 &= f_+'(z), & u_3 &= f_-(z), & u_4 &= f_-'(z), \\ v_1 &= f_+(\lambda z), & v_2 &= \tfrac{d}{dz} f_+(\lambda z), & v_3 &= f_-(\lambda z), & v_4 &= \tfrac{d}{dz} f_-(\lambda z) \end{aligned} \tag{53}$$

(derivatives in z throughout). Write $u = (u_1, \dots, u_4)$, $v = (v_1, \dots, v_4)$, and $\overline{\mathbb{Q}}[u, v]$ for the polynomial ring in the eight corresponding variables. The functions satisfy the block-diagonal first-order system

$$\begin{aligned} u_1' &= u_2, & u_2' &= -\tfrac{5}{3z} u_2 - u_1, & u_3' &= u_4, & u_4' &= -\tfrac{1}{3z} u_4 - u_3, \\ v_1' &= v_2, & v_2' &= -\tfrac{5}{3z} v_2 - \lambda^2 v_1, & v_3' &= v_4, & v_4' &= -\tfrac{1}{3z} v_4 - \lambda^2 v_3, \end{aligned} \tag{54}$$

with coefficients in $\mathbb{Q}(\lambda)(z)$ and common denominator $T(z) = z$. Define the two *Wronskian polynomials*

$$G_1 := z(u_1 u_4 - u_2 u_3) - \tfrac{2}{3}(u_1 u_3 - 1), \qquad G_2 := z(v_1 v_4 - v_2 v_3) - \tfrac{2}{3}(v_1 v_3 - 1). \tag{55}$$

Both vanish identically on the functions: G_1 is Lemma 5, and for G_2 one computes $v_1 v_4 - v_2 v_3 = \lambda\,(f_+ f_-' - f_+' f_-)(\lambda z)$, so that $z(v_1 v_4 - v_2 v_3) = (\lambda z)(f_+ f_-' - f_+' f_-)(\lambda z) = \tfrac{2}{3}(v_1 v_3 - 1)$ by Lemma 5 applied at the argument λz.

Theorem 7. *For $\lambda \in \mathbb{C}^\times$ with $\lambda^2 \neq 1$, the eight functions (53) have transcendence degree exactly 6 over $\mathbb{C}(z)$, and also over $\overline{\mathbb{Q}}(z)$ when $\lambda \in \overline{\mathbb{Q}}$.*[5]

Proof. Normalization. Set $j_+ = z^{1/3} f_+$ and $j_- = z^{-1/3} f_-$ on a simply connected domain $\Omega \subset \mathbb{C} \setminus \{0\}$ on which branches of $z^{1/3}$ and $(\lambda z)^{1/3}$ are fixed ($\lambda\Omega$ is again simply connected and omits 0, so such branches exist; below only the identity $(\lambda z)^{1/3} = \omega^m \lambda^{1/3} z^{1/3}$, ω a cube root of unity, is used, which holds for every branch choice). All eight functions of (53), and both fundamental systems below, are regarded as meromorphic functions on this single domain Ω. A direct computation from the two scalar equations above shows both $j_\pm$ satisfy the *same* equation

$$z^2 j'' + z j' + (z^2 - \tfrac{1}{9}) j = 0 \qquad \text{(Bessel, } \nu = \tfrac{1}{3}\text{)}.$$

Their Wronskian: a direct computation gives $W\{j_+, j_-\} = f_+ f_-' - f_+' f_- - \tfrac{2}{3z} f_+ f_-$, and Lemma 5 evaluates this to $-\tfrac{2}{3z}$; so (j_+, j_-) is a fundamental system. The trace of the companion matrix is $-1/z$, and $y' = -y/z$ has the rational solution z^{-1}: the determinant module is trivial. The j's are Γ-free multiples of the f's; no transcendental constant appears in any generator below.

Galois groups. Work over $\mathbb{C}(z)$; the constants of the field of meromorphic functions on Ω are $\mathbb{C}$, so the Picard–Vessiot extensions below have no new constants. Let N_1 be the

[5] Corroborated by Gorelov [96], whose specialization to $\nu = \tfrac{1}{3}$ confirms $\lambda^2 \neq 1$ is sharp; the proof below is self-contained.

Bessel-$(\frac{1}{3})$ module with fundamental system $(j_+, j_-)(z)$ and Picard–Vessiot field K_1, and let N_2, K_2 be their pullbacks under $z \mapsto \lambda z$ (again Bessel-$(\frac{1}{3})$ in the variable λz; rescaling the derivation by λ changes no group). Since $\frac{1}{3} \notin \frac{1}{2} + \mathbb{Z}$ and the determinants are trivial, $\mathrm{Gal}(K_i/\mathbb{C}(z)) = \mathrm{SL}_2(\mathbb{C})$. This is Kolchin's computation [71, Appendix], whose Appendix is titled "The Galois group of the Bessel equation"; it is a statement about one equation at a time, which suffices here. The Goursat argument below supplies the interaction between scalings.

Goursat. Let $K = K_1K_2$ and $G = \mathrm{Gal}(K/\mathbb{C}(z)) \leq \mathrm{SL}_2(\mathbb{C}) \times \mathrm{SL}_2(\mathbb{C})$, both projections surjective (restriction of automorphisms of the compositum of normal extensions). If G is a proper subdirect subgroup, Goursat's lemma yields a nontrivial common quotient Q of the two factors with an induced isomorphism. Since G is Zariski closed, all groups here are algebraic, and the induced isomorphism is one of algebraic groups: its graph is closed, so both projections from it are bijective homomorphisms of algebraic groups, hence isomorphisms in characteristic 0. The normal closed subgroups of $\mathrm{SL}_2(\mathbb{C})$ are 1, $\{\pm I\}$, $\mathrm{SL}_2(\mathbb{C})$, so $Q = \mathrm{SL}_2(\mathbb{C})$ or $Q = \mathrm{PSL}_2$; there is no finite option.

Case $Q = \mathrm{SL}_2(\mathbb{C})$. Both quotient maps are isomorphisms, so G is the graph of an automorphism of $\mathrm{SL}_2(\mathbb{C})$ *as an algebraic group*; all such are inner, so as G-representations $V_2 \cong V_1$, i.e. $N_2 \cong N_1$ as differential modules (the dictionary between G-representations and differential modules, [102, Thm. 2.33], applied to the Tannakian category generated by $M = N_1 \oplus N_2$, whose Picard–Vessiot field is K and whose differential Galois group is G). Compare formal data at $z = \infty$, where the exponential parts are isomorphism invariants of the formal classification [102, Thm. 3.1(3), Remarks 3.20(5), Exercise 3.27]: by the Hankel asymptotics [97, 10.17.2, 10.17.5–6] the Levelt–Turrittin exponential parts of N_1 are $\{iz, -iz\}$ (distinct, slope 1), and those of N_2 are $\{i\lambda z, -i\lambda z\}$. Multiset equality forces, on subtracting the members of each pair, $2iz = \pm 2i\lambda z$, i.e. $\lambda = \pm 1$ — excluded.

Case $Q = \mathrm{PSL}_2$. The induced automorphism of $\mathrm{PSL}_2 = \mathrm{PGL}_2$ is inner (type A_1); absorbing it, the two representations $\rho_1, \rho_2 \colon G \to \mathrm{GL}_2$ induce the same map to PGL_2, so $\rho_2 = \chi \cdot \rho_1$ for a character $\chi \colon G \to \mathbb{G}_m$ (Schur; V_1 is irreducible since pr_1 is onto $\mathrm{SL}_2(\mathbb{C})$). Hence $N_2 \cong N_1 \otimes L$ for a rank-1 module L with $\det N_2 \cong \det N_1 \otimes L^{\otimes 2}$; triviality of both determinants gives $L^{\otimes 2} \cong \mathbf{1}$. A 2-torsion rank-1 module is generated by an algebraic function $w^{1/2}$, so its exponential part at ∞ vanishes, and the exponential-part comparison $\{iz + \rho_L, -iz + \rho_L\} = \{i\lambda z, -i\lambda z\}$ again forces $\lambda = \pm 1$ by the same difference argument (which is ρ_L-invariant for *any* ρ_L). Excluded.

Conclusion over $\mathbb{C}(z)$. Hence $G = \mathrm{SL}_2(\mathbb{C}) \times \mathrm{SL}_2(\mathbb{C})$, of dimension 6. Since $\det = -\frac{2}{3z}$ is a unit in $\mathbb{C}(z)$, inverses of the fundamental matrices are polynomial in their entries over the base, so $K = \mathbb{C}(z)(j\text{-functions})$ exactly, and the transcendence degree of the eight j-functions over $\mathbb{C}(z)$ equals $\dim G = 6$ [102, Cor. 1.30(3)].

Transfer and descent. The fields $\mathbb{C}(z)(f\text{-block})(z^{1/3}, \lambda^{1/3})$ and $\mathbb{C}(z)(j\text{-block})(z^{1/3}, \lambda^{1/3})$ coincide (both inclusions use only $j_\pm = z^{\pm 1/3} f_\pm$ and the chain rule), and finite algebraic extensions preserve transcendence degree; hence $\mathrm{tr.deg}_{\mathbb{C}(z)}(f\text{-block}) = 6$. For $\lambda \in \overline{\mathbb{Q}}$: algebraic independence over $\mathbb{C}(z)$ implies algebraic independence over the subfield $\overline{\mathbb{Q}}(z)$, giving transcendence degree ≥ 6 over $\overline{\mathbb{Q}}(z)$; and ≤ 6 because (G_1, G_2) lies in the relation ideal and is a regular sequence in $\overline{\mathbb{Q}}(z)[u, v]$ — proved in Lemma 8 below without refer-

ence to the present bound — hence of height 2, so the relation variety has dimension at most $8 - 2 = 6$. □

Lemma 8. *For $\lambda \in \overline{\mathbb{Q}}^{\times}$, $\lambda^2 \neq 1$, the prime ideal $I \subset \overline{\mathbb{Q}}(z)[u, v]$ of all algebraic relations of the eight functions (53) over $\overline{\mathbb{Q}}(z)$ is generated by G_1 and G_2 of (55).*[6]

Proof. First, independently of Theorem 7: each G_i is geometrically irreducible — G_1 is of degree 1 in u_4 with coefficient zu_1 (z a unit in $\overline{\mathbb{Q}}(z)$), and its content is trivial since setting $u_1 = 0$ leaves $-zu_2u_3 + \frac{2}{3} \neq 0$; this argument is valid over every field extension. The two live in disjoint variable blocks, so the quotient by (G_1, G_2) is a tensor product of a geometrically integral algebra with a domain, hence a domain: (G_1, G_2) is prime. It is a regular sequence, so of height 2 and dimension $8 - 2 = 6$. (This much is what the descent step of Theorem 7 uses.)

Now I is prime (kernel of evaluation into a domain), contains (G_1, G_2), and its variety has dimension $\operatorname{tr.deg}_{\overline{\mathbb{Q}}(z)}(u, v) = 6$ by Theorem 7. In the catenary, equidimensional ring $\overline{\mathbb{Q}}(z)[u, v]$, a prime contained in a prime of equal dimension equals it. □

N.4 Arithmetic transfer and exclusion

Lemma 9. *Let $\zeta_0 \in \overline{\mathbb{Q}}^{\times}$ and $\lambda \in \overline{\mathbb{Q}}^{\times}$ with $\lambda^2 \neq 1$. Every* homogeneous *polynomial $P \in \overline{\mathbb{Q}}[u, v]$ vanishing on the eight values of (53) at $z = \zeta_0$ lies in the specialized ideal $(g_1, g_2) \subset \overline{\mathbb{Q}}[u, v]$, where $g_i := G_i|_{z=\zeta_0}$ with G_i as in (55). The point may be complex: [70, Thms. 1.2 and 1.3] require $\xi T(\xi) \neq 0$, which here is just $\zeta_0 \neq 0$.*

Proof. The eight functions are E-functions in the sense of [70, §1]: each is entire, each satisfies a linear differential equation over $\overline{\mathbb{Q}}[z]$ (clear denominators in (54)), and, writing $f = \sum_k a_k z^k/k!$, the a_k are algebraic and satisfy Beukers' condition on the height of the *tuple*, $h(a_0, \dots, a_k) = O(k)$. For ${}_0F_1$ with rational parameters this is the case Siegel treated. Two estimates give the tuple bound: the common denominators grow like C^k (the Mayer–Siegel hypergeometric lemma, [98, Lemma 2.2.1]), and the house of the a_k is bounded (immediate here, since $|a_k| \leq 1$). Clearing denominators leaves projective height unchanged and makes the coordinates algebraic integers of house $\leq C^k B$, so there is no finite contribution and the archimedean one is $\leq k \log C + \log B = O(k)$. Scaling $z \mapsto \lambda z$ replaces (a_k) by $(\lambda^k a_k)$, and $h(a_0, \lambda a_1, \dots, \lambda^k a_k) \leq h(a_0, \dots, a_k) + k\, h(\lambda)$, so the class is preserved; derivatives of E-functions are E-functions. They solve the first-order system (54) with $T(z) = z$, and $\zeta_0 T(\zeta_0) \neq 0$. By the refined Siegel–Shidlovskii theorem [70, Thms. 1.2 and 1.3] there exists $Q \in \overline{\mathbb{Q}}[z, u, v]$, homogeneous in (u, v), with $Q(z, \cdot) \equiv 0$ on the functions and $Q(\zeta_0, \cdot) = P$ as polynomials. Then $Q \in I = (G_1, G_2)$ over $\overline{\mathbb{Q}}(z)$ (Lemma 8): $d(z)Q = pG_1 + qG_2$ with $p, q \in \overline{\mathbb{Q}}[z, u, v]$, $d \in \overline{\mathbb{Q}}[z]$. If $d(\zeta_0) = 0$, evaluate at

[6] The identities of Lemma 5, the connection formulae and levels of Lemma 4, the vanishing of G_1, G_2, and the contiguity relations of Lemma 22 also hold in exact symbolic computation. An exhaustive mod-p search through monomial degree ≤ 3, at $\lambda = 2, \sqrt{2}$ and a primitive cube root of unity, gives exactly the nullity of (G_1, G_2)-multiples; the excluded case $\lambda = 1$ collapses. These computational checks are independent of the proofs.

ζ_0: $0 = p(\zeta_0)g_1 + q(\zeta_0)g_2$. Now (g_1, g_2) is a regular sequence in $\overline{\mathbb{Q}}[u, v]$ (g_1 is irreducible by the content argument of Lemma 8 with $z = \zeta_0 \neq 0$, so $\overline{\mathbb{Q}}[u]/(g_1)$ — the quotient of the polynomial ring in $u_1, \ldots, u_4$ — is a domain, $(\overline{\mathbb{Q}}[u]/(g_1))[v]$ is a domain, and $g_2 \neq 0$ there), so the syzygy is Koszul: $p(\zeta_0) = hg_2$, $q(\zeta_0) = -hg_1$ with $h \in \overline{\mathbb{Q}}[u, v]$. Replacing (p, q) by $(p - hG_2,\ q + hG_1)$ leaves $pG_1 + qG_2$ unchanged while the new p, q vanish at ζ_0, so $(z - \zeta_0)$ cancels against d. Induction on $\mathrm{ord}_{\zeta_0} d$ therefore permits $d(\zeta_0) \neq 0$, and $P = Q(\zeta_0, \cdot) \in (g_1, g_2)$. □

Lemma 10. *For $c_j, c_k \in \overline{\mathbb{Q}}^\times$ and either sign, $r_\pm = c_j u_1 v_3 \mp c_k v_1 u_3 \notin (g_1, g_2)$.*

Proof. Apply the ring homomorphism φ sending the derivative variables u_2, u_4, v_2, v_4 to 0. Then $\varphi(g_1) = -\frac{2}{3}(u_1 u_3 - 1)$, $\varphi(g_2) = -\frac{2}{3}(v_1 v_3 - 1)$, and $\varphi(r_\pm) = r_\pm$. Membership would give $r_\pm \in (u_1 u_3 - 1,\ v_1 v_3 - 1)$, which is exactly the kernel of the localization $\overline{\mathbb{Q}}[u_1, u_3, v_1, v_3] \to \overline{\mathbb{Q}}[u_1^{\pm 1}, v_1^{\pm 1}]$, $u_3 \mapsto u_1^{-1}$, $v_3 \mapsto v_1^{-1}$; but $r_\pm$ maps to $(c_j u_1^2 \mp c_k v_1^2)/(u_1 v_1) \neq 0$. □

N.5 Proof of the theorem

Proof of Theorem 2 and Corollary 3. Suppose $-a_j \neq -a_k$ are negative real zeros, each of Ai or of Bi, with a_j, a_k algebraic. Then $\zeta_i = \frac{2}{3} a_i^{3/2}$ are algebraic, positive and distinct (Lemma 4: the map $a \mapsto \zeta$ is strictly increasing); $\lambda := \zeta_k/\zeta_j$ is algebraic, positive, $\neq 1$, hence $\lambda^2 \neq 1$ (and the configuration is symmetric under $j \leftrightarrow k$, which replaces λ by λ^{-1}). By Lemma 4,

$$(\zeta_i/2)^{2/3} f_+(\zeta_i) = \varepsilon_i\, \gamma_*\, f_-(\zeta_i), \qquad \varepsilon_i \in \{\pm 1\} \quad (\varepsilon = -1 \text{ for Ai}, +1 \text{ for Bi}),$$

with $f_-(\zeta_i) \neq 0$ by Lemma 6. Dividing the two relations eliminates γ_*:

$$c_j\, f_+(\zeta_j) f_-(\zeta_k) \;-\; \varepsilon_j \varepsilon_k\, c_k\, f_+(\zeta_k) f_-(\zeta_j) \;=\; 0, \qquad c_i = (\zeta_i/2)^{2/3} \in \overline{\mathbb{Q}}^\times,$$

a nontrivial homogeneous degree-2 relation over $\overline{\mathbb{Q}}$ among the eight values of (53) at $\zeta_0 = \zeta_j$ — it is the relation of Lemma 10 carrying the sign $-\varepsilon_j \varepsilon_k$ on its second term, and that lemma excludes both signs, so which one occurs (same-type or mixed Ai/Bi pairs) does not matter. By Lemma 9 the relation lies in (g_1, g_2); by Lemma 10 it does not. This contradiction proves that no two such zeros are algebraic. Since all zeros of Ai are negative real, Theorem 2 follows. For Corollary 3 note also that Ai and Bi have no common zero, by $W\{\mathrm{Ai}, \mathrm{Bi}\} = 1/\pi$ [97, 9.2.7], so distinct zeros of the union are distinct numbers and every pair is covered. □

N.6 The remaining exception, and extensions

The following one-block statement isolates what the method does and does not say about a *single* zero.

Lemma 11. *For every algebraic $\zeta > 0$, $f_-(\zeta) \neq 0$ and $h(\zeta) = (\zeta/2)^{2/3} f_+(\zeta)/f_-(\zeta)$ is transcendental. The same holds on the second level: $\hat{f}_-(\zeta) \neq 0$ and $\hat{h}(\zeta) = (\zeta/2)^{4/3} \hat{f}_+(\zeta)/\hat{f}_-(\zeta)$ is transcendental, where $\hat{f}_\pm$ are as in Lemma 22.*

Proof. Apply the argument of Theorem 7 to the single module N_1: $\mathrm{Gal}(K_1/\mathbb{C}(z)) = \mathrm{SL}_2(\mathbb{C})$ gives $\mathrm{tr.deg}_{\mathbb{C}(z)}(f_+, f_+', f_-, f_-') = 3$, and the one-block case of Lemma 8 identifies the relation ideal as (G_1) (the argument is the same, with one variable block and one Wronskian polynomial). Lemma 9 then applies verbatim in four variables: every homogeneous $\overline{\mathbb{Q}}$-relation among the four values at ζ lies in (g_1), $g_1 = G_1|_{z=\zeta}$. Now $f_-(\zeta) = 0$ is a homogeneous relation of degree 1, and so is $(\zeta/2)^{2/3} f_+(\zeta) - \beta f_-(\zeta) = 0$ for $\beta \in \overline{\mathbb{Q}}$, which is what $h(\zeta) = \beta$ asserts (the coefficient $(\zeta/2)^{2/3}$ is algebraic). But degrees add in $\overline{\mathbb{Q}}[u]$ and $\deg g_1 = 2$, so every nonzero element of (g_1) has degree ≥ 2. Neither relation can hold. For the second level, Lemma 22 writes $\hat{f}_+ = -\frac{4}{3z} f_-'$ and $\hat{f}_- = f_+ + \frac{3z}{2} f_+'$, so $\hat{f}_-(\zeta) = 0$ and $(\zeta/2)^{4/3} \hat{f}_+(\zeta) - \beta \hat{f}_-(\zeta) = 0$ are again homogeneous relations of degree 1 in u_1, u_2, u_4, excluded by the same degree count. □

Remark 12. The obstruction to removing the final exception is precise. A single algebraic zero ζ asserts, by (52), that $h(\zeta) = (\zeta/2)^{2/3} f_+(\zeta)/f_-(\zeta)$ lies on the line $\overline{\mathbb{Q}} \cdot \Gamma(\frac{1}{3})^2/\pi$ (the line, rather than the single value $-\gamma_*$, because the Bi level is $+\gamma_*$ and the machinery is insensitive to the algebraic factor). By Lemma 11 the value $h(\zeta)$ is transcendental for every algebraic $\zeta > 0$ — so the method does constrain the single-zero case, just not enough. For *which* transcendental number $h(\zeta)$ is, the E-function machinery cannot say; whether it can be an algebraic multiple of $\Gamma(\frac{1}{3})^2/\pi$ requires an arithmetic independence statement not established here. For Ai alone the exact target is the single level $-\gamma_*$; excluding the whole algebraic-multiple line is sufficient but stronger than necessary.

Sufficient arithmetic hypothesis. Suppose that for every positive algebraic ζ,

$$\mathrm{tr.deg}_{\overline{\mathbb{Q}}}\, \overline{\mathbb{Q}}\big(f_+(\zeta), f_+'(\zeta), f_-(\zeta), f_-'(\zeta), \Gamma(\tfrac{1}{3}), \pi\big) = 5.$$

Then every zero of Ai is transcendental. Indeed, an algebraic zero gives, by (52), $\Gamma(\frac{1}{3})^2/\pi \in \overline{\mathbb{Q}}(f_+(\zeta), f_-(\zeta))$. The four E-values have transcendence degree 3 by the one-block argument; adjoining π adds at most one, and the displayed relation then makes $\Gamma(\frac{1}{3})$ algebraic over that field. The resulting joint transcendence degree is at most 4, a contradiction. □

The joint-degree hypothesis is not proved here. A route through the period conjecture for exponential motives [99] would also require an explicit realization of these normalized values as periods and the joint motivic group at each specified algebraic point. The differential Galois group of the family alone does not supply those fixed-point arithmetic statements.

The obstruction's geometry. The zero condition is a linear equation in $(\pi, \Gamma(\frac{1}{3})^2)$ with E-function-value coefficients at the algebraic point; two such equations share the target and dividing eliminates it — the at-most-one mechanism. The stronger sufficient exclusion asks whether the target can lie on any of the lines generated by the Bessel ratios above.

Unconditionally the question remains open. With Proposition 25 the boundary of the method is a pair of lines rather than one: a single algebraic element of $\mathcal{Z}$ asserts either $h(\zeta) \in \overline{\mathbb{Q}} \cdot \Gamma(\frac{1}{3})^2/\pi$ or $\hat{h}(\zeta) \in \overline{\mathbb{Q}} \cdot \pi/\Gamma(\frac{1}{3})^2$, and Lemma 11 shows both values transcendental without deciding either membership. The two lines are genuinely distinct — their ratio is γ_*/Λ, transcendental by Lemma 23 — so excluding one would not exclude the other.

Algebraic endpoints

Two finite boundary conditions permit a stronger conclusion than decay at infinity. Define the canonical Airy basis

$$Y_0(t) = {}_0F_1(;\tfrac{2}{3};t^3/9), \qquad Y_1(t) = t\,{}_0F_1(;\tfrac{4}{3};t^3/9).$$

It satisfies $y'' = ty$ with initial vectors $(1,0)$ and $(0,1)$ at zero, so its fundamental matrix has determinant 1. The basis identities and the algebraic exclusion witnesses below have exact checks (supplementary code).

Theorem 13. *Every nonzero solution of $y'' = ty$, with arbitrary complex initial coefficients, has at most one distinct algebraic zero in $\mathbb{C}$.*

Proof. Two zeros at distinct algebraic t_0, t_1 would force $\Delta = Y_0(t_0)Y_1(t_1) - Y_1(t_0)Y_0(t_1) = 0$. If both points are nonzero, choose algebraic ζ_j with $\zeta_j^2 = -4t_j^3/9$. Then $Y_0(t_j) = f_-(\zeta_j)$ and $Y_1(t_j) = t_j f_+(\zeta_j)$. When $t_0^3 \neq t_1^3$, put $\lambda = \zeta_1/\zeta_0$; its square differs from 1. Lemma 9 would put $P = t_1u_3v_1 - t_0u_1v_3$ in (g_1, g_2). But setting $u_1 = u_3 = v_1 = v_3 = 1$ and all derivative variables to zero annuls both g_i while giving $P = t_1 - t_0 \neq 0$.

If $t_0^3 = t_1^3$, evenness of f_+, f_- gives $\Delta = (t_1 - t_0)f_+(\zeta_0)f_-(\zeta_0) \neq 0$. Here both normalized Bessel functions are nonzero at every nonzero algebraic argument: the one-block transfer and degree-one exclusion in Lemma 11 apply unchanged at complex algebraic points. Finally, if one point is zero, Δ is, up to sign, $Y_1(t) = tf_+(\zeta) \neq 0$ at the other point. All cases contradict the assumed two zeros. □

The same bound has an exact address-centered form. For each $a \in \overline{\mathbb{Q}}$, put

$$S_a(x) = Y_0(-a)Y_1(x-a) - Y_1(-a)Y_0(x-a).$$

The unit Wronskian gives $S_a(0) = 0$ and $S_a'(0) = 1$, while $S_a'' = (x-a)S_a$. Algebraic translation in Theorem 13 makes 0 its only algebraic zero. The fixed address therefore supplies the unique allowed algebraic node in this family. Decay remains an additional condition: fixing the address alone does not select a or make S_a a decaying mode.

Theorem 14. *Let A, B, L be positive real algebraic numbers. Every eigenvalue of*

$$-Au''(r) + Br\,u(r) = Eu(r), \qquad 0 \le r \le L,$$

with separated boundary conditions $p_0u(0) + q_0u'(0) = p_1u(L) + q_1u'(L) = 0$ is transcendental, provided each fixed pair (p_j, q_j) is nonzero and real algebraic. This includes Dirichlet, Neumann and mixed boundary conditions.

Proof. Green's identity and the real separated conditions make every eigenvalue real. If E were algebraic, set $\beta = (B/A)^{1/3}$ and $t = \beta(r - E/B)$. The equation becomes $y'' = ty$ at two distinct real algebraic endpoints, and the boundary covectors become $(p_j, \beta q_j)$. Dirichlet data are excluded immediately by Theorem 13.

For general data, let M_j be the fundamental matrix of Y_0, Y_1 at endpoint t_j. When both endpoints are nonzero their cubes differ. Each M_j is an invertible algebraic linear transformation of its four normalized Bessel values, with determinant 1; in the second block the derivative variables of (53) are divided by λ. The specialized relation ideal is therefore exactly the two separate determinant-one relations. A nonzero covector multiplied by an $\mathrm{SL}_2(\mathbb{C})$ matrix can be any nonzero row. Thus the two boundary rows can independently be $(1, 0)$ and $(0, 1)$ on this variety. Their determinant is not identically zero, whereas a common nonzero solution would force precisely that homogeneous quadratic relation. Lemma 9 excludes it.

If one endpoint is zero, its fundamental matrix is the identity. The boundary determinant is then a nonzero algebraic linear form in the other block, excluded by one-block transfer. This covers every case. □

Corollary 15. *Let A, B, L be positive real algebraic numbers. The finite loop*

$$-A\psi''(s) + B\min(s, 2L - s)\psi(s) = E\psi(s), \qquad 0 \le s \le 2L,$$

with continuous value and first derivative at $s = L$ and periodic value and first derivative at $0, 2L$ has only transcendental eigenvalues. The signed return $\psi(2L - s) = -\psi(s)$ selects exactly the Dirichlet spectrum on the outgoing interval $[0, L]$.

Proof. Reflection $s \mapsto 2L - s$ commutes with the operator and preserves its boundary conditions. Every eigenspace therefore has a nonzero even or odd reflection component. Its restriction to $[0, L]$ satisfies Neumann conditions at both ends in the even case and Dirichlet conditions at both ends in the odd case. Theorem 14 excludes algebraic eigenvalues in both sectors. Conversely, even or odd reflection extends every such interval eigenfunction to the loop.

For the signed return explicitly, write $\psi(s) = u(s)$ on the outgoing leg and $\psi(s) = -u(2L - s)$ on the return. Continuity at the turn gives $u(L) = 0$, and periodic value matching at the address gives $u(0) = 0$. The return derivative is $+u'(2L - s)$, so both derivative matches hold automatically. This is odd reflection between the legs, with periodic matching over the complete loop. □

These finite boundary problems differ from the source problem of Eq. 51, which has one origin node and a decaying tail. The figures use $\mathrm{Ai}(r/\ell - a_n)$, $\ell = R/a_{17}$; R is the seventeenth soft-turning reach and a display cutoff. At that turn $\mathrm{Ai}(0) \neq 0$. The signed (odd) finite loop above instead requires a node at its turn. Neither the drawn tips nor the separation of return legs imposes that second boundary on the source.

Local elastic return. A sign reversal at the address itself has an exact realization within the original half-line problem. For $E > 0$, set $k = \sqrt{E/A}$ and choose exact initial-value solutions with endpoint values 1 and slopes $+ik$ and $-ik$. Their amplitudes satisfy

$$u(0) = a_{\mathrm{out}} + a_{\mathrm{in}}, \qquad u'(0) = ik(a_{\mathrm{out}} - a_{\mathrm{in}}).$$

The conserved scalar current is $\mathrm{Im}(\bar{u}u') = k(|a_{\mathrm{out}}|^2 - |a_{\mathrm{in}}|^2)$. Thus the local rule $a_{\mathrm{out}} = -a_{\mathrm{in}}$ is perfectly elastic and equivalent to the origin node. Elasticity alone fixes the reflection modulus, while this sign fixes its phase.

Put $\beta = (B/A)^{1/3}$, $\epsilon = \beta E/B$, $f = \mathrm{Ai}(-\epsilon)$ and $g = \beta\,\mathrm{Ai}'(-\epsilon)$. Decay into the increasing-cost region selects the exact return factor

$$R_{\mathrm{soft}}(E) = \frac{f + ig/k}{f - ig/k}, \qquad |R_{\mathrm{soft}}| = 1, \qquad (-1)R_{\mathrm{soft}} = 1 \iff f = 0.$$

Here f, g are real and cannot vanish together, so the denominator is nonzero. These are exact endpoint identities, with algebraic checks (supplementary code). The local signed reflection therefore preserves the existing Ai spectrum without imposing a node at a finite soft turnaround. Equivalently, when $f = 0$, the odd extension $\Psi(x) = \mathrm{sgn}(x)\,\mathrm{Ai}(|x| - \epsilon)$, with $\Psi(0) = 0$, has continuous value and derivative through the address and solves $(-d^2/dx^2 + |x|)\Psi = \epsilon\Psi$ on the full line. This unfolds through the address rather than joining two legs at a finite outer point. The scalar sign is independent of the orientation basis: $U(-I)U^{-1} = -I$. These facts realize the local boundary rule; their round-trip equation repeats the original Ai zero condition and supplies no additional arithmetic exclusion.

Proposition 16. *Let $\beta > 0$ and $u_n(r) = \mathrm{Ai}(\beta r - a_n)$ for $r \geq 0$, where $0 < a_1 < a_2 < \cdots$ are the magnitudes of the* Ai *zeros. For every $N \geq 1$, the only common zero of $u_1, \ldots, u_N$ is $r = 0$. Among the first seventeen modes, any two distinct modes already have $r = 0$ as their only common zero.*

Proof. Every mode vanishes at the origin. The ground profile $u_1(r)$ is strictly positive for $r > 0$, since its argument lies to the right of the largest real Ai zero. This proves the first assertion. For the second, the positive nodes of mode n are $(a_n - a_k)/\beta$, $1 \leq k < n$. Certified rational enclosures of the first seventeen zeros give pairwise disjoint intervals for all 136 positive differences $a_n - a_k$. The minimum certified clearance exceeds 0.0040132 before division by β. The certificate reruns the root enclosures with explicit series-tail and root-index bounds, then compares these differences in exact rational arithmetic (supplementary code). Thus no positive node is shared by two of these modes. □

For the same supplied operator, put $\beta = (B/A)^{1/3}$. The modes unit-normalized in dr are

$$\psi_n(r) = \sqrt{\beta}\,\frac{\mathrm{Ai}(\beta r - a_n)}{\mathrm{Ai}'(-a_n)}, \qquad |\psi_n'(0)|^2 = \beta^3 = \frac{B}{A}.$$

Indeed, integrating $d[\mathrm{Ai}'(z)^2 - z\,\mathrm{Ai}(z)^2]/dz = -\mathrm{Ai}(z)^2$ gives $\int_0^\infty \mathrm{Ai}(\beta r - a_n)^2\,dr = \mathrm{Ai}'(-a_n)^2/\beta$. Thus whole-wave normalization preserves the common individual-mode boundary weight at the one address. The energies $E_n = Ba_n/\beta$ and profiles remain distinct; this identity adds no spectral selection condition (supplementary code).

The source family therefore has one shared radial node: the preserved address. All its profiles use the same origin in Eq. 51; an excited profile's additional nodes are not additional addresses. The arithmetic theorem permits at most one algebraic level a_n of that fixed operator. If such a level exists, Theorem 13, translated by algebraic a_n, makes $x = 0$ its mode's only algebraic node. This is consistent with the common address; it neither proves nor excludes such a level.

Remark 17. *The function-field counterpart.* Over $\mathbb{F}_q(\theta)$, Papanikolas's theorem [108] relates period transcendence degree to the Galois group of a rigid analytically trivial Frobenius-difference module. The lifting criterion of Anderson–Brownawell–Papanikolas [109] supplies the arithmetic transfer when its hypotheses hold. A three-strand construction in this setting has a computable character lattice (supplementary code). For $q = 4$, interpolating the fractional order $\frac{1}{3}$ through $\nu = \theta^{1/3}$, with $\omega \in \mathbb{F}_4^\times$ of order 3, yields a rank-3 Frobenius-difference equation on the cube cover $w^3 = t$ whose three solution ladders — the strands — are Kummer conjugates of a single rank-1 object. The Moore determinants of the three ladder coefficient vectors satisfy a first-order twisted recurrence with rational multiplier, a coefficient-level counterpart of Lemma 5. Each strand privately owns the Frobenius orbits of its cubic $w^3 + \omega^k w + \theta$ while all three share the cover's orbit, so the character lattice is trivial, the motive's Galois group is the full torus, and the Carlitz period $\tilde{\pi}$ and the three strand periods are algebraically independent. Each strand satisfies a constant-coefficient equation of order six in the ν-difference operator $\delta g(z) = g(\nu z) - \nu g(z)$, and the flow matrix's characteristic polynomial *is* the strand's Frobenius denominator: six eigenvalues in two zero-sum triads — the three cube roots of θ, shared, and the three roots of the strand's own cubic, private — the deck's 120° turn permuting the shared triad within each flow and handing the private triads from strand to strand. Six eigen-directions for a cell with six primitive address ports; the count matches the effective packing's isostatic $z = 2D$ (Fig. 20), but ports and contacts are not identified. This count does not identify the flow's coordinate-zero set with the period lattice of an Anderson t-module. The t-side period results therefore do not by themselves prove transcendence of zeros in the argument z. For the explicit $q = 4$ strand, direct coefficient and norm arguments give the following more limited results.

A proved exclusion in characteristic two

Put $\nu^3 = \theta$, $x = \nu^{-1}$, and $F = \mathbb{F}_4(\nu)$. Work in the completed algebraic closure C_∞ of $\mathbb{F}_4((x))$, with $v(x) = 1$. Define

$$D_0 = 1, \qquad D_i = D_{i-1}^4(\theta^{4^i} + \theta)(\theta^{4^i} + \theta + \nu^{4^i}), \qquad f(z) = \sum_{i \geq 0} \frac{z^{4^i}}{D_i}.$$

These monic polynomials satisfy $D_i \mid D_m$ for $i \leq m$ and $\deg_\nu D_i = 6i4^i$. Thus f is entire and $\mathbb{F}_4$-linear, with $f' = 1$. In particular 0 is an algebraic zero. Exact finite coefficient checks corroborate the uniform arguments below (supplementary code).

Lemma 18. *For $i \geq 1$, the first odd nonzero coefficient of $u_i = x^{-6i4^i}/D_i$ is the coefficient 1 at degree $5 \cdot 4^i - 3$. For $t \geq 1$, let w_t be the zero of $f_t(z) = \sum_{i=0}^t z^{4^i}/D_i$ with $x^8 w_t \in 1 + x\mathbb{F}_4[[x]]$. It exists uniquely, converges to a zero w of f, and*

$$v(w_{t+1} - w_t) = v(w - w_t) = (6t - 2)4^{t+1}.$$

Proof. Write $Q = 4^i$. The normalized newest factor is

$$h_i = \frac{1}{(1 + x^{3Q-3})(1 + x^{3Q-3} + x^{2Q})} = \frac{1}{1 + x^{2Q} + x^{6Q-6} + x^{5Q-3}}.$$

The exponents $2Q$ and $6Q - 6$ are even, and the least odd exponent is $5Q - 3$. Expanding the reciprocal shows that its first odd coefficient is 1 at that degree. Since $u_i = u_{i-1}^4 h_i$, the preceding factor has constant term 1 and support in $4\mathbb{Z}$. Only its constant term can contribute to that first odd coefficient.

Substitute $z = x^{-8}U$ in f_t and multiply by x^8. Modulo x the equation is $U + U^4 = 0$, whose derivative is 1. Hensel lifting selects the unique solution with $U \equiv 1 \pmod{x}$. Additivity gives $f_{t+1}(w_{t+1} - w_t) = w_t^{4^{t+1}}/D_{t+1}$. The right side has valuation $(6t - 2)4^{t+1} > 0$. The shared leading term gives $v(\delta) > -8$. On this ball all nonlinear terms in $f_{t+1}(\delta)$ have valuation strictly greater than $v(\delta)$, so the correction has exactly that valuation. Successive correction valuations strictly increase. Their convergent sum gives w, and later corrections cannot cancel the leading difference from w_t. □

Proposition 19. *For every $i \geq 1$, f has exactly $3 \cdot 4^{i-1}$ nonzero zeros of valuation $-(6i + 2)$, and no other nonzero zeros. All are simple, and $Z(f) \cap \theta Z(f) = \{0\}$.*

Proof. The Newton polygon has vertices $(4^i, 6i4^i)$ and successive slopes $6i + 2$, strictly increasing to infinity. The analytic Newton polygon theorem gives the stated valuations and multiplicities; $f' = 1$ gives simplicity. Multiplication by $\theta = x^{-3}$ shifts a root valuation by -3, which misses every permitted shell. □

Theorem 20. *For $\mathcal{L} = Z(f) \subset C_\infty$,*

$$\{c \in C_\infty : c\mathcal{L} \subseteq \mathcal{L}\} = \mathbb{F}_4.$$

Proof. First suppose $f(cz) = P(f(z))$ for a finite $\mathbb{F}_4$-linear polynomial $P(Y) = \sum_{j=0}^{r} p_j Y^{4^j}$, with $p_r \neq 0$. The case $c = 0$ is immediate. Comparing coefficients for $i \geq r$ gives

$$c^{4^i} = R(\nu^{4^i}), \qquad R(X) = \sum_{j=0}^{r} p_j R_j(X),$$

$$R_j(X) = \prod_{h=0}^{j-1} (X^6 + X^4 + \nu^{3\cdot 4^h} X + \nu^{6\cdot 4^h}),$$

where $R_0 = 1$. This follows from $D_i / D_{i-j}^{4^j} = R_j(\nu^{4^i})$. Comparing the fourth power with the next identity shows $R(X^4) = R(X)^4$ at infinitely many distinct points, hence identically. Every coefficient of R lies in $\mathbb{F}_4$; injectivity of Frobenius then gives $c = R(\nu)$.

If $r \geq 1$, $\deg R = 6r$ and $p_r \in \mathbb{F}_4^\times$. The coefficient of X^{6r-5} is $p_r \sum_{h=0}^{r-1} \nu^{3\cdot 4^h}$: a loss of five degrees in the top product requires exactly one degree-one term. Degree-four terms lose two each, and constant terms lose six. Lower R_j have degree at most $6r - 6$ and cannot cancel this nonconstant coefficient. This contradicts $R \in \mathbb{F}_4[X]$. Thus $r = 0$, $c \in \mathbb{F}_4$ and $P(Y) = cY$. Exact coefficient checks corroborate the all-degree obstruction (supplementary code).

Now let $c \neq 0$ preserve $\mathcal{L}$. The smallest-root shell forces $v(c) = -6k$ for an integer $k \geq 0$. At $k = 0$, c preserves the one-dimensional smallest-root space, so $c \in \mathbb{F}_4$. For $k \geq 1$, put $\mathcal{M} = c^{-1}\mathcal{L} \supset \mathcal{L}$. At the cutoff $v \geq -(6s + 2)$, these spaces have dimensions $s + k$ and s over $\mathbb{F}_4$, respectively. Their quotient images are nested, have dimension k, and exhaust $\mathcal{M}/\mathcal{L}$. Consequently $V = f(\mathcal{M})$ is a finite vector space of size 4^k. Its root polynomial $Q(Y) = \prod_{v \in V}(Y - v)$ is $\mathbb{F}_4$-linear and separable. For example, adjoining a one-dimensional complement replaces $Q_0(Y)$ by $Q_0(Y)^4 - Q_0(w)^3 Q_0(Y)$.

The entire functions $f(cz)$ and $Q(f(z))$ have the same simple zeros, exactly $\mathcal{M}$. On each closed disk their zero factors cancel, so their quotient is entire and zero-free. The analytic Newton polygon theorem makes such a function constant (the same quotient argument is used in [115, §2.2]). Hence $f(cz) = P(f(z))$ for a constant multiple P of Q, contradicting the first part. The $\mathbb{F}_4$ scalars preserve $\mathcal{L}$ by linearity. □

This theorem concerns preservation of the whole zero set. It does not exclude every individual ratio of two roots, or an isolated algebraic root.

Theorem 21. *Let $\alpha \neq 0$ be algebraic over F, let $K = F(\alpha)$ have degree d, and let n be the local degree of the chosen infinity place of K at which $f(\alpha)$ is evaluated. If $f(\alpha) = 0$, then $d \geq 4n$. Consequently f has no nonzero algebraic zero of degree at most three over F, and none whose generated field has only one infinity place.*

Proof. Normalize all valuations above infinity by $v_w(\nu) = -1$, and put $n_w = e_w f_w$, so $\sum_{w|\infty} n_w = d$; these are local degrees including inseparability. Choose a nonzero polynomial $b \in \mathbb{F}_4[\nu]$ with $b\alpha$ integral. Put

$$S_m = \sum_{i=0}^{m} \frac{\alpha^{4^i}}{D_i}, \qquad T_m = D_m b^{4^m} S_m.$$

Each summand of T_m is integral, since $D_i \mid D_m$ and $b^{4^m}\alpha^{4^i} = b^{4^m - 4^i}(b\alpha)^{4^i}$. For every infinity place, the term valuations $(6i + v_w(\alpha))4^i$ tend to infinity, so $v_w(S_m)$ is bounded below independently of m.

At the selected place w_0, the identity $f(\alpha) = 0$ writes S_m as the negative tail. For sufficiently large m that tail has a unique term of least valuation. Hence $S_m \neq 0$ and

$$v_{w_0}(S_m) = (6(m+1) + v_{w_0}(\alpha))4^{m+1}.$$

The norm of the nonzero integral element T_m is a nonzero polynomial in ν. Its infinity valuation is therefore nonpositive, giving

$$\sum_{w|\infty} n_w v_w(S_m) \leq d(6m + \deg b)4^m.$$

The other places contribute terms bounded below. Divide by $6m4^m$ and let $m \to \infty$ to obtain $4n \leq d$. If there is only one infinity place, $n = d$, a contradiction. □

More generally, if $\mathcal{R}$ is the set of infinity places of K at which $f(\alpha) = 0$, the same proof sums the exact tail valuations at all these places and gives $4\sum_{w\in\mathcal{R}} n_w \leq d$. This exclusion does not settle candidates with $d/n \geq 4$. Nor does Lemma 18 prove an infinite digit kernel: nonzero corrections at every truncation depth can also converge to a rational series. Christol's criterion [110] would exclude algebraicity of an $\mathbb{F}_4$-coefficient root if infinitely many distinct decimations of its full limiting digit sequence were proved. That separation, its extension to every root, and the unrestricted nonzero-zero transcendence claim remain open. These characteristic-two results supply no transfer theorem for classical Ai.

The derivative families

The derivative families require no new functions — only a new exclusion certificate. They sit on a *second* level, and the product of the two levels is rational, which is what lets the same argument reach them.

Lemma 22. *Put* $\hat{f}_+(z) = {}_0F_1(;\frac{5}{3};-\frac{z^2}{4})$, $\hat{f}_-(z) = {}_0F_1(;\frac{1}{3};-\frac{z^2}{4})$ *and* $\Lambda := \Gamma(\frac{5}{3})/\Gamma(\frac{1}{3})$. *For* $a > 0$ *and* $\zeta = \frac{2}{3}a^{3/2}$,

$$\mathrm{Ai}'(-a) = 0 \iff (\zeta/2)^{4/3}\hat{f}_+(\zeta) = \Lambda\,\hat{f}_-(\zeta),$$

$$\mathrm{Bi}'(-a) = 0 \iff (\zeta/2)^{4/3}\hat{f}_+(\zeta) = -\Lambda\,\hat{f}_-(\zeta).$$

Moreover

$$\hat{f}_+ = -\frac{4}{3z}\,f'_-, \qquad \hat{f}_- = f_+ + \frac{3z}{2}\,f'_+,$$

so $\hat{f}_\pm$ *lie in the* $\overline{\mathbb{Q}}(z)$*-span of a single variable block of* (53)*: the derivative families introduce no new functions.*

Proof. Insert $J_{\pm 2/3}(z) = (z/2)^{\pm 2/3}\hat{f}_\pm(z)/\Gamma(1 \pm \frac{2}{3})$ into $\mathrm{Ai}'(-x) = \frac{x}{3}\big(J_{2/3} - J_{-2/3}\big)(\zeta)$ and $\mathrm{Bi}'(-x) = \frac{x}{\sqrt{3}}\big(J_{2/3} + J_{-2/3}\big)(\zeta)$ [97, 9.6.7, 9.6.9] (again the two prefactors differ, by the same factor $\sqrt{3}$ as for Ai, Bi) and multiply by $\Gamma(\frac{5}{3})(\zeta/2)^{2/3}$; the outer prefactors multiply a vanishing bracket and drop out.

For the contiguities write $F_b(z) = {}_0F_1(;b;-\frac{z^2}{4}) = \sum_{k\ge 0}(-1)^k z^{2k}/(4^k(b)_k k!)$, so that $f_+ = F_{4/3}$, $f_- = F_{2/3}$, $\hat{f}_+ = F_{5/3}$, $\hat{f}_- = F_{1/3}$. The first identity is the derivative rule $F'_b = -\frac{z}{2b}F_{b+1}$ at $b = \frac{2}{3}$. For the second,

$$f_+ + \tfrac{3z}{2}f'_+ = \sum_{k\ge 0}\frac{(-1)^k(1+3k)\,z^{2k}}{4^k\,(\frac{4}{3})_k\,k!},$$

and $(\frac{4}{3})_k = (3k+1)(\frac{1}{3})_k$ — both are products of k consecutive terms of $\frac{1}{3}, \frac{4}{3}, \frac{7}{3}, \ldots$ offset by one step, so their ratio is $\frac{(3k+1)/3}{1/3} = 3k+1$ — which turns the summand into $(-1)^k z^{2k}/(4^k(\frac{1}{3})_k k!)$, the series of $\hat{f}_-$. □

Lemma 23. $\gamma_*\Lambda = \frac{2}{9}$, *while* $\gamma_*/\Lambda = 3\Gamma(\frac{1}{3})^4/(8\pi^2)$ *is transcendental.*

Proof. Both statements are the Γ recurrence. For the product, more generally, writing $\gamma_*(\nu) = \Gamma(1+\nu)/\Gamma(1-\nu)$ and $\Lambda(\nu) = \Gamma(2-\nu)/\Gamma(\nu)$ — which at $\nu = \frac{1}{3}$ are the two levels above — one has $\Gamma(1+\nu) = \nu\Gamma(\nu)$ and $\Gamma(2-\nu) = (1-\nu)\Gamma(1-\nu)$, so the product telescopes to $\gamma_*(\nu)\Lambda(\nu) = \nu(1-\nu)$, which is $\frac{2}{9}$ at $\nu = \frac{1}{3}$. No transcendence input is used and $\Gamma(\frac{1}{3})^2$ is never recognized; that the level product is rational is forced by the recurrence alone, not by anything special about the value $\frac{2}{9}$. (The identity is used only at $\nu = \frac{1}{3}$. No general-ν derivative family is defined here, so $\Lambda(\nu)$ is not asserted to be its level at other orders.) For the quotient the same recurrence gives $\gamma_*/\Lambda = \frac{1}{2}\big(\Gamma(\frac{1}{3})/\Gamma(\frac{2}{3})\big)^2$, and $\Gamma(\frac{1}{3})\Gamma(\frac{2}{3}) = 2\pi/\sqrt{3}$ turns this into $3\Gamma(\frac{1}{3})^4/(8\pi^2)$, transcendental because $\Gamma(\frac{1}{3})$ and π are algebraically independent ([69], at $q = e^{-2\pi\sqrt{3}}$). □

So the four classical families populate the two levels two and two: Ai and Bi at $\mp\gamma_*$, and Ai′ and Bi′ at $\pm\Lambda$. Two zeros on the *same* level cancel it by cross-multiplication, as in

the proof of Theorem 2; two zeros on *opposite* levels cancel it by multiplication, since $\gamma_*\Lambda$ is rational. (Lemma 23 also shows the two routes are not interchangeable, γ_*/Λ being transcendental while $\gamma_*\Lambda$ is not. This distinction explains the two eliminations; the proof uses their displayed forms directly.)

The slice φ of Lemma 10 annihilates a relation between two Λ-level zeros: each monomial contains a derivative variable through $\hat{f}_\pm$ (Lemma 22). It still excludes mixed-level relations. The following bidegree statement covers all cases.

Lemma 24. *Let $\zeta_0 \in \overline{\mathbb{Q}}^\times$ and let $g_i = G_i|_{z=\zeta_0}$ be as in Lemma 9. Call a monomial of $\overline{\mathbb{Q}}[u,v]$ of* bidegree $(1,1)$ *if it is $u_a v_b$ for some a, b. Then a polynomial all of whose monomials have bidegree $(1,1)$ lies in (g_1, g_2) if and only if it is zero.*

Proof. By (55), $g_1 = \left[\zeta_0(u_1u_4 - u_2u_3) - \frac{2}{3}u_1u_3\right] + \frac{2}{3}$ is a form of bidegree $(2,0)$ plus a constant, and g_2 is a form of bidegree $(0,2)$ plus the same constant. Order the variables $u_1 > u_2 > u_3 > u_4 > v_1 > \cdots > v_4$ lexicographically. The leading monomial of g_1 is u_1u_3, with coefficient $-\frac{2}{3}$, and that of g_2 is v_1v_3: both are *pure-block* of degree 2, and neither depends on ζ_0. They are coprime, so by Buchberger's first criterion the S-polynomial reduces to zero and $\{g_1, g_2\}$ is a Gröbner basis of the ideal it generates — for *every* $\zeta_0 \neq 0$. A monomial $u_a v_b$ has u-degree 1 and v-degree 1, so it is divisible by neither leading monomial; hence a polynomial supported on bidegree-$(1,1)$ monomials is its own normal form, and it lies in the ideal exactly when that normal form vanishes. □

This includes Lemma 10: $r_\pm = c_j u_1 v_3 \mp c_k v_1 u_3$ is nonzero and has bidegree $(1,1)$. That lemma also has the independent proof above without Gröbner theory.

Proposition 25. *Let $\mathcal{Z}$ be the set of $a > 0$ for which $y(-a) = 0$ or $y'(-a) = 0$ for some $y = \alpha\,\mathrm{Ai} + \beta\,\mathrm{Bi}$ with $\alpha, \beta \in \overline{\mathbb{Q}}$ not both zero. Then at most one element of $\mathcal{Z}$ is algebraic. In particular at most one algebraic number occurs among the negative real zeros of* Ai, Bi, Ai′ *and* Bi′ *taken together.*

Proof. Write $A = \alpha - \sqrt{3}\beta$ and $B = \alpha + \sqrt{3}\beta$, so that by [97, 9.6.6, 9.6.8]

$$y(-x) = \tfrac{\sqrt{x}}{3}\big(AJ_{1/3} + BJ_{-1/3}\big)(\zeta), \qquad y'(-x) = \tfrac{x}{3}\big(BJ_{2/3} - AJ_{-2/3}\big)(\zeta),$$

the second by [97, 9.6.7, 9.6.9] together with $\alpha + \sqrt{3}\beta = B$ and $-\alpha + \sqrt{3}\beta = -A$. So a single pair (A, B) governs both, and $AB = \alpha^2 - 3\beta^2$.

The degenerate rays. If $AB = 0$ then y and y' each degenerate to a single Bessel function — for $A = 0$, $y \propto J_{-1/3}$ and $y' \propto J_{2/3}$; for $B = 0$, $y \propto J_{1/3}$ and $y' \propto J_{-2/3}$ (note the two degenerate to *different* orders) — and all four orders lie outside $\frac{1}{2} + \mathbb{Z}$, so Siegel's theorem [68] gives every nonzero zero transcendental. Such rays therefore contribute no algebraic element to $\mathcal{Z}$ at all, and may be ignored; assume $AB \neq 0$ from here.

Levels. Clearing as in Lemmas 4 and 22, a zero of y at ζ gives

$$(\zeta/2)^{2/3} f_+(\zeta) = \varepsilon\,\gamma_*\, f_-(\zeta), \qquad \varepsilon = -B/A \in \overline{\mathbb{Q}}^\times,$$

and a zero of y' gives $(\zeta/2)^{4/3}\hat{f}_+(\zeta) = \delta\,\Lambda\,\hat{f}_-(\zeta)$ with $\delta = A/B \in \overline{\mathbb{Q}}^\times$. For $y = \mathrm{Ai}, \mathrm{Bi}$ this returns $\varepsilon = \mp 1$ and for $y' = \mathrm{Ai}', \mathrm{Bi}'$ it returns $\delta = \pm 1$; note that ε and δ now range over all of $\overline{\mathbb{Q}}^\times$ rather than over $\{\pm 1\}$, which is exactly what admits the whole pencil.

The pairing. Suppose $a_j \neq a_k$ both lie in $\mathcal{Z}$ and are algebraic. Then $\zeta_i = \frac{2}{3}a_i^{3/2}$ are algebraic, positive and distinct (Lemma 4: $a \mapsto \zeta$ is strictly increasing), so $\lambda := \zeta_k/\zeta_j$ is algebraic with $\lambda^2 \neq 1$, and Lemmas 9 and 24 apply with base point $\zeta_0 = \zeta_j$. Set $c_i = (\zeta_i/2)^{2/3}$, $d_i = (\zeta_i/2)^{4/3}$ and, using Lemma 22 together with $f_+'(\zeta_k) = v_2/\lambda$, $f_-'(\zeta_k) = v_4/\lambda$,

$$\Pi_j = -\tfrac{4d_j}{3\zeta_j}u_4, \quad \Theta_j = u_1 + \tfrac{3}{2}\zeta_j u_2, \qquad \Pi_k = -\tfrac{4d_k\zeta_j}{3\zeta_k^2}v_4, \quad \Theta_k = v_1 + \tfrac{3}{2}\zeta_j v_2,$$

so that Π_i is the value of $d_i\hat{f}_+$ and Θ_i the value of $\hat{f}_-$ at ζ_i, each expressed in the eight variables. Here ζ_j in Θ_k follows because the ζ_k from Lemma 22 cancels against λ. Four cases arise, and in each the level cancels, leaving a relation that holds at the eight values of (53) at $z = \zeta_j$ and is supported on bidegree-$(1,1)$ monomials — in particular homogeneous of degree 2 over $\overline{\mathbb{Q}}$:

$$\begin{aligned}
&\text{both on } \gamma_* : && R_1 = \varepsilon_k c_j\, u_1 v_3 - \varepsilon_j c_k\, v_1 u_3,\\
&\text{both on } \Lambda : && R_2 = \delta_k\,\Pi_j\Theta_k - \delta_j\,\Pi_k\Theta_j,\\
&\gamma_* \text{ at } \zeta_j,\ \Lambda \text{ at } \zeta_k : && R_3 = c_j\,u_1\Pi_k - \tfrac{2}{9}\varepsilon_j\delta_k\,u_3\Theta_k,\\
&\Lambda \text{ at } \zeta_j,\ \gamma_* \text{ at } \zeta_k : && R_4 = c_k\,v_1\Pi_j - \tfrac{2}{9}\varepsilon_k\delta_j\,v_3\Theta_j.
\end{aligned}$$

Each is a cross-multiplication, so no nonvanishing hypothesis is needed anywhere: $\varepsilon_k c_j u_1 v_3 = \varepsilon_j\varepsilon_k\gamma_* u_3 v_3 = \varepsilon_j c_k v_1 u_3$ for R_1; $\delta_k\Pi_j\Theta_k = \delta_j\delta_k\Lambda\Theta_j\Theta_k = \delta_j\Pi_k\Theta_j$ for R_2; and for R_3, $(c_j u_1)\Pi_k = (\varepsilon_j\gamma_* u_3)(\delta_k\Lambda\Theta_k) = \frac{2}{9}\varepsilon_j\delta_k u_3\Theta_k$ by $\gamma_*\Lambda = \frac{2}{9}$ (Lemma 23), and symmetrically for R_4.

Each R_m is nonzero: its four displayed forms have, respectively, the monomials u_1v_3, u_4v_1, u_1v_4 and v_1u_4 with coefficients $\varepsilon_k c_j$, $-\frac{4\delta_k d_j}{3\zeta_j}$, $-\frac{4c_j d_k\zeta_j}{3\zeta_k^2}$ and $-\frac{4c_k d_j}{3\zeta_j}$, all nonzero because $\zeta_i, c_i, d_i \neq 0$ and $\varepsilon_i, \delta_i \in \overline{\mathbb{Q}}^\times$. So no R_m lies in (g_1, g_2), by Lemma 24. But each R_m is homogeneous and vanishes at the eight values, so each *does* lie in (g_1, g_2), by Lemma 9. This contradiction shows that no two distinct elements of $\mathcal{Z}$ are algebraic.

For the final sentence, Ai and Bi are the members $(\alpha, \beta) = (1,0)$ and $(0,1)$, and $\mathrm{Ai}', \mathrm{Bi}'$ their derivatives; note that if two of the four happen to vanish at a common point — which $W\{\mathrm{Ai}, \mathrm{Bi}\} = 1/\pi$ forbids for the pairs Ai, Bi and $\mathrm{Ai}', \mathrm{Bi}'$ but not for $\mathrm{Ai}, \mathrm{Bi}'$ — that point is a single element of $\mathcal{Z}$ and the count is unaffected. □

Remark 26. Specialization can invalidate a Gröbner normal form for (G_1, G_2) over $\overline{\mathbb{Q}}(z)$ if leading coefficients vanish at the algebraic point. In Lemma 24, the leading monomials of g_1, g_2 are pure-block and *independent of* ζ_0; the only place ζ_0 appears is as a coefficient of non-leading terms, and the basis property follows from coprimality of the leading monomials rather than from any computation. An independent check, which avoids Gröbner theory entirely, is available: the line $u = (1, 0, 1 + \frac{3}{2}\zeta_j s, s)$, $v = (1, 0, 1, 0)$ lies in $V(g_1, g_2)$

for every s — both g_i being specialized at the same base point ζ_j — and each R_m restricts along it to a polynomial of degree exactly 1 in s, hence is nonzero somewhere on the variety.

The entire initial-value basis avoids sector-dependent connection factors and extends the joint bound to both real axes and the complex plane.

Corollary 27. *Let $\mathcal{A} \subset \overline{\mathbb{Q}}$ be the set of all algebraic zeros of y or y' as nonzero $y = \alpha\,\mathrm{Ai} + \beta\,\mathrm{Bi}$ ranges over algebraic α, β. For a primitive cube root of unity ω, either*

$$\mathcal{A} = \{0\}, \qquad \text{or} \qquad \mathcal{A} = \{0, r, \omega r, \omega^2 r\} \quad (r \in \overline{\mathbb{Q}} \cap \mathbb{R},\ r \neq 0).$$

Thus at most one nonzero real algebraic point occurs across both signs and all these function and derivative families. If the nonzero orbit exists, it belongs exclusively to the function-zero family or exclusively to the derivative-zero family.

Proof. Put $c = \mathrm{Ai}(0)$ and $d = \mathrm{Ai}'(0)$, both nonzero. The initial values [97, 9.2.3–9.2.6] give $y = cBY_0 + dAY_1$, where $A = \alpha - \sqrt{3}\beta$ and $B = \alpha + \sqrt{3}\beta$. At nonzero algebraic t, the canonical fundamental matrix $M(t) = \left(\begin{smallmatrix} Y_0 & Y_1 \\ Y_0' & Y_1' \end{smallmatrix}\right)$ is an invertible algebraic linear transformation of the one-block Bessel values, with relation ideal $\det M - 1$, as in Theorem 14. One-block homogeneous transfer excludes a zero in any of its four entries. Hence a nonzero algebraic zero of y or y' requires $AB \neq 0$.

Suppose t_j, t_k are two such points with unequal cubes. For $e_i \in \{0, 1\}$ put $X_i = Y_0^{(e_i)}(t_i)$ and $Z_i = Y_1^{(e_i)}(t_i)$. Eliminating c, d from the zero equations gives

$$B_j A_k X_j Z_k - B_k A_j X_k Z_j = 0.$$

This is a nonzero homogeneous bilinear relation over $\overline{\mathbb{Q}}$. Choose $\zeta_i^2 = -4t_i^3/9$; unequal cubes give $(\zeta_k/\zeta_j)^2 \neq 1$, so Lemma 9 applies. In canonical coordinates its variety is $\mathrm{SL}_2(\mathbb{C}) \times \mathrm{SL}_2(\mathbb{C})$. The selected row of the first matrix can be $(1, 0)$ and that of the second $(0, 1)$, giving $B_j A_k \neq 0$. The polynomial therefore cannot belong to the relation ideal, a contradiction.

All nonzero points have the same cube. The identities $Y_0(\omega t) = Y_0(t)$ and $Y_1(\omega t) = \omega Y_1(t)$ show that $y(\omega t)$ is another algebraic pencil, replacing (B, A) by $(B, \omega A)$. Differentiation contributes only the nonzero factor ω, so each zero type is invariant under cube rotation. Complex conjugation also preserves each family. A nonempty orbit thus has real cube and contains its unique real algebraic cube root r. The origin belongs to both families through the distinct rays $B = 0$ and $A = 0$.

Finally, function and derivative occurrences in one nonzero orbit can be rotated to the same point. Eliminating c/d then gives $M_{00}M_{11} - kM_{01}M_{10} = 0$ for some $k \in \overline{\mathbb{Q}}^\times$. One-block homogeneous transfer excludes this relation: at $M = I$ its value is 1. This proves the type assertion. □

The remaining possible nonzero orbit is not excluded by this bound; exceptionlessness of the classical Airy spectrum remains open (supplementary code).

O The growth constraints

O.1 Addresses and the two clocks

Space grows by discovery. Expansion activates stable content in the fixed, unbounded address field and gives it a place in the spatial atlas. An address is an eventually-zero radix-six word. Every branch gains one digit per global tick; padding with zeros leaves an established address fixed. Its last nonzero depth records discovery. Prefix ancestry is refinement history. Spatial adjacency instead consists of six primitive reciprocal neighbour ports per active $D{=}3$ address. These ports are not Euclidean sphere contacts, and their count alone chooses no dimension, metric, or atlas. That decoder remains open (Appendix O).

Activation is free and irreversible. The resolution rule admits dormant addresses without a reversible state injection from the preceding spatial slice. Each address receives one activation attempt per void tick, independently of the load: the substrate has $d{=}0$, no cargo and no dilation. A far observer uses standing-wave clocks. Dense regions then appear to discover slowly, voids rapidly; fewer resident ticks elapse per void tick near a mass.

One idle region through refinement. The void tick is the substrate's unthrottled clock. A fully idle connected region has one resolved interior occupancy state, $\binom{M}{0} = 1$, for any number M of addresses; subdivision adds no resident pattern to update. The region's size and boundary belong to its geometry, with any resolved interface already specified by the surrounding content. A passing read retains its own state and travel time. For a supplied homogeneous, transparent one-dimensional interval of fixed proper length L, the two boundary channels have delay transfer $T_L(s) = \exp(-sL/c)$, where s is the Laplace frequency. A partition into lengths L_j gives $\prod_j T_{L_j} = T_{\sum_j L_j}$ exactly. Finer description therefore preserves the completed read; physical growth changes it through the growing length. For a supplied evolving metric $d\ell = a(t)\,dX$, define $\eta(t) = \int^t c\,du/a(u)$. For a crossing whose target $\eta(t) + \Delta X$ lies in the range of η, the arrival is

$$F_{\Delta X}(t) = \eta^{-1}(\eta(t) + \Delta X)\,, \qquad F_{X_2} \circ F_{X_1} = F_{X_1+X_2}. \tag{56}$$

Where defined, this composition remains exact under repeated sixfold refinement. If a separator empties, the joined region includes its length, and the changed boundary response reaches each endpoint causally. In-flight reads keep their histories (supplementary code). The region construction compresses an idle interior on supplied geometry; the intrinsic admission and evolving-atlas law remain the separate growth problem below.

For a completed unobstructed read across N primitive hops, $L = N\ell_P$ and $T = N\ell_P/c$; inserting one additional hop adds one Planck diameter and one void tick (supplementary code).

On a supplied active path with retained directional ports, prescribed local unitary refractions can distribute one endpoint input over $N \geq 2$ alternate sites in $2(N-1)$ unit-hop ticks. A nonzero amplitude at the farthest site attains the causal time bound. Successive ticks alternate the occupied parity. The reverse refractions restore the full input in

the same number of ticks, including its internal state and external-reference correlations. Spatial transport preserves the expectation of an unchanged internal demand operator $I_{\text{spatial}} \otimes D_{\text{int}}$. For a uniform prepared profile, each site's contribution is $1/N$ of that unchanged demand. The prescribed refractions specify neither the binding lens nor the cost of their spatial control (supplementary code).

The two clocks supply the spatial metric. Per resident tick, the substrate has subdivided $(1+|\Phi|)$ times as much as in flat space, putting $(1+|\Phi|)$ more cells beneath a rod. The rod itself stretches by $(1+|\Phi|)$ with its slower clock. Light crosses one primitive neighbour relation per void tick and crosses both factors: the isotropic index is $(1+|\Phi|)^2 \approx 1 + 2|\Phi|$. The factor of two is one load, two clocks. Most ticks update waves and registers. Occasionally an address becomes active (Fig. 35).

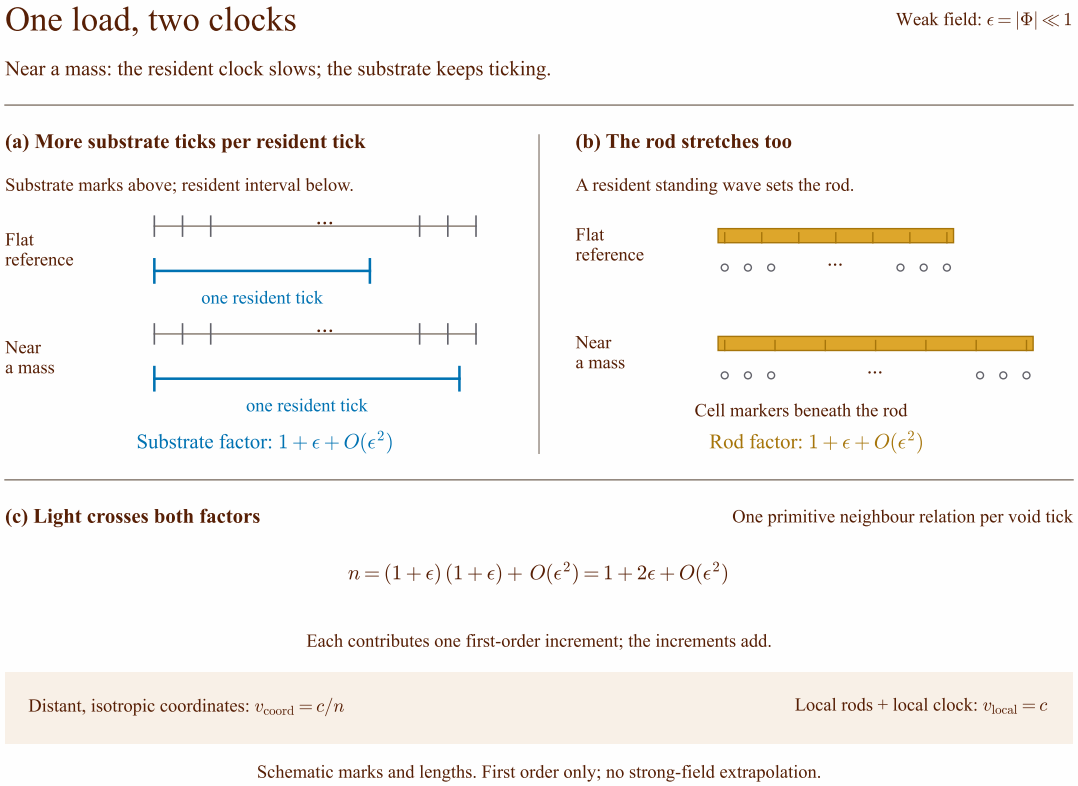


Figure 34: *One load, two clocks. In a static weak gravitational field, $\epsilon = |\Phi| \ll 1$ is the dimensionless potential depth. Relative to flat space, a slower resident standing-wave clock allows $1 + \epsilon + O(\epsilon^2)$ substrate (void) ticks per resident tick. The paper identifies this ratio with the additional substrate factor beneath a rod; the rod's own stretch supplies a second $1 + \epsilon + O(\epsilon^2)$ factor (Appendix O.1). Their product is the isotropic coordinate index $n = (1+\epsilon)^2 + O(\epsilon^2) = 1 + 2\epsilon + O(\epsilon^2)$, with $v_{\text{coord}} = c/n$. A nearby observer using local rods and proper time still measures c (Section 4.2). Tick marks, interval widths and rod/cell icons are schematic, with changes exaggerated for visibility; they specify neither an integer cell count nor an activation schedule. The factors retain only first-order information and do not determine a second-order coefficient or a strong-field metric. The continuum qualifications of Section 4.2 continue to apply.*

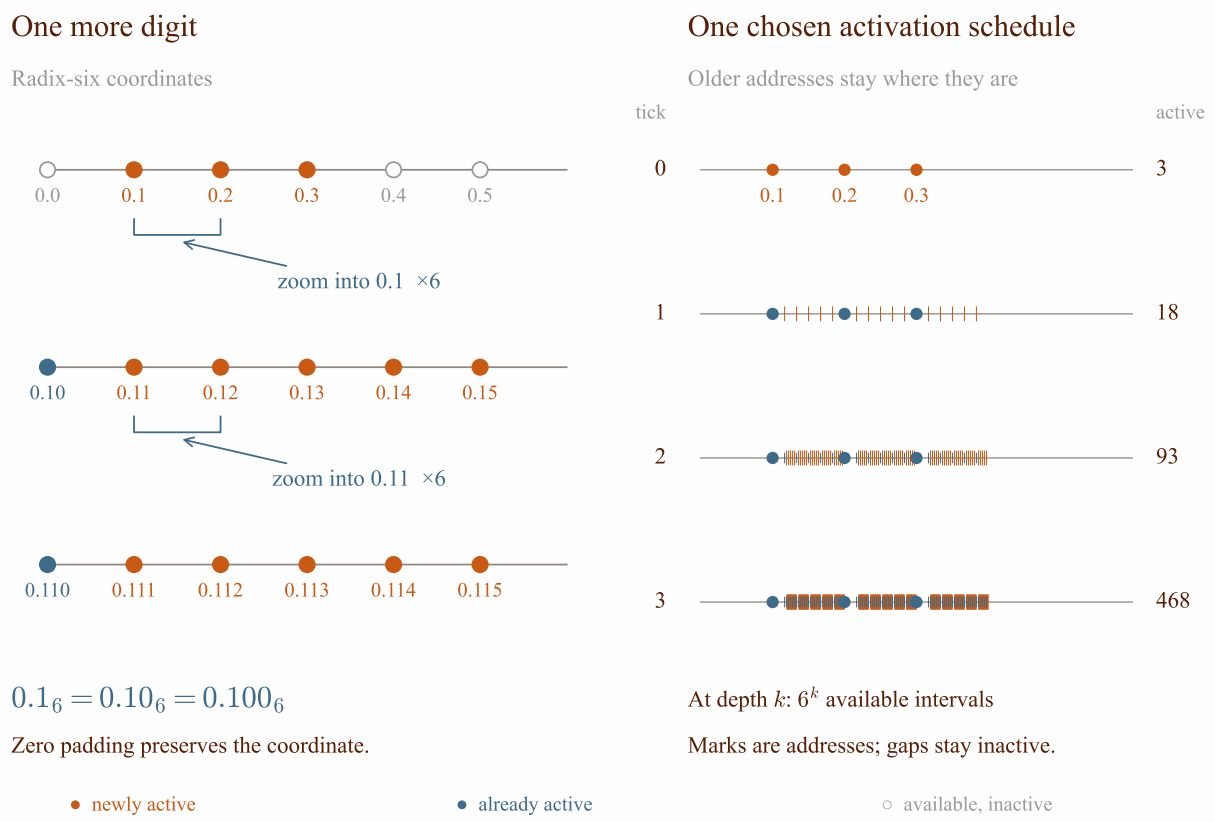


Figure 35: *Discovery drawn on a line. Left: successive sixfold zooms show radix-six refinement. Each branch gains a digit per global tick. Zero padding preserves an established endpoint:* $0.1_6 = 0.10_6 = 0.100_6$*; its canonical last nonzero depth records discovery. Right: one illustrative schedule starts at* $0.1_6, 0.2_6, 0.3_6$ *and admits the five nonzero extensions of each newly activated word at the next tick, retaining all earlier addresses. This gives* $3 \to 18 \to 93 \to 468$ *distinct active cells. Only the newest words branch, so gaps beside older addresses persist; the marks denote individual addresses. The newest words at tick t have depth* $k = t + 1$*. The* 6^k *available address cylinders at depth k are a separate count; the active population depends on the admission rule, and this schedule is not a physical growth law. Prefix parenthood records refinement, not spatial adjacency. In the D=3 atlas each active address has six primitive reciprocal neighbour ports; this line is only a coordinate illustration.*

O.2 Constraints on the intrinsic growth law

The event-read network grows by activating stable addresses. Four exact or finite constraints, together with the address and transport results below, narrow its law. The intrinsic exact-six atlas and continuum transport limit remain open.

A *thread* is a persisting computation line: a worldline chain of events in the causal order (Section 4.2). A *splice* joins two threads and consumes one unit of each allowance; k is the number of meetings a thread can afford. Growth adds one splice per step, with birth order among incomparable events gauge.

The thread/splice construction tests criticality and covariance on a bare relational state. It is a surrogate for these constraints, not the fixed transistor field's ontology. It proves the k=3 budget identity and gives a covariant equal-action history sum. At a multiply reachable endpoint, however, equal nonzero route amplitudes cannot coexist with

quotient-state row isometry (supplementary code). These results do not yet form one growth law.

On the complete content-bearing Planck-cell state, a synchronous reversible local update is a finite A2 existence witness with exact evaluation-order covariance and no splice scheduler. With a spinor action and a manifold limit on this same causal history, local Lorentz invariance, gauge structure, and locality would constrain interaction vertices through standard uniqueness theorems (Appendix E). The finite reversible witness does not establish those premises.

I. Classical covariance obstruction. A classical law assigns real weights to candidate splices and selects proportionally. Every candidate precursor has exactly two maximal elements, the participating threads' latest events. This is definitional and independent of the selected pair. The Rideout–Sorkin transitive-percolation weights [56] also depend on past cardinality and give unequal probabilities to birth orders of the same final splice web (supplementary code).

Exhaustive small-order checks, covering every tested weight functional and both eligibility conventions, give the same obstruction for every non-degenerate classical rule. Independent code and a Gröbner-basis check agree (supplementary code). Local weights do not remove it: fixed order-independent candidate properties are exhausted by the two maximal elements, while neighborhood contents depend on fill order. This dependence is directly checked at the tested radius and follows at any radius because neighborhood content records the filling order.

II. Covariant amplitude witness. Assign each path amplitude e^{iS}, with S the reference cost accumulated by counting links formed at each step. The sum telescopes to the finished order's link count, identical for every construction path (exact identity; supplementary code). Paths to one unlabeled order therefore interfere at equal phase, removing Constraint I's covariance obstruction exactly.

This gives one successful amplitude dynamics, not uniqueness among non-classical structures. Constraint I is the exhaustive classical exclusion. Amplitudes alone remain insufficient: without additional structure they weight unlabeled orders by squared extension counts and favor layered, non-manifoldlike orders. The remaining constraints are necessary. The complex amplitudes of Appendix I.1 thus suffice for covariance where the classical alternatives fail.

III. Criticality and budget neutrality. A splice recruits one new thread. With allowance k, the branching ratio is $\mu(k) = (k-1)/2$: subcritical below $k = 3$, critical at $k = 3$, supercritical above. Each splice consumes two allowance units and gives the newborn $k-1$, leaving $k-3$ per event. Population therefore decays below three, is budget-neutral at three, and inflates above. The two descriptions are one condition, $\mu(k) - 1 = (k-3)/2$. In the growing construction, the recruited thread is the newborn and receives exactly $k-1$, making this an exact mechanism rather than a mean-field assumption.

Both accountings give the same numerical behavior. At $k = 2$, fragmentation limits causal cones to a handful of events regardless of web size; at $k \geq 4$, the web inflates toward rope; only at $k = 3$ is the population eternal, without decay or explosion (supple-

mentary code). A3 selects this minimum viable allowance, the unique scale-free eternal point.

The selection does not fix dimension. On a manifoldlike control, the ordering-fraction estimator [64, 67] varies by one to two dimension units with window shape and reach into the genesis boundary. Size-matched sampling is stable to a few hundredths, so dimension-drift comparisons require matched protocols (supplementary code). The critical web's comparability fraction also depends on an underived construction choice.

Exact neutrality freezes the population: $k = 3$ cannot create a wide universe. Width must come from a free-growth era and freeze at the transition. An inflationary era is therefore a structural requirement of this allowance (supplementary code).

IV. Bare-state isometry. The surrogate quantum growth map admits exact isometric solutions on small crossing webs, satisfying A2 norm preservation at that dynamical tier (supplementary code). The result concerns amplitudes on a fixed relational state. Activation is free and irreversible; the photon is the completed endpoint read. Neither requires reversible injection between completed spatial slices, and norm preservation alone does not select address or transfer laws.

The read layer. An event's read set is the support of the winning amplitude: events on contributing paths of the growth walk that forms it. Reads are relations, as in Section 4.2, where a read is the route it completes. The construction has no tunable local weight, but its fixed horizon remains a named, underived ingredient (supplementary code). Read relations constrain causal order without changing genealogy. In the prototype, they invert the bare genealogical interval profile toward the manifold family across all tested seeds; the derived rule's own profile remains open.

Two diagnostics apply: ordering-fraction dimension and interval abundance. Both are calibrated empirically against known-dimensional Poisson sprinklings with identical code, rather than formulas. Before an embedding exists, they measure dimension and within-family comparisons; absolute isotropy requires the embedding whose existence is at issue.

Activation in the tested class. The possible-address field is fixed; expansion grows its active graph. Nine constructions across three families partition fixed-population webs into all width without depth, all depth without width, or neither (supplementary code). None grows space and time together. Thread birth does so in the surrogate, with exact budget neutrality at $k = 3$. The result establishes activation's necessity across these tested families, not thread birth's uniqueness or identity with the microscopic law.

A finite dimensional-lifecycle witness. A discrete implementation with separate address admission runs the cosmological history end to end. Subdivision grows size at fixed measured dimension; unrestricted attachment grows dimension with scale. Boundary-limited attachment can also satisfy dimension stability, so subdivision is an existence witness only.

The toy's fresh global axis and sheet-to-slab lift are anisotropic controls. The physical $D{=}2 \to 3$ transition instead has no inherited $D = 2$ normal: every address gains a complete isotropic local $D = 3$ chart, with graph-ball growth r^3 after gluing. The toy

demonstrates ballistic address growth with diffusive occupancy. Under causal smoothing it deposits a near-scale-invariant fossil of tilt -2.08 $[-2.10, -2.04]$, against pure freeze -1.87 (supplementary code). These remain toy-tier existence results, not an A1–A4 atlas derivation.

The address atlas. Radix-six digits encode stable identity and discovery time. Every branch gains a digit in lockstep; eventual zero-padding fixes established coordinates; the last nonzero depth marks discovery. Adjacency remains separate. One finite label set supports simple, connected, reciprocal, twin-free exact-six graphs with one-, two-, or three-dimensional ball growth. Degree six alone selects no opposite pairs, faces, cells, connected overlaps, isotropy, or dimension.

A unique square-corner and cube-horn rule would give every vertex an octahedral S^2 link and a connected seed a boundaryless cubical three-manifold. This exact sufficient construction yields a crystalline cubic atlas. An intrinsic confluent law with indefinitely many varying spherical links, exact degree six, and locality remains open.

Stable identity constrains gluing exactly. Two supplied congruent local $D = 3$ metric charts sharing four affinely independent addresses admit at most one $O(3)$ transition: equal address content fixes the correspondence. Rank-sufficient triple overlaps then satisfy the cocycle by uniqueness. This does not construct charts, choose overlaps, or prove congruence.

In the finite $N = 400$ cover, every adjacent radius-two overlap has affine rank three and residual below 1.31×10^{-15}. Two larger toroidal intersections split into $3 + 1$ deck branches, each fitting below 4.49×10^{-16}. Their ranks two and zero leave the full $O(3)$ map nonunique, so componentwise transition selection belongs to the open atlas law.

The equal-radius shell through six selected neighbors reconstructs the saved center to median 0.0008–0.0010 cell diameters and p99 0.047–0.052; the arithmetic centroid misses by 0.196–0.210. This inverse check establishes consistency with the supplied embedding, not its derivation from the six-set.

Finite packing evidence gives $d = 3.31 \pm 0.08$ across seeds, past-cone crossover from $\sim$2 to $\sim$4, a Laplacian spectrum within 3.3% of the cubic control, and winding ratio about 1.3, isotropic to 0.3% and scale-stable to 1% (supplementary code). These are mechanical-contact-graph measurements at mean isostatic degree, not proofs for the primitive exact-six graph. Selected finite exact-six graphs also support an exactly unitary scalar walk and positive nearly isotropic low-twist tensor. Their global finite selection does not define a stationary, ergodic thermodynamic law.

The open construction. A content-local law on one history must generate and indefinitely refine the reciprocal exact-six amorphous atlas; select A4 activation and the endpoint-read instrument, including dormant frontier ports; assign timestamp-respecting local transfer weights consistent under causal extension of completed reads; and prove nondegenerate isotropic homogenization with first-receipt delay, attenuation, and ballistic scaling. The fixed-graph spectral map transfers the final graph theorem to a scalar Klein–Gordon endpoint cone; a spinor/Clifford Dirac operator remains a separate field-theory problem. Until then, $k = 3$ diagnoses the realized ledger rather than selecting it microscopically, and manifoldlike order remains conditional.

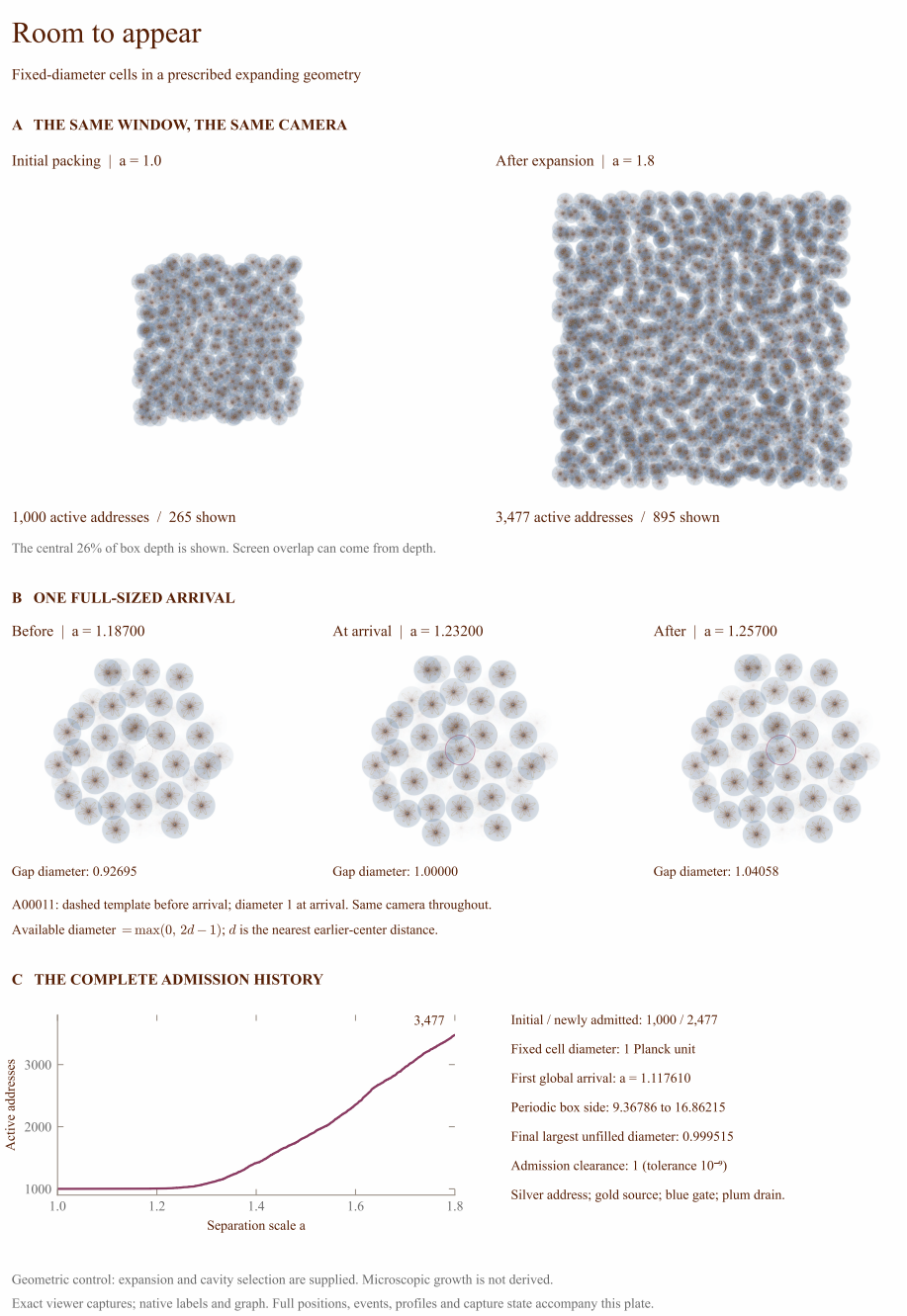


Figure 36: *A prescribed geometric growth control. Existing centres separate under a supplied expansion while cell diameters remain fixed; a global largest-empty-ball search admits a centre when the control can fit a complete unit-diameter cell. The displayed data describe this periodic packing control, not a derived local activation law or a physical tick calibration. Projected overlaps do not imply three-dimensional intersections. The data bundle accompanying the simulation contains every admitted centre and its admission scale.*

P Horizon boundaries, escape, and radiation

The following constructions develop the black-hole discussion in Section 4.8. Their supplied boundary, transport, and continuum conditions are stated separately; no one construction supplies the complete evaporation history.

P.1 Stationary horizon and exterior domain

The horizon condition $\mathcal{D}(\mathbf{x}) = -1/2$ makes the stationary exterior clock rate vanish. Under a physical frozen-boundary rule, its labels persist without internal updates; an exterior boundary map specifies their exchange with the surroundings. Three transmitted values match no-hair: D=3 carries mass and charge, D=2 conserves angular momentum, and D=1 transmits nothing, giving $2{+}1{+}0 = 3$. Freezing stops the register's internal update bill without erasing its state (Section 2.5); the stationary exterior is read through these aggregate quantities. The radius obeys $r_s/\ell_P = 2M/m_P$, one cell diameter per half Planck mass. Here r_s is the areal radius of the supplied Schwarzschild horizon, with area $4\pi r_s^2$. Across that family, changing M changes the radius selected by the horizon condition; it need not track a fixed set of addresses. Activation and the evolving spatial atlas retain the separate construction in Appendix O. For a supplied Schwarzschild exterior, $R_{abcd}R^{abcd} = 12r_s^2/r^6 \le 12/r_s^4$ on $r \ge r_s > 0$. This bounds curvature at fixed positive horizon size; finite total capacity alone does not establish an invariant interior continuation or its exclusion. A horizon-regular ingoing chart is related to the stationary chart by a unit-Jacobian time shear on their overlap, so restricting coordinate changes to unit Jacobian alone does not exclude continuation (supplementary code).

P.2 Stretching and return paths

Physical stretching can change the resident/exterior clock condition at a marked location while its address label stays fixed. For a supplied drain-only read with the same normalization, $\mathcal{D}_0 = -\sum_j w_j/r_j = -1/2$, take $w_j \ge 0$ and resolved separations $r_j \ge \ell_P$. Replace r_j by $a_j r_j$ while retaining the weights. Setting $p_j = 2w_j/r_j$ gives $\sum_j p_j = 1$ and

$$\alpha_{\mathrm{new}}^2 = 1 + 2\mathcal{D}_{\mathrm{new}} = 1 - \sum_j \frac{p_j}{a_j} = \sum_j p_j \left(1 - \frac{1}{a_j}\right).$$

Thus $a_j \ge 1$, with strict stretch at a positively weighted pair, gives a positive clock factor at the old marked location. Uniform stretch gives $\alpha_{\mathrm{new}}^2 = 1 - 1/a$. This compares specified causal reads of the kernel-weighted load; the physical growth history and outgoing flux require the common state and metric evolution (supplementary code).

Nonincreasing ratios q_j/a_j, where q_j scales each source weight, preserve any positive clock factor reached under this same complete drain read. For a supplied whole-return metric $ds^2 = -c^2\alpha(t)^2dt^2 + a(t)^2dX^2$, define $s(t) = \int^t \alpha(u)/a(u)\,du$. If $0 < \alpha_* \le \alpha \le 1$ holds along the whole route after a finite time, accumulated null travel distance $cs(t)$

permits infinitely many traversals of a fixed reference length exactly when $\int^{\infty} dt/a(t)$ diverges. Linear growth satisfies this condition; quadratic growth can give unbounded proper time but finite total traversal distance. These conditions on the full path accompany the boundary response and transported energy/work balance in a release calculation (supplementary code).

P.3 Correlation transfer and return

A frozen register can release its correlations without internal updates. For a pure register with a supplied tensor factorization into N binary gate factors, a coherent boundary-transfer isometry gives radiation entropy

$$S_R(k) \leq \min(k, N-k)\ln 2, \qquad S_R = -\operatorname{tr}(\rho_R \ln \rho_R), \tag{57}$$

after k factors have been transferred. For even N, fixed initial paired phases with separated-partner release saturate this bound, giving a deterministic Page-like turnover without Haar sampling [51]. Releasing the same partners consecutively instead gives $S_R/\ln 2 = k \bmod 2$, while every single factor remains maximally mixed. These are coherent outgoing encoding factors, not classical outcome strings; read-basis dephasing gives a different entropy. State-blind K=1 label prices alone do not select the initial correlations or release order (supplementary code). The construction supplies neither the Hawking frequency spectrum nor an emission rate. Complete transfer here leaves a residual register with a single available state. Identifying it with physical vacuum, and reaching it as the horizon shrinks, requires an evaporation law; A4's distinguishability threshold alone does not supply that law.

For supplied residual-energy sectors and exterior field encodings degenerate at each required energy gap, the transfer can satisfy $V^\dagger V = I$ and $H_{\text{out}} V = V H_{\text{in}}$ while preserving an arbitrary outside reference. A lossless return with constant single-pass escape probability $\mathcal{T}$, no incident field and orthogonal outgoing encounter modes retains mean energy fraction $(1-\mathcal{T})^J$ after J encounters. For a specified finite exterior round-trip time τ, its sampled energy decay rate is $\gamma = -\ln(1-\mathcal{T})/\tau$, for $0 < \mathcal{T} < 1$. The elastic point supplies the return phase; the travelling wave supplies the time. In a stationary Schwarzschild exterior the optical round-trip time diverges as a reflecting surface approaches the horizon. A decomposition of a given wave into retained and escaping amplitudes introduces no independent source reservoir; its energy balance retains any coupling terms. These constructions require one common physical carrier and evolving return domain before describing evaporation (supplementary code).

P.4 Escape from finite networks

A fixed finite scalar chain $\mathbf{M}\ddot{q} + \mathbf{D}\dot{q} + \mathbf{K}q = 0$, with positive diagonal $\mathbf{M}$, symmetric positive-definite tridiagonal $\mathbf{K}$, every adjacent coupling nonzero, and diagonal damping positive only at its output endpoint, releases every initial state's energy asymptotically through that output. Energy dissipativity excludes growing modes. Positive $\mathbf{K}$ excludes

a zero mode; any other undamped mode vanishes at the output, and the endpoint equations force it to vanish throughout the chain. This finite-chain result supplies neither a uniform continuum decay law nor a physical evaporation endpoint (supplementary code).

The same proof covers a finite branching network when its exits force every mode to vanish. For nonnegative diagonal damping, mark the vertices with positive damping in the graph defined by the nonzero off-diagonal entries of $\mathbf{K}$. Whenever a marked vertex has exactly one unmarked neighbour, mark that neighbour. If this reaches the whole graph, an undamped eigenmode's equation propagates its zero values along the same sequence [117]. For a finite tree with at least two vertices, damping all leaves except any one suffices; a singleton requires its own exit. Connectedness alone permits weighted branch amplitudes to cancel at an exit. The marking test is sufficient, not necessary for particular weights.

Write $x = (q, \dot{q})$, $H = \operatorname{diag}(\mathbf{K}, \mathbf{M})$, $Q = \operatorname{diag}(0, \mathbf{D})$, and $\dot{x} = Ax$. Then $A^T H + HA = -2Q$ and every fixed return interval $T > 0$ has a positive-definite outgoing-energy matrix

$$G(T) = 2\int_0^T e^{A^T t} Q e^{At}\, dt = H - e^{A^T T} H e^{AT} > 0.$$

Thus $\epsilon = \lambda_{\min}(H^{-1/2} G(T) H^{-1/2}) > 0$ bounds energy loss for every input. A compact coefficient family on the same finite graph has a uniform positive ϵ when masses, stiffness spectrum, exit damping and nonzero edges stay nondegenerate and bounded. This includes an upper damping bound: unlimited damping can slow escape arbitrarily (supplementary code).

Bounded changes during a return can preserve this margin. Use the initial energy metric H_a throughout the comparison, and let $B_0 = H_a^{1/2} A_0 H_a^{-1/2}$ be a dissipative reference with return gap ϵ. If the actual normalized generator differs from B_0 by operator norm at most δ throughout the interval and $H_{a+T} \le (1+\rho) H_a$, $\rho \ge 0$, variation of constants gives

$$E_{a+T} \le \zeta E_a, \qquad \zeta = (1+\rho)\left[\sqrt{1-\epsilon} + e^{\delta T} - 1\right]^2.$$

Uniform $\zeta < 1$ on successive intervals gives geometric decay; a uniform intermediate bound $H_{a+t} \le C_H H_a$ also controls energy between them. For the specified equation $\mathbf{M}(t)\ddot{q} + \mathbf{D}(t)\dot{q} + \mathbf{K}(t)q = 0$, the exact work term is $(q^T \dot{\mathbf{K}} q + \dot{q}^T \dot{\mathbf{M}} \dot{q})/2$. It accompanies the outgoing power $\dot{q}^T \mathbf{D} \dot{q}$. Arbitrary switching among individually decaying networks need not satisfy this bound. These results concern a supplied finite scalar instrument; its physical network, boundary and evolving geometry remain to be selected (supplementary code).

P.5 Changing transport and energy balance

For changing unitary retained/outgoing updates on a fixed finite retained space, let T_t and R_t be the retained and outgoing blocks, with fresh unoccupied outgoing input and earlier output retained in orthogonal modes that never return. Write

$\Phi_{a,0} = I$ and $\Phi_{a,j} = T_{a+j-1}\cdots T_a$. The complete-wave balance over L updates is $G_{a,L} = \sum_{j=0}^{L-1} \Phi_{a,j}^\dagger R_{a+j}^\dagger R_{a+j}\Phi_{a,j} = I - \Phi_{a,L}^\dagger \Phi_{a,L}$. If each successive window satisfies $G_{a,L} \geq \epsilon I$ for fixed $0 < \epsilon < 1$, the remaining norm squared is at most $(1-\epsilon)^{\lfloor N/L \rfloor}$ times its initial value after N updates. The preceding transport matters: instantaneous openings can cover every direction while internal turning preserves a moving dark mode. For a full-block scalar opening $T_t^\dagger T_t = (1-\eta_t)I$, $0 \leq \eta_t < 1$, complete escape is equivalent to $\sum_t \eta_t = \infty$; $\eta_t = 1/(t+2)^2$ instead retains exactly half the initial norm squared. These are norm and information statements; changing mode energies require the common field/resident energy balance. The physical geometry must supply the response history, encounter times and output boundary (supplementary code).

The changing demand has an exact transported balance when that amplitude decomposition also separates the physical retained and radiated energies at each read. Let $H_t > 0$ and $K_t \geq 0$ be their supplied energy operators, with earlier output energies fixed in the same account after release, and put $W_t = T_t^\dagger H_{t+1} T_t + R_t^\dagger K_t R_t - H_t$. This Hermitian term is the work or exchange that a common geometry/resident evolution must supply. Define

$$G_{a,L}^E = \sum_{j=0}^{L-1} \Phi_{a,j}^\dagger R_{a+j}^\dagger K_{a+j} R_{a+j} \Phi_{a,j}, \qquad G_{a,L}^W = \sum_{j=0}^{L-1} \Phi_{a,j}^\dagger W_{a+j} \Phi_{a,j},$$

$$\Phi_{a,L}^\dagger H_{a+L} \Phi_{a,L} + G_{a,L}^E = H_a + G_{a,L}^W. \tag{58}$$

The identity follows by telescoping and holds for every input, including one entangled with an outside reference. If successive windows satisfy $G_{a,L}^E \geq \epsilon H_a$, $G_{a,L}^W \leq \delta H_a$ and $0 \leq q = 1 - \epsilon + \delta < 1$, retained energy falls by at least the factor q per window. Decay at every tick additionally follows from a uniform bound $\Phi_{a,j}^\dagger H_{a+j}\Phi_{a,j} \leq C H_a$ inside each window; $W_t \leq 0$ at every tick gives $C = 1$. Energy depletion includes signed work and is not automatically all radiation.

A stronger sufficient bridge to the norm bound is $H_{t+1}T_t = T_t H_t$ and $K_t R_t = R_t H_t$ on this same separated amplitude carrier. These imply $W_t = 0$ and transfer $\Phi_{0,N}^\dagger \Phi_{0,N} \leq p_N I$ directly to $\Phi_{0,N}^\dagger H_N \Phi_{0,N} \leq p_N H_0$. Mean-energy balance alone does not imply this spectral intertwining. If retained and outgoing fields remain coupled at the read, their cross energy must be included in the complete balance, as must any later work on released radiation. A tensor emission branch may still contain a massive resident; its total branch energy cannot all be counted as radiation. Thus the physical energy decomposition, response history and geometry remain essential inputs to an evaporation model (supplementary code).

P.6 Thermal flux and conditional mass loss

Along a supplied quasistatic Schwarzschild family, $d(Mc^2) = \Theta_H \, dS_{\text{geom}}$, where $S_{\text{geom}} = A/(4\ell_P^2)$ is dimensionless thermodynamic entropy and $\Theta_H = k_B T_H = \hbar c/(4\pi r_s)$. This entropy need not equal S_R. On each stationary member, supply massless bosonic channels with thermal occupation before transmission and the one-direction unit-flux mea-

sure $d\omega/(2\pi)$. If the transmission functions $\Gamma_j(x)$ are unchanged across the family, with $x = \omega r_s/c$ measured by the exterior clock and a convergent channel sum, their transmitted thermal luminosity is

$$P_\infty = \frac{\hbar c^2}{r_s^2}\, C_\Gamma, \qquad C_\Gamma = \frac{1}{2\pi} \sum_j \int_0^\infty \frac{x\, \Gamma_j(x)}{e^{4\pi x} - 1}\, dx. \tag{59}$$

The index j counts all channel degeneracies and polarizations. This fixes the inverse-square size dependence under the stated scale covariance. Finite boundary-exchange controls preserve the same equilibrium temperature while varying transfer probability per tick (supplementary code). Net mass loss also requires the incoming-flux balance. An evolving field and geometry must share one energy balance, including work done on the wave by changes of its operator or any moving reflecting boundary (supplementary code). The shrinking endpoint needs that common evolution.

For a supplied reciprocal exchange between resident energy sectors, let d_i, d_f be their degeneracies and B the transition block for one specified exterior field mode carrying the energy difference. Uniform density operators in the two sectors give $P_{\text{em}} = \operatorname{tr}(B^\dagger B)/d_i$ and $P_{\text{abs}} = \operatorname{tr}(BB^\dagger)/d_f$. Nonzero exchange therefore fixes $P_{\text{em}}/P_{\text{abs}} = d_f/d_i$. If both admitted mass sectors additionally satisfy $\ln d(M) = S_{\text{geom}}(M) = 4\pi M^2$ in Planck units, a transition $M \to M - \epsilon$, $0 < \epsilon \leq M$, gives $\exp[-8\pi M\epsilon + 4\pi\epsilon^2]$. Its small fractional-loss limit recovers the Hawking detailed-balance exponent. For an isolated source initially at rest, equating this energy gap with its rest-mass loss also requires zero net emitted momentum. The comparison reverses a transition between different masses; it does not fix an absolute rate or an outgoing correlation history. Pre-existing entanglement can supply uniform sector marginals without internal updates, but equal label prices alone do not prepare them (supplementary code).

For the supplied minimally coupled massless scalar equation on that exterior, separation gives the radial potential $V_\ell = (1 - r_s/r)[\ell(\ell+1)/r^2 + r_s/r^3]$, in $u_{r_* r_*} + [\omega^2/c^2 - V_\ell]u = 0$. An ingoing horizon condition and asymptotic incident-flux normalization determine its transmission. Numerical integration gives $C_{\text{scalar}} \simeq 2.9755 \times 10^{-4}$, including the $(2\ell+1)$ degeneracies (supplementary code). This coefficient belongs to the absorbing scalar problem; it does not specify photon channels or the return law of an elastic frozen boundary. The physical boundary coupling, channel content and formation state must select C_Γ and its evolution.

In the leading source-free Maxwell continuum description on the same supplied exterior, the two radiative parity sectors have $V_\ell^{(\gamma)} = (1 - r_s/r)\ell(\ell+1)/r^2$, $\ell \geq 1$ [116]. With the same absorbing-horizon condition and thermal occupation, their $2(2\ell+1)$ channel sum gives $C_\gamma \simeq 1.3455 \times 10^{-4}$ (supplementary code). These are external photon polarizations, not internal d=0 cargo. This conditional continuum coefficient does not select the elastic boundary's exchange or formation law. With negligible incoming flux, using this photon-only luminosity along an adiabatic Schwarzschild family gives $r_s(t)^3 = r_s(t_0)^3 - 6C_\gamma \ell_P^2 c(t - t_0)$. Its adiabaticity parameter is $|\dot\kappa|/\kappa^2 = C_\gamma (m_P/M)^2$, where $\kappa = c/(2r_s)$. This is a conditional macroscopic evolution for $M \gg m_P$, not a continuation to its formal zero-size endpoint (supplementary code).

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